GRADUATE TRACER SYSTEM WITH PREDICTIVE EMPLOYABILITY TREND ANALYSIS

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CHAPTER I

Introduction

Graduate tracer studies are one of the main tools universities use to check whether their programs produce graduates who are ready for work. These studies gather data on where graduates work, how long it takes them to find jobs, and whether their training matched what employers needed, which helps institutions decide what to improve in their curriculum (Dzomeku et al., 2024). Many universities in the Philippines still collect this information through paper surveys, spreadsheets, and social media messages, which produces incomplete or outdated records. Common problems include low response rates, poor data quality, and the lack of a single platform for monitoring graduate employment (Cieciora et al., 2025).

Web-based systems offer a practical way to solve these problems. Automated platforms reduce the manual workload of tracking graduates, keep records accurate, and make data easier to update (Babu et al., 2022). When analytics tools are added, institutions can move beyond storing data and start identifying employment trends that support curriculum decisions (Nabablit & Dajao, 2023). Despite this progress, no single solution has been widely adopted across higher education institutions, which leaves a clear need for institution-specific systems.

At Carlos Hilado Memorial State University (CHMSU) – Talisay Campus, the Bachelor of Science in Information Systems (BSIS) program has no formal platform for tracking its graduates. Without structured tracking, institutions struggle to see the gap between what students learned and what employers need (Albina & Sumagaysay, 2020). To address this, the study developed a Graduate Tracer System with Predictive

Employability Trend Analysis for BSIS graduates of CHMSU – Talisay Campus. The system includes face verification with a liveness check, geographic employment mapping, employment verification by employers through one-time links, and a predictive module that forecasts employment trends for future batches.

1.1 Background of the Study

The BSIS program at Carlos Hilado Memorial State University – Talisay Campus prepares students for careers in software development, systems analysis, and information technology management. The program is expected to regularly check whether its graduates find relevant work and whether the curriculum meets industry demands. Graduate tracer studies are the standard way of doing this, providing data on employment status, time-to-hire, job alignment, and graduate skills (Lagatic et al., 2024).

At present, the BSIS department has no dedicated system for gathering this data. Employment information is collected through paper surveys at graduate events, personal follow-ups by faculty, and messages over Facebook Messenger and group chats. The data is typed into spreadsheets without a consistent format, which makes it nearly impossible to compare batches or spot trends. Records from older batches are often incomplete or unavailable.

To address these gaps, the study developed a Graduate Tracer System with Predictive Employability Trend Analysis covering BSIS graduates from batches 2020 to 2025, and accepting graduating students ahead of their graduation month for future tracing. Graduates register, pass a face and liveness check, give their data privacy consent, and keep their employment and skills records current. Employers confirm a graduate's employment through a one-time link without creating an account.

Administrators get batch-based analytics, a consent-based geographic employment map, six structured report types, and a predictive module that forecasts employment rates, time-to-hire, and skills in demand for the next one to two batches. A centralized platform has been shown to improve graduate data completeness and reduce the burden of manual tracer studies (Babaran Jr., 2025).

1.2 Objectives of the Study

The general objective of this study is to develop a Web-Based Graduate Tracer System with Predictive Employability Trend Analysis for the Bachelor of Science in Information Systems (BSIS) program of Carlos Hilado Memorial State University - Talisay Campus that will centralize, manage, and analyze graduate employment data to support tracer activities and curriculum improvement.

Specifically, this study aimed to:

1. Design and develop a Graduate Tracer System with the following features:
    1.1. face recognition verification;
    1.2. predictive employability trend analysis;
    1.3. geographic mapping of graduate employment distribution;
    1.4. employer feedback portal;
    1.5. automatic 2 year re-tracking gate;
2. Test the functionality of the features mentioned above.
3. Evaluate the usability and reliability of the system..
4. Develop a User Manual for the developed system.

1.3 Scope and Limitations of the Study

This study covers the development of a Web-Based Graduate Tracer System with Predictive Employability Trend Analysis for BSIS graduates of Carlos Hilado Memorial State University – Talisay Campus, batches 2020 to 2025. Graduates and administrators (the Program Chairperson and BSIS faculty) hold accounts; employers take part without one. Development followed the Agile model, and the system was evaluated through a structured questionnaire measuring functionality, usability, and reliability.

Graduates register by matching their name and batch year against an admin-uploaded master list, pass a face capture with a liveness check and GPS stamp, and accept the terms and data privacy notice under Republic Act 10173, with a separate opt-in for the geomap. Data validation refuses invalid entries, and an administrator approves or rejects each account. Graduates manage their personal data, a PSGC-based home address, and their employment and skills records. They can send employers a one-time verification link to confirm the job and submit an optional confidential evaluation. Suspicious answers are flagged for review. Administrators get a consent-based geomap, a predictive analytics dashboard, and six exportable reports. The predictive module reports each batch's observed employment rate, time-to-first-job bands, and skills with 95 percent confidence intervals. It also estimates the chance of employment within twelve months of graduation using a logistic regression on four facts known at graduation: Latin honors, paid work besides the OJT, a portfolio, and a scholarship. The model is shown only after it passes eight acceptance rules fixed before training. The system is deployed as a website with a mobile app wrapper.

Data completeness depends on graduates updating their records. The model requires at least 150 graduates with a known outcome, 40 of them in the smaller group; until then, the dashboard runs on clearly labeled simulated graduates, so its figures are a demonstration, not a finding. The system reports factors associated with employment, not causes, and does not forecast an individual's job title or hiring date or use external labor market data. Because employers have no accounts, the system cannot fully prove who answered a verification link and can only flag suspicious answers. The study excludes event management, alumni networking, and fundraising.

1.4 Significance of the Study

The system addresses the lack of a centralized employment tracking platform for BSIS graduates at CHMSU - Talisay Campus. The results of the study would benefit the following:

Carlos Hilado Memorial State University. The university gains a structured, data-driven way to monitor the employment outcomes of its BSIS graduates, analyze employability trends across batches, and support evidence-based curriculum improvement.

Graduate Office. The office gains a centralized platform for managing graduate records and employment outcomes across batches, with validated data that stays accurate and up to date.

BSIS Dean. The Dean gains employment statistics and trend analyses that show how well the program prepares graduates for the workforce and where the curriculum can improve.

BSIS Chairperson. The Chairperson gains batch-based employability data, curriculum alignment and skills reports, and predictive forecasts that guide curriculum review.

Employers. Employers can confirm a graduate's employment and give confidential feedback on eleven performance items and two open-ended questions through a single link, without creating or maintaining an account.

BSIS Graduates. Graduates get one platform to update their employment records and skills, control whether they appear on the geomap, and document their career progress. The two-year re-tracking reminder keeps their data current.

Future Researchers. The system design, liveness-checked face verification, link-based employer verification, and predictive analytics approach can serve as a reference for similar tracer systems in other programs.

1.5 Definition of Terms

The following key terms were defined operationally based on their use in this study.

Acceptance Gate. The eight checks a trained model must pass before it may be shown on the dashboard: no input measured after the outcome, enough graduates, a Brier score better than the no-model baseline, an area under the curve of at least 0.65, agreement on the two most recent batches the model never saw, a calibration slope between 0.7 and 1.3, stable reported factors, and comparable accuracy for female and male graduates. The rules are fixed before training, and every run is recorded whether it passes or fails.

Administrative Dashboard. A feature for administrators that shows the number and percentage of employed, self-employed, and unemployed graduates through KPI cards, employment rate and time-to-hire charts, and links to the pending account queue, batch upload, geomap, and analytics, filterable by batch, industry, region, and employment status.

Bachelor of Science in Information Systems (BSIS). A four-year program covering information technology, systems analysis and design, and software development. In this study it refers to the program at CHMSU - Talisay Campus whose graduates from batches 2020 to 2025 are the subjects of the system.

Biometric Face Verification. A feature that captures one frontal face photo with a GPS stamp during registration and compares a live frontal scan against it on every sign-in using the InsightFace recognition model. Blurry frames are rejected.

Data Privacy Consent. The required acceptance of the system's terms and data privacy notice under Republic Act 10173, recorded with a timestamp at registration, together with a separate optional consent to appear on the geomap.

Data Validation. The checks applied at every entry point that refuse invalid data before it is saved, such as names containing digits, impossible dates, nonsensical job titles, and master list files with any invalid row.

Effectiveness. The accuracy and completeness with which users achieve their goals using the system. It is one of the measures of usability.

Efficiency. The time and effort users spend in relation to the accuracy and completeness of the goals they achieve. It is one of the measures of usability.

Employer Verification Link. A one-time link, valid for seven days by default, that a graduate sends to their employer. The employer confirms or denies the employment, gives their name, work email, and position, and may complete a confidential evaluation form. Answers that look suspicious are flagged for administrator review.

Functionality Testing. An evaluation method that determines whether the system performs its intended features correctly.

Geographic Mapping. A feature powered by Leaflet that plots graduates who gave geomap consent on an interactive map, filterable by batch year, region, and employment status.

Graduate Profile and Employment Record Management. A feature that lets verified graduates update their employment status, sector, job title chosen from the admin list or typed when not listed, time-to-hire, work address, BSIS alignment, and skills. Survey answers are copied into structured rows for analytics.

Graduate Tracer System. A web-based information system that collects, monitors, and analyzes the employment outcomes of BSIS graduates at CHMSU Talisay Campus.

Liveness Detection. A check performed with the face scan that asks the graduate to blink or turn their head in a randomly chosen direction, which prevents sign-in with a printed photo or a recorded video.

Master List. The roster of graduates uploaded by the administrator per batch year. Registrations are matched against it by name and batch year, and the result is shown to the administrator who approves or rejects the account.

Philippine Standard Geographic Code (PSGC). The official list of regions, provinces, cities, municipalities, and barangays published by the Philippine Statistics Authority, used for all address selections in the system.

Predictive Employability Trend Analysis. A feature that reports the employment rate, time-to-first-job bands, and skills of each graduating batch with ninety-five percent confidence intervals, estimates the chance of being employed within twelve months of graduation using a logistic regression fitted on four facts known at graduation, and gives a likely range for the employment rate of the next batch. It is released only after passing the acceptance gate.

Re-Tracking Gate. A mechanism triggered when a graduate's employment record is more than two years old. The system sends an email reminder and redirects the graduate to the employment update form until the survey is completed.

Report Generation. A feature that produces six report types: Batch Summary, Employment Outcomes, Skills Inventory, Further Studies, Data Quality, and Predictive Employability Trend, filterable by batch and exportable in PDF, XLSX, or CSV depending on the report.

Satisfaction. The degree to which users are comfortable and content with the system. It is one of the measures of usability.

Skills Management. A feature that lets graduates record technical and soft skills from a categorized reference list. Skill data feeds the analytics and the skill demand forecast.

Target Leakage. The use of an answer that only becomes available after the outcome being predicted, such as a current job title, which makes a model appear

accurate in testing while being useless on a graduate who has not yet been hired. The earlier model of this study was replaced for this reason.

Usability Testing. An evaluation method that measures how well intended users achieve their goals with the system in terms of effectiveness, efficiency, and satisfaction.

User Manual. A document that guides graduates, employers, and administrators in using the features of the system.

Web-Based System. A software application accessed through a web browser over the internet. In this study it is the platform through which graduates, administrators, and employers use the Graduate Tracer System.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of related literature, studies, and interrelated systems that serve as the foundation for developing the Graduate Tracer System with Predictive Employability Trend Analysis. The discussion is organized into foreign studies, local studies, foreign systems, and local systems, followed by a synthesis that highlights the connections between the reviewed works and the proposed system. The review covers both the tracer study literature that establishes what such a system should collect and report, and the prediction methodology literature that governs how the predictive component of the system may be built, validated, and presented.

2.1 Foreign Studies

Observation of Imbalance Tracer Study Data for Graduates Employability Prediction in Indonesia

Examined the problem of class imbalance in tracer study datasets used for predicting the employability of university graduates in Indonesia. The study found that imbalanced data distributions significantly affect the performance of machine learning classifiers, resulting in biased predictions that favor the majority class. By applying oversampling techniques such as SMOTE alongside several classification algorithms, the researchers demonstrated that balancing the dataset improved predictive accuracy and reliability. The findings are directly relevant to the proposed Graduate Tracer System, which similarly relies on tracer study data that may be unevenly distributed across graduate cohorts. The study underscores the importance of preprocessing steps and algorithm selection when building an employability prediction model embedded within a tracer system (Abdulloh et al., 2022).

Soft Skills and Graduate Employability: Evidence from Malaysian Tracer Study

Analyzed the relationship between soft skills and the employability of graduates using tracer study data collected in Malaysia. The study identified communication, teamwork, and problem-solving as the soft skill dimensions most strongly associated with employment outcomes. Using regression-based analysis, the researchers confirmed that soft skill competencies are significant predictors of graduate employability beyond academic qualifications alone. This finding supports the inclusion of non-academic indicators in the proposed system's data collection module, particularly for graduate profile fields that capture perceived competency levels. The study also provides methodological reference for designing survey instruments that assess both hard and soft skill dimensions within a tracer study framework (Basir et al., 2022).

A Tracer Study of Psychosocial Counselling Graduates Working in Different Healthcare Facilities across Malawi

Conducted a tracer study among graduates of a psychosocial counseling program in Malawi to assess their employment outcomes and career distribution across healthcare settings. Using a descriptive cross-sectional design, the study tracked where graduates were employed, the roles they occupied, and the extent to which their positions aligned with their academic training. Findings revealed that a substantial proportion of graduates were working in roles that matched their field of study, although gaps in career guidance and program-to-market alignment were noted. The methodological approach and data structure of this study served as a reference for the proposed system's graduate tracking and career outcome documentation modules,

which aim to monitor employment placement and job alignment across graduate cohorts (Chima et al., 2023).

Graduate Career Tracking Systems of Higher Education Institutions: A Systematic Literature Review

Conducted a systematic literature review of graduate career tracking systems deployed across higher education institutions worldwide. The review synthesized findings from multiple studies to identify the most commonly implemented features, data collection strategies, and technological approaches used in institutional tracer systems. Key findings indicated that effective tracking systems integrate online survey platforms, graduate databases, employment analytics dashboards, and longitudinal follow-up mechanisms. The review also highlighted persistent challenges such as low graduate response rates, data quality issues, and limited system interoperability. The proposed Graduate Tracer System draws from this review to benchmark its feature set and to anticipate common implementation barriers in the implementation of the proposed system (Cieciora et al., 2025).

Information Technology Graduates Employability Prediction Model in a Low-Income Country Using Tree-Based Machine Learning Classifiers

Developed an employability prediction model for information technology graduates in a low-income country context, employing tree-based machine learning classifiers including Decision Tree, Random Forest, and Gradient Boosting. The study demonstrated that Random Forest consistently outperformed other classifiers in terms of accuracy, precision, recall, and F1-score, making it particularly suitable for datasets with mixed feature types common in tracer study datasets. The contextual similarity

between the study setting and the Philippine higher education environment makes its findings directly applicable to the proposed system. The study informed the initial choice of tree-based ensemble methods for the predictive employability trend analysis component of the Graduate Tracer System, and its reported accuracies later served as the point of comparison against which the leakage-free model of this study was assessed (Dake, 2025).

Tracer Study to Assess the Employability of Graduates and Quality of a Nursing Program: A Descriptive Cross-Sectional Survey

Assessed the employability outcomes of nursing graduates and the quality of their training program through a descriptive cross-sectional tracer study. The study collected data on employment status, job alignment, and graduates' self-assessed preparedness, then linked these outcomes to program-level attributes such as curriculum design and clinical exposure. Findings confirmed that employment outcomes can serve as indirect indicators of program quality, suggesting that tracer data should feed back into institutional program review processes. This dual function of tracer studies as both a graduate monitoring tool and a program quality assessment mechanism informed the analytics dashboard design of the proposed system, which aims to provide actionable insights for institutional program improvement (Dzomeku et al., 2024).

Employability Prediction of Information Technology Graduates Using Machine Learning Algorithms

Compared multiple machine learning algorithms for predicting the employability of information technology graduates, including Naive Bayes, k-Nearest Neighbor, Support

Vector Machine, and Random Forest. The study evaluated classifier performance using accuracy, precision, and recall metrics derived from a real-world dataset of IT graduate employment records. Random Forest and SVM produced the highest classification accuracy, with feature importance analysis identifying GPA, specialization track, and internship experience as the most influential predictors. These results directly informed the feature engineering and algorithm selection process for the predictive analytics module of the Graduate Tracer System, and the observation that the strongest predictors are attributes fixed at graduation guided the final selection of inputs used by this study (ElSharkawy et al., 2022).

Predicting University Graduates' Employability Using Support Vector Machine Classification

Applied Support Vector Machine classification to predict the employability of university graduates using a dataset of student academic and demographic records. The study optimized SVM kernel parameters to maximize classification accuracy and validated the model using k-fold cross-validation. Results demonstrated that SVM achieved competitive predictive performance for binary employability classification, distinguishing employed graduates from those who remained unemployed after a defined period following graduation. The study supports the inclusion of SVM as one of the candidate models evaluated during the development of the predictive trend analysis module of the proposed system, providing a basis for model comparison and selection (Haikal and Palupi, 2024).

Classification Techniques Using Machine Learning for Graduate Student Employability Predictions

Benchmarked a range of machine learning classification techniques for predicting graduate student employability, including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The study applied feature importance analysis to identify the variables most strongly associated with employability outcomes and used accuracy, AUC-ROC, and F1-score as evaluation metrics. Random Forest and Gradient Boosting consistently produced the highest performance across all metrics, while feature importance rankings highlighted academic achievement, program of study, and co-curricular participation as key predictors. The multi-model evaluation approach adopted in this study was adapted for the predictive analytics component of the Graduate Tracer System to ensure that the most appropriate algorithm is selected through empirical comparison (Haque et al., 2024).

Predicting Employability of Congolese Information Technology Graduates Using Contextual Factors: Towards Sustainable Employability

Investigated the role of contextual factors, including geographic location, economic conditions, and institutional characteristics, in predicting the employability of information technology graduates in the Democratic Republic of Congo. Using machine learning models trained on locally collected graduate data, the study demonstrated that contextual variables significantly improved prediction accuracy when combined with academic indicators. This perspective informed the design of the Graduate Tracer System's data collection instruments, ensuring that graduate profiles capture contextual information such as geographic location, employment sector, and industry of placement alongside academic performance indicators (Mpia et al., 2022).

Predicting Graduate Employability Using Hybrid AHP-TOPSIS and Machine Learning: A Moroccan Case Study

Developed a hybrid employability prediction framework that combined the Analytic Hierarchy Process and the Technique for Order of Preference by Similarity to Ideal Solution with machine learning classifiers. The AHP-TOPSIS component assigned weighted scores to multiple employability criteria, which were then used as engineered features in machine learning models. This multi-criteria decision-making approach resulted in improved classification performance compared to models relying on raw feature values alone. The study inspired the consideration of weighted scoring mechanisms within the Graduate Tracer System's predictive analytics module, particularly for ranking graduate employability likelihood across cohorts and academic programs (Nouib et al., 2025).

Leakage in Data Mining: Formulation, Detection, and Avoidance

Formalized target leakage as the presence, among a model's inputs, of information that would not legitimately be available at the moment a prediction has to be made, and showed that leakage is the most common cause of models that score highly during development yet fail in deployment. The authors documented cases in which a single leaking column produced near-perfect accuracy while carrying no predictive value whatsoever, and proposed detecting leakage by examining when each variable becomes observable relative to the outcome being predicted. This study is the direct basis for the redesign of the predictive module of the Graduate Tracer System. The earlier model of this study read current job sector, job source, and location type, which only a graduate who has already been hired can answer, and it is precisely this that

produced its inflated headline accuracy. Applying the detection criterion described here, those fifteen columns were removed and a rule that refuses any input measured after the outcome was written into the acceptance gate of the system (Kaufman et al., 2012).

Machine Learning Algorithm Validation with a Limited Sample Size

Examined how machine learning performance estimates behave when the number of observations is small, a condition typical of single-program tracer studies. Through simulation on data containing no real signal, the authors showed that ordinary k-fold cross-validation produces strongly optimistic accuracy estimates at small sample sizes, and that the bias grows as the sample shrinks. They recommended repeated nested or repeated stratified cross-validation, together with reporting the spread of the fold results rather than a single mean, so that the instability of a small-sample estimate is visible to the reader. The proposed Graduate Tracer System applies this recommendation directly: every candidate model is scored under ten-times-repeated stratified five-fold cross-validation, and the acceptance gate examines both the mean area under the curve and its fifth percentile, so that a model whose apparent accuracy rests on a favourable split cannot be released (Vabalas et al., 2019).

Calculating the Sample Size Required for Developing a Clinical Prediction Model

Derived the minimum number of observations and outcome events needed before a prediction model can be developed without overfitting, and showed that the widely used rule of ten events per candidate predictor is a lower bound rather than a target. The authors demonstrated that models fitted below this threshold produce coefficients that do not reproduce on new data and prediction intervals that understate their own uncertainty. Although developed for clinical prediction, the requirement is a

property of the estimation problem rather than of the subject matter, and it applies equally to a tracer study of one academic program. The proposed Graduate Tracer System encodes the rule as a hard precondition: a model may not be shown at all until at least one hundred fifty graduates have a known twelve-month outcome, with at least forty in the smaller outcome group and at least ten such cases for every input used (Riley et al., 2020).

Calibration: The Achilles Heel of Predictive Analytics

Argued that the evaluation of prediction models concentrates almost exclusively on discrimination, the ability to rank one case above another, while neglecting calibration, the agreement between the probabilities a model reports and the frequencies actually observed. The authors showed that a model with acceptable discrimination can still be badly miscalibrated, systematically overstating or understating risk, and that such a model is misleading when its output is read as a probability rather than as a ranking. They proposed the calibration slope as the minimum reportable measure. The proposed Graduate Tracer System reports the chance of employment as a probability rather than a rank, and therefore adopts this recommendation as an acceptance criterion: a candidate model is rejected unless its calibration slope falls between 0.7 and 1.3, both in cross-validation and on the two most recent batches held out from training (Van Calster et al., 2019).

Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis: The TRIPOD Statement

Established a twenty-two item reporting standard for studies that develop, validate, or update a prediction model, covering the definition of the outcome, the source

and completeness of the data, the handling of candidate predictors, the method of model selection, the measures of performance reported, and the limitations of the resulting model. The central principle of the standard is that the criteria by which a model is judged acceptable should be stated before the results are known, so that a model is not retrospectively justified by whichever metric happens to favour it. The proposed Graduate Tracer System follows this principle in the design of its acceptance gate: eight checks with fixed thresholds are defined in advance, every training run is stored with its metadata whether it passes or fails, and only a run that satisfies all eight may be activated for display (Collins et al., 2015).

Two-Sided Confidence Intervals for the Single Proportion: Comparison of Seven Methods

Compared seven methods of placing a confidence interval around an observed proportion and evaluated them on coverage, interval location, expected width, and freedom from aberrations such as limits falling outside the zero-to-one range. The normal approximation, the method most commonly applied, was shown to perform poorly at small sample sizes and at proportions near zero or one, exactly the conditions under which a single graduating batch is reported. The score method attributed to Wilson was recommended as the best general-purpose choice. The proposed Graduate Tracer System applies the Wilson interval to every rate it displays, including the employment rate of each batch, and applies the same author's hybrid score method to the difference between the employment rates of graduates who do and do not hold a given skill, so that a figure computed from a handful of respondents is never presented as though it were precise (Newcombe, 1998).

Scikit-learn: Machine Learning in Python

Described the design and implementation of the scikit-learn library, which provides standard implementations of classification, regression, clustering, and model selection routines behind a uniform interface, together with the pipeline abstraction that binds preprocessing steps to an estimator so that the same transformations are applied during training and at prediction time. The library is the machine learning toolkit of the proposed Graduate Tracer System. The employment model is a scikit-learn pipeline combining median imputation, standard scaling, and logistic regression; the repeated stratified cross-validation, the area under the curve, the Brier score, and the calibration fit used by the acceptance gate are all taken from the same library; and binding the preprocessing into the pipeline is what guarantees that a graduate scored on the dashboard is transformed exactly as the training data was (Pedregosa et al., 2011).

2.2 Local Studies

Employability Tracer Study of Information Technology Education Graduates from a State University in the Philippines

Conducted a tracer study among information technology graduates of a Philippine state university to assess their employment status, job alignment, and career progression. The study found that the majority of graduates were employed within six months of graduation, with a significant proportion working in IT-related positions. However, a notable gap was identified between the skills acquired during formal education and the competencies demanded by industry employers. These findings are directly applicable to the proposed Graduate Tracer System, as they establish local benchmarks for employment rates, job alignment metrics, and the types of data fields

needed to capture meaningful graduate outcome information. The study also underscores the importance of embedding analytical capabilities within the tracer system to enable program-level feedback (Albina and Sumagaysay, 2020).

Employability of Computing Students in the Age of Disruption: A Graduate Tracer Study

Examined the employability of computing graduates in the context of rapid technological disruption, using tracer study data collected across multiple cohorts. The study assessed how well graduates were prepared for emerging roles in artificial intelligence, data analytics, cloud computing, and cybersecurity. Findings indicated that while general computing skills remained in demand, graduates with specializations in high-growth technology areas demonstrated higher employment rates and faster career progression. This study highlighted the need for the Graduate Tracer System's analytics dashboard to track industry-sector distribution and emerging employment trends over time, enabling the institution to align its curricula with evolving labor market demands (Andaya et al., 2024).

An Employability Tracer Study of Secondary Education Mathematics Graduates from 2022 to 2023

Conducted an employability tracer study covering secondary education mathematics graduates across two academic years, providing a cross-cohort analysis of employment status, job alignment, and career outcomes. The study employed a descriptive survey design with structured questionnaires administered to graduates and compared outcomes between cohorts to identify trends. Findings confirmed that cross-cohort tracking is essential for detecting changes in employability patterns over

time. This temporal analysis approach directly informed the trend visualization functionality of the proposed Graduate Tracer System, which is designed to display employment trend lines across multiple graduation batches, supporting evidence-based academic program reviews (Asoy et al., 2024).

Development of a Web-Based Platform for Graduate Employability and Career Tracking: A Tracer Study at St. Paul University Philippines

Developed and evaluated a web-based graduate tracking platform at St. Paul University Philippines as a practical solution for conducting tracer studies and monitoring graduate career progression. The system featured an online graduate registration module, a tracer survey instrument, career tracking records, and a reporting dashboard accessible to institutional administrators. The study demonstrated that a centralized web-based platform significantly improved graduate data completeness and reduced the administrative burden of manual tracer study implementation. The architecture and functional scope of this system closely parallel those of the proposed Graduate Tracer System, making it the most directly relevant local study reference in terms of system design, deployment context, and institutional objectives (Babaran Jr., 2025).

From Challenges to Solutions: Development and Evaluation of Graduate Engagement Portal System for Tracer Studies

Documented the development and evaluation of an graduate engagement portal designed to address common challenges in tracer study implementation, including low response rates, outdated graduate contact information, and fragmented data management. The system incorporated automated reminder notifications, graduate

profile update mechanisms, and a dashboard for monitoring survey completion progress. Evaluation results indicated that the portal significantly improved graduate participation and data quality compared to traditional manual survey methods. The challenges and solutions identified in this study directly informed the design decisions of the proposed Graduate Tracer System, particularly the graduate engagement features and automated notification mechanisms intended to maximize data collection rates among graduates (Cresencio, 2026).

Graduate's Employability: A Tracer Study for Davao de Oro State College Graduates

Conducted a tracer study among graduates of Davao de Oro State College to document their employment outcomes, career pathways, and the relevance of their academic training to their current occupations. The study used a standardized tracer study questionnaire adapted from the CHED Graduate Tracer Study framework and analyzed data across multiple degree programs. Findings revealed considerable variation in employment rates and job alignment across programs, underscoring the importance of program-disaggregated reporting in institutional tracer systems. This study's alignment with the Philippine state college context and its use of CHED-compliant survey instruments provided valuable guidance for the proposed system's data collection design and reporting structure (Cubero et al., 2024).

A Tracer Study on Employment Outcomes and Career Progression of BSIT Graduates: Basis for Enhancing Institutional Reputation and Graduate Employability

Conducted a tracer study specifically targeting BSIT graduates to examine their employment outcomes, career progression trajectories, and the influence of institutional factors on graduate employability. The study found that graduates from institutions with strong industry partnerships and updated curricula demonstrated better employment outcomes and faster career advancement. Importantly, the study linked tracer study findings to institutional reputation management, arguing that regularly reported graduate employment data can enhance institutional credibility with accrediting bodies and industry stakeholders. This framing is highly relevant to the proposed Graduate Tracer System, which serves not only as a monitoring tool but also as an institutional reporting mechanism that supports program accreditation and quality assurance processes (Demafeliz, 2025).

Tracer Study on the Profile and Employability of Bachelor of Science Graduates

Documented the demographic profiles, employment status, and career outcomes of Bachelor of Science graduates across multiple programs at a Philippine higher education institution. The study used a descriptive research design with questionnaires covering educational background, first employment details, time-to-employment, and job relevance. Findings highlighted significant differences in employability rates across programs, with graduates in technical and applied science fields generally reporting faster employment timelines. The comprehensive graduate profile documentation approach used in this study shaped the data fields and graduate registration form design of the proposed Graduate Tracer System, ensuring that the system captures employment-relevant demographic and academic information for each registered graduate (Lagatic et al., 2024).

Development and Evaluation of a Senior High School Graduate Tracer System

Developed and evaluated a graduate tracer system for senior high school graduates, assessing the system's acceptability and functionality using ISO 25010 software quality criteria. The study employed usability testing with faculty evaluators and graduate respondents, reporting high satisfaction scores for functionality, reliability, and ease of use. The development process followed an iterative design approach that incorporated user feedback from early prototype stages. The evaluation methodology and ISO 25010 framework used in this study directly influenced the acceptance testing and evaluation design of the proposed Graduate Tracer System, providing a locally validated approach for assessing system quality within the Philippine educational context (Maitem and Canono, 2025).

Tracer Study of Bachelor of Science in Industrial Technology Graduates

Conducted a tracer study among BSIT Industrial Technology graduates to assess employment outcomes, job satisfaction, and the alignment of academic training with occupational demands. The study collected data on employment status, industry sector, salary ranges, and graduate satisfaction with their academic preparation. Findings indicated that while most graduates were employed, a notable proportion were working outside their field of specialization, suggesting curriculum gaps that required institutional attention. The emphasis on multi-indicator graduate outcome documentation reinforced the proposed system's data model, which captures not only employment status but also job alignment, salary tier, and graduate satisfaction data to support holistic program evaluation (Relloso and Aure, 2024).

The 4th Philippine Graduate Tracer Study

Reported the fourth national graduate tracer study of the Philippines, covering graduates of academic years 2009 to 2011 across the higher education system, and examined employment, citizenship, and life satisfaction outcomes from the perspective of the graduates themselves. Of the total sample, eleven thousand five hundred forty-seven graduates responded, a response rate of thirty-two and seven-tenths percent, and the authors reported this figure alongside every outcome so that readers could judge each estimate against the share of graduates it actually represents. The study reports observed rates disaggregated by program, region, and cohort, and it makes no attempt to forecast the employment of an individual graduate. This is the national reporting standard the proposed Graduate Tracer System aligns itself with. The analytics dashboard of the system therefore leads with observed per-batch rates, states the number of respondents behind each figure, displays the response rate measured against the master list, and warns the administrator when a batch falls below a usable response rate rather than presenting a rate computed from a handful of answers as a finding (Tutor et al., 2021).

Predicting Students' Employability Using Machine Learning Approach

Built an employability prediction model for graduates of a Philippine higher education institution using three thousand observations drawn from mock job interview evaluations, on-the-job training performance ratings, and general point averages recorded between 2015 and 2018, and compared Decision Tree, Random Forest, and Support Vector Machine classifiers, of which the Support Vector Machine performed best at ninety-one and twenty-two hundredths percent accuracy. What distinguishes the

study for the present work is not the accuracy figure but the composition of its predictor set: every variable it used was recorded before the graduate left the institution, so the model could in principle be applied to a student who has not yet graduated. This is the same constraint the redesigned predictive module of the Graduate Tracer System imposes on itself, and the study demonstrates that a usable model can be built within it. The reliance of the study on interview and training evaluations, which a tracer system administered years after graduation cannot recover, also explains why the present study restricts itself to four durable facts that a graduate can still report accurately long after the fact (Casuat and Festijo, 2019).

2.3 Foreign Systems

Web-Based Graduate Information System for Tracer Study at STMIK Abulyatama

Developed and deployed a web-based graduate information system at STMIK Abulyatama in Indonesia specifically designed to support tracer study data collection and graduate record management. The system featured online graduate registration, a web-based survey questionnaire, and basic reporting capabilities for institutional administrators. Evaluation results confirmed the system's functional adequacy for managing graduate contact information and employment status data. The straightforward architecture of this system provided a foundational design reference for the proposed Graduate Tracer System, particularly for the graduate registration portal and online survey components, and its implementation demonstrated the feasibility of transitioning from paper-based tracer study processes to a fully online platform in a developing country higher education context (Ardiansyah, 2021).

Automated Graduate Data Tracking System

Presented an automated graduate data tracking system that streamlined the collection, storage, and retrieval of graduate employment records through automated data capture mechanisms integrated with institutional registrar systems. The system reduced manual data entry through form automation and periodic reminder notifications sent to graduates for profile updates. By aggregating employment data from multiple sources, the system maintained up-to-date graduate records with minimal administrative intervention. The automation features described in this system informed the proposed Graduate Tracer System's approach to graduate data management, particularly the design of automated follow-up notifications and employment status update prompts to sustain data currency across graduate cohorts (Babu et al., 2022).

A Web-Based Graduate Career Prediction and Analysis System Using J48 Algorithm

Developed a web-based graduate career prediction and analysis system that integrated the J48 decision tree algorithm to forecast graduate career outcomes based on historical tracer study data. The system combined a graduate database, an online survey module, and a prediction engine that classified graduates according to projected career trajectories. Evaluation confirmed that the J48-based prediction component achieved acceptable accuracy rates, and the integrated dashboard enabled institutional decision-makers to visualize career trends across programs and graduation years. This system is the most technically analogous foreign reference for the proposed Graduate Tracer System, as it combines tracer data collection with machine learning-based career prediction in a unified web platform, with J48 serving as a baseline comparison model during algorithm selection (Kurniawan et al., 2024).

Development of a Web-Based Graduate Tracer Information System with Data Analytics

Developed a web-based graduate tracer information system that incorporated data analytics capabilities to generate employment trend reports and program performance summaries for institutional stakeholders. The system supported online survey administration, graduate record storage, and an analytics dashboard with visualization components including bar charts and trend lines for employment rates across cohorts. The system was evaluated using faculty and administrator respondents, reporting high acceptability scores for usability and functional adequacy. The architecture and analytics dashboard design of this system served as a direct structural reference for the proposed Graduate Tracer System, particularly for the trend visualization module, which aims to display employment and career outcome patterns across graduate cohorts (Nabablit and Dajao, 2023).

Improving the Performance of Graduate Achievement Assessment by Integrating Website-Based Tracer Study Information Systems and Telegram API

Enhanced an existing website-based tracer study system by integrating the Telegram API as an automated notification and graduate engagement channel. The integration enabled the system to send real-time survey reminders, completion notifications, and employment update prompts directly to graduates. The study reported a significant improvement in survey response rates and graduate data completeness following the integration. While the proposed Graduate Tracer System does not adopt Telegram specifically, this study's approach to leveraging messaging APIs for graduate engagement informed the system's notification design, which incorporates email-based

automated reminders and in-system notifications to encourage graduate participation and profile updates (Sucipto et al., 2020).

2.4 Local Systems

Development of a Web-Based Platform for Graduate Employability and Career Tracking at St. Paul University Philippines

From a systems perspective, the platform developed by (Babaran Jr., 2025) at St. Paul University Philippines represents the most contextually aligned local system reference for the proposed Graduate Tracer System. The platform's technical implementation using a web-based stack with an online graduate portal, structured survey instrument, and administrative reporting dashboard closely mirrors the proposed system's architecture. The system's deployment within a Philippine private higher education institution also provides practical guidance on institutional onboarding, graduate data governance, and stakeholder adoption strategies. The proposed Graduate Tracer System extends the scope of this reference system by incorporating predictive analytics and employability trend visualization capabilities not present in the SPUP platform.

From Challenges to Solutions: Development and Evaluation of Graduate Engagement Portal System for Tracer Studies

From a systems perspective, the graduate engagement portal developed by (Cresencio, 2026) demonstrated the technical feasibility of building a scalable, database-driven graduate tracking platform capable of managing longitudinal data from multiple graduate cohorts. The system employed structured data storage for graduate profiles, employment records, and survey responses, with an administrative dashboard

for monitoring data completeness and generating institutional reports. Its design for sustained graduate engagement through automated communication tools is a technical feature replicated in the proposed system's notification module. The portal's evaluation methodology using ISO-aligned criteria also provided a validated framework for conducting acceptance testing of the Graduate Tracer System.

Graduate Tracer with Analytics for Ilocos Sur Polytechnic State College

Developed a graduate tracer system with integrated analytics capabilities for Ilocos Sur Polytechnic State College, a Philippine state institution comparable in scale and academic context to the target institution. The system supported online graduate registration, tracer survey administration, and an analytics module that generated employment rate summaries and program-level outcome reports. The study confirmed that graduate tracer systems with built-in analytics improve the timeliness and depth of institutional data available for program review and accreditation purposes. As a Philippine state college system deployment, this reference directly validates the technical and institutional feasibility of the proposed Graduate Tracer System, confirming that similar systems have been successfully implemented and evaluated within comparable local higher education environments (Guiling, 2024).

An Online Graduate Tracer System for the Graduates of Cavite State University

Developed and evaluated an online graduate tracer system for Cavite State University, managing graduate registration, survey data collection, employment status tracking, and report generation across multiple colleges and degree programs. A key design consideration was the system's scalability to accommodate a large and geographically dispersed graduate population. Evaluation using the ISO 25010 quality

model yielded high scores for functional suitability, usability, and performance efficiency. The proposed Graduate Tracer System draws from this reference to ensure that its architecture is designed for scalability and multi-program data management, aligning with the proposed system's requirements for multi-program institutional deployment (Lacuesta et al., 2025).

Graduate Tracer and Management System with Data Analytics: Enhancing Graduate Engagement and Data Management for South East Asian Institute of Technology, Inc.

Developed a graduate tracer and management system with integrated data analytics for the South East Asian Institute of Technology, Inc., combining graduate record management, tracer survey administration, and an analytics dashboard that visualized employment trends and graduate demographic distributions. The study reported significant improvements in graduate engagement rates and data completeness following system deployment, attributing these gains to automated notification features and an intuitive user interface. The system's combined focus on data management and analytics aligns closely with the dual objectives of the proposed Graduate Tracer System: to provide a reliable data collection and management platform while simultaneously enabling evidence-based institutional decision-making through predictive employability trend analysis (Trillano et al., 2025).

2.5 Synthesis

The following tables synthesize the reviewed foreign and local studies and systems, identifying the features and functionalities of each related work and describing

their specific contributions to the development of the Graduate Tracer System with Predictive Employability Trend Analysis.

Table 1

Synthesis of Related Literature and Studies

Related Studies	Features and Functionalities	Developed System
Observation of Imbalance Tracer Study Data for Graduates Employability Prediction in Indonesia	• Imbalanced dataset handling • ML classifiers for employability prediction • tracer study data analysis	Informed preprocessing strategies and algorithm selection for the prediction module, particularly for handling class imbalance common in tracer datasets
Soft Skills and Graduate Employability: Evidence from Malaysian Tracer Study	• Soft skill profiling • tracer survey methodology • employability factor analysis	Supported inclusion of non-academic indicators in the graduate profile module and informed survey instrument design
A Tracer Study of Psychosocial Counselling Graduates across Malawi	• Descriptive cross-sectional tracer methodology • career outcome monitoring • workplace tracking	Provided a methodological model for systematic career tracking across employment sectors, informing the graduate tracking module
Graduate Career Tracking Systems: A Systematic Literature Review	• Systematic review of tracer system features • benchmarking functionalities • institutional tracking frameworks	Served as a comprehensive reference for identifying standard features expected in graduate career tracking systems
IT Graduates Employability Prediction Using Tree-Based Machine Learning	• Decision tree classifiers • Random Forest and Gradient Boosting comparison • developing country context	Informed the initial choice of tree-based ensemble methods and later served as the accuracy benchmark against which the leakage-free model was compared
Tracer Study on Employability of Nursing Graduates	• Graduate employability assessment • program quality evaluation • descriptive survey methodology	Highlighted the dual use of tracer data for graduate monitoring and program quality review, shaping the analytics dashboard
Employability Prediction of IT Graduates Using Machine Learning	• ML algorithm comparison (Naive Bayes, KNN, SVM, Random Forest) • feature importance analysis • IT-specific indicators	Guided algorithm selection and showed that the strongest predictors are attributes fixed at graduation, guiding the final input set
Predicting University Graduates' Employability Using SVM	• SVM classification • kernel parameter optimization • k-fold cross-validation	Informed the inclusion of SVM as a candidate model in the prediction engine and provided a cross-validation benchmark
Classification Techniques for Graduate Student Employability Predictions	• Multi-classifier benchmarking • AUC-ROC and F1-score evaluation	Supported a multi-model evaluation approach for the prediction engine and identified

Related Studies	Features and Functionalities	Developed System
	• feature importance ranking	key employability predictor variables
Predicting Employability of Congolese IT Graduates Using Contextual Factors	• Contextual predictor variables • geographic and socioeconomic factors • sustainable employability framework	Expanded the data model to include contextual indicators such as region, employment sector, and economic environment
Predicting Graduate Employability Using Hybrid AHP-TOPSIS and ML	• AHP-TOPSIS multi-criteria weighting • hybrid ML integration • Moroccan case study	Inspired weighted multi-criteria scoring mechanisms for the predictive analytics module
Leakage in Data Mining: Formulation, Detection, and Avoidance	• Formal definition of target leakage • detection by observability of each input • documented inflated accuracy cases	Basis for replacing the original model. Fifteen post-outcome columns were removed and a leakage check was written into the acceptance gate
Machine Learning Algorithm Validation with a Limited Sample Size	• Bias of k-fold CV at small n • repeated and nested cross-validation • reporting the spread of fold results	Set the validation design: ten-times-repeated stratified five-fold cross-validation, judged on both the mean AUC and its fifth percentile
Calculating the Sample Size Required for Developing a Clinical Prediction Model	• Minimum events per candidate predictor • overfitting below threshold • pre-development sample size rules	Set the minimum data rule of the acceptance gate: 150 graduates with a known outcome, 40 in the smaller group, and 10 per input
Calibration: The Achilles Heel of Predictive Analytics	• Discrimination versus calibration • calibration slope as minimum reportable measure • risk of miscalibrated probabilities	Added the calibration slope (0.7 to 1.3) as an acceptance criterion, in cross-validation and on the two held-out batches
TRIPOD: Transparent Reporting of a Multivariable Prediction Model	• 22-item reporting checklist • pre-specified acceptance criteria • full disclosure of data and limitations	Shaped the acceptance gate as eight fixed checks declared before training, with every run stored whether it passes or fails
Two-Sided Confidence Intervals for the Single Proportion	• Comparison of seven interval methods • failure of the normal approximation at small n • Wilson score interval recommended	Set the interval method for every rate the dashboard shows, and the hybrid score method for skill-to-skill rate differences
Scikit-learn: Machine Learning in Python	• Uniform estimator interface • pipeline binding preprocessing to the model • model selection and metrics	The toolkit of the predictive module. The pipeline abstraction guarantees train and serve parity for every scored graduate
Employability Tracer Study of IT Education Graduates in the Philippines	• Philippine IT graduate employment data • state university tracer methodology • job alignment analysis	Provided local benchmarks for employment rates and job-course alignment, directly contextualizing the system's scope

Related Studies	Features and Functionalities	Developed System
Employability of Computing Students in the Age of Disruption	• Disruption-era employability tracking • multi-cohort analysis • career readiness indicators	Informed the analytics dashboard's tracking of emerging employment sectors and technology-related career trends
An Employability Tracer Study of Secondary Education Mathematics Graduates (2022–2023)	• Cross-cohort employability comparison • employment trend documentation • descriptive survey design	Reinforced the importance of cohort-based trend tracking, embedded in the system's trend visualization functionality
Web-Based Platform for Graduate Employability and Career Tracking at SPUP	• Online graduate registration • tracer survey instrument • career tracking and reporting dashboard	Served as the primary local reference for system architecture, directly informing module design and institutional deployment approach
Development and Evaluation of Graduate Engagement Portal System for Tracer Studies	• Graduate engagement portal • automated reminder notifications • survey completion monitoring dashboard	Addressed common tracer study challenges, informing the notification and graduate engagement features of the proposed system
Graduate's Employability: A Tracer Study for Davao de Oro State College	• Philippine state college tracer study • CHED-compliant survey instrument • program-disaggregated reporting	Provided a comparable Philippine institutional tracer model and guided the system's reporting structure and data collection design
A Tracer Study on Employment Outcomes and Career Progression of BSIT Graduates	• BSIT-specific employment and career progression tracking • institutional reputation linkage • accreditation relevance	Directly aligned with the target population and institutional objectives of the proposed Graduate Tracer System
Tracer Study on the Profile and Employability of BS Graduates	• Graduate profile documentation • time-to-employment analysis • program-level employment comparisons	Shaped the graduate registration form design and data fields to capture employment-relevant demographic and academic information
Development and Evaluation of a Senior High School Graduate Tracer System	• ISO 25010-based evaluation • iterative system development • usability and functionality testing	Provided a validated evaluation framework and development methodology applicable to the proposed system's acceptance testing
Tracer Study of BSIT Industrial Technology Graduates	• Multi-indicator outcome documentation • job alignment and satisfaction data • program improvement linkage	Reinforced the system's multi-indicator data model covering employment status, job alignment, and graduate satisfaction
The 4th Philippine Graduate Tracer Study	• National tracer study of 11,547 graduates • 32.7% response rate reported with every outcome • observed rates disaggregated by program and region	The national reporting standard the dashboard follows: observed per-batch rates with n, the response rate against the master list, and a warning on thin batches
Predicting Students' Employability Using	• Philippine HEI dataset of 3,000 observations	Local evidence that a model built only from pre-graduation

Related Studies	Features and Functionalities	Developed System
Machine Learning Approach	• DT, RF and SVM compared (SVM best at 91.22%) • all predictors recorded before graduation	attributes is workable, and that interview and OJT ratings are unrecoverable years later

The reviewed foreign and local studies collectively establish the theoretical and empirical foundation for the Graduate Tracer System with Predictive Employability Trend Analysis. Foreign studies demonstrated that machine learning algorithms, particularly Random Forest, Support Vector Machine, and Gradient Boosting, are effective tools for predicting graduate employability when applied to tracer study datasets containing both academic and contextual variables. Studies conducted in Indonesia, Malaysia, Morocco, and Ghana confirmed that predictive models perform best when trained on balanced, multi-indicator datasets that capture soft skills, specialization tracks, internship experience, and socioeconomic context alongside academic achievement, directly informing the feature engineering and algorithm selection process for the proposed system's predictive analytics module.

The prediction methodology literature reviewed alongside these studies governs how that module may legitimately be built and reported. Kaufman et al. (2012) supplied the definition of target leakage under which the first model of this study was found unsound and replaced, since its strongest inputs were job details that only a graduate who had already been hired could supply. Riley et al. (2020) and Vabalas et al. (2019) together set the conditions under which a model fitted on a single program's graduates may be believed at all, namely a minimum quantity of outcome data and a validation design that does not flatter a small sample. Van Calster et al. (2019) established that a system reporting a probability rather than a ranking must demonstrate calibration, and Collins et al. (2015) established that the criteria for accepting a model must be fixed

before the results are seen. Newcombe (1998) supplied the interval method by which every rate the system displays is qualified. Taken together these works are the reason the predictive component of this study is presented as a narrow, gated claim about the chance of employment within twelve months rather than as a forecast of individual careers.

Local studies provided contextually grounded evidence for the relevance and feasibility of the proposed system within the Philippine higher education landscape. Studies conducted at state universities and colleges in the Philippines confirmed that tracer study data is an underutilized resource for program quality improvement and institutional decision-making, and that web-based tracer systems significantly improve data collection rates compared to traditional manual approaches. The findings of Albina and Sumagaysay (2020), Demafeliz (2025), and Babaran Jr. (2025), among others, validated the need for a dedicated Graduate Tracer System that not only collects graduate employment data but also generates actionable insights through trend analysis and predictive modeling. Tutor et al. (2021) established the reporting convention the system follows, in which observed rates are published with the number of respondents and the response rate behind them, while Casuat and Festijo (2019) demonstrated within a Philippine institution that an employability model restricted to attributes recorded before graduation is workable.

Table 2

Synthesis of Related Systems

Related Systems	Features and Functionalities	Developed System
Web-Based Graduate Information System for	• Online graduate registration	Provided a foundational architectural reference for the

Related Systems	Features and Functionalities	Developed System
Tracer Study at STMIK Abulyatama	• web-based survey questionnaire • basic administrative reporting	graduate registration portal and online survey components
Automated Graduate Data Tracking System	• Automated data capture • graduate record management • employment data aggregation	Informed the system's automated graduate data management features and employment status update mechanisms
Web-Based Graduate Career Prediction and Analysis System Using J48 Algorithm	• J48 decision tree career prediction • integrated prediction engine • career trend visualization dashboard	Directly influenced the predictive analytics module; J48 served as a baseline comparison model for algorithm selection
Web-Based Graduate Tracer Information System with Data Analytics	• Graduate tracer system with analytics • employment trend reporting • visualization dashboard with charts	Served as a direct structural reference for the analytics dashboard and trend visualization module of the proposed system
Website-Based Tracer Study Integrated with Telegram API	• Automated notification via Telegram API • survey reminder system • improved graduate engagement	Inspired the automated notification and graduate engagement features designed to maximize survey response rates
Web-Based Platform for Graduate Employability and Career Tracking at SPUP (Local)	• Philippine HEI web-based graduate tracking • career progression monitoring • online survey and reporting	Provided the most contextually aligned local system reference for architecture, feature scope, and institutional deployment
Graduate Engagement Portal System for Tracer Studies (Local)	• Database-driven graduate portal • longitudinal cohort data management • automated communication tools	Validated scalable portal design and informed the notification module and ISO-based evaluation framework
Graduate Tracer with Analytics for Ilocos Sur Polytechnic State College	• Philippine state college tracer system • analytics module • employment rate summaries and program reports	Confirmed feasibility of analytics-enabled tracer systems in comparable Philippine state college contexts
Online Graduate Tracer System for Cavite State University	• University-wide online tracer system • multi-program management • ISO 25010 quality evaluation	Validated scalability and multi-program data management design, informing the system's institutional deployment architecture
Graduate Tracer and Management System with Data Analytics at SEAIT	• Graduate tracer and management system • analytics dashboard • employment trend visualization	Confirmed that combined data management and analytics improves engagement and data quality, aligned with the proposed system's core objectives

CHAPTER III

METHODOLOGY

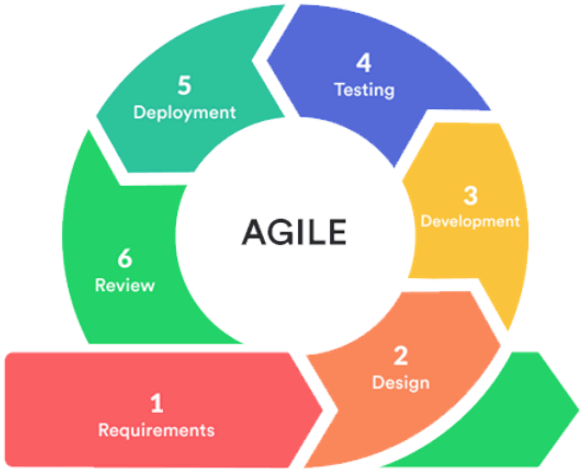
This chapter presents the development method, operational framework, use case diagram and scenarios, the design of the predictive employability model, data flow diagrams, entity relationship diagram, data dictionary, and hardware and software requirements of the Graduate Tracer System with Predictive Employability Trend Analysis.

3.1 System Development Method

The system was developed using the Agile model, an iterative approach that cycles through requirements, design, development, testing, deployment, and review (Pressman & Maxim, 2020). Agile was chosen because the modules, namely graduate registration, face recognition verification, employment monitoring, employer verification, geomapping, and predictive analytics, depend on each other and needed repeated refinement from stakeholder and panel feedback.

Figure 1

Agile Development Model



The figure above illustrates the Agile development model used in this study. The model consists of six iterative phases: Requirements, Design, Development, Testing, Deployment, and Review. Each phase was applied cyclically to ensure that the developed Graduate Tracer System with Predictive Employability Trend Analysis met the functional and operational requirements of the BSIS Program Chair, faculty, and the graduate of Carlos Hilado Memorial State University – Talisay Campus.

Phase 1: Requirements. The researchers consulted the Graduate Office and the BSIS Program Chair on the current tracing process. The CHED Graduate Tracer Survey was adapted into a survey that also collects the variables needed by the predictive module. Panel revisions added three requirements: employers verify through one-time links instead of accounts, graduates give recorded data privacy consent with a separate geomap opt-in, and the system refuses invalid data at entry.

Phase 2: Design. Figma mockups followed the look of the CHMSU Student Information System. The architecture is client-server: a Next.js frontend calls a Django REST backend, and data is stored in PostgreSQL. The database design is shown in the Entity Relationship Diagram.

Phase 3: Development. Modules were built in this order: registration and verification, face recognition verification, profile and employment monitoring, employer verification, geomapping, predictive analytics, the administrative dashboard, and reporting, with the analytics built last on top of the collected data, as in similar tracer systems (Nabablit & Dajao, 2023; Trillano et al., 2025). Face recognition uses InsightFace (Deng et al., 2019), with a blink or head-turn challenge against photos and videos (Soukupová & Čech, 2016). Addresses follow the Philippine Standard Geographic Code (Philippine Statistics Authority, 2024), and validation in the forms, API, and database refuses invalid data.

The predictive model, built with scikit-learn (Pedregosa et al., 2011), estimates how likely a graduate is to be employed within twelve months of graduation. An earlier Random Forest scored highly only because half its inputs could only come from graduates who were already employed, a problem called target leakage (Kaufman et al., 2012). It was replaced with a logistic regression on four facts known at graduation: Latin honors, paid work besides the OJT, a portfolio, and a scholarship. This follows Casuat and Festijo (2019) and Cenita and Lagman (2026). The model is tested only on graduates it did not learn from, over fifty repeated five-part splits (Vabalas et al., 2019). It is shown on the dashboard only after passing eight checks fixed before training (Collins et al., 2015), including at least 150 graduates with a known outcome (Riley et

al., 2020) and percentages that match reality (Van Calster et al., 2019). Otherwise the dashboard shows only observed rates, each with its sample size and a 95 percent range (Newcombe, 1998; Tutor et al., 2021).

Phase 4: Testing. Functional testing used black-box and white-box test cases across all modules, and a security scan was run with OWASP ZAP. User acceptance testing with graduates, employers, and administrators uses an instrument based on ISO/IEC 25010 (International Organization for Standardization, 2011) to measure functionality, usability, and reliability.

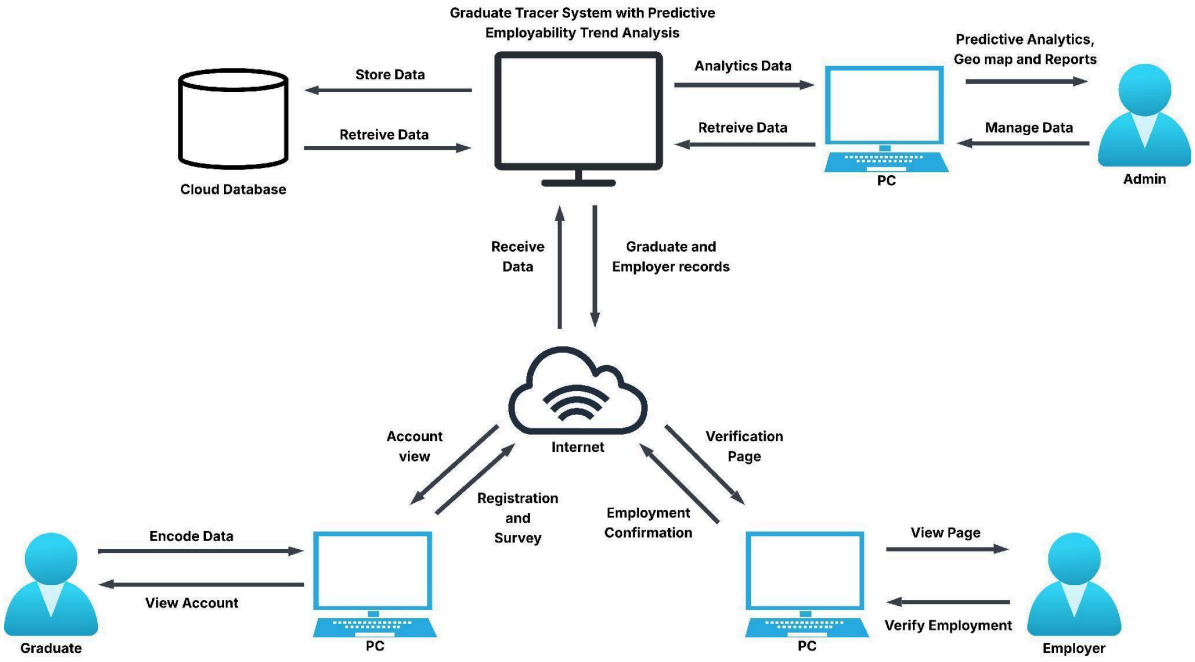
Phase 5: Deployment. The system runs on a Hostinger virtual private server using Docker Compose, with Caddy providing HTTPS. The database and face images are hosted on Supabase, and emails are sent through Gmail SMTP. The website is also packaged as a mobile application that wraps the site. A user orientation and a User Manual support the move from manual tracing.

Phase 6: Review. Each cycle ended with a review against the requirements. After deployment, the BSIS Program Chair and faculty review the system for functionality, usability, and reliability, and the predictive model is retrained as real graduate data accumulates.

3.2 Operational Framework

Figure 2

Operational Framework of Graduate Tracer System with Predictive Employability Trend Analysis

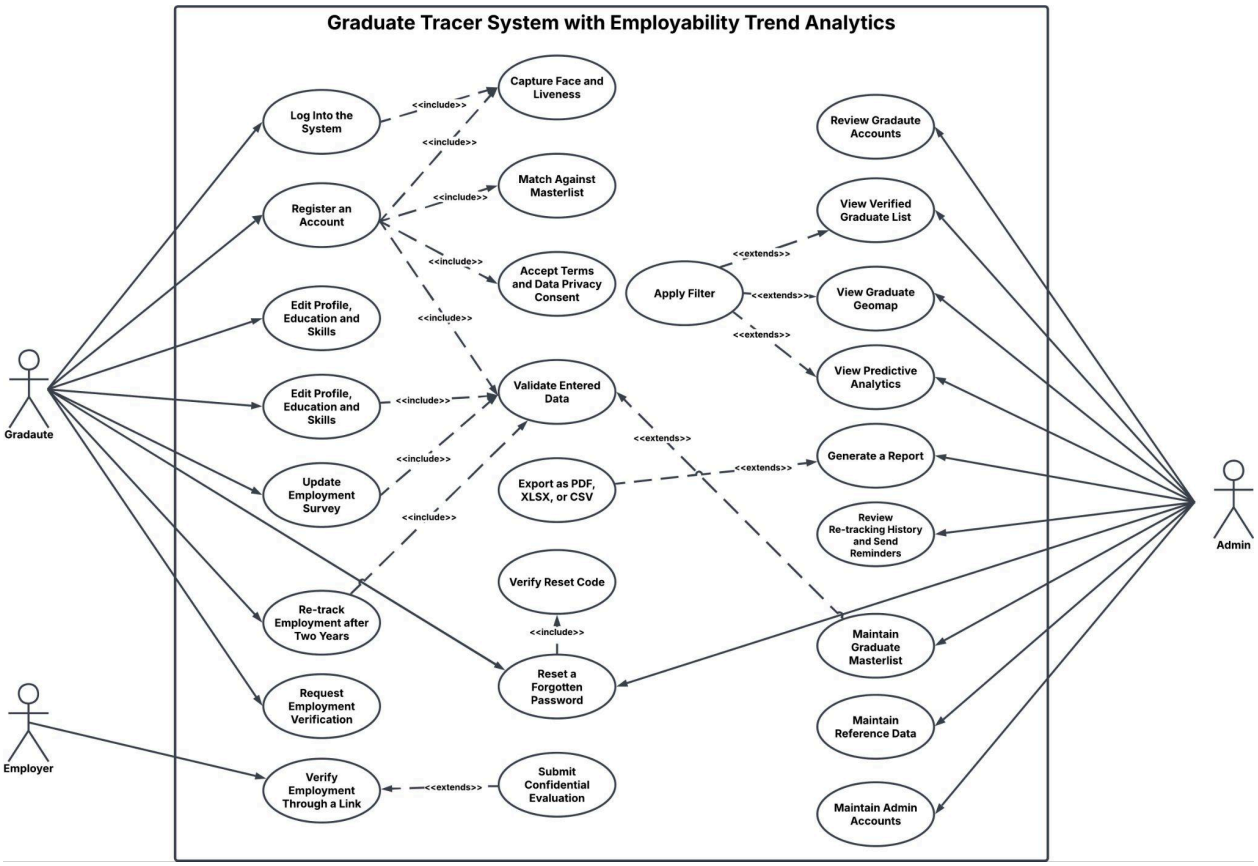


Graduates access the system through a browser or the mobile application to register, pass the face and liveness check, give consent, and submit their personal and employment data. Employers do not register; they receive a one-time link from a graduate by email and use it to confirm the graduate's employment and submit an evaluation. Data is validated before it is stored in the Supabase PostgreSQL database and is used by the analytics and predictive modules. The BSIS administrators upload the master list, approve or reject accounts, maintain reference data, and review the dashboard, geomap, forecasts, and reports.

3.3 Use Case Diagram

Figure 3

Use Case Diagram



Three actors use the system. The Graduate can register, log in, manage the profile, update the employment survey, and request employment verification. The Employer has no account and can only verify employment through a link, with the evaluation form as an optional step. The Admin can review pending graduates, view verified graduates, the geomap, and predictive analytics, generate reports, and maintain the master list, reference data, and admin accounts. Reset Password is shared by the Graduate and the Admin. Capture Face and Liveness, Accept Terms and Consent, and

Validate Survey Data are included in registration; Apply Filter and Export Report extend the admin views.

3.4 Use Case Scenarios

Each table walks through one use case. User actions and system responses are listed in order.

Table 3

Graduate Registers an Account

Use Case 1: Graduate Registers an Account
Use Case Goal: To let a BSIS graduate create an account using a master list match, a face and liveness check, and recorded consent.
Scenario 1: Graduate completes registration
Actor: Graduate (BSIS Graduate)
Pre-condition: The graduate is on the registration page with a working camera. The master list contains the graduate's record.
Main Flow: (user's action only)
Step 1: Fill in account, personal, and education details
Step 2: Capture the frontal face photo and perform the liveness gesture
Step 3: Answer the employment survey
Step 4: Accept the terms and choose whether to allow the geomap
Step 5: Submit the registration
User Action: Fills in email, password, name, birth month and year, contact details, and home address.
System Response: Loads the region, province, city, and barangay lists and checks the name and batch year against the master list.
User Action: Clicks "Use my current location".
System Response: Reads the GPS reading and fills in the matching address and pin.
User Action: Holds a frontal pose, then blinks or turns the head as asked.
System Response: Rejects blurry frames, keeps one frontal photo, and confirms liveness.
User Action: Answers the employment survey and picks a job title from the list.
System Response: Skips the steps that do not apply to the chosen employment status.
User Action: Accepts the terms, sets the geomap choice, and clicks Submit.
System Response: Validates every answer, saves a pending account with the consent time, face template, and survey rows, and shows the "waiting for admin" page.
Scenario 2: Registration has an invalid answer
Actor: Graduate (BSIS Graduate)
Pre-condition: The graduate submits a form containing an impossible or invalid value.
Main Flow: (user's action only)
Step 1: Submit the registration
Step 2: Correct the flagged field
Step 3: Submit again

User Action: Clicks Submit.
System Response: Refuses the registration, saves nothing, and returns the graduate to the form that owns the invalid field.
User Action: Corrects the field and submits again.
System Response: Keeps all other answers and the face capture, validates again, and saves the pending account.

Table 4

Graduate Logs Into the System

Use Case 2: Graduate Logs Into the System
Use Case Goal: To let an approved graduate sign in with a password, a live face scan, and a liveness check.
Scenario 1: Graduate signs in
Actor: Graduate (BSIS Graduate)
Pre-condition: The graduate has an active account with an enrolled face template. The browser allows camera access.
Main Flow: (user's action only)
Step 1: Enter email and password
Step 2: Face the camera
Step 3: Perform the liveness gesture
User Action: Enters email and password and clicks Sign In.
System Response: Checks the password and lockout state, then opens the camera.
User Action: Faces the camera straight on.
System Response: Captures the frontal frame used for matching.
User Action: Blinks, or turns the head in the direction shown if no blink is detected.
System Response: Confirms liveness, compares the face with the enrolled template, saves the login audit, and opens the dashboard.
Scenario 2: Five wrong passwords trigger a lockout
Actor: Graduate (BSIS Graduate)
Pre-condition: The graduate has an active account.
Main Flow: (user's action only)
Step 1: Enter a wrong password five times
Step 2: Try again during the lockout
User Action: Enters a wrong password for the fifth time.
System Response: Locks sign-in for 15 seconds; later lockouts grow to 30, 60, 120, 300, and 600 seconds.
User Action: Tries to sign in during the lockout.
System Response: Disables the Sign In button and shows the remaining seconds.

Table 5

Graduate Updates Employment Survey

Use Case 3: Graduate Updates Employment Survey

Use Case Goal: To let a graduate keep employment, job title, work address, and skills current.
Scenario 1: Graduate updates employment
Actor: Graduate (BSIS Graduate)
Pre-condition: The graduate is signed in with an active account.
Main Flow: (user's action only)
Step 1: Open Employment
Step 2: Update status, job details, and work address
Step 3: Pick skills
Step 4: Save
User Action: Opens Employment and picks an employment status.
System Response: Shows only the steps that apply to that status.
User Action: Picks a job title from the list and fills in the work address.
System Response: Loads the admin job title list and the PSGC location lists.
User Action: Picks technical and soft skills and clicks Save.
System Response: Validates the answers, saves the structured employment, work address, and skill rows, and shows the saved record.
Scenario 2: Job title is not in the list
Actor: Graduate (BSIS Graduate)
Pre-condition: The graduate's job title is missing from the list.
Main Flow: (user's action only)
Step 1: Choose "My job isn't listed"
Step 2: Type the job title
Step 3: Save
User Action: Types the job title.
System Response: Refuses the title if it is vulgar or nonsense and asks for a real title.
User Action: Saves a valid typed title.
System Response: Saves the record and adds the title to the admin's "Job titles to review" list.

Table 6

Graduate Requests Employment Verification

Use Case 4: Graduate Requests Employment Verification
Use Case Goal: To let a graduate send a one-time verification link to their employer.
Scenario 1: Graduate sends a verification link
Actor: Graduate (BSIS Graduate)
Pre-condition: The graduate is signed in and has a current employment record.
Main Flow: (user's action only)
Step 1: Open Employment
Step 2: Enter the employer's work email
Step 3: Send the link
User Action: Enters the employer's work email and clicks Send.
System Response: Creates a one-time link valid for seven days, emails it to the employer, and shows a copyable link.
Scenario 2: Graduate enters their own email
Actor: Graduate (BSIS Graduate)
Pre-condition: The graduate is on the verification request form.
Main Flow: (user's action only)

Step 1: Enter own email address Step 2: Click Send User Action: Enters their own email and clicks Send. System Response: Refuses the request and asks for the employer's address.

Table 7

Employer Verifies Employment Through a Link

Use Case 5: Employer Verifies Employment Through a Link
Use Case Goal: To let an employer confirm or deny a graduate's employment without creating an account. **Scenario 1: Employer confirms employment and submits an evaluation** Actor: Employer (no account) Pre-condition: The employer has received a valid, unused link. **Main Flow: (user's action only)** Step 1: Open the link Step 2: Confirm the employment Step 3: Enter verifier details Step 4: Complete the evaluation form Step 5: Submit User Action: Opens the link. System Response: Shows only the graduate's name, program, and batch, with the data privacy notice. User Action: Clicks Confirm and enters job details, name, work email, and position. System Response: Validates the fields and refuses a verifier email that matches the graduate's. User Action: Fills in the eleven ratings and two written answers and clicks Submit. System Response: Saves the decision and evaluation, flags the answer if it came from a different address, the same device, or within one minute, marks the link used, and sets the record to verified. **Scenario 2: Link is expired or already used** Actor: Employer (no account) Pre-condition: The link has passed its expiry or was already answered. **Main Flow: (user's action only)** Step 1: Open the link User Action: Opens the link. System Response: Refuses the link and tells the employer to ask the graduate for a new one.

Table 8

Admin Reviews Pending Graduate Accounts

Use Case 6: Admin Reviews Pending Graduate Accounts
Use Case Goal: To let the admin approve or reject new registrations. **Scenario 1: Admin approves a graduate** Actor: Admin (Program Chair / Faculty) Pre-condition: The admin is signed in and at least one pending account exists.

Main Flow: (user's action only) Step 1: Open Pending Verification Step 2: Review a graduate Step 3: Click Approve User Action: Opens Pending Verification and selects a graduate. System Response: Shows the profile, frontal photo, capture time, GPS stamp, and master list match. User Action: Clicks Approve. System Response: Activates the account and emails the graduate. **Scenario 2: Admin rejects a graduate with a reason** Actor: Admin (Program Chair / Faculty) Pre-condition: The admin is reviewing a pending graduate. **Main Flow: (user's action only)** Step 1: Click Reject Step 2: Type the reason Step 3: Confirm User Action: Clicks Reject, types the reason, and confirms. System Response: Emails the reason to the graduate and removes the account so the email can register again.

Table 9

Admin Views Verified Graduate List

Use Case 7: Admin Views Verified Graduate List
Use Case Goal: To give the admin a filterable list of approved graduates and their employment. **Scenario 1: Admin filters the verified list** Actor: Admin (Program Chair / Faculty) Pre-condition: The admin is signed in. **Main Flow: (user's action only)** Step 1: Open Verified Graduates Step 2: Set filters Step 3: Open a record User Action: Opens Verified Graduates and sets batch, name, or status filters. System Response: Lists matching graduates with employment and verification status. User Action: Opens a record. System Response: Shows the full profile, face photo, and employment history.

Table 10

Admin Views the Graduate Geomap

Use Case 8: Admin Views the Graduate Geomap
Use Case Goal: To show where consenting graduates are located on an interactive map. **Scenario 1: Admin views and filters the geomap** Actor: Admin (Program Chair / Faculty) Pre-condition: The admin is signed in. **Main Flow: (user's action only)**

Step 1: Open Geomap
Step 2: Set filters
Step 3: Click a marker
User Action: Opens Geomap.
System Response: Plots only graduates who gave geomap consent.
User Action: Sets batch, region, or status filters.
System Response: Redraws the markers for the matching graduates.
User Action: Clicks a marker.
System Response: Shows the graduate's name, batch, company, and job title.

Table 11

Admin Views Predictive Analytics

Use Case 9: Admin Views Predictive Analytics
Use Case Goal: To show observed employability figures per batch, the expected range for the next batch, and the factors from a model that passed its acceptance checks.
Scenario 1: Admin reviews employability analytics
Actor: Admin (Program Chair / Faculty)
Pre-condition: The admin is signed in.
Main Flow: (user's action only)
Step 1: Open Analytics
Step 2: Pick a batch
User Action: Opens Analytics.
System Response: Shows the employment rate of each batch and share employed within twelve months with 95 percent ranges, the expected range for the next batch, and, if a model passed every acceptance check, its factors and check results; states when the figures come from simulated graduates.
User Action: Picks a batch.
System Response: Updates the summary figures and the time-to-first-job chart for that batch, hiding any group of fewer than five graduates.

Table 12

Admin Generates and Exports a Report

Use Case 10: Admin Generates and Exports a Report
Use Case Goal: To produce structured reports and export them.
Scenario 1: Admin exports a report
Actor: Admin (Program Chair / Faculty)
Pre-condition: The admin is signed in.
Main Flow: (user's action only)
Step 1: Open Reports
Step 2: Pick a report and batch range
Step 3: Export

User Action: Picks one of the six reports and a batch range.
System Response: Builds the report sections, including curriculum alignment and IS field distribution where relevant.
User Action: Clicks PDF, XLSX, or CSV.
System Response: Downloads the file in the chosen format; Data Quality exports to PDF only and Predictive Trend to PDF and XLSX.

Table 13

Admin Maintains Graduate Master List

Use Case 11: Admin Maintains Graduate Master List
Use Case Goal: To load and maintain the official roster used for registration matching.
Scenario 1: Admin uploads a valid CSV
Actor: Admin (Program Chair / Faculty)
Pre-condition: The admin is signed in and has a CSV file in the template format.
Main Flow: (user's action only)
Step 1: Open Batch Upload
Step 2: Choose the CSV file
Step 3: Review and save
User Action: Chooses the CSV file.
System Response: Checks the header and every row and shows a preview.
User Action: Clicks Save.
System Response: Saves the records and shows how many were added.
Scenario 2: CSV file contains an invalid row
Actor: Admin (Program Chair / Faculty)
Pre-condition: The file has a wrong header, a bad row, or a duplicate.
Main Flow: (user's action only)
Step 1: Choose the CSV file
User Action: Chooses the file.
System Response: Refuses the whole file, saves nothing, and lists the rows to fix.
Scenario 3: Admin corrects an entry
Actor: Admin (Program Chair / Faculty)
Pre-condition: The admin is signed in and the master list is shown.
Main Flow: (user's action only)
Step 1: Open Batch Upload and view the master list
Step 2: Click the edit button on the entry to correct
Step 3: Correct the name or the batch year and save
User Action: Clicks the edit button on the entry.
System Response: Opens the name and the batch year of that entry for editing.
User Action: Corrects the entry and clicks save.
System Response: Checks that the name is a full name, that the batch year is within range, and that no other entry holds the same name and batch. Saves the correction, matches the roster again, and reports how many graduates were linked by it.
Scenario 4: Admin retires an entry
Actor: Admin (Program Chair / Faculty)

Pre-condition: The admin is viewing the master list.
Main Flow: (user's action only)
Step 1: Click the retire button on the entry
User Action: Clicks the retire button.
System Response: Marks the entry inactive. It stays on the list, shown as retired, and no longer matches a registration. Clicking the button again restores it.
Scenario 5: Admin deletes an unused entry
Actor: Admin (Program Chair / Faculty)
Pre-condition: The admin is viewing the master list and no graduate has registered against the entry.
Main Flow: (user's action only)
Step 1: Click the delete button on the entry
Step 2: Confirm the deletion
User Action: Clicks the delete button and confirms.
System Response: Removes the entry. If a graduate has registered against it, the deletion is refused and the admin is told to retire the entry instead, so the graduate keeps the link to the roster.
Scenario 6: Admin exports the master list
Actor: Admin (Program Chair / Faculty)
Pre-condition: The admin is viewing the master list.
Main Flow: (user's action only)
Step 1: Click Export CSV
User Action: Clicks Export CSV.
System Response: Downloads the roster as a CSV file holding each graduate's name, batch year, whether the graduate has registered, and whether the entry is active or retired.

Table 14

Admin Maintains Reference Data

Use Case 12: Admin Maintains Reference Data
Use Case Goal: To keep industries, job titles, skills, and locations current.
Scenario 1: Admin adds a job title
Actor: Admin (Program Chair / Faculty)
Pre-condition: The admin is signed in.
Main Flow: (user's action only)
Step 1: Open Settings, Industries and Jobs
Step 2: Add the title with its industry and IS field
Step 3: Save
User Action: Enters the title, industry, and IS field and clicks Save.
System Response: Saves the title and sets its BSIS alignment from the IS field.
Scenario 2: Admin adds a typed job title from the review list
Actor: Admin (Program Chair / Faculty)
Pre-condition: Graduates have typed job titles that are not in the list.
Main Flow: (user's action only)
Step 1: Open Job titles to review
Step 2: Add a title
User Action: Opens the review list and adds a title.

System Response: Adds it to the list and links every employment record that used that title.

Table 15

Graduate Re-tracks Employment After Two Years

Use Case 13: Graduate Re-tracks Employment After Two Years
Use Case Goal: To keep employment data current for graduates whose record is over two years old.
Scenario 1: Graduate completes the re-track
Actor: Graduate (BSIS Graduate)
Pre-condition: The graduate's employment record is more than 730 days old.
Main Flow: (user's action only)
Step 1: Sign in
Step 2: Update the employment survey
Step 3: Save
User Action: Signs in.
System Response: Opens the employment form in re-tracking mode instead of the dashboard.
User Action: Updates the survey and clicks Save.
System Response: Saves the record and unlocks the dashboard.
Scenario 2: System sends the reminder email
Actor: System (scheduled command)
Pre-condition: The reminder command runs daily.
Main Flow: (user's action only)
Step 1: Run the reminder command
User Action: The scheduled command runs.
System Response: Emails graduates whose record is over two years old and who were not reminded in the last 30 days, then saves the reminder time.

Table 16

Admin Maintains Admin Accounts

Use Case 14: Admin Maintains Admin Accounts
Use Case Goal: To add, deactivate, or delete administrator accounts.
Scenario 1: Admin adds an admin
Actor: Admin (Program Chair / Faculty)
Pre-condition: The admin is signed in.
Main Flow: (user's action only)
Step 1: Open Settings, Users
Step 2: Enter email and password
Step 3: Save
User Action: Enters the new admin's email and password and clicks Save.
System Response: Checks that the email is unique and creates the admin account.
Scenario 2: Admin deactivates an admin

Actor: Admin (Program Chair / Faculty) Pre-condition: The admin is viewing the admin list. **Main Flow: (user's action only)** Step 1: Turn off the active flag Step 2: Save User Action: Turns off the active flag. System Response: Blocks that admin from signing in.

Table 17

User Resets a Forgotten Password

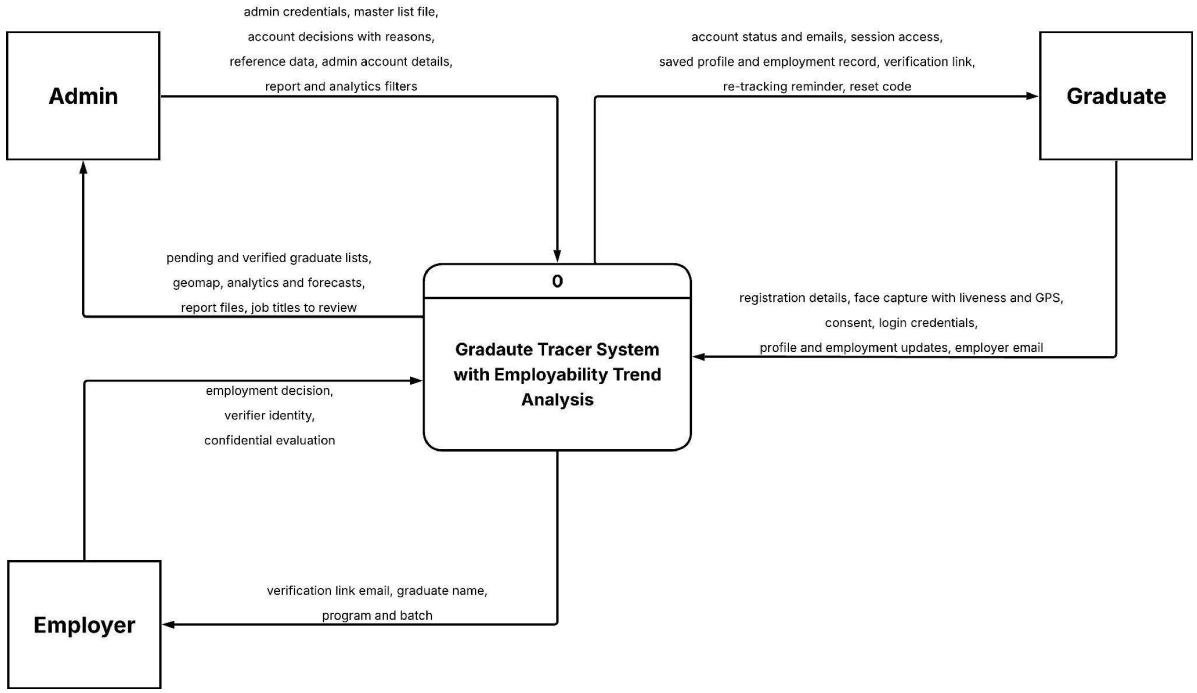
Use Case 15: User Resets a Forgotten Password
Use Case Goal: To let a graduate or admin recover access with an emailed code. **Scenario 1: User resets the password** Actor: Graduate or Admin Pre-condition: The user clicked "Forgot password?" on the sign-in page. **Main Flow: (user's action only)** Step 1: Request a code Step 2: Enter the code Step 3: Set a new password User Action: Enters the email and clicks Send code. System Response: Saves a hash of a 12-digit code that expires in 15 minutes and emails the code. User Action: Enters the code. System Response: Checks the code and issues a reset ticket valid for five minutes. User Action: Enters and confirms the new password. System Response: Saves the new password, marks the code used, and clears the login lockout. **Scenario 2: User enters wrong codes** Actor: Graduate or Admin Pre-condition: A reset code was issued and has not expired. **Main Flow: (user's action only)** Step 1: Enter a wrong code five times User Action: Enters a wrong code for the fifth time. System Response: Voids the code and asks the user to request a new one.

3.5 Data Flow Diagrams

The data flow diagrams use three external entities, seven Level 1 processes, and ten data stores: D1 Graduate Master List, D2 Graduate Accounts, D3 Employment and Skills, D4 Admin Accounts, D5 Reference Data, D6 Verification Records, D7 Biometric and Audit Logs, D8 Model Artifacts, D9 Password Reset Codes, and D10 Login Throttle.

Figure 4

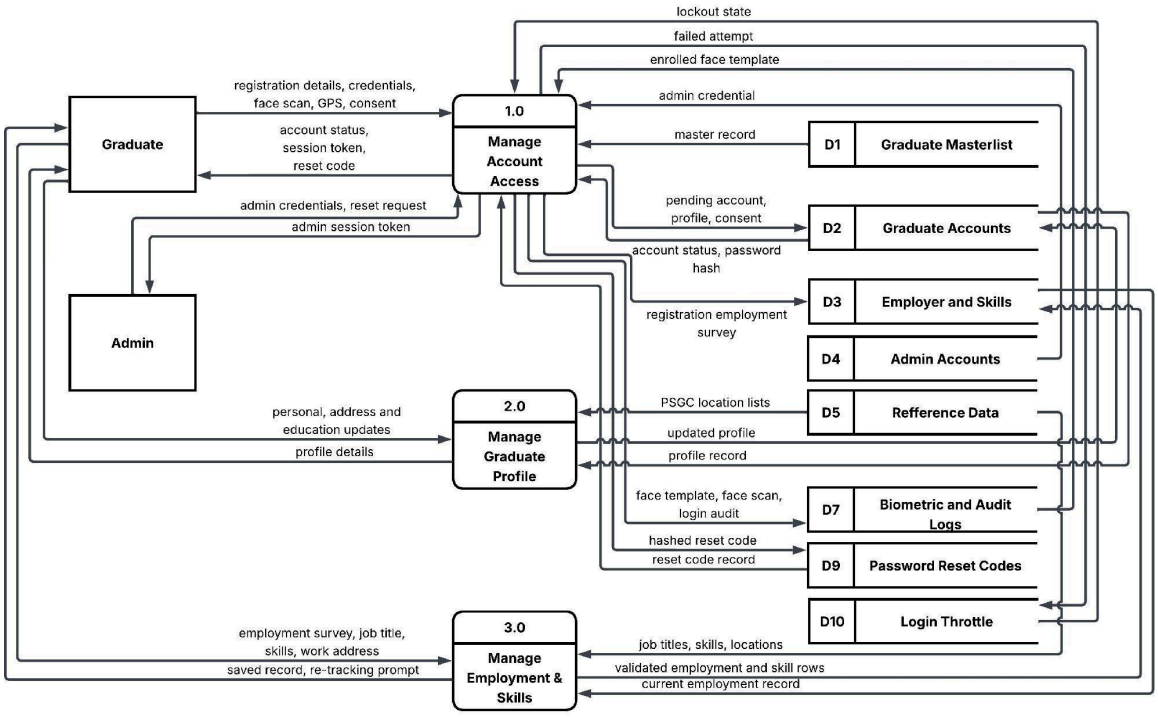
Context Diagram of Graduate Tracer System with Predictive Employability Trend Analysis



The context diagram shows the system as one process with three external entities. Graduates hold accounts and send registration details, face captures, consent, and employment updates. Employers have no account; they receive a one-time verification link and return a decision with their identity and an optional evaluation. Admins manage the master list, accounts, reference data, and reports, and receive the lists, geomap, forecasts, and report files.

Figure 5

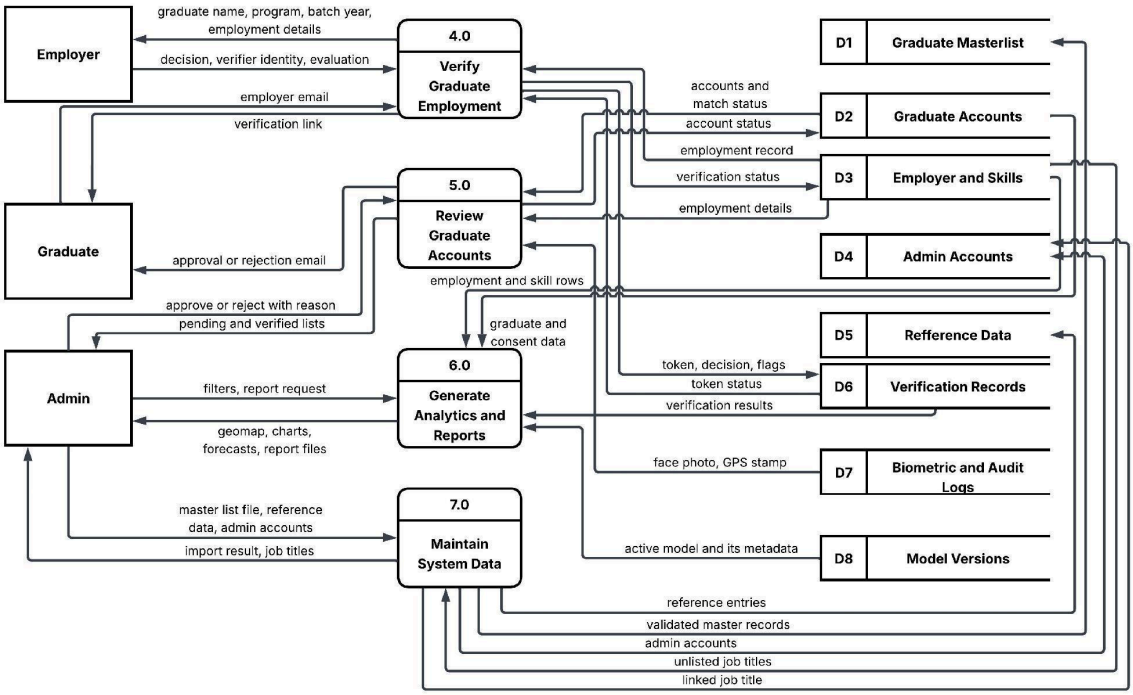
Level 1 Diagram (Part 1 of 2): Graduate-Facing Processes



Part 1 of the Level 1 diagram covers the processes a graduate uses directly. Manage Account Access (1.0) handles registration, face and liveness sign-in for graduates, admin sign-in, password reset, the login throttle, and the login audit. Manage Graduate Profile (2.0) loads and updates personal, home address, and education data using the PSGC location lists. Manage Employment and Skills (3.0) validates and saves the employment survey, job title, skills, and work address.

Figure 6

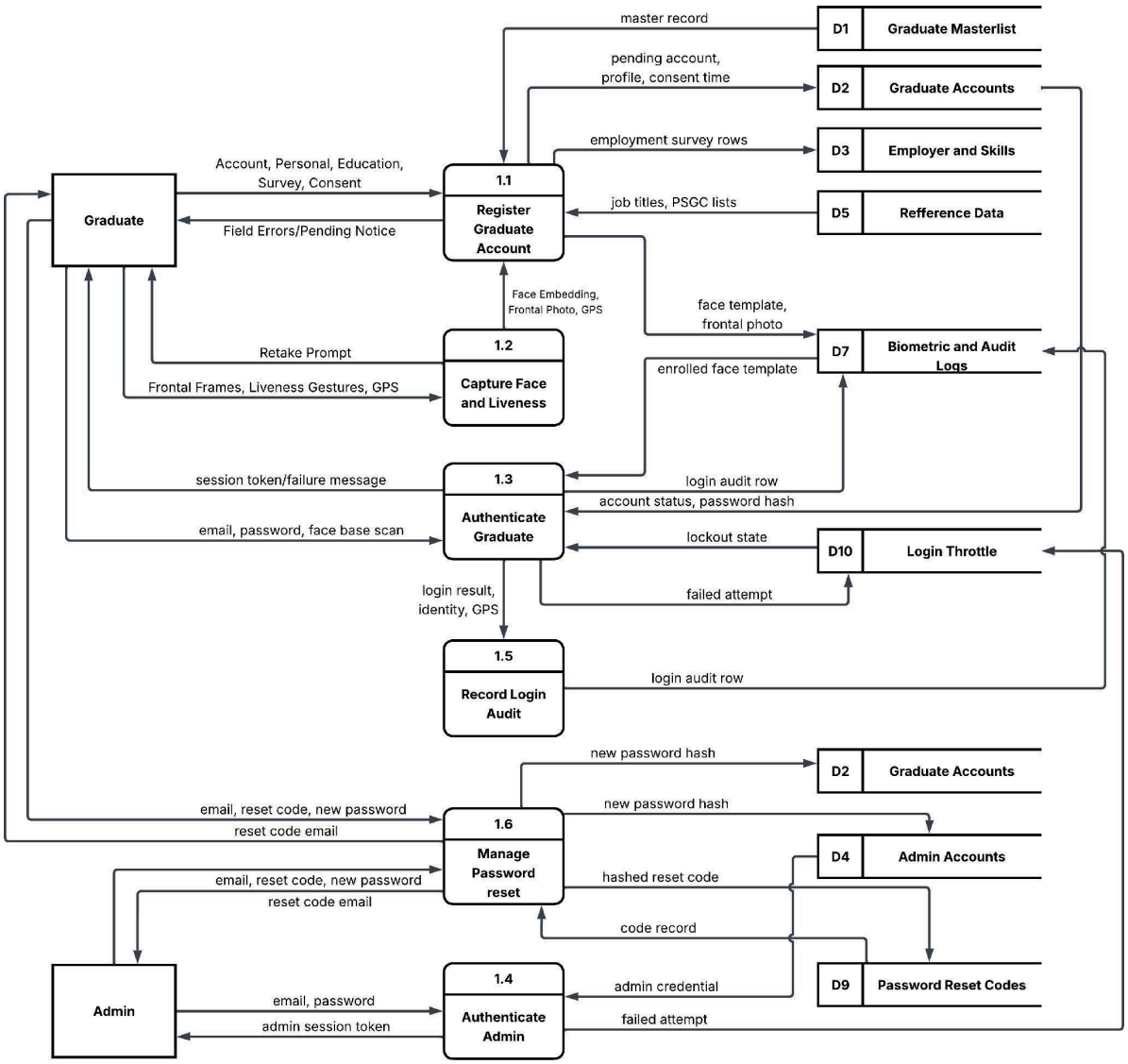
Level 1 Diagram (Part 2 of 2): Verification and Administration Processes



Part 2 covers the remaining processes. Verify Graduate Employment (4.0) issues one-time links and records the employer's decision and flags. Review Graduate Accounts (5.0) lets the admin approve or reject registrations, which sends an email, and view verified graduates. Generate Analytics and Reports (6.0) produces the geomap, dashboard, forecasts, and report files. Maintain System Data (7.0) keeps the master list, reference data, and admin accounts current, including the review of typed job titles.

Figure 7

Level 2 Diagram of Manage Account Access

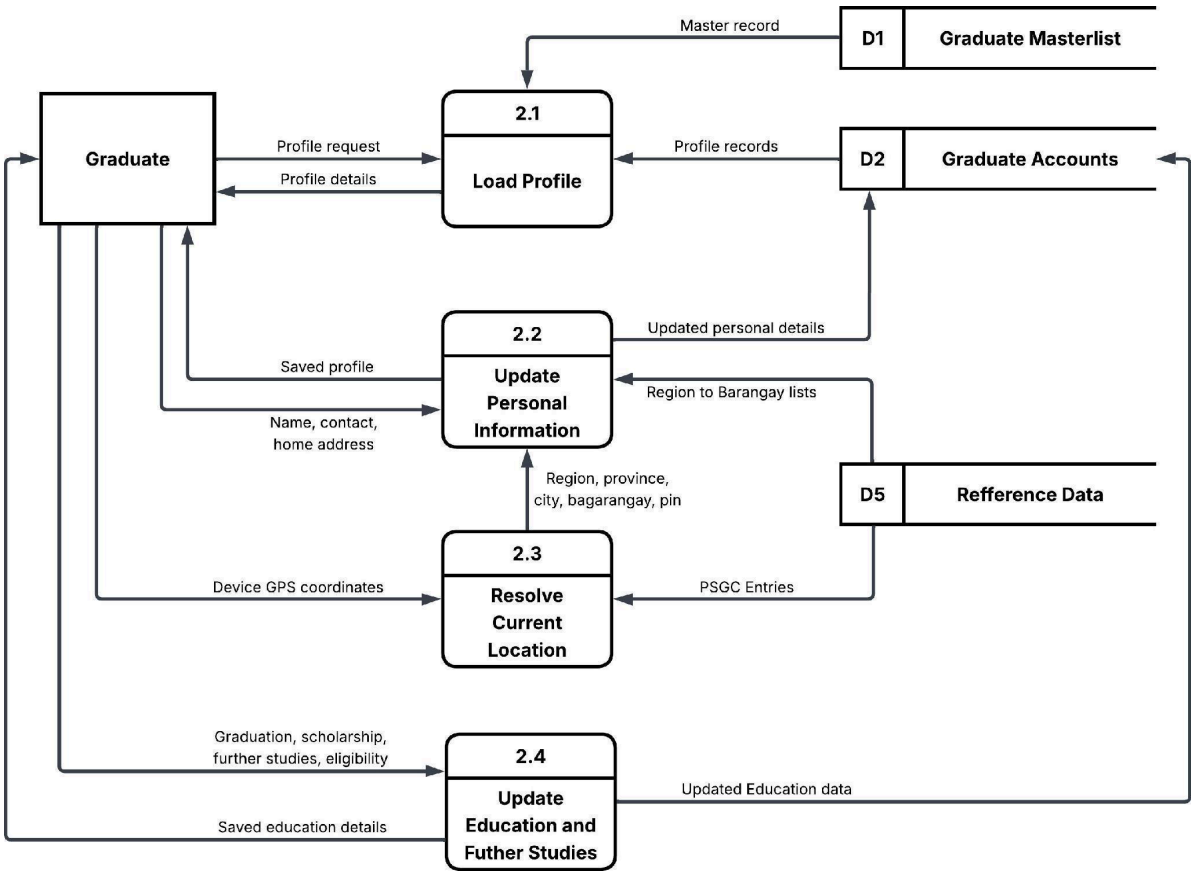


Capture Face and Liveness (1.2) takes one frontal photo, asks for a blink or head turn, and reads the GPS. Register Graduate Account (1.1) checks the name and batch year against the master list, validates every answer, and saves a pending account with the consent time, survey rows, and face template. Authenticate Graduate (1.3) checks the password and lockout, then compares a live face scan with the enrolled template; Record Login Audit (1.5) saves the result. Authenticate Admin (1.4) checks admin

credentials against the same throttle. Manage Password Reset (1.6) emails a hashed one-time code and saves the new password.

Figure 8

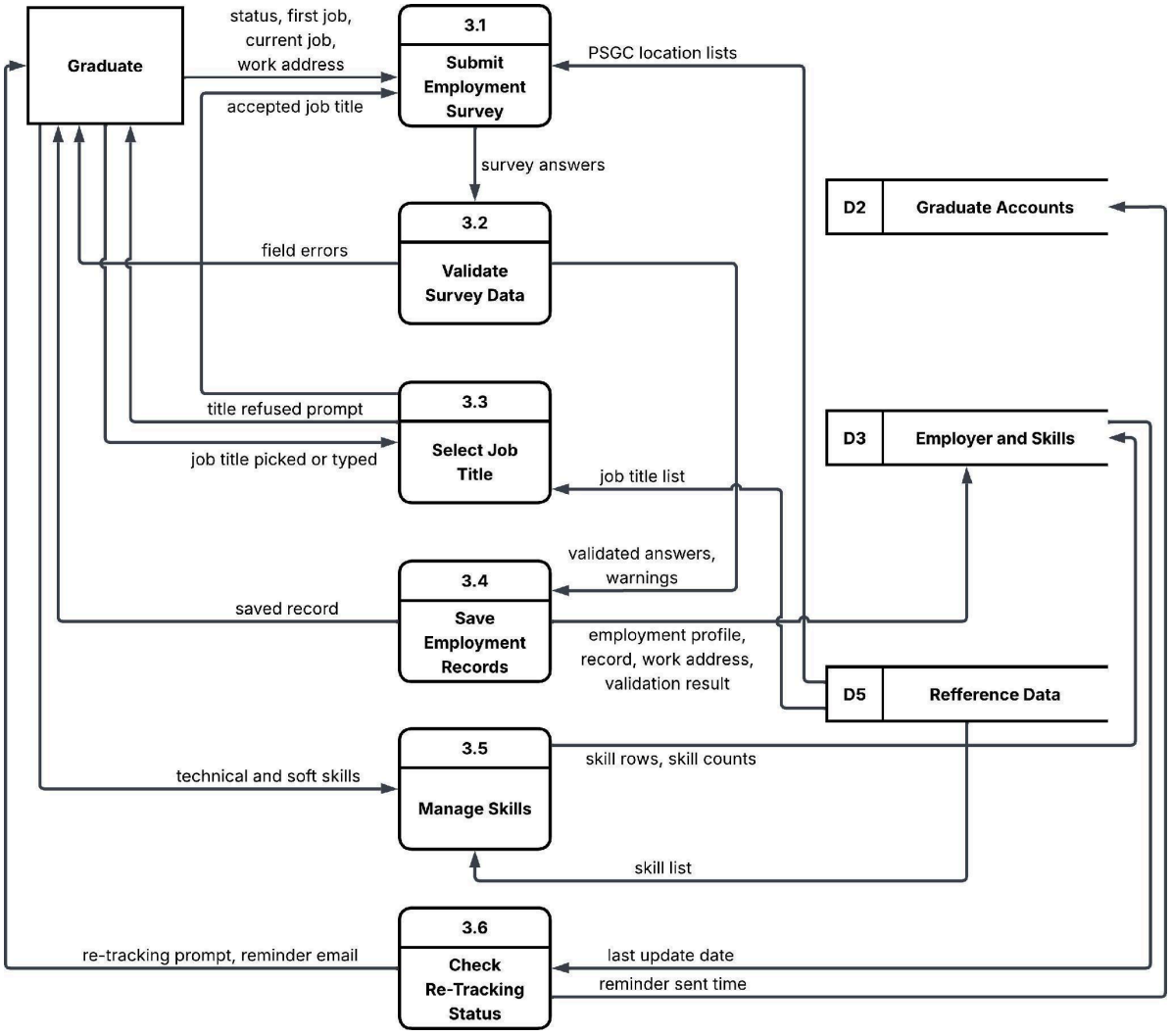
Level 2 Diagram of Manage Graduate Profile



Load Profile (2.1) returns the graduate's saved profile with its master list context. Update Personal Information (2.2) saves names, contact details, and the home address picked from region down to barangay. Resolve Current Location (2.3) turns the device GPS reading into a matching PSGC address and pin that fill the address fields. Update Education and Further Studies (2.4) saves graduation, scholarship, post-graduate, and eligibility details.

Figure 9

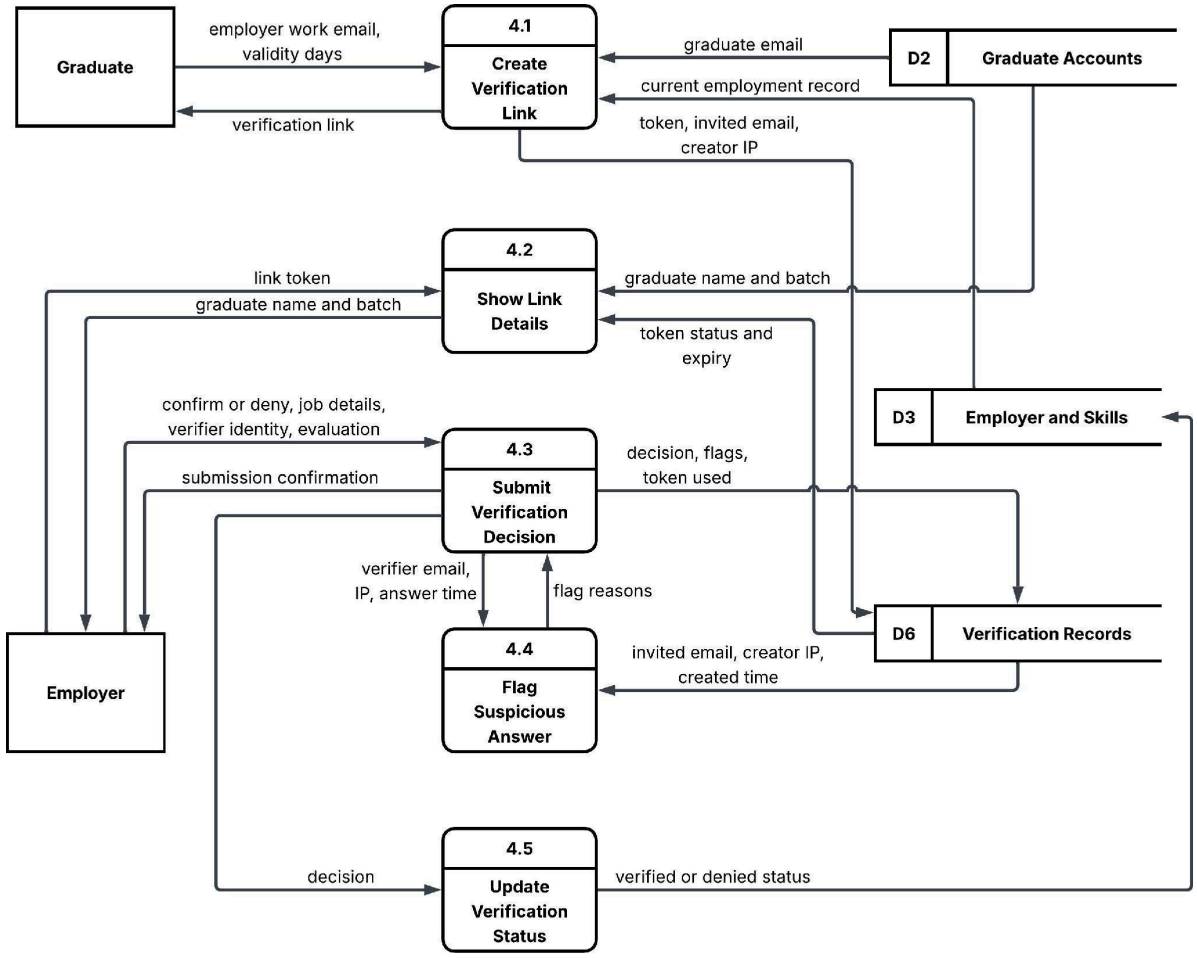
Level 2 Diagram of Manage Employment and Skills



Select Job Title (3.3) lets the graduate pick from the admin list or type a title, which is refused if it is vulgar or nonsense. Submit Employment Survey (3.1) collects status, first job, current job, and work address, skipping steps that do not apply. Validate Survey Data (3.2) rejects impossible answers and returns the graduate to the field at fault. Save Employment Records (3.4) writes the structured rows and the validation result. Manage Skills (3.5) saves the selected skills and counts. Check Re-Tracking Status (3.6) prompts and emails graduates whose record is over two years old.

Figure 10

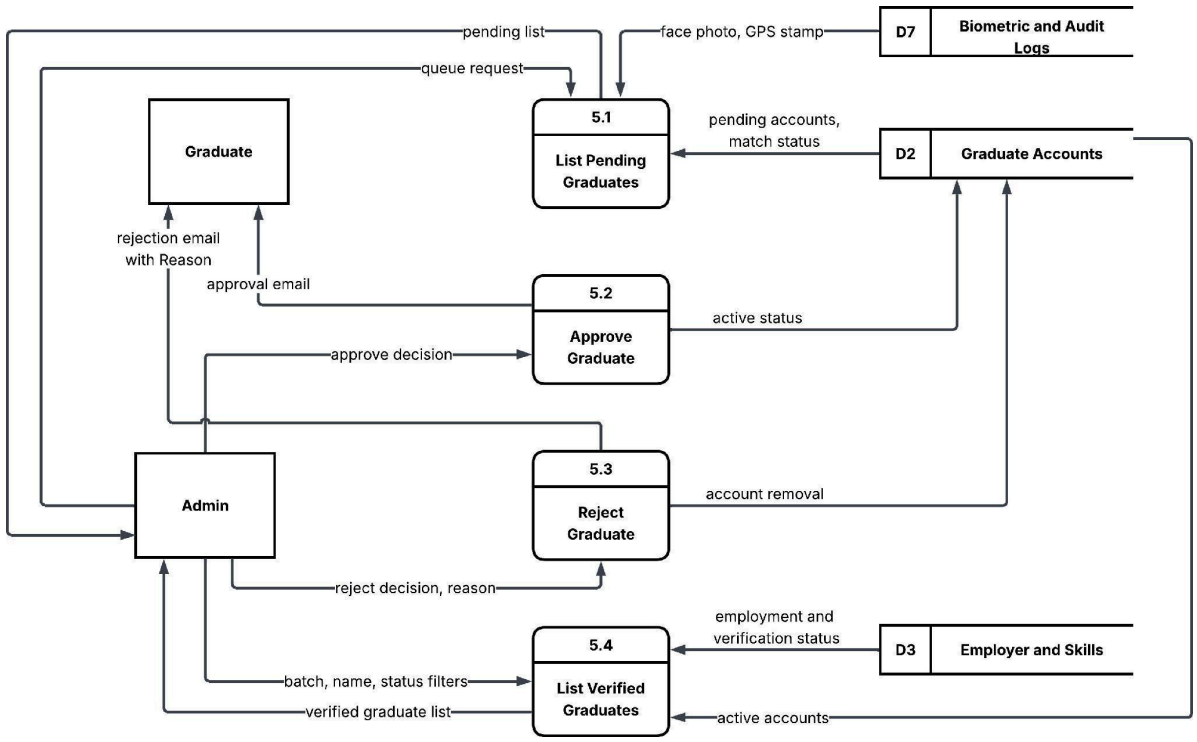
Level 2 Diagram of Verify Graduate Employment



Create Verification Link (4.1) issues a one-time token for the graduate's current record, valid for seven days by default, refuses the graduate's own email, and sends the link. Show Link Details (4.2) shows the employer only the graduate's name, program, and batch, and refuses used or expired links. Submit Verification Decision (4.3) records the answer, the verifier's identity, and the optional evaluation, then marks the token used. Flag Suspicious Answer (4.4) marks answers from a different address than invited, from the device that created the link, or within one minute. Update Verification Status (4.5) sets the record to verified or denied.

Figure 11

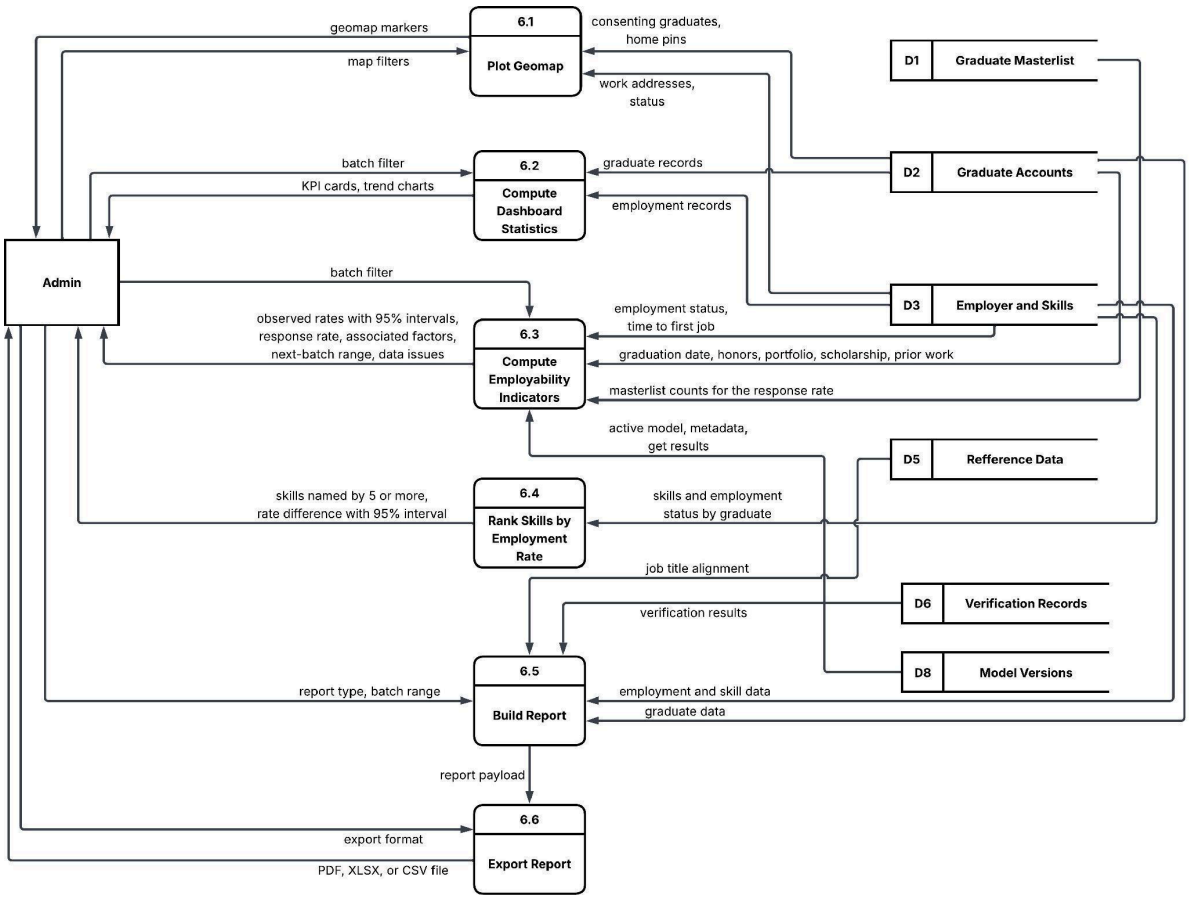
Level 2 Diagram of Review Graduate Accounts



List Pending Graduates (5.1) shows each registration with its photo, GPS stamp, and master list match. Approve Graduate (5.2) activates the account and emails the graduate. Reject Graduate (5.3) emails the reason and removes the account so the email address can register again. List Verified Graduates (5.4) returns active graduates with their employment and verification status, filtered by batch, name, and status.

Figure 12

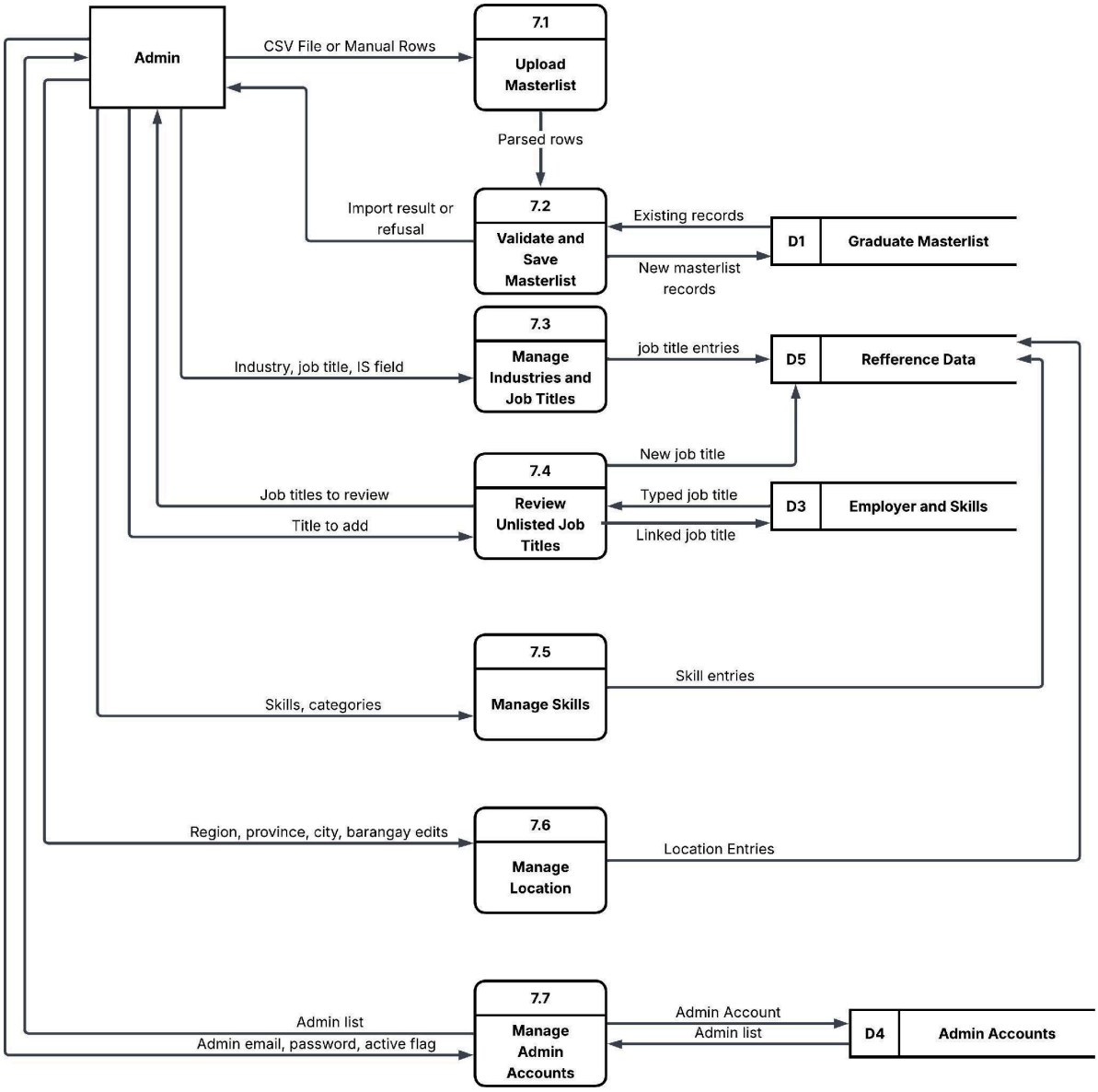
Level 2 Diagram of Generate Analytics and Reports



Plot Geomap (6.1) places only consenting graduates on the map. Compute Dashboard Statistics (6.2) builds the KPI cards and trend charts. Compute Employability Forecast (6.3) runs the trained models and shows predicted beside actual values with a forecast for the next one or two batches. Compute Skill Demand Forecast (6.4) projects the top skills. Build Report (6.5) assembles one of six reports, including curriculum alignment from the job title classification, and Export Report (6.6) saves it as PDF, XLSX, or CSV depending on the report.

Figure 13

Level 2 Diagram of Maintain System Data

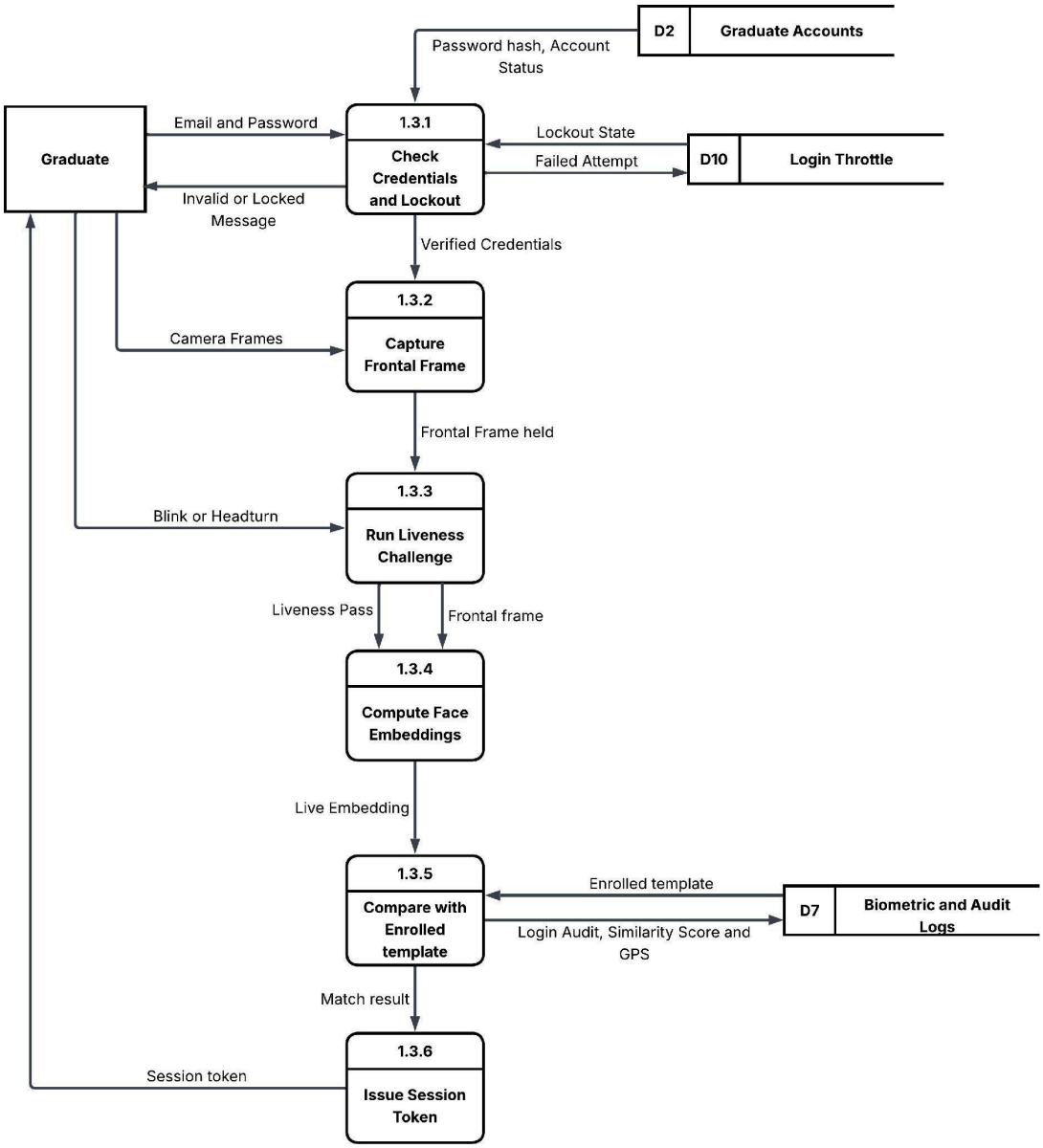


Upload Master List (7.1) reads a CSV or typed rows. Validate and Save (7.2) rejects the whole file if the header, any row, or a duplicate is invalid; otherwise saves. Maintain Master List (7.7) corrects a name or batch year (re-running the roster match so earlier registrants link), retires an entry so it no longer matches registrations, deletes entries no graduate registered against, and exports the roster as CSV. Manage

Industries and Job Titles (7.3) maintains titles and each title's IS field. Review Unlisted Job Titles (7.4) shows graduate-typed titles; adding one links matching employment records. Manage Skills (7.5) and Locations (7.6) maintain the skill and PSGC lists. Manage Admin Accounts (7.7) adds, deactivates, and deletes admins.

Figure 14

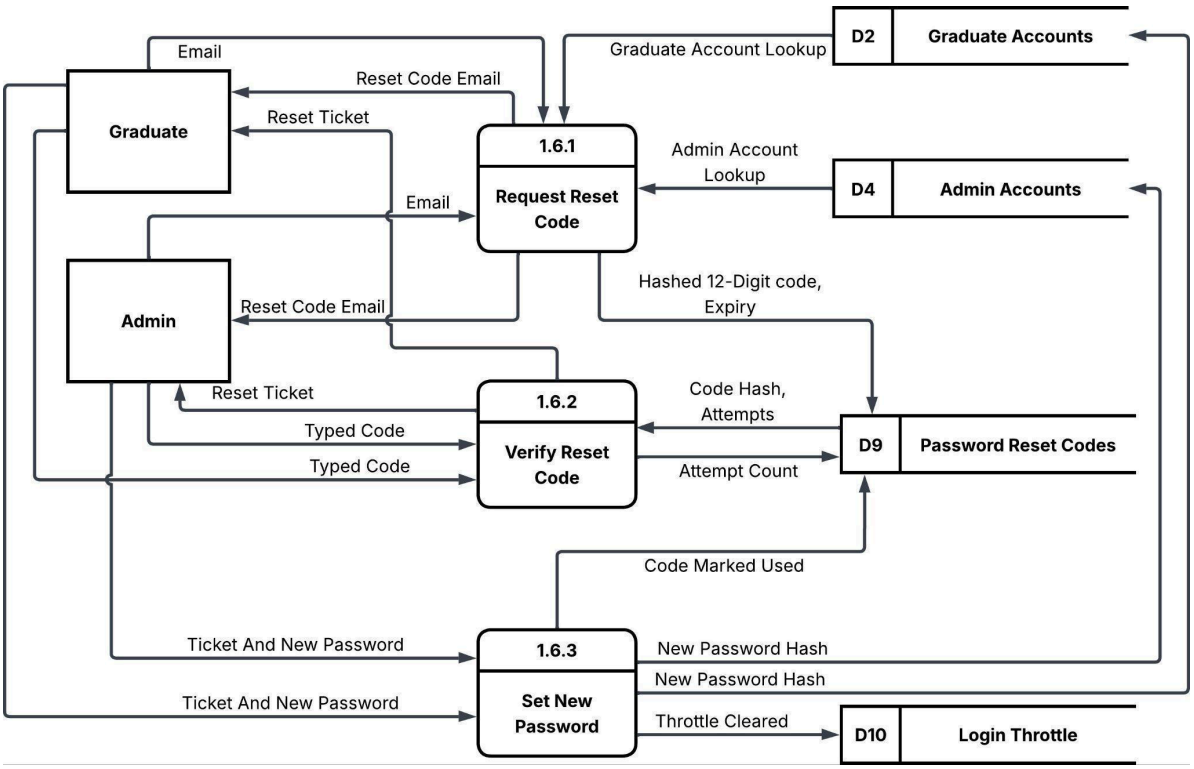
Level 3 Diagram of Authenticate Graduate



Check Credentials and Lockout (1.3.1) validates the password and refuses sign-in during a lockout, which starts after five failures and grows from 15 seconds to 10 minutes. Capture Frontal Frame (1.3.2) holds a straight-on frame before any gesture. Run Liveness Challenge (1.3.3) asks for a blink, then a head turn in a random direction if no blink is detected. Compute Face Embedding (1.3.4) turns the frontal frame into an InsightFace embedding on the server. Compare with Enrolled Template (1.3.5) measures the cosine distance, where 0.40 or less is a match, and saves the login audit. Issue Session Token (1.3.6) signs the graduate in.

Figure 15

Level 3 Diagram of Manage Password Reset

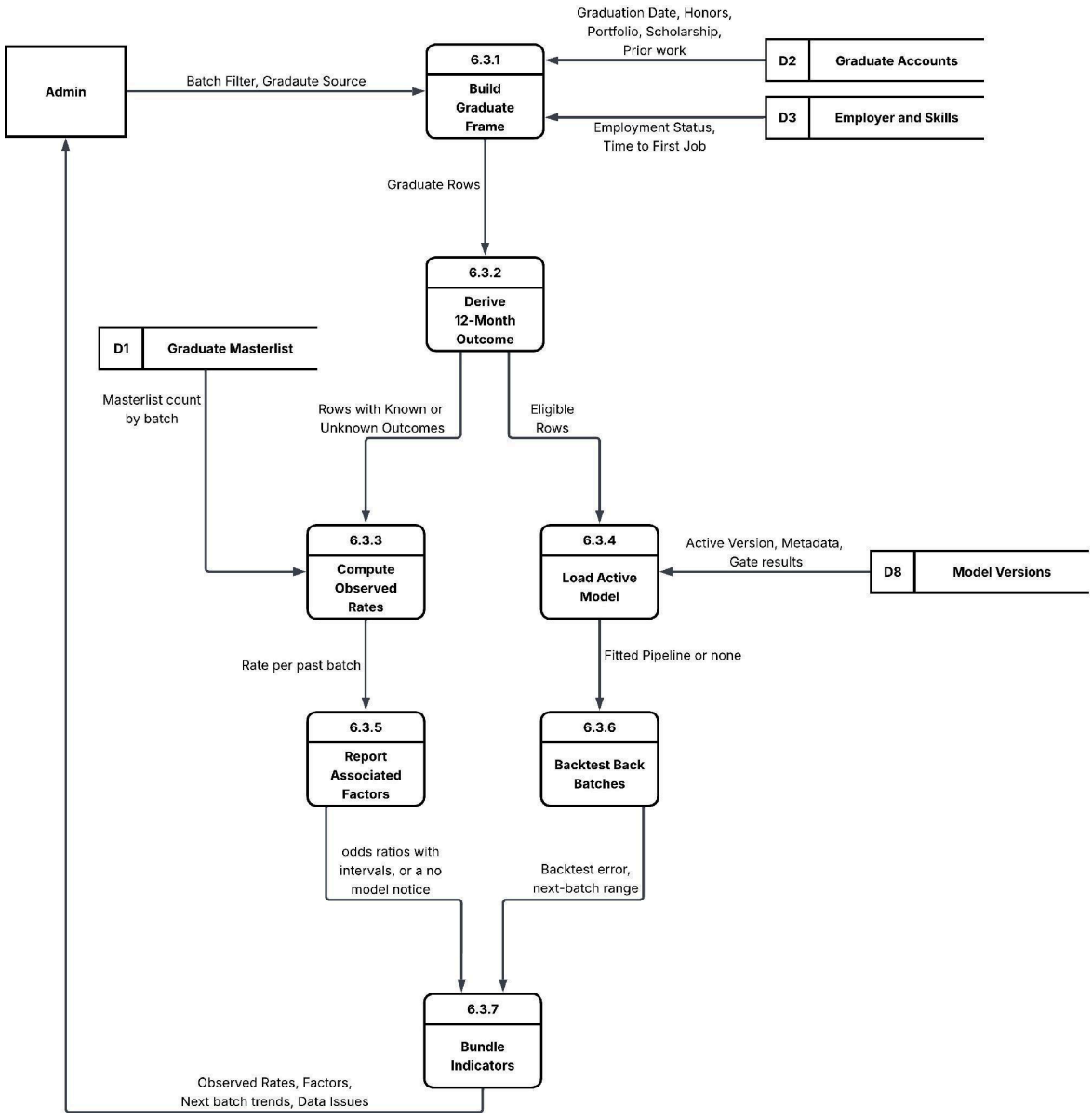


Request Reset Code (1.6.1) finds the graduate or admin account, saves only a hash of a 12-digit code that expires in 15 minutes, and emails the code, with a

60-second resend cooldown. Verify Reset Code (1.6.2) compares the typed code, voids it after five wrong attempts, and returns a reset ticket valid for five minutes. Set New Password (1.6.3) checks the ticket, saves the new password, marks the code used, and clears the login throttle.

Figure 16

Level 3 Diagram of Compute Employability Forecast

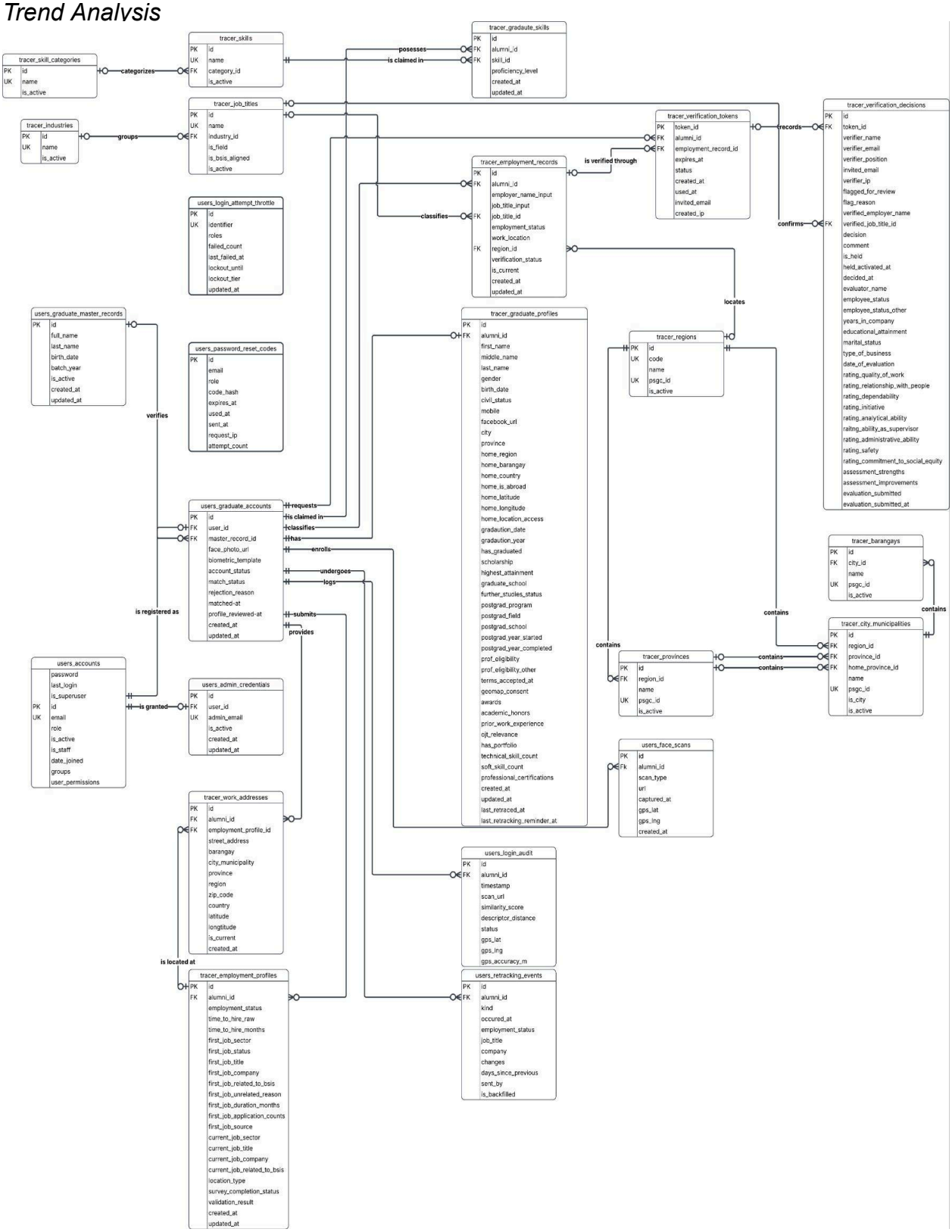


Build Graduate Frame (6.3.1) takes the batch filter and graduate source chosen by the administrator and builds one row per graduate from Graduate Accounts (graduation date, honors, portfolio, scholarship, and prior work) and Employer and Skills (employment status and time to first job). Derive 12-Month Outcome (6.3.2) marks each graduate as employed or not employed within twelve months of graduation, or as unknown while that year has not yet passed. Compute Observed Rates (6.3.3) uses these rows and the master list count of each batch to compute the observed rate of every past batch with a ninety-five percent confidence interval. Load Active Model (6.3.4) reads the active version, its metadata, and its gate results from Model Versions, and returns the fitted pipeline, or none when no model has passed the acceptance gate. Report Associated Factors (6.3.5) gives the odds ratio of each factor with its interval, or a notice that no model is active. Backtest Back Batches (6.3.6) measures the error of the model on earlier batches and gives the likely range for the next batch. Bundle Indicators (6.3.7) returns the observed rates, factors, next-batch trends, and any data issues to the administrator.

3.6 Entity Relationship Diagram

Figure 17

Entity Relationship Diagram of Graduate Tracer System with Predictive Employability Trend Analysis



3.7 Data Dictionary

The data dictionary lists every field of the system's tables with its data type, format, size, description, and example. Table names use the graduate wording of this manuscript; the physical names use "alumni" in place of "graduate".

Table 18

users_accounts

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
email	varchar	Email	254	Login email. Unique.	juan@gmail.com
password	varchar	PBKDF2 hash	128	Hashed password. Plaintext is never stored.	pbkdf2_sha256$...
role	varchar	Enum	20	alumni or admin.	alumni
is_active	boolean	true / false	—	False blocks sign-in.	true
is_staff	boolean	true / false	—	True for admin accounts.	false
is_superuser	boolean	true / false	—	Grants every permission.	false
date_joined	timestamptz	ISO 8601	—	When the account was created.	2026-09-15 T...
last_login	timestamptz	ISO 8601	—	Most recent successful sign-in. Nullable.	2026-09-15 T...

Table 19

users_graduate_master_records

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
full_name	varchar	Text	255	Full name as printed on the roster. Must contain letters and no digits (DB check).	JUAN S. DELA CRUZ
last_name	varchar	Text	120	Last name used with first name and batch year for matching.	DELA CRUZ
birth_date	date	YYYY-MM-DD	—	Optional and informational only. No longer a match key. Nullable.	2000-06-15

batch_year	smallint	Integer	—	Graduation batch year. Must be 2000 to 2100 (DB check). Indexed.	2024
is_active	boolean	true / false	—	Soft-delete flag. An administrator retires an entry from the Batch Upload page; retired entries stay on the list but are hidden from registration matching.	true
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15T ...
updated_at	timestamptz	ISO 8601	—	Auto-updated modification timestamp.	2026-09-15T ...

Table 20

users_graduate_accounts

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
user_id	uuid	UUID v4	36	FK → users_accounts.id. Owning login row. Unique.	550e8400-...
master_record_id	uuid	UUID v4	36	FK → users_graduate_master_records.id. Matched roster row. Nullable when unmatched.	550e8400-...
face_photo_url	varchar	URL	500	Supabase Storage URL of the single frontal registration photo.	https://...
biometric_template	jsonb	JSON object	—	Face embedding tagged with the engine that produced it (InsightFace: 512 values).	{"engine":"insightface", ...}
account_status	varchar	Enum	20	pending, active, rejected, or suspended.	pending
match_status	varchar	Enum	20	pending, matched, unmatched, or broken.	matched
rejection_reason	text	Text	—	Admin reason sent in the rejection email. Blank when not rejected.	Not on the roster
matched_at	timestamptz	ISO 8601	—	When the roster match was confirmed. Nullable.	2026-09-15T...
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15T...
updated_at	timestamptz	ISO 8601	—	Auto-updated modification timestamp.	2026-09-15T...

Table 21

users_graduate_profiles

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
graduate_id	uuid	UUID v4	36	FK → users_graduate_accounts.id. One profile per account.	550e8400-...
first_name	varchar	Text	120	Given name. No digits (DB check).	Juan
middle_name	varchar	Text	120	Full middle name, not an initial. No digits (DB check).	Santos
last_name	varchar	Text	120	Family name. No digits (DB check).	Dela Cruz
gender	varchar	Enum	20	Male, Female, or Other.	Male
birth_date	varchar	YYYY-MM	10	Birth month and year (DB check).	2000-06
civil_status	varchar	Enum	20	Single, Married, Widowed, or Separated.	Single
mobile	varchar	Digits	20	Philippine mobile number.	9171234567
facebook_url	varchar	URL	500	Facebook profile link.	https://fb.com/...
home_region	varchar	Text	120	Home region (PSGC name).	Western Visayas
province	varchar	Text	120	Home province.	Negros Occidental
city	varchar	Text	120	Home city or municipality.	Talisay City
home_barangay	varchar	Text	160	Home barangay.	Zone 12
home_country	varchar	Text	120	Country when living abroad.	Philippines
home_is_abroad	boolean	true / false	—	True when the home address is outside the Philippines.	false
home_latitude	decimal	Decimal(9,6)	—	Exact home pin from "Use my current location". Nullable.	10.736512
home_longitude	decimal	Decimal(9,6)	—	Exact home pin longitude. Nullable.	122.967841
home_location_accuracy_m	float	Meters	—	GPS accuracy of the home pin. Nullable.	18.0
graduation_date	varchar	YYYY-MM	10	Graduation month and year (DB check).	2024-06
graduation_year	smallint	Integer	—	Year derived from graduation_date.	2024

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
has_graduated	boolean	true / false		False only for a graduating student who registered before graduating. Defaults to True	true
scholarship	varchar	Text	120	Scholarship held in college, or blank.	DOST Scholar
highest_attainment	varchar	Enum	20	NA, Graduate, PostGrad, or Doctorate.	Graduate
graduate_school	varchar	Text	255	Graduate school, if any.	CHMSU
further_studies_status	varchar	Enum	20	none, enrolled, or completed.	enrolled
postgrad_program	varchar	Text	200	Post-graduate program name.	MIT
postgrad_field	varchar	Text	200	Post-graduate field of study.	Information Technology
postgrad_school	varchar	Text	255	Post-graduate school.	UP Diliman
postgrad_year_started	smallint	Integer	—	Year started. Nullable.	2025
postgrad_year_completed	smallint	Integer	—	Year completed. Nullable.	2027
prof_eligibility	varchar	Comma list	255	Professional eligibilities.	CSE, TESDA
prof_eligibility_other	varchar	Text	255	Other eligibility not in the list.	CCNA
terms_accepted_at	timestamptz	ISO 8601	—	When the graduate accepted the Terms and Data Privacy notice.	2026-09-15 T...
geomap_consent	boolean	true / false	—	Separate opt-in to appear on the admin geomap. Default false.	true
awards	text	Text	—	Awards received.	Dean's Lister
academic_honors	integer	Encoded 1-4	—	1 None, 2 Cum Laude, 3 Magna, 4 Summa.	2
prior_work_experience	boolean	true / false	—	Worked before graduating.	false
ojt_relevance	integer	Encoded 0-3	—	0 N/A, 1 not related, 2 somewhat, 3 directly related.	3
has_portfolio	boolean	true / false	—	Keeps a professional portfolio.	true
technical_skill_count	integer	0 to 12	—	Count of technical skills picked. Model input.	5
soft_skill_count	integer	0 to 10	—	Count of soft skills picked. Model input.	4
professional_certifications	jsonb	JSON array	—	Certification names.	["AWS CCP"]
last_retracking_reminder_at	timestamptz	ISO 8601	—	Last two-year re-tracking reminder email. Nullable.	2026-09-15 T...

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15 T...
updated_at	timestamptz	ISO 8601	—	Auto-updated modification timestamp.	2026-09-15 T...

Table 22

users_face_scans

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
graduate_id	uuid	UUID v4	36	FK → users_graduate_accounts.id. Owner of the scan.	550e8400-...
scan_type	varchar	Enum	20	face_front (registration) or login. face_left and face_right remain on older rows.	face_front
url	varchar	URL	500	Supabase Storage URL of the image.	https://...
captured_at	timestamptz	ISO 8601	—	Browser capture time. Nullable.	2026-09-15 T...
gps_lat	decimal	Decimal(9,6)	—	Latitude at capture. Nullable.	10.719300
gps_lng	decimal	Decimal(9,6)	—	Longitude at capture. Nullable.	122.962400
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15 T...

Table 23

users_login_audits

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
graduate_id	uuid	UUID v4	36	FK → users_graduate_accounts.id. Graduate who signed in.	550e8400-...
timestamp	timestamptz	ISO 8601	—	Date and time of the attempt.	2026-09-15 T...

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
scan_url	varchar	URL	500	Login scan image. Nullable.	https://...
similarity_score	float	0.0 to 1.0	—	Similarity between live and enrolled face. Nullable.	0.82
descriptor_distance	float	Distance	—	Engine distance (cosine for InsightFace). Nullable.	0.21
status	varchar	Enum	20	success or failed.	success
gps_lat	decimal	Decimal(9,6)	—	Latitude at sign-in. Nullable.	10.719300
gps_lng	decimal	Decimal(9,6)	—	Longitude at sign-in. Nullable.	122.962400
gps_accuracy_m	float	Meters	—	Reported GPS accuracy. Nullable.	12.5

Table 24

users_admin_credentials

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
user_id	uuid	UUID v4	36	FK → users_accounts.id. Owning login row. Unique.	550e8400-...
admin_email	varchar	Email	254	Admin email. Unique.	chair@chmsu.edu.ph
is_active	boolean	true / false	—	Disabled admins cannot sign in.	true
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15T...
updated_at	timestamptz	ISO 8601	—	Auto-updated modification timestamp.	2026-09-15T...

Table 25

users_password_reset_codes

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
email	varchar	Email	254	Address the code was sent to. Indexed.	juan@gmail.com
role	varchar	Enum	10	graduate or admin	graduate

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
code_hash	varchar	SHA-256 hex	128	Hash of the 12-digit code.	a3f5...
expires_at	timestamptz	ISO 8601	—	Code expiry, 15 minutes after issue.	2026-09-15 T...
used_at	timestamptz	ISO 8601	—	When the code was consumed. Nullable.	2026-09-15 T...
sent_at	timestamptz	ISO 8601	—	When the email was sent. 60-second resend cooldown.	2026-09-15 T...
request_ip	inet	IPv4 / IPv6	39	Requesting IP. Nullable.	203.0.113.4
attempt_count	smallint	Integer	—	Wrong attempts. The code is voided after 5.	0

Table 26

users_login_attempt_throttle

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	bigint	Auto increment	—	PK.	1024
identifier	varchar	role:email	320	Throttle key. Unique.	graduate:juan@gmail.com
role	varchar	Enum	10	graduate or admin.	graduate
failed_count	integer	Integer	—	Consecutive failures. Lockout starts at 5.	3
last_failed_at	timestamptz	ISO 8601	—	Most recent failure. Nullable.	2026-09-15 T...
lockout_until	timestamptz	ISO 8601	—	When the lockout lifts. Nullable.	2026-09-15 T...
lockout_tier	smallint	0 to 6	—	1 to 6 map to 15, 30, 60, 120, 300, 600 seconds.	2
updated_at	timestamptz	ISO 8601	—	Auto-updated modification timestamp.	2026-09-15 T...

Table 27

tracer_industries

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
name	varchar	Text	120	Industry name. Unique. Must contain letters (DB check).	IT and BPO
is_active	boolean	true / false	—	Soft-delete flag. Inactive rows are hidden from selectors.	true

Table 28

tracer_job_titles

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
name	varchar	Text	150	Job title. Unique. Must contain letters (DB check).	Systems Analyst
industry_id	uuid	UUID v4	36	FK → tracer_industries.id. Parent industry. Nullable.	550e8400-...
is_field	varchar	Enum	20	software_dev, data_db, network_infra, it_support, systems_analysis, or non_is. Null = unclassified.	systems_analysis
is_bsis_aligned	boolean	true / false / null	—	Set automatically from is_field. Null = unknown.	true
is_active	boolean	true / false	—	Soft-delete flag. Inactive rows are hidden from selectors.	true

Table 29

tracer_skill_categories

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
name	varchar	Text	120	Category name. Unique.	Web Development
is_active	boolean	true / false	—	Soft-delete flag. Inactive rows are hidden from selectors.	true

Table 30

tracer_skills

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
name	varchar	Text	120	Skill name. Unique.	Python
category_id	uuid	UUID v4	36	FK → tracer_skill_categories.id. Parent category. Nullable.	550e8400-...
is_active	boolean	true / false	—	Soft-delete flag. Inactive rows are hidden from selectors.	true

Table 31

tracer_regions

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
code	varchar	Text	40	Short region code. Unique.	R6
name	varchar	Text	120	PSGC region name. Unique.	Western Visayas
psgc_id	varchar	PSGC code	12	PSA identifier used by the in-place PSGC sync.	060000000
is_active	boolean	true / false	—	Soft-delete flag. Inactive rows are hidden from selectors.	true

Table 32

tracer_provinces

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
region_id	uuid	UUID v4	36	FK → tracer_regions.id. Parent region.	550e8400-...
name	varchar	Text	120	Province name. Unique within its region.	Negros Occidental

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
psgc_id	varchar	PSGC code	12	PSA identifier. Unique.	0604500000
is_active	boolean	true / false	—	Soft-delete flag. Inactive rows are hidden from selectors.	true

Table 33

tracer_city_municipalities

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
region_id	uuid	UUID v4	36	FK → tracer_regions.id. Parent region.	550e8400-...
province_id	uuid	UUID v4	36	FK → tracer_provinces.id. Parent province. Null for NCR and highly urbanized cities.	550e8400-...
home_province_id	uuid	UUID v4	36	FK → tracer_provinces.id. Province a highly urbanized city is listed under. Nullable.	550e8400-...
name	varchar	Text	160	City or municipality name.	City of Talisay
psgc_id	varchar	PSGC code	12	PSA identifier. Unique.	0604529000
is_city	boolean	true / false	—	True for a city, false for a municipality.	true
is_active	boolean	true / false	—	Soft-delete flag. Inactive rows are hidden from selectors.	true

Table 34

tracer_barangays

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
city_id	uuid	UUID v4	36	FK → tracer_city_municipalities.i	550e8400-...

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
				d. Parent city or municipality.	
name	varchar	Text	160	Barangay name.	Zone 12
psgc_id	varchar	PSGC code	12	PSA identifier. Unique.	0604529012
is_active	boolean	true / false	—	Soft-delete flag. Inactive rows are hidden from selectors.	true

Table 35

tracer_employment_records

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
graduate_id	uuid	UUID v4	36	FK → users_graduate_accounts.id. Owner of the record.	550e8400-...
employer_name_input	varchar	Text	255	Company name typed by the graduate.	ABC Tech Corp
job_title_input	varchar	Text	255	Job title as picked or typed. Typed titles pass a quality check.	Systems Analyst
job_title_id	uuid	UUID v4	36	FK → tracer_job_titles.id. Linked list title. Set later when an admin adds an unlisted title. Nullable.	550e8400-...
employment_status	varchar	Enum	20	employed, self_employed, or unemployed.	employed
work_location	varchar	Text	255	Display text of the work location.	Bacolod City
region_id	uuid	UUID v4	36	FK → tracer_regions.id. Work region. Nullable.	550e8400-...
verification_status	varchar	Enum	20	pending, verified, or denied.	verified
is_current	boolean	true / false	—	One current record per graduate (unique constraint).	true
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15T...
updated_at	timestamptz	ISO 8601	—	Auto-updated modification timestamp.	2026-09-15T...

Table 36

tracer_graduate_skills

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
graduate_id	uuid	UUID v4	36	FK → users_graduate_accounts.id. Owner.	550e8400-...
skill_id	uuid	UUID v4	36	FK → tracer_skills.id. Skill held. Unique per graduate.	550e8400-...
proficiency_level	varchar	Enum	20	beginner, intermediate, advanced, or expert.	intermediate
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15T...
updated_at	timestamptz	ISO 8601	—	Auto-updated modification timestamp.	2026-09-15T...

Table 37

tracer_verification_tokens

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
token_id	uuid	UUID v4	36	PK. Appears in the /verify/<token> link.	550e8400-...
graduate_id	uuid	UUID v4	36	FK → users_graduate_accounts.id. Graduate being verified.	550e8400-...
employment_record_id	uuid	UUID v4	36	FK → tracer_employment_records.id. Record being verified. Nullable.	550e8400-...
expires_at	timestamptz	ISO 8601	—	Expiry. Default 7 days, allowed 1 to 30.	2026-09-15T...
status	varchar	Enum	20	pending, used, expired, or revoked.	pending
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15T...
used_at	timestamptz	ISO 8601	—	When a decision consumed the link. Nullable.	2026-09-15T...
invited_email	varchar	Email	254	Employer address the graduate entered. Cannot be the graduate's own.	hr@company.com

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
created_ip	inet	IPv4 / IPv6	39	IP that created the link. Used for the same-device flag. Nullable.	203.0.113.4

Table 38

tracer_verification_decisions

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
token_id	uuid	UUID v4	36	FK → tracer_verification_tokens.token_id. Link that was answered.	550e8400-...
verifier_name	varchar	Text	200	Name of the person who answered.	Maria Santos
verifier_email	varchar	Email	254	Work email of the verifier. Must differ from the graduate's. Indexed.	maria@company.com
verifier_position	varchar	Text	200	Position of the verifier.	HR Manager
invited_email	varchar	Email	254	Address the link was sent to.	hr@company.com
verifier_ip	inet	IPv4 / IPv6	39	IP of the answer. Nullable.	198.51.100.7
flagged_for_review	boolean	true / false	—	True when any soft flag fired. Indexed.	false
flag_reason	varchar	Text	255	Different address than the invite; same device as the link; answered under one minute.	—
verified_employer_name	varchar	Text	255	Employer name as confirmed.	ABC Tech Corp
verified_job_title_id	uuid	UUID v4	36	FK → tracer_job_titles.id. Job title confirmed. Nullable.	550e8400-...
decision	varchar	Enum	20	confirm or deny.	confirm
comment	text	Text	—	Optional comment.	Confirmed.
is_held	boolean	true / false	—	Legacy hold flag. Always false for link-based decisions.	false
held_activated_at	timestamptz	ISO 8601	—	Legacy. Nullable.	2026-09-15T...
decided_at	timestamptz	ISO 8601	—	Auto-set when the decision is saved.	2026-09-15T...

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
evaluator_name	varchar	Text	255	Evaluator named on the feedback form.	Maria Santos
employee_status	varchar	Enum	30	regular, probationary_casual_jo, or other.	regular
employee_status_other	varchar	Text	100	Used when employee_status is 'other'.	Consultant
years_in_company	smallint	Integer	—	Graduate's years with the company. Nullable.	2
educational_attainment	varchar	Text	100	From the feedback form.	Bachelor's
marital_status	varchar	Text	20	From the feedback form.	Single
type_of_business	varchar	Text	150	Employer's type of business.	IT services
date_of_evaluation	date	YYYY-MM-DD	—	Date of the evaluation. Nullable.	2026-09-15
rating_* (11 fields)	varchar	Enum	20	quality_of_work, work_habits, relationship_with_people, dependability, quantity_of_work, initiative, analytical_ability, ability_as_supervisor, administrative_ability, safety, commitment_to_social_equity. Scale: excellent, very_good, good, fair, unsatisfactory.	very_good
assessment_strengths	text	Text	—	Open-ended strengths.	Reliable...
assessment_improvements	text	Text	—	Open-ended improvements.	Could improve...
evaluation_submitted	boolean	true / false	—	True when the evaluation form was filled in.	true
evaluation_submitted_at	timestamptz	ISO 8601	—	When the evaluation was submitted. Nullable.	2026-09-15T...

Table 39

tracer_employment_profiles

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
graduate_id	uuid	UUID v4	36	FK → users_graduate_accounts. id. Owner.	550e8400-. ..
employment_status	varchar	Enum	20	employed_full_time, employed_part_time, self_employed, seeking, or not_seeking.	employed_f ull_time
time_to_hire_raw	varchar	Text	50	Answer as picked.	1-3 months
time_to_hire_month s	float	Months	—	Must be 1, 3, 4.5, 9, 18, or 30 (DB check). Null if never employed.	3
first_job_sector	varchar	Enum	20	government, private, or entrepreneurial.	private
first_job_status	varchar	Enum	30	regular, probationary, contractual, or self_employed.	regular
first_job_title	varchar	Text	150	First job title.	Junior Developer
first_job_related_to_ bsis	boolean	true / false	—	Graduate-reported relevance. Nullable.	true
first_job_unrelated_r eason	varchar	Text	200	Reason when unrelated.	Salary
first_job_duration_m onths	integer	Integer	—	Months in the first job.	18
first_job_application s_count	integer	Encoded 1-4	—	1 = 1-5, 2 = 6-15, 3 = 16-30, 4 = 31+.	1
first_job_source	varchar	Enum	50	personal_network, online_portal, career_fair, walk_in, social_media, entrepreneurship, or other.	online_port al
current_job_sector	varchar	Text	20	Sector of the current job.	private
current_job_title	varchar	Text	150	Current job title.	Systems Analyst
current_job_compan y	varchar	Text	200	Current company.	ABC Tech Corp
current_job_related_ to_bsis	boolean	true / false	—	Graduate-reported relevance. Nullable.	true
location_type	boolean	true / false	—	True = Philippines, false = abroad. Nullable.	true
survey_completion_ status	varchar	Enum	20	pending or completed.	completed
validation_result	jsonb	JSON object	—	Intake check result: status, completeness score, warnings. Nullable.	{"status":"v alid",...}
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15 T...
updated_at	timestamptz	ISO 8601	—	Auto-updated modification timestamp.	2026-09-15 T...

Table 40

tracer_work_addresses

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
graduate_id	uuid	UUID v4	36	FK → users_graduate_accounts.id. Owner.	550e8400-...
employment_profile_id	uuid	UUID v4	36	FK → tracer_employment_profiles.id. Linked profile. Nullable.	550e8400-...
street_address	varchar	Text	200	Street and number.	123 Lacson St
barangay	varchar	Text	100	Barangay from the PSGC list.	Villamonte
city_municipality	varchar	Text	100	City or municipality.	Bacolod City
province	varchar	Text	100	Province.	Negros Occidental
region	varchar	Enum	50	Philippine region or Abroad.	Region VI
zip_code	varchar	Postal code	20	ZIP code.	6100
country	varchar	Text	100	Country. Default Philippines.	Philippines
latitude	float	Degrees	—	-90 to 90 (DB check). Nullable.	10.6765
longitude	float	Degrees	—	-180 to 180 (DB check). Nullable.	122.9509
is_current	boolean	true / false	—	Current work address.	true
created_at	timestamptz	ISO 8601	—	Auto-set creation timestamp.	2026-09-15 T...

Table 41

tracer_competency_profiles

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
id	uuid	UUID v4	36	PK. Auto-generated unique identifier.	550e8400-...
graduate_id	uuid	UUID v4	36	FK → users_graduate_accounts.id. Owner.	550e8400-...

FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
technical_skills	jsonb	JSON array	—	Technical skills picked.	[{"name":"Python",...}]
soft_skills	jsonb	JSON array	—	Soft skills picked.	[{"name":"Teamwork",...}]
technical_skill_count	integer	0 to 12	—	Count of technical skills.	5
soft_skill_count	integer	0 to 10	—	Count of soft skills.	4
professional_certifications	varchar	Text	500	Comma-separated certifications.	AWS CCP
assessment_date	timestamptz	ISO 8601	—	Auto-updated on save.	2026-09-15T...

Hardware Requirements

1. Processor: Dual-core 1.6 GHz or higher (Intel Core i3, AMD Ryzen 3, or equivalent); quad-core for smartphones
2. Memory (RAM): 4 GB for desktop or laptop; 2 GB for tablet or smartphone
3. Storage: 500 MB free space (browser cache and the Android app)
4. Graphics (GPU): Integrated graphics; no dedicated GPU needed
5. Display: 1366 × 768 for desktop or laptop; 5-inch HD screen for smartphone
6. Camera: 720p front-facing camera or webcam (graduate registration and sign-in)
7. Location: GPS or browser geolocation (optional)
8. Network: Wi-Fi or mobile data, at least 5 Mbps

Software Requirements

1. Operating System: Windows 10 or later, macOS 11 or later, Android 8.0 or later, iOS 15 or later
2. Web Browser: Latest Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari
3. Email: A working email account for verification, notifications, and password reset.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results and discussions based on the development of the Graduate Tracer System with Predictive Employability Trend Analysis, aligned with the research objectives. It includes screenshots of the system's interfaces and modules, the results of functionality testing conducted through black-box and white-box methods at both the alpha and beta testing stages, as well as the evaluation results using the ISO/IEC 25010:2011 Software Quality Model, which outlines key software quality characteristics and their sub-characteristics. Data was gathered from the identified respondents using a five-point Likert scale, with responses ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The evaluation covers the eight quality characteristics defined by the standard: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

This chapter also discusses the findings of the system evaluation conducted for the Graduate Tracer System with Predictive Employability Trend Analysis. The assessment utilized a standardized questionnaire based on the ISO/IEC 25010:2011 model, administered to the intended user groups including graduates and IT experts who interacted with the developed system. The results provide an overview of the system's effectiveness, usability, and overall quality. Furthermore, this chapter explains the implications of the findings, highlighting areas of strength and identifying opportunities for future improvements and system enhancements.

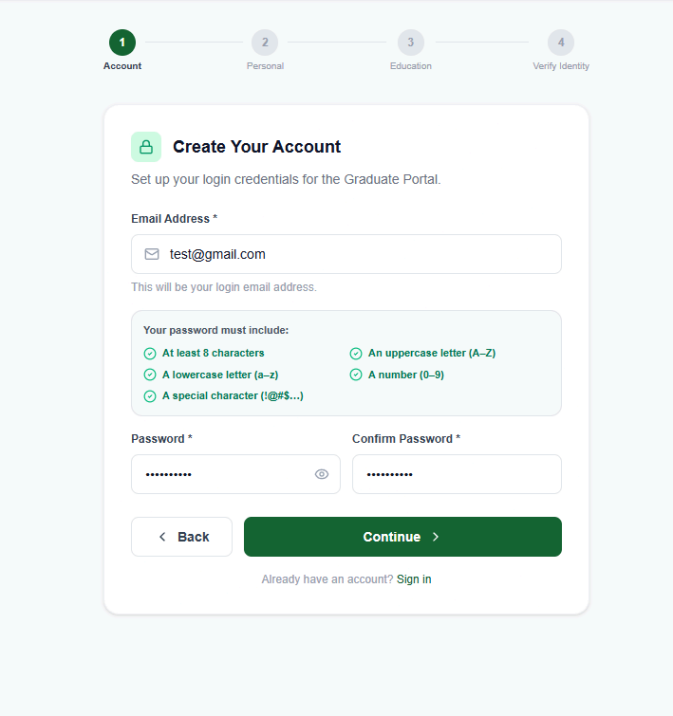
4.1 Results of the Study

The first objective of this study was to design and develop a Graduate Tracer System with Predictive Employability Trend Analysis with the following features: face recognition verification, predictive employability trend analysis, geographic mapping of graduate employment distribution, an employer verification portal, and an automatic two-year re-tracking gate.

The result of this objective was the successful development of the Graduate Tracer System with Predictive Employability Trend Analysis, which integrated the intended features. The system is capable of registering and verifying graduate accounts through biometric face verification, recording and updating employment and skills information, visualizing graduate employment distribution on a geographic map, allowing employers to confirm employment through a link-based portal, and enforcing re-tracking of graduates after two years.

Figure 18

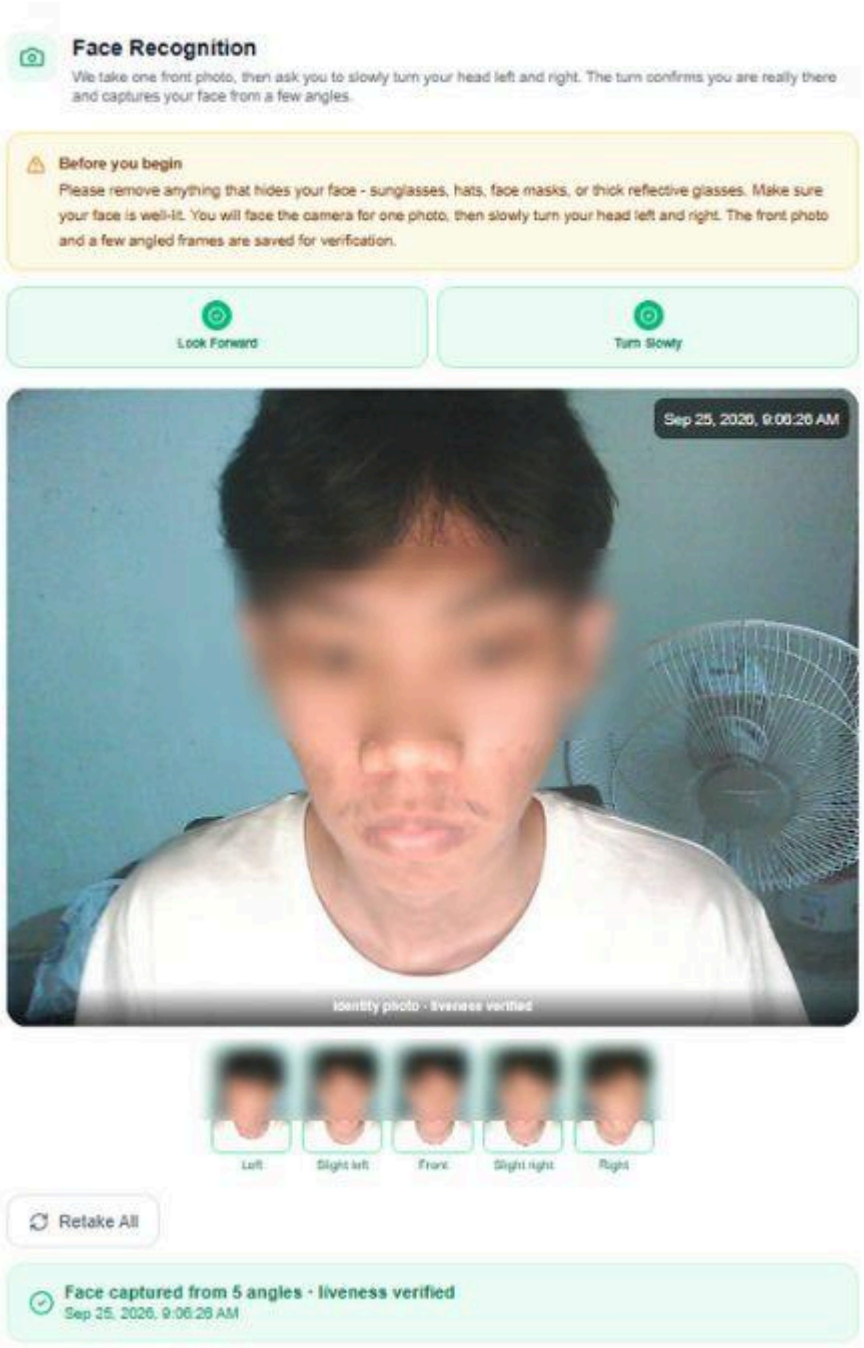
Graduate Registration Interface



This figure shows the Graduate Registration interface of the Graduate Tracer System with Predictive Employability Trend Analysis. The entry point for BSIS graduates. It collects personal, educational, and contact information through cascading address dropdowns, enforces password and mobile number validation, requires terms acceptance, offers a separate geographic mapping consent, and captures the graduate's face for biometric enrollment.

Figure 19

Face Verification Interface

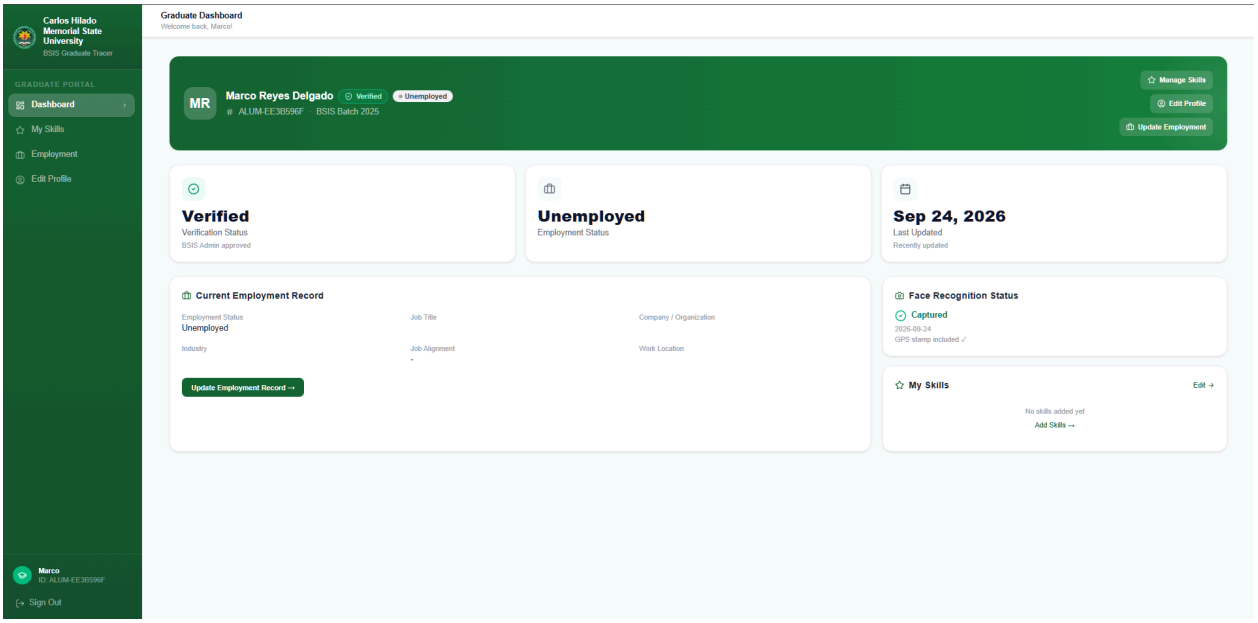


This figure displays the Face Recognition Sign-In interface of the Graduate Tracer System with Predictive Employability Trend Analysis. This page allows a

registered graduate to sign in by confirming their identity through biometric face verification rather than by password alone. After the graduate's credentials are accepted, the interface activates the device camera and guides the user through a short liveness challenge that requires a frontal pose, a blink, and a head turn, ensuring that a live person is present rather than a photograph. The interface refuses blurred or low-quality frames with a prompt to retake, and an unanswered challenge times out cleanly instead of leaving the screen unresponsive. Once the captured face matches the stored descriptor, the graduate is granted access to their account, providing a secure and convenient method of authentication suited to the tracer system's purpose of confirming graduate identity.

Figure 20

Graduate Dashboard Interface

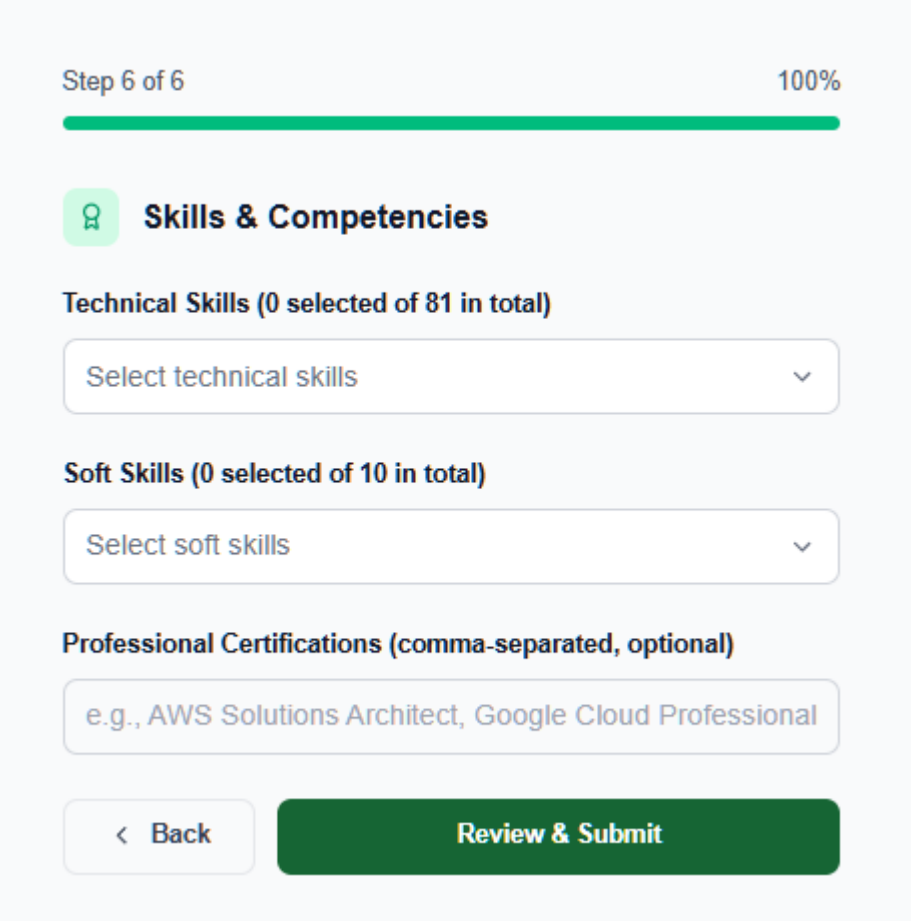


This figure shows the Graduate Dashboard interface of the Graduate Tracer System with Predictive Employability Trend Analysis. This page is the main area a

graduate reaches after signing in, presenting a summary of their profile and providing access to the actions available to them. From this dashboard, the graduate can view and update their personal and educational information, complete or revise the employment survey, and generate a verification link for an employer. Through its organized layout, the dashboard serves as the central point from which graduates manage their participation in the tracer system without navigating through unrelated pages.

Figure 21

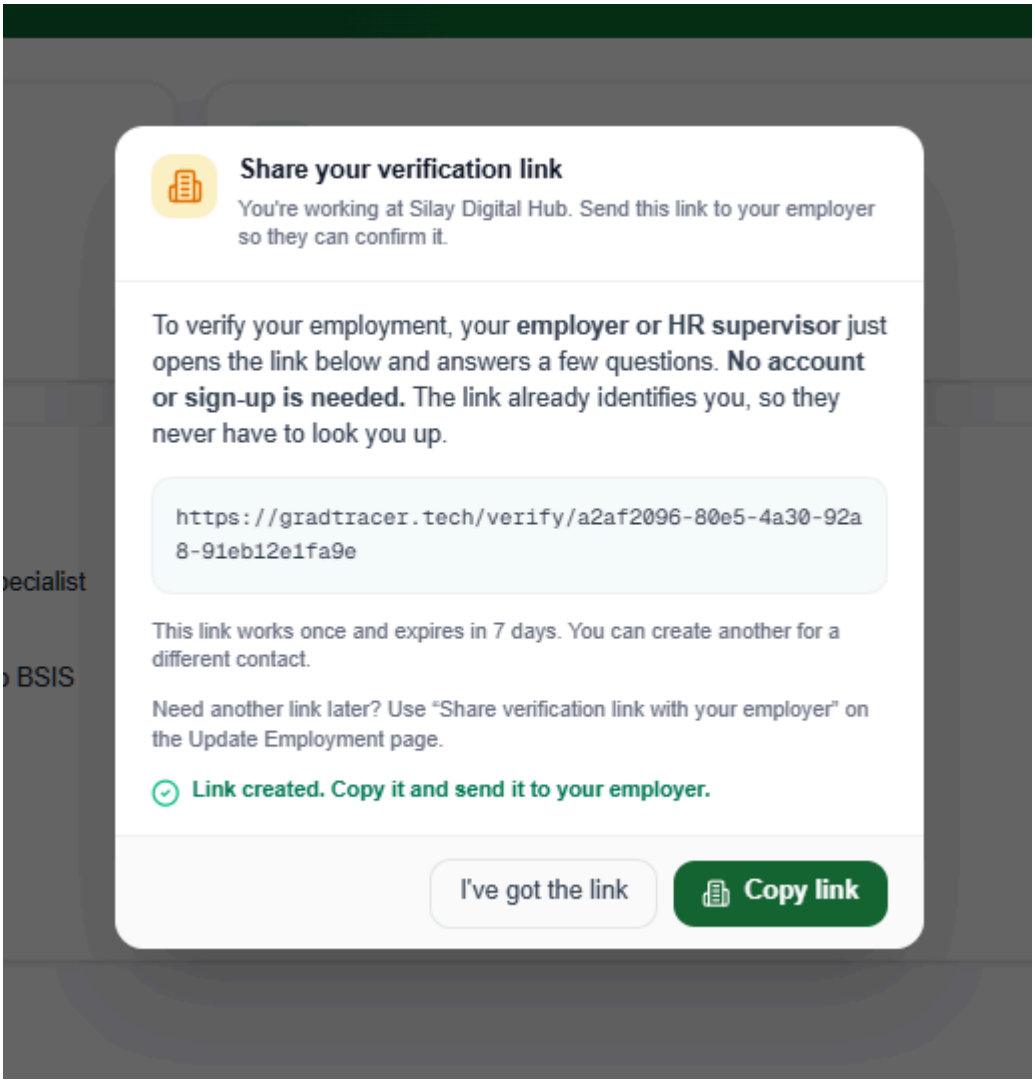
Employment Survey Interface



This figure presents the Employment Survey interface of the Graduate Tracer System with Predictive Employability Trend Analysis. This module records the employment information used by the system's analytics and forecasting features. The survey is presented as a series of steps, and the questions shown adapt to the graduate's employment status: a graduate who reports being employed is presented with all steps covering first job, current job, and work address, while a graduate who reports never having been employed is taken directly to the skills step, omitting the questions for which no data could exist. The interface allows the graduate to select every technical and soft skill that applies, with no limit on the number chosen. Should the graduate return to an earlier answer and change their employment status, the steps that were previously skipped are correctly reopened, ensuring the collected data remains consistent with the graduate's actual situation.

Figure 22

Employer Verification Landing Page Interface

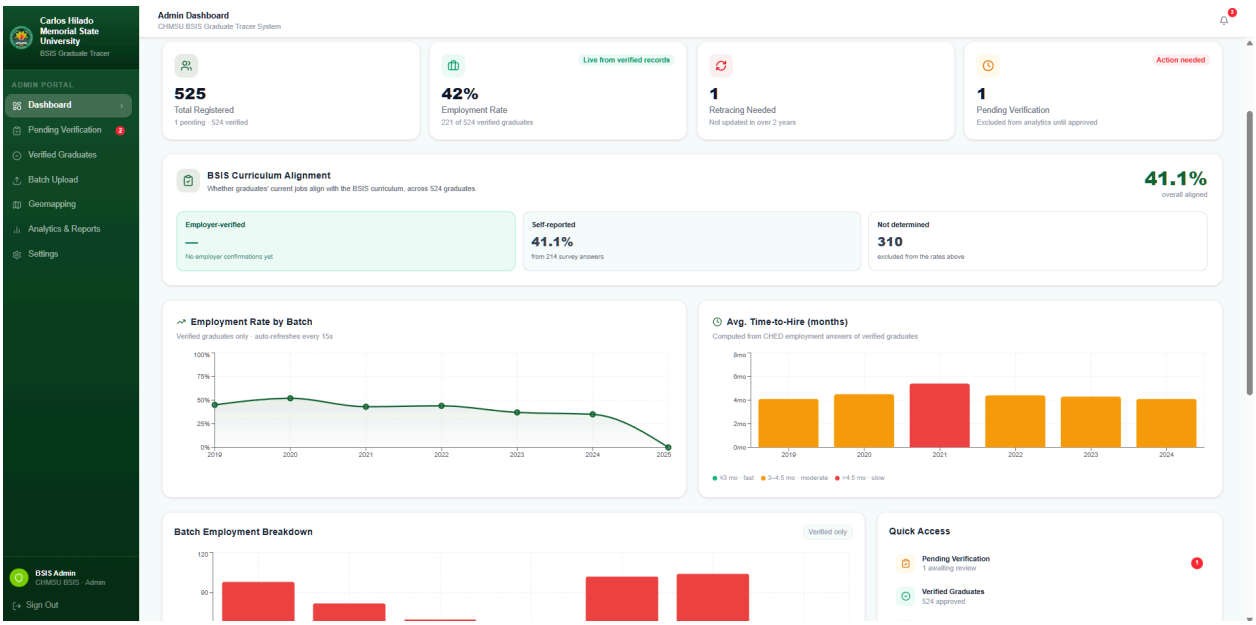


This figure displays the Employer Verification Landing Page of the Graduate Tracer System with Predictive Employability Trend Analysis. This page is opened by an employer through a single-use link generated by the graduate, and it does not require the employer to create an account or sign in. The page presents the employment claim the graduate has made and provides a form through which the employer confirms whether the graduate is employed with them, along with an optional evaluation of the

graduate's performance. The interface requires the verifier to identify themselves before a decision can be submitted, does not expose the graduate's email address or other private contact details, and prevents a link from being used more than once. This design allows employers to participate in verifying graduate employment as an external party while keeping graduate information protected and the verification tied to a controlled, single-use token.

Figure 23

Administrator Dashboard Interface

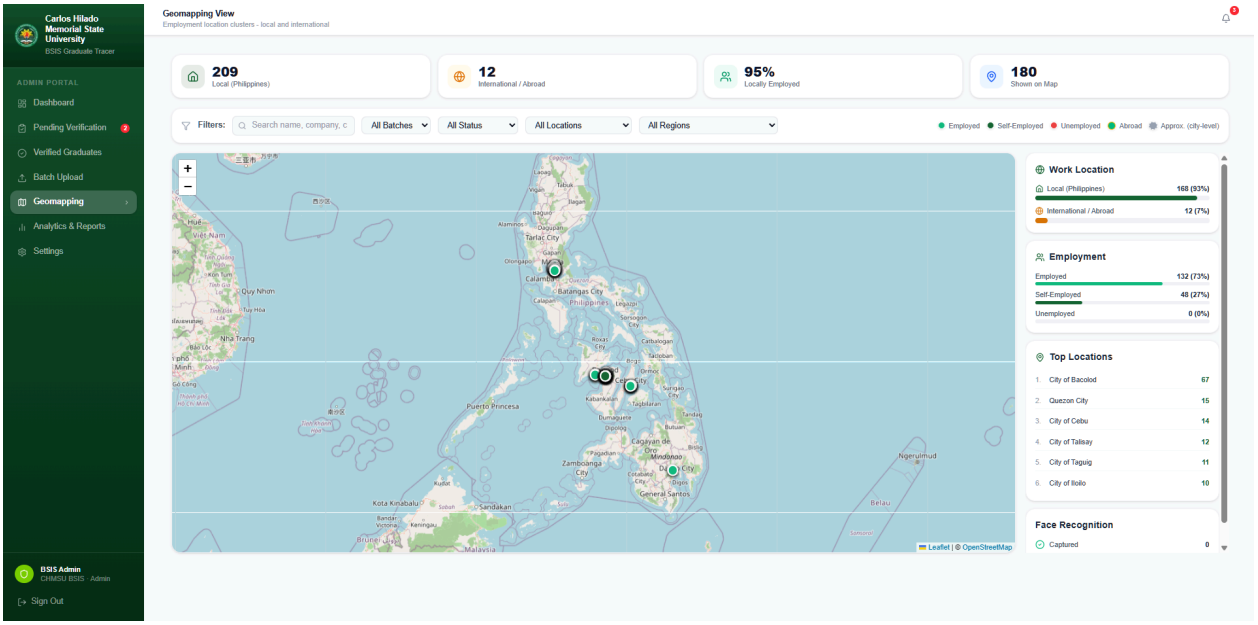


This figure shows the Administrator Dashboard interface of the Graduate Tracer System with Predictive Employability Trend Analysis. This page provides the administrator with an overview of the system and access to account review and reporting functions. The dashboard lists graduate accounts pending review, allowing the administrator to approve or reject each request, with decisions reflected in the affected accounts. It also provides access to the verified alumni list and to the analytics and

reporting features of the system. Through its arrangement, the dashboard becomes the primary workspace from which the administrator manages graduate participation and monitors the overall state of the tracer system.

Figure 24

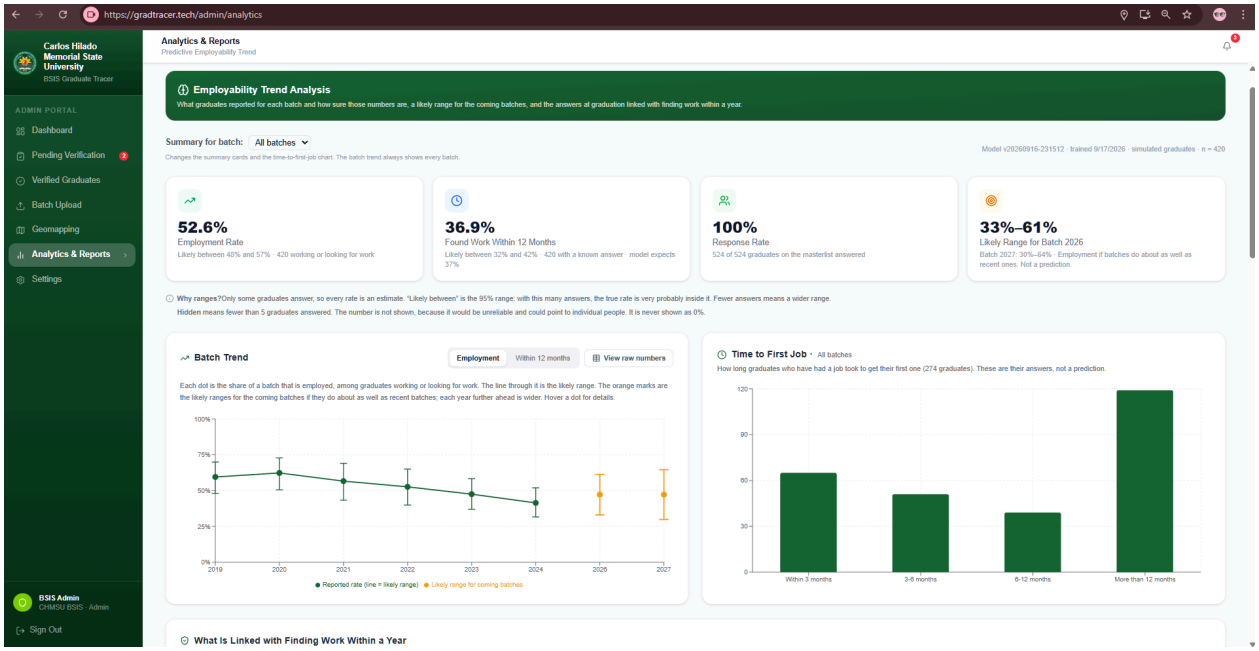
Graduate Geomap Interface



This figure presents the Graduate Geomap of the Graduate Tracer System with Predictive Employability Trend Analysis. This page visualizes the geographic distribution of graduate employment on an interactive map, plotting the work locations of graduates who consented to location disclosure during registration. The map allows the administrator to see at a glance where BSIS graduates are employed, supporting the tracer study's goal of understanding graduate outcomes across regions. Because location plotting depends on the consent given at registration, the map reflects only the graduates who agreed to share their location, keeping the visualization consistent with the system's privacy handling.

Figure 25

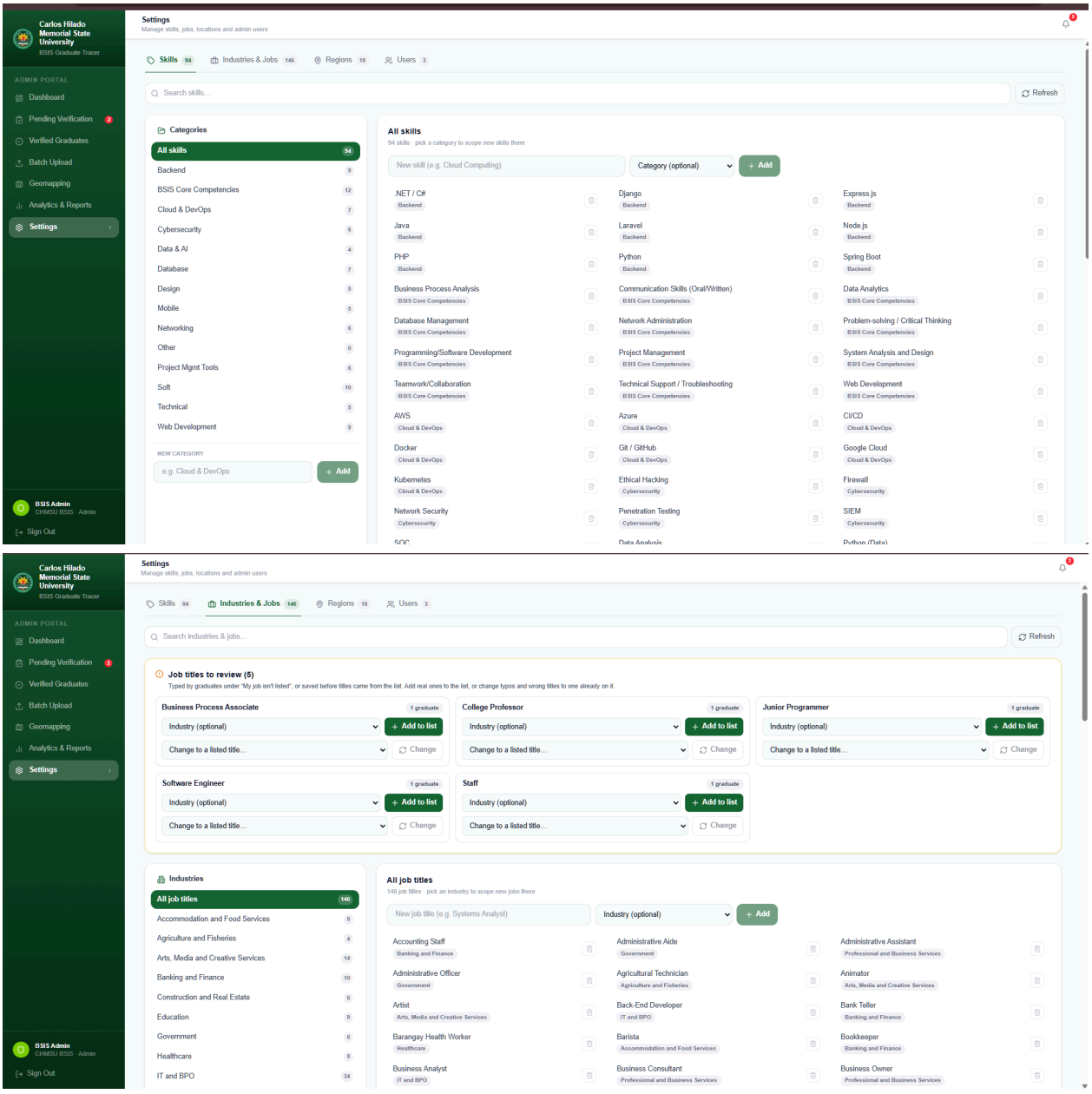
Predictive Analytics Dashboard



This figure displays the Predictive Analytics Dashboard of the Graduate Tracer System with Predictive Employability Trend Analysis, which is the feature central to the study's purpose. The page leads with the observed employment rate, time-to-first-job bands and skills of each graduating batch, each reported with a ninety-five percent confidence interval and withheld when a group is too small to report. Once a model has passed the acceptance gate, the page adds the four factors associated with being employed within twelve months of graduation, with their odds ratios, and a single likely range for the employment rate of the next batch. The page also states which graduate source, real or simulated, is being shown and whether a model has been accepted, so that a demonstration figure is never mistaken for a finding, and it allows the administrator to inform curriculum and program decisions with data drawn from the tracer system.

Figure 26

Administrative Reference Data Settings



This figure shows the Reference Data Management interface of the Graduate Tracer System with Predictive Employability Trend Analysis. This page allows the administrator to maintain the controlled lists used throughout the system, including skills, industries, job titles, and the region, province, and city selections used during

registration. By managing these values through the interface, the administrator can keep the system's options current without requiring developer intervention. The page presents each category in an organized form for adding, editing, and removing entries, ensuring that the data offered to graduates during registration and used in analytics remains accurate and up to date.

The second objective of this study was to test the functionality of the features mentioned above.

The result of this objective was that the system successfully underwent alpha and beta testing, confirming that all modules and functions operated as intended. During alpha testing, the researchers verified the core features of the system including graduate registration, face verification, the employment survey, employer verification, the graduate geomap, profile and skills editing, re-tracking, master list matching and profile review, administrator functions, and the predictive analytics module. A total of one hundred thirty-two (132) test cases were executed, of which one hundred fifteen (115) passed and seventeen (17) failed, yielding an initial pass rate of eighty-seven and one-tenth percent (87.1%). The seventeen defects identified during alpha testing were corrected prior to beta testing.

Beta testing was subsequently conducted with the intended end users, who were able to navigate the system, register their accounts, complete the employment survey, generate verification links for their employers, and view their profile information successfully. A total of one hundred forty-two (142) test cases were executed, and all one hundred forty-two (142) test cases passed, producing a final pass rate of one hundred percent (100%) and leaving no open defects at the close of functionality

testing. These results demonstrate that the developed system meets its functional requirements and was ready for real-world use.

Of the one hundred thirty-two (132) test cases executed during alpha testing, seventy-five used black-box methods and fifty-seven used white-box methods. Sixty of the seventy-five black-box cases passed and fifteen failed, while fifty-five of the fifty-seven white-box cases passed and two failed. Two of these failures were found by the original test suite and the other fifteen by the cases added in the 19 September 2026 revision; all seventeen are reported in full in the tables and discussions that follow, together with the modules in which they occurred. Black-box testing results are presented first, grouped by module.

Table 42

Black-Box Testing of the Alpha Testing of the Graduate Registration Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-GradRegis-001	Register successfully with valid details	Registration succeeds. A waiting-for-admin screen is shown. The account appears under Pending Verification in the admin portal.	Matches expected outcome	Pass	No issues encountered
TC-GradRegis-002	Reject a weak password	Cannot continue. A message states the full policy: minimum length, uppercase, lowercase, number and special character.	Matches expected outcome	Pass	No issues encountered
TC-GradRegis-003	Reject mismatched password confirmation	Cannot continue. A passwords-do-not-match message is shown.	Matches expected outcome	Pass	No issues encountered
TC-GradRegis-004	Reject an invalid Philippine	Cannot continue. A message states Philippine numbers	Matches expected outcome	Pass	No issues encountered

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
	mobile number	must start with 9 or 09.			
TC-GradRegis-005	Accept both valid mobile formats	Both formats are accepted.	Matches expected outcome	Pass	No issues encountered
TC-GradRegis-006	Reject a non-Facebook social link	Cannot continue. A message names the accepted hosts: facebook.com, fb.com, fb.me.	Matches expected outcome	Pass	No issues encountered
TC-GradRegis-007	Address dropdowns cascade correctly	Province options are limited to the chosen region. City options are limited to the chosen province. Address is chosen from lists, not typed.	Matches expected outcome	Pass	No issues encountered
TC-GradRegis-008	Further studies is a dropdown, and reveals detail fields	The control is a dropdown, not radio buttons. Selecting enrolled or completed reveals the postgraduate detail panel. Selecting none hides it.	Matches expected outcome	Pass	No issues encountered
TC-GradRegis-009	No GPA or General Average is requested anywhere	No such field exists at any step. It was removed in the revision.	Matches expected outcome	Pass	No issues encountered
TC-Grad Regis-010	Terms must be scrolled before they can be accepted	The checkbox is disabled until the terms are scrolled through, then becomes enabled.	Matches expected outcome	Pass	No issues encountered
TC-Grad Regis-011	Cannot submit without accepting the terms	Submission is blocked with a message that the terms must be accepted.	Matches expected outcome	Pass	No issues encountered
TC-Grad Regis-012	Can register while declining Geomap consent	Registration completes. The two consents are independent. The graduate is later absent from the admin Geomap but still counted in reports.	Matches expected outcome	Pass	No issues encountered

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Grad Regis-013	Rejected answer does not destroy the form or the face scan	An error names the specific field. You return to the form that owns it. All other answers are still filled in and the face capture is NOT requested again. Correcting the one field and resubmitting succeeds.	Matches expected outcome	Pass	No issues encountered

Table 43

Black-Box Testing of the Alpha Testing of the Face Recognition and Sign-In Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-FaceRecog-001	Face capture asks for blink and turn, never open mouth	The prompts are: hold a frontal pose, blink once, and one head turn. No open-mouth gesture is ever requested.	Matches expected outcome	Pass	No issues encountered
TC-FaceRecog-002	A still photograph does not pass the blink check	The blink stage never completes. A photograph cannot produce the required open-closed-open eye movement.	Matches expected outcome	Pass	No issues encountered
TC-FaceRecog-003	A blurry capture is refused	The frame is refused with a retake prompt. Nothing is saved.	Matches expected outcome	Pass	No issues encountered
TC-FaceRecog-004	Sign in successfully with the enrolled face	Sign-in succeeds and the graduate dashboard opens.	Matches expected outcome	Pass	No issues encountered
TC-FaceRecog-005	Reject a different person	Access is refused with a face-verification message. The dashboard does not open.	Access verified	Fail	Recalibrate the face-matching similarity threshold. Passed on re-test.
TC-FaceRecog-006	Reject a wrong password	Sign-in is refused before the camera step.	Matches expected outcome	Pass	No issues encountered

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-FaceRecog-007	Liveness challenge times out	The attempt fails with a timeout message and a retry is offered. The system does not hang.	Matches expected outcome	Pass	No issues encountered
TC-FaceRecog-008	Password reset by emailed code	The code arrives. The password changes. The new password works and the old one does not. NOTE: if no email arrives, check the mail transport before failing this case.	Matches expected outcome	Pass	No issues encountered
TC-FaceRecog-009	Reject an incorrect reset code	The code is refused and the password is unchanged.	Matches expected outcome	Pass	No issues encountered

Table 44

Black-Box Testing of the Alpha Testing of the Employment Survey Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-EmpSurvey-001	Employed graduate answers all six steps	All six steps are presented in order: academic profile, status, first job, current job, work address, skills.	Matches expected outcome	Pass	No issues encountered
TC-EmpSurvey-002	Never employed skips job and address steps	The survey jumps straight to Skills. First job, current job and work address are not asked, because there is no job history to record.	Matches expected outcome	Pass	No issues encountered
TC-EmpSurvey-003	Seeking asks about the first job but not the current one	First job details ARE asked. The survey then jumps to Skills, skipping current job and work address.	Matches expected outcome	Pass	No issues encountered
TC-EmpSurvey-004	Employment status is a dropdown	It is a dropdown list, not a set of clickable cards.	Matches expected outcome	Pass	No issues encountered
TC-EmpSurvey-005	Skills selection accepts every listed skill	Every listed skill can be selected and none is silently	Matches expected outcome	Pass	No issues encountered

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
		ignored. The count shown matches the number chosen.			
TC-EmpSurvey-006	Changing status re-opens skipped steps	The previously skipped steps are presented again.	Matches expected outcome	Pass	No issues encountered

Table 45

Black-Box Testing of the Alpha Testing of the Administrator Functions Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-AdminFunc-001	Admin approves a pending graduate	The account moves to Verified Graduates and the graduate can sign in.	Matches expected outcome	Pass	No issues encountered
TC-AdminFunc-002	Admin rejects with a reason	The account is marked rejected and the reason is recorded.	Matches expected outcome	Pass	No issues encountered
TC-AdminFunc-003	Filter verified graduates	Results narrow correctly. The empty case shows an empty-state message, not a blank or broken table.	Matches expected outcome	Pass	No issues encountered
TC-AdminFunc-004	Geomap respects consent	That graduate does NOT appear on the map, while consenting graduates do.	Matches expected outcome	Pass	No issues encountered
TC-AdminFunc-005	Analytics names the graduate source and the model status	A clearly visible notice names which graduate source, real or simulated, is being shown and states whether a model has been accepted. Observed and predicted figures are labelled distinctly.	Matches expected outcome	Pass	No issues encountered
TC-AdminFunc-006	Missing figures show a dash, never zero	Absent values display as a dash. They must NOT display as 0 percent	Matches expected outcome	Pass	No issues encountered

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
		or 0.0 months, which would assert a real zero.			
TC-AdminFunc-007	Generate and export a report	The report renders with per-batch rows. Curriculum Alignment shows verified and self-reported rates separately plus an unknown count. Both exports download and open.	Matches expected outcome	Pass	No issues encountered
TC-AdminFunc-008	Upload a master list CSV	Rows are parsed and previewed before saving. After saving they appear in the master list.	Matches expected outcome	Pass	No issues encountered
TC-AdminFunc-009	Re-uploading the same file creates no duplicates	No duplicate records are created. The import is idempotent.	Matches expected outcome	Pass	No issues encountered
TC-AdminFunc-010	Malformed rows are reported, not silently dropped	The broken rows are reported so the coordinator knows what did not land. Valid rows still import.	Matches expected outcome	Pass	No issues encountered
TC-AdminFunc-011	Manage reference data	The new skill appears in the registration form. After deactivation it no longer appears as a choice.	Skill added did not reflect on the form	Fail	Update the form to read reference data at point of use, not a cached copy. Passed on re-test.
TC-AdminFunc-012	Manage administrator accounts	The new administrator can sign in. After deactivation they cannot.	Matches expected outcome	Pass	No issues encountered

Table 46

White-Box Testing of the Alpha Testing of the Graduate Registration and Employment Survey Modules

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-DataValReg-001	validate_comprehensive_survey()	Flat payload passed WITHOUT the section adapter	Flat dict with employment_status as a string	AttributeError raised. The flat payload cannot be validated directly.	Matches expected result	Pass
TC-DataValReg-002	flat_to_sections()	Full payload, then sparse payload	15 survey fields + 9 personal fields, then a single field	8 sections built from the full payload. Absent sections omitted from the sparse one, not fabricated as empty.	Matches expected result	Pass
TC-DataValReg-003	validate_registration_payload()	Happy path, no rule violated	Complete valid registration payload	is_valid true, no blocking errors, step is null.	Matches expected result	Pass
TC-DataValReg-004	validate_registration_payload()	Blocking branch, three simultaneous violations	time_to_hire_months = -5, employment_status = banana, first_job_sector = nonsense	is_valid false. All three fields listed in field_errors. step = employment.	Matches expected result	Pass
TC-DataValReg-005	_calculate_completeness()	Absent-value branch and completeness scoring	Only employment_status = never_employed, then a full payload	Sparse scores below 60 percent (43.5). Full scores 100.	Matches expected result	Pass
TC-DataValReg-006	FieldValidationRules.BATCH_RANGE	Lower bound, upper bound, and beyond range	graduation_year = 2019, then 2026, then 2999	2019 accepted. 2026 accepted. 2999 rejected.	Matches expected result	Pass
TC-DataValReg-007	_parse_birth_date()	All stored formats plus unparseable input	2000-05, 2000-05-14, 03/05, nonsense	None of these block a registration.	Matches expected result	Pass

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-DataValReg-008	Clean-data gate	Blocking branch	Multipart registration with time_to_hire_months = -5	HTTP 400. field_errors names the field. step = employment.	Matches expected result	Pass
TC-DataValReg-009	Clean-data gate	Post-rejection state	Multipart registration with employment_status = banana	No User row and no AlumniAccount row created.	Matches expected result	Pass

Table 47

White-Box Testing of the Alpha Testing of the Face Recognition and Sign-In Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-AuthName-001	AdminLoginView	Database error during authentication	Simulated OperationalError	HTTP 503 with retryable true.	Matches expected result	Pass
TC-AuthName-002	AlumniLoginView	Database error during authentication	Simulated OperationalError	HTTP 503 with retryable true.	Matches expected result	Pass
TC-AuthName-003	AdminLoginView	Happy path	Valid administrator credentials	HTTP 200 with accessToken present in the response.	Matches expected result	Pass
TC-AuthName-004	require_admin()	Protected endpoint with no Authorization header	GET /api/admin/alumni/pending/ with no token	HTTP 401.	Matches expected result	Pass
TC-AuthName-005	_mark_logged_in()	Success path for each role	Valid admin, employer and graduate sign-ins	User.last_login stamped for all three. Graduate sign-in also writes a LoginAudit row.	Matches expected result	Pass
TC-AuthName-006	derive_last_name()	Generational suffix present	Rolly Lerit Samson Jr. / Jose Rizal Jr / Pedro Penduko Sr.	Samson / Rizal / Penduko.	Matches expected result	Pass

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-AuthName-007	derive_last_name()	Compound surname particles	Jose Dela Cruz / Regie Calago De La Torre / Sarah Joy De Los Santos / Luis Del Rosario	Dela Cruz / De La Torre / De Los Santos / Del Rosario.	Matches expected result	Pass
TC-AuthName-008	derive_last_name()	Suffix and particle together	Juan Dela Cruz III	Dela Cruz.	Matches expected result	Pass
TC-AuthName-009	derive_last_name()	Comma format	Samson, Rolly Lerit / Dela Cruz, Juan	Samson / Dela Cruz.	Matches expected result	Pass
TC-AuthName-010	derive_last_name()	Ordinary and degenerate names	Maria Santos / extra whitespace / Madonna / empty string	Santos / Reyes / Madonna / empty.	Matches expected result	Pass

Table 48

White-Box Testing of the Alpha Testing of the Employer Verification Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-EmpVerify-001	VerificationTokenDecisionView	Full flow with NO auth header	Mint a token, then submit a decision with verifier identity	HTTP 200. Token status becomes used. Record becomes verified.	Matches expected result	Pass
TC-EmpVerify-002	VerificationTokenDecisionView	Second decision on a spent token	Re-submit against the same token	Refused. Exactly one decision row exists.	Matches expected result	Pass
TC-EmpVerify-003	VerificationTokenDecisionView	Verifier identity missing	Decision submitted with no verifier_name or verifier_email	HTTP 400.	Matches expected result	Pass
TC-EmpVerify-004	VerificationTokenDecisionView	Self-verification attempt	verifier_email equals the	HTTP 400. No decision recorded.	Matches expected result	Pass

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
			graduate own address			
TC-EmpVerify-005	AlumniVerificationInviteView	Two invites for one graduate	Mint two tokens for the same graduate	Both remain pending and both are answerable.	Matches expected result	Pass
TC-EmpVerify-006	VerificationTokenDetailView	Public landing payload	GET the token detail endpoint with no credentials	Name, program and batchYear present. Email ABSENT.	Matches expected result	Pass
TC-EmpVerify-007	AlumniVerificationInviteView	Self-invitation at invite time	employer_email equals the graduate own address	HTTP 400.	Matches expected result	Pass

Table 49

White-Box Testing of the Alpha Testing of the Analytics and Prediction Modules

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-Analytics Pred-001	JobTitle.save()	Classified and unclassified branches	Software Development, Outside Information Systems, and an unclassified title	True, False, and None respectively.	Matches expected result	Pass
TC-Analytics Pred-002	resolve_alignment()	Verified contradicts self-report	Graduate self-reports aligned, employer-verified title is non-IS	Resolved False with source verified.	Matches expected result	Pass
TC-Analytics Pred-003	summarize() and BatchSummaryReportView	Mixed sources	One verified, one self-reported, one with no data	Verified 100 percent (n=1), self-reported 100 percent (n=1), unknown 1.	Matches expected result	Pass

Table 50

White-Box Testing of the Alpha Testing of the Administrator Functions Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-AdminEndpoint-001	Reference list view	Unauthenticated read	GET /api/reference/regions/ with no token	HTTP 200.	Matches expected result	Pass
TC-AdminEndpoint-002	Reference write views	Unauthenticated write	POST /api/reference/regions/ with no token	HTTP 401.	Matches expected result	Pass
TC-AdminEndpoint-003	Reference CRUD	Full lifecycle with a valid admin token	Create, patch, then delete a region	201, 200, 204. Deletion is soft.	Matches expected result	Pass
TC-AdminEndpoint-004	_admin_alumni_payload()	GPS fallback branch	FaceScan row carries GPS but the capture-meta blob does not	lat and lng surfaced to the admin payload from the FaceScan row.	Matches expected result	Pass

Table 51

White-Box Testing of the Alpha Testing of the Front End Logic

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-FrontEnd-001	reducer NEEDS_CORRECTION	Server rejected a field	Submit with an impossible answer.	Stage returns to the OWNING form. fieldErrors populated. Data still held.	Matches expected result	Pass
TC-FrontEnd-002	reducer SET_ERROR	Non-validation failure	Submit while the backend is unreachable	Error screen with a retry affordance.	Matches expected result	Pass
TC-FrontEnd-003	Error branch discrimination	400 WITHOUT field_errors	Server returns 400 with no field_errors key	Falls through to the generic error screen.	Matches expected result	Pass

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-FrontEnd-004	nextStep()	Path: employed_full_time	employment_status = employed_full_time	Steps 1 to 6 in order, none skipped.	Matches expected result	Pass
TC-FrontEnd-005	nextStep()	Path: seeking	employment_status = seeking	1, 2, 3, then jumps to 6.	Matches expected result	Pass
TC-FrontEnd-006	nextStep()	Path: never_employed and not_seeking	employment_status = never_employed, then not_seeking	1, 2, then jumps to 6.	Matches expected result	Pass
TC-FrontEnd-007	nextStep()	Changing status after skipping	Choose never_employed, reach step 6, go back, change to employed_full_time	The previously skipped steps become reachable again.	Matches expected result	Pass
TC-FrontEnd-008	register-terms.tsx scroll gate	Terms not yet scrolled	Open the terms step without scrolling	The agreement checkbox is disabled until the terms are scrolled to the end.	Matches expected result	Pass
TC-FrontEnd-009	register-terms.tsx	Short terms that do not overflow	View on a large screen where the terms fit without scrolling	The checkbox is enabled immediately.	Matches expected result	Pass
TC-FrontEnd-010	register-terms.tsx consent matrix	Terms accepted, Geomap declined	Tick terms only	Submit is ALLOWED. The two consents are independent.	Matches expected result	Pass
TC-FrontEnd-011	register-alumni-personal.tsx	Correction pass with an existing capture	initialBiometric supplied by the orchestrator	Capture stage starts complete, the stored photo is shown.	Matches expected result	Pass
TC-FrontEnd-012	Null-safe formatters	actual values are null	A batch with no recorded answers	Renders a dash, not 0.0 percent or 0.0 months.	Matches expected result	Pass

Table 52

Black-Box Testing of the Alpha Testing of the Employer Verification Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-EmpVerif-001	Graduate generates a verification link for an employer	A single-use link is created and sent to the employer address given. The graduate sees that the invitation is pending.	Link created but not sent; no pending status shown to the graduate	Fail	Add an employer email field that sends the link, and show the pending status on the dashboard. Passed at beta testing.
TC-EmpVerif-002	Graduate cannot invite their own address	The invitation is refused with a message. No link is created.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-003	The link opens without signing in	The landing page opens for an employer with no account and shows the name, program and batch year of the graduate. The email address of the graduate is not shown.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-004	Verifier identity is required before a decision can be submitted	Submission is blocked until a name, work email and position are supplied.	Decision accepted without the verifier's position	Fail	Make the position a required field on the page and in the API. Passed at beta testing.
TC-EmpVerif-005	Employer confirms the employment claim	The record is marked verified, the decision is recorded with the identity of the verifier, and the link is spent.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-006	Employer denies the employment claim	The record is marked denied with the reason recorded. The graduate self-report is kept but no longer counts as verified.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-007	The evaluation form is optional	A decision can be submitted without the eleven rating items. Submitting them records the evaluation confidentially.	Confirmation blocked until the feedback form was completed	Fail	Make the feedback form optional when confirming. Passed at beta testing.

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-EmpVerif-008	A spent link cannot be used again	The second attempt is refused. Exactly one decision exists for that link.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-009	An expired link is refused	A link older than the seven-day validity is refused with an expiry message rather than an error page.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-010	A suspicious reply is flagged	A reply submitted from the same device that created the link is accepted but flagged for administrator review.	Matches expected outcome	Pass	No issues encountered

Table 53

Black-Box Testing of the Beta Testing of the Employer Verification Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-EmpVerif-001	Graduate generates a verification link for an employer	A single-use link is created and sent to the employer address given. The graduate sees that the invitation is pending.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.
TC-EmpVerif-002	Graduate cannot invite their own address	The invitation is refused with a message. No link is created.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-003	The link opens without signing in	The landing page opens for an employer with no account and shows the name, program and batch year of the graduate. The email address of the graduate is not shown.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-004	Verifier identity is required before a decision can be submitted	Submission is blocked until a name, work email and position are supplied.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-EmpVerif-005	Employer confirms the employment claim	The record is marked verified, the decision is recorded with the identity of the verifier, and the link is spent.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-006	Employer denies the employment claim	The record is marked denied with the reason recorded. The graduate self-report is kept but no longer counts as verified.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-007	The evaluation form is optional	A decision can be submitted without the eleven rating items. Submitting them records the evaluation confidentially.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.
TC-EmpVerif-008	A spent link cannot be used again	The second attempt is refused. Exactly one decision exists for that link.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-009	An expired link is refused	A link older than the seven-day validity is refused with an expiry message rather than an error page.	Matches expected outcome	Pass	No issues encountered
TC-EmpVerif-010	A suspicious reply is flagged	A reply submitted from the same device that created the link is accepted but flagged for administrator review.	Matches expected outcome	Pass	No issues encountered

Table 54

Black-Box Testing of the Alpha Testing of the Graduate Geomap Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Geomap-001	Map plots consenting graduates only	Graduates who gave geomap consent appear as pins. A graduate who declined does not appear, while	Matches expected outcome	Pass	No issues encountered

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
		still being counted in the reports.			
TC-Geomap-002	Filter by batch year	Only the pins of the selected batch remain. Clearing the filter restores the full set.	Matches expected outcome	Pass	No issues encountered
TC-Geomap-003	Filter by employment status and region	The pins narrow to the selection. An empty result shows an empty-state message, not a blank map.	No region filter; an empty result showed a blank map	Fail	Add a region filter and an empty-state message. Passed at beta testing.
TC-Geomap-004	A graduate with no work address produces no pin	No pin is drawn at the map origin or at a default location.	Never-employed graduate pinned at the home location	Fail	Plot only graduates who have a work address. Passed at beta testing.
TC-Geomap-005	Pin detail withholds private contact data	The pin shows the name, batch, employment status and work location. It does not show the email address, mobile number or home address.	Matches expected outcome	Pass	No issues encountered

Table 55

Black-Box Testing of the Beta Testing of the Graduate Geomap Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Geomap-001	Map plots consenting graduates only	Graduates who gave geomap consent appear as pins. A graduate who declined does not appear, while still being counted in the reports.	Matches expected outcome	Pass	No issues encountered
TC-Geomap-002	Filter by batch year	Only the pins of the selected batch remain. Clearing the filter restores the full set.	Matches expected outcome	Pass	No issues encountered

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Geomap-003	Filter by employment status and region	The pins narrow to the selection. An empty result shows an empty-state message, not a blank map.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.
TC-Geomap-004	A graduate with no work address produces no pin	No pin is drawn at the map origin or at a default location.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.
TC-Geomap-005	Pin detail withholds private contact data	The pin shows the name, batch, employment status and work location. It does not show the email address, mobile number or home address.	Matches expected outcome	Pass	No issues encountered

Table 56

Black-Box Testing of the Alpha Testing of the Graduate Profile and Skills Editing Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-ProfileSkills-001	Graduate edits personal details	The change is saved and is still present after signing out and back in.	Family name reverted after signing back in	Fail	Save the family name field to the graduate profile. Passed at beta testing.
TC-ProfileSkills-002	A graduation month later than the current month is refused	The edit is blocked with a message naming the field. The stored value is unchanged.	Matches expected outcome	Pass	No issues encountered
TC-ProfileSkills-003	Graduate edits the skills list	Skills can be added and removed. After saving and reloading, the list is exactly what was selected, with none silently dropped.	Two BSIS skills dropped after saving and reloading	Fail	Keep BSIS skills that share a name with a soft skill. Passed at beta testing.
TC-ProfileSkills-004	An invalid edit preserves the rest of the form	An error names the offending field. Every other answer is still filled in, and correcting the one field saves successfully.	Matches expected outcome	Pass	No issues encountered

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-ProfileSkills-005	Geomap consent can be changed after registration	Turning consent off removes the graduate from the map. Turning it on restores the pin. Reports are unaffected either way.	No option to change geomap consent after registration	Fail	Add a geomap consent switch to Edit Profile. Passed at beta testing.
TC-ProfileSkills-006	Home address can be re-selected or filled from the current location	Region, province, city and barangay cascade correctly. Filling from the current location populates the same fields and remains editable.	Free-text address; no cascading lists or current-location option	Fail	Add cascading address lists and a current-location button. Passed at beta testing.

Table 57

Black-Box Testing of the Beta Testing of the Graduate Profile and Skills Editing Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-ProfileSkills-001	Graduate edits personal details	The change is saved and is still present after signing out and back in.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.
TC-ProfileSkills-002	A graduation month later than the current month is refused	The edit is blocked with a message naming the field. The stored value is unchanged.	Matches expected outcome	Pass	No issues encountered
TC-ProfileSkills-003	Graduate edits the skills list	Skills can be added and removed. After saving and reloading, the list is exactly what was selected, with none silently dropped.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.
TC-ProfileSkills-004	An invalid edit preserves the rest of the form	An error names the offending field. Every other answer is still filled in, and correcting the one field saves successfully.	Matches expected outcome	Pass	No issues encountered
TC-ProfileSkills-005	Geomap consent can be changed	Turning consent off removes the graduate from the map. Turning it on	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
	after registration	restores the pin. Reports are unaffected either way.			
TC-ProfileSkills-006	Home address can be re-selected or filled from the current location	Region, province, city and barangay cascade correctly. Filling from the current location populates the same fields and remains editable.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.

Table 58

Black-Box Testing of the Alpha Testing of the Re-tracking History and Reminders Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Retrack-001	History lists every re-tracking event in order	Each event appears with its type and timestamp, most recent first.	Matches expected outcome	Pass	No issues encountered
TC-Retrack-002	Registration is the first entry of the history	A newly registered graduate has exactly one entry, recorded at registration.	Matches expected outcome	Pass	No issues encountered
TC-Retrack-003	Administrator sends a reminder on demand	The email is sent to the graduate and the reminder timestamp is recorded and shown.	Matches expected outcome	Pass	No issues encountered
TC-Retrack-004	A graduate overdue by more than two years is flagged	The verified graduate list marks the account as due, with the number of days overdue.	Flag showed time since last update, not days overdue	Fail	Show the number of days overdue on the flag. Passed at beta testing.
TC-Retrack-005	Completing the survey closes the gate	The graduate regains the full dashboard, the due date moves forward two years, and a new history entry is written.	Matches expected outcome	Pass	No issues encountered

Table 59

Black-Box Testing of the Beta Testing of the Re-tracking History and Reminders Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Retrack-001	History lists every re-tracking event in order	Each event appears with its type and timestamp, most recent first.	Matches expected outcome	Pass	No issues encountered
TC-Retrack-002	Registration is the first entry of the history	A newly registered graduate has exactly one entry, recorded at registration.	Matches expected outcome	Pass	No issues encountered
TC-Retrack-003	Administrator sends a reminder on demand	The email is sent to the graduate and the reminder timestamp is recorded and shown.	Matches expected outcome	Pass	No issues encountered
TC-Retrack-004	A graduate overdue by more than two years is flagged	The verified graduate list marks the account as due, with the number of days overdue.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.
TC-Retrack-005	Completing the survey closes the gate	The graduate regains the full dashboard, the due date moves forward two years, and a new history entry is written.	Matches expected outcome	Pass	No issues encountered

Table 60

Black-Box Testing of the Alpha Testing of the Master List Matching and Profile Review Module

Test Case ID	***Description***	***Expected Outcome***	***Actual Outcome***	***Pass/ Fail***	***Comments***
TC-Master List-001	*A registration matching the master list is active at signup*	*The graduate can sign in immediately and is listed for the administrator as matched and awaiting review.*	*Matches expected outcome*	*Pass*	*No issues encountered*

TC-Master List-002	*A registration with no master list match is held*	*The account is listed as unmatched and cannot sign in until an administrator decides.*	*Unmatched pending account could sign in*	*Fail*	*Refuse sign-in until the account is approved. Passed at beta testing.*
TC-Master List-003	*Administrator confirms a profile review*	*The account becomes active, the review timestamp is recorded, and the graduate is notified by email.*	*No email sent to the graduate*	*Fail*	*Email the graduate when a profile review is confirmed. Passed at beta testing.*
TC-Master List-004	*Administrator rejects a profile review with a reason*	*The account is marked rejected, the reason is stored, and the graduate is notified by email.*	*Account deleted; no rejected status or reason stored*	*Fail*	*Keep the account as rejected with the reason stored. Passed at beta testing.*
TC-Master List-005	*A master list CSV with any invalid row imports nothing*	*The whole file is refused and the offending rows are named. No partial import occurs, and re-uploading a corrected file imports every row once.*	*Matches expected outcome*	*Pass*	*No issues encountered*
TC-Master List-006	*Administrator corrects a misspelled entry*	*The entry is saved with the corrected name and surname, and a graduate who registered earlier with the correct spelling is linked to it.*	*Matches expected outcome*	*Pass*	*No issues encountered*
TC-Master List-007	*Administrator retires an entry*	*The entry is shown as retired and no longer matches a registration with that name and batch. Restoring it makes it match again.*	*Matches expected outcome*	*Pass*	*No issues encountered*

TC-Master List-008	*Deleting an entry a graduate registered against is refused*	*The deletion is refused with a message to retire the entry instead, and the graduate keeps the link to the roster. An entry nobody registered against is deleted.*	*Matches expected outcome*	*Pass*	*No issues encountered*
TC-Master List-009	*Administrator exports the master list*	*A CSV file downloads holding every entry with its name, batch year, whether the graduate has registered, and whether the entry is active or retired.*	*Matches expected outcome*	*Pass*	*No issues encountered*

Table 61

Black-Box Testing of the Beta Testing of the Master List Matching and Profile Review Module

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-Master List-001	A registration matching the master list is active at signup	The graduate can sign in immediately and is listed for the administrator as matched and awaiting review.	Matches expected outcome	Pass	No issues encountered
TC-Master List-002	A registration with no master list match is held	The account is listed as unmatched and cannot sign in until an administrator decides.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.
TC-Master List-003	Administrator confirms a profile review	The account becomes active, the review timestamp is recorded, and the graduate is notified by email.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.

TC-Master List-004	Administrator rejects a profile review with a reason	The account is marked rejected, the reason is stored, and the graduate is notified by email.	Matches expected outcome	Pass	Corrected after alpha testing. No issues encountered.
TC-Master List-005	A master list CSV with any invalid row imports nothing	The whole file is refused and the offending rows are named. No partial import occurs, and re-uploading a corrected file imports every row once.	Matches expected outcome	Pass	No issues encountered
TC-Master List-006	Administrator corrects a misspelled entry	The entry is saved with the corrected name and surname, and a graduate who registered earlier with the correct spelling is linked to it.	Matches expected outcome	Pass	No issues encountered
TC-Master List-007	Administrator retires an entry	The entry is shown as retired and no longer matches a registration with that name and batch. Restoring it makes it match again.	Matches expected outcome	Pass	No issues encountered
TC-Master List-008	Deleting an entry a graduate registered against is refused	The deletion is refused with a message to retire the entry instead, and the graduate keeps the link to the roster. An entry nobody registered against is deleted.	Matches expected outcome	Pass	No issues encountered
TC-Master List-009	Administrator exports the master list	A CSV file downloads holding every entry with its name, batch year, whether the graduate has registered, and whether the entry is active or retired.	Matches expected outcome	Pass	No issues encountered

At beta testing, the nine cases of the Master List Matching and Profile Review module were executed again on the corrected build, and all nine passed. An account awaiting verification is now refused at sign-in with a message that it is awaiting the administrator's approval, confirming a profile review emails the graduate, and a rejected account is kept with a rejected status and the stored reason. So that a rejection does not lock the graduate's email address, registering again with the same address replaces the rejected account.

The white-box cases added in the revision exercise the rebuilt Predictive Employability Module directly, calling the functions that derive the outcome, estimate the rates, apply the acceptance gate and serve the analytics endpoint. They are presented next, for alpha testing and then for beta testing.

Table 62

White-Box Testing of the Alpha Testing of the Predictive Employability Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-PredEmploy-001	leaked_features()	A post-outcome column is offered as an input	Feature list including current_job_sector	The column is returned as leaked and the acceptance gate fails the run.	Matches expected result	Pass
TC-PredEmploy-002	employed_within_12_months()	Every time-to-hire bracket and the still-looking case	Each bracket, a seeking graduate past 12 months, and an employed graduate with no hire time	Up to 6 months to 1 year returns 1, longer returns 0, seeking past 12 months returns 0, no hire time returns unknown.	Matches expected result	Pass
TC-PredEmploy-003	build_graduate_frame()	A graduate inside the first twelve months	Graduation date four months ago	The outcome stays unknown and the row is excluded from	Matches expected result	Pass

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
				training rather than counted as not employed.		
TC-PredEmployoy-004	wilson_interval()	Boundary proportions and no data	k=0 of n=20, k=20 of n=20, and n=0	The interval stays inside 0 to 1 at both boundaries and returns null rather than 0 when n is 0.	Matches expected result	Pass
TC-PredEmployoy-005	group_indicators()	A group below the suppression threshold	A batch with 4 respondents	The rate is suppressed rather than displayed, and is reported as unknown, never as 0 percent.	Matches expected result	Pass
TC-PredEmployoy-006	evaluate_candidate()	Inputs carrying no signal	Random inputs unrelated to the outcome	The baseline and discrimination checks fail and the run is not eligible for activation.	Discrimination failed but the baseline check passed (Brier 0.248 vs 0.248)	Fail
TC-PredEmployoy-007	evaluate_candidate()	Inputs carrying signal, with enough data	A frame meeting the sample size rule with an informative input	All eight checks pass, the factors carry odds ratios with intervals, and the run is eligible.	Matches expected result	Pass
TC-PredEmployoy-008	activate_version()	Activating a failed run	A saved run with at least one failed check	Activation is refused. The previously active model, if any, stays active.	Matches expected result	Pass
TC-PredEmployoy-009	employment_outlook()	Fewer than three backtest errors available	Two batches with reportable rates	The range falls back to plus or minus 20 points rather than reporting a narrow interval.	Matches expected result	Pass
TC-PredEmployoy-010	AdminAnalyticsPredictionsView	No active model	Request the analytics endpoint with no model activated	Observed indicators are returned with a statement that no model has been accepted. No simulated	Matches expected result	Pass

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
				rows are substituted.		
TC-PredEmploy-011	is_future_graduation()	A graduation month later than the current month	A record dated next year	The record is excluded from every figure, counted under data issues, and flagged on the dashboard and in the report.	Excluded from the figures but not counted under data issues (0 instead of 1)	Fail
TC-PredEmploy-012	skill_summary()	A skill held by fewer than five graduates	One skill named by 3 graduates, one by 12	The rare skill is omitted. The common one is listed with its rate difference and a 95 percent interval, shown only when both groups reach five.	Matches expected result	Pass

The results show that the rebuilt Predictive Employability Module largely functioned as designed, with ten of twelve test cases passing. The leakage check rejected post-outcome inputs, the twelve-month outcome was derived correctly for all time-to-hire brackets, the Wilson interval stayed within bounds, small groups were suppressed, and the acceptance gate, analytics endpoint, and skill summary all behaved as expected. Two cases failed. TC-PredEmploy-006 passed the baseline check despite having no signal, because its Brier score was lower only in the fourth decimal place (0.248 against 0.248). TC-PredEmploy-011 excluded a future-dated record from all figures but failed to count it under data issues, so the dashboard and report gave no indication the record existed.

Table 63

White-Box Testing of the Beta Testing of the Predictive Employability Module

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-PredEmploy-001	leaked_features()	A post-outcome column is offered as an input	Feature list including current_job_sector	The column is returned as leaked and the acceptance gate fails the run.	Matches expected result	Pass
TC-PredEmploy-002	employed_within_12_months()	Every time-to-hire bracket and the still-looking case	Each bracket, a seeking graduate past 12 months, and an employed graduate with no hire time	Up to 6 months to 1 year returns 1, longer returns 0, seeking past 12 months returns 0, no hire time returns unknown.	Matches expected result	Pass
TC-PredEmploy-003	build_graduate_frame()	A graduate inside the first twelve months	Graduation date four months ago	The outcome stays unknown and the row is excluded from training rather than counted as not employed.	Matches expected result	Pass
TC-PredEmploy-004	wilson_interval()	Boundary proportions and no data	k=0 of n=20, k=20 of n=20, and n=0	The interval stays inside 0 to 1 at both boundaries and returns null rather than 0 when n is 0.	Matches expected result	Pass
TC-PredEmploy-005	group_indicators()	A group below the suppression threshold	A batch with 4 respondents	The rate is suppressed rather than displayed, and is reported as unknown, never as 0 percent.	Matches expected result	Pass
TC-PredEmploy-006	evaluate_candidate()	Inputs carrying no signal	Random inputs unrelated to the outcome	The baseline and discrimination checks fail and the run is not eligible for activation.	Matches expected result	Pass
TC-PredEmploy-007	evaluate_candidate()	Inputs carrying	A frame meeting the sample size	All eight checks pass, the factors carry odds ratios	Matches expected result	Pass

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
		signal, with enough data	rule with an informative input	with intervals, and the run is eligible.		
TC-PredEmploy-008	activate_version()	Activating a failed run	A saved run with at least one failed check	Activation is refused. The previously active model, if any, stays active.	Matches expected result	Pass
TC-PredEmploy-009	employment_outlook()	Fewer than three backtest errors available	Two batches with reportable rates	The range falls back to plus or minus 20 points rather than reporting a narrow interval.	Matches expected result	Pass
TC-PredEmploy-010	AdminAnalyticsPredictionsView	No active model	Request the analytics endpoint with no model activated	Observed indicators are returned with a statement that no model has been accepted. No simulated rows are substituted.	Matches expected result	Pass
TC-PredEmploy-011	is_future_graduation()	A graduation month later than the current month	A record dated next year	The record is excluded from every figure, counted under data issues, and flagged on the dashboard and in the report.	Matches expected result	Pass
TC-PredEmploy-012	skill_summary()	A skill held by fewer than five graduates	One skill named by 3 graduates, one by 12	The rare skill is omitted. The common one is listed with its rate difference and a 95 percent interval, shown only when both groups reach five.	Matches expected result	Pass

Table 64 lists the seventeen defects found during alpha testing, the correction made to each, and its result when the case was executed again at beta testing. Most were found by the cases added in the revision and concerned the employer verification

flow, the geomap, profile editing and master list review. With every defect corrected and every case passing at beta testing, the results demonstrate that the developed system meets its functional requirements.

Table 64

Defects Found During Alpha Testing and Their Correction

Test Case ID	Module	Defect Found in Alpha Testing	Correction Before Beta Testing	Beta Result
TC-FaceRecog-005	Face Recognition and Sign-In	A second person who completed the face verification step on another graduate's account was granted access.	The face-matching similarity threshold was recalibrated, so a non-matching face is refused.	Pass
TC-AdminFunc-011	Administrator Functions	A skill added in Settings did not appear on the registration form.	The registration form reads the reference data when it opens instead of a cached copy.	Pass
TC-EmpVerif-001	Employer Verification	The invitation sheet collected no employer address, so the graduate had to forward the link personally and no pending invitation was shown.	The sheet asks for the employer's email and sends the single-use link to it; the dashboard shows the verification as pending.	Pass
TC-EmpVerif-004	Employer Verification	The verifier's position was optional, and a decision with only a name and a work email was accepted.	Name, work email and position are required before a decision can be submitted.	Pass
TC-EmpVerif-007	Employer Verification	The page blocked the confirmation until all eleven evaluation items were rated.	The evaluation form is optional; employment can be confirmed with or without it.	Pass
TC-Geomap-003	Graduate Geomap	There was no region filter, and a filter combination with no match left a blank map without a message.	A region filter was added, and an empty result shows “No graduates match these filters.” with an option to clear them.	Pass
TC-Geomap-004	Graduate Geomap	A graduate who was never employed was plotted at the home location as though it were a workplace.	Only graduates with a work address are plotted.	Pass
TC-ProfileSkills-001	Graduate Profile and Skills Editing	A change to the family name reverted after signing out and back in.	The family name is saved to the graduate profile.	Pass

TC-ProfileSkills-003	Graduate Profile and Skills Editing	Two BSIS skills that share a name with a soft skill were dropped on reload and deleted on the next save.	The skills page keeps every selected BSIS program skill.	Pass
TC-ProfileSkills-005	Graduate Profile and Skills Editing	Geomap consent could not be changed after registration.	A geomap consent switch was added to Edit Profile.	Pass
TC-ProfileSkills-006	Graduate Profile and Skills Editing	The permanent address was two free-text fields, with no location lists and no current-location option.	The address is chosen from cascading region, province, city and barangay lists, and “Use my current location” fills the same fields.	Pass
TC-Retrack-004	Re-tracking History and Reminders	The overdue flag showed the time since the last confirmation instead of the days overdue.	The flag states the days overdue, for example “Needs retracing · 70 days overdue”.	Pass
TC-MasterList-002	Master List Matching and Profile Review	An account without a master list match could sign in before the administrator decided.	An account awaiting verification is refused at sign-in with a message explaining why.	Pass
TC-MasterList-003	Master List Matching and Profile Review	Confirming a profile review sent no email to the graduate.	Confirming a review emails the graduate.	Pass
TC-MasterList-004	Master List Matching and Profile Review	Rejecting a profile review deleted the account, leaving no record of the reason.	The account is kept with a rejected status and the reason; registering again with the same email replaces it.	Pass
TC-PredEmploy-006	Predictive Employability	The baseline check passed a model without signal, whose Brier score was lower only in the fourth decimal place.	The Brier score must now be at least one percent below the baseline's.	Pass
TC-PredEmploy-011	Predictive Employability	A record dated next year was excluded from the figures but not counted under data issues.	Future-dated records are counted under data issues and flagged on the dashboard and report, and stay excluded from every figure.	Pass

The third objective of this study was to evaluate the overall quality of the Graduate Tracer System with Predictive Employability Trend Analysis using the eight characteristics of the ISO/IEC 25010:2011 Software Quality Model. These characteristics include Functional Suitability, Performance Efficiency, Compatibility,

Usability, Reliability, Security, Maintainability, and Portability, which collectively determine whether the developed system meets software quality expectations.

The result of this objective was the successful validation of the system's overall software quality, obtaining an Average Mean Score of 4.46, interpreted as Very High. The evaluation was administered to a total of thirty (30) respondents, comprising three (3) IT Experts and twenty-seven (27) BSIS Alumni graduates, who assessed the system across the thirty-one (31) sub-characteristics of the ISO/IEC 25010:2011 model. This indicates that the developed system meets software quality standards across all eight characteristics. Among the evaluated criteria, Performance Efficiency received the highest rating (4.56), confirming that the system responds efficiently in terms of time behaviour, resource utilization, and capacity, while Portability obtained the lowest rating (4.40), suggesting that future enhancements may focus on improved installability and replaceability, particularly the development of a mobile-native counterpart to complement the current web-based build. Overall, the results demonstrate that the system was reliable, user friendly, and technically sound based on the ISO/IEC 25010:2011 framework.

Table 65

ISO/IEC 25010:2011 Software Quality Model Characteristics

Criteria	Mean Score	Verbal Interpretation
Functional Suitability Characteristics	4.42	Very High
Performance Efficiency Characteristics	4.56	Very High
Compatibility Characteristics	4.53	Very High
Usability Characteristics	4.47	Very High
Reliability Characteristics	4.43	Very High
Security Characteristics	4.49	Very High
Maintainability Characteristics	4.44	Very High
Portability Characteristics	4.40	Very High

Criteria	Mean Score	Verbal Interpretation
Average Mean	4.46	Very High

The quality evaluation of the Graduate Tracer System with Predictive Employability Trend Analysis shows strong performance based on the ISO/IEC 25010:2011 Software Quality Model. Among the eight criteria, Performance Efficiency received the highest mean score of 4.56, indicating that the system operates efficiently in terms of response time and resource management. Portability obtained the lowest mean score of 4.40, suggesting that while the system functions effectively across web-based environments, improvements can still be made to enhance installability and replaceability across different platforms and devices.

These results were consistent with the study by Wirasasmita et al. (2025), in which the ISO/IEC 25010 framework was similarly applied to evaluate a web-based alumni information system. Their study reported that Performance Efficiency (96.5%) and Usability (96%) received the strongest ratings among the evaluated characteristics, confirming that web-based tracer and alumni systems commonly excel in these two areas. The present study reflects a similar pattern, with Performance Efficiency emerging as the highest-rated characteristic. Both studies also identified a specific area for continued improvement, indicating that even highly rated tracer systems retain characteristics that warrant refinement in subsequent development cycles. Overall, the evaluation confirms that the developed Graduate Tracer System with Predictive Employability Trend Analysis meets high software quality standards and performs consistently with existing tracer study research, with Portability identified as a meaningful area for future enhancement.

Figure 27

User's Guide Cover Page

Figure 27 shows the cover page of the User Guide for the Graduate Tracer System with Predictive Employability Trend Analysis, which provides an overview of how the system operates for its different user roles. Graduates have access to their own account and were primarily responsible for registering their profile, completing the employment survey, updating their career progress, and generating verification links for their employers. Employers accessed the system only through a single-use verification link and were responsible for confirming the employment claims of the graduates who invited them, without the need to create an account. System Administrators, together with the BSIS Program Chair, had full system access and were responsible for reviewing pending graduate registrations, managing the master list of BSIS graduates, maintaining the controlled reference lists, and monitoring the analytics and reporting features of the system. (See Appendix J, User Manual)

CHAPTER V
CONCLUSION AND RECOMMENDATION

This chapter presents the conclusions drawn from the development, testing, and evaluation of the Graduate Tracer System with Predictive Employability Trend Analysis developed for the Bachelor of Science in Information Systems (BSIS) program of Carlos Hilado Memorial State University – Talisay Campus. It consolidates the key results of the study in relation to its objectives and confirms that the developed system addressed the major challenges in BSIS graduate tracing, particularly the informal collection of employment data, its inconsistent storage across spreadsheets, and the difficulty of comparing outcomes across batches. Furthermore, this chapter includes practical recommendations based on the study's outcomes to sustain and enhance the system's implementation.

5.1 Summary of Findings

The findings of this study confirmed that the developed Graduate Tracer System with Predictive Employability Trend Analysis was both technically capable and highly acceptable to its users based on the ISO/IEC 25010:2011 Software Quality Model. The system obtained an Overall Average Mean Score of 4.46, interpreted as Very High, indicating that the respondents strongly agreed that the system performs effectively and meets its intended purpose. Performance Efficiency received the highest rating of 4.56, showing that the system responds efficiently in terms of time behaviour, resource utilization, and capacity, while Portability attained the lowest rating of 4.40, still interpreted as Very High, suggesting that only minimal improvements may be needed for smoother use across different platforms and devices in the future. In summary, the

results demonstrate that the system successfully achieved its objectives and complies with international software quality standards, proving it to be a reliable, functional, and user-friendly tool that supports efficient documentation, monitoring, and analysis of BSIS graduate employment outcomes.

5.2 Conclusions

This study successfully developed, tested, and evaluated a web-based Graduate Tracer System with Predictive Employability Trend Analysis for the BSIS program of Carlos Hilado Memorial State University – Talisay Campus. The system addressed the research problem of informal and inconsistent graduate tracing by providing face verification, predictive employability trend analysis, geographic mapping of graduate employment, an employer feedback portal, and an automatic two-year re-tracking gate, replacing the paper surveys and scattered spreadsheets previously used with a single platform on which graduate records for batches 2020 to 2025 are collected, verified, and analyzed. Functionality testing confirmed that these features perform as designed, with the defects found during alpha testing resolved before beta testing rather than carried forward.

The evaluation using the ISO/IEC 25010:2011 Software Quality Model resulted in an Overall Average Mean Score of 4.46, interpreted as Very High, indicating that users strongly agree with the system's performance, usability, and reliability. Performance Efficiency received the highest rating, indicating that the system responds promptly in the tasks graduates and administrators actually perform, while Portability received the lowest score, still rated Very High, pointing toward a mobile-native counterpart to the

present web-based build rather than to any deficiency in its core functions. The findings confirm that the system is not only technically functional but also beneficial to its intended users: BSIS graduates gain a single place to record and update their employment history, employers gain a straightforward means of confirming employment and giving feedback, and the BSIS Program Chairperson and faculty gain batch-level employability data and forecasts that can be carried directly into curriculum review. In summary, the system has achieved its objectives and has demonstrated its value as a tool for strengthening BSIS graduate tracing and supporting evidence-based curriculum improvement.

5.3 Recommendations

Based on the findings of this study, it is recommended that the BSIS Program Chairperson and the BSIS faculty, as the administrators of the system, adopt it as the standing instrument for graduate tracing and schedule the release of the verification and registration links to coincide with graduation rites and alumni gatherings, when graduates are most reachable. They are encouraged to maintain the graduate master list and the controlled reference lists and to review the batch analytics and skills inventory at each curriculum review cycle, so that the gaps identified between graduate competencies and industry demand are translated into concrete course decisions. BSIS graduates, in turn, are encouraged to complete their registration and update their records whenever the two-year re-tracking gate prompts them, and employers to respond to the verification links and complete the confidential evaluation, since the usefulness of the system rests on this participation. It is also advised that the University

explore integrating the system with existing registrar and program records to support a more unified reporting process and reduce duplicate records.

Future improvements may include retraining the predictive model on real graduate responses once a sufficient number has accumulated, along with automated notifications for the two-year re-tracking cycle, an offline mode for graduates with limited internet access, and a mobile-native counterpart addressing the lowest-rated quality characteristic. The face verification may also be strengthened by requiring a submitted photograph to be matched against the live capture when confirming a graduate's identity at sign-in, adding a further layer of assurance beyond the present liveness sequence. Extending the system beyond the BSIS program to the other programs of the College of Computer Studies, and integrating external labor market data so that forecasts can be read against wider industry conditions, are further recommended. Continuous review and updates are advised to ensure the system remains useful, secure, and adaptable to future needs.

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APPENDICES

Appendix A
ISO/IEC 25010:2011 Software Quality Requirements and Evaluation Questionnaire

QUESTIONNAIRE
ISO/IEC 25010:2011
Software Product Quality Model

Name: (Optional) _____
Date: _____

Evaluators (check all that apply):

☐ BSIS Program Chair
☐ Faculty Administrator
☐ IT Expert / Evaluator
☐ System Administrator
☐ Graduate (BSIS Alumni)
☐ Employer

Software Evaluation: CRITERIA-BASED ASSESSMENT (ISO/IEC 25010:2011 for the "Graduate Tracer System with Predictive Employability Trend Analysis") for evaluating the software quality of the developed system. (International Organization for Standardization, 2011)

Instructions:

Please rate the Software Product Quality Model of the "Graduate Tracer System with Predictive Employability Trend Analysis" using the scale below by checking (✓) the level that corresponds to your answer.

Scale:

1 = Strongly Disagree 2 = Disagree 3 = Undecided 4 = Agree 5 = Strongly Agree

Functional Suitability	**1**	**2**	**3**	**4**	**5**
Functional Completeness *The system provides all the functions needed to accomplish the specified tasks and objectives of its intended users.*	☐	☐	☐	☐	☐
Functional Correctness *The system produces accurate and correct results when used by its intended users.*	☐	☐	☐	☐	☐
Functional Appropriateness *The functions provided by the system facilitate the accomplishment of the users' specified tasks and objectives.*	☐	☐	☐	☐	☐
Performance Efficiency	**1**	**2**	**3**	**4**	**5**
Time Behavior *The system's response time and throughput rates meet requirements when performing its functions.*	☐	☐	☐	☐	☐
Resource Utilization *The amount and types of resources used by the system meet requirements.*	☐	☐	☐	☐	☐
Capacity *The system's maximum limits (e.g., number of users, records, storage) meet the requirements of its intended use.*	☐	☐	☐	☐	☐
Compatibility	**1**	**2**	**3**	**4**	**5**
Co-existence *The system performs its required functions efficiently while sharing a common environment and resources with other products.*	☐	☐	☐	☐	☐
Interoperability *The system can exchange information with other products and mutually use the information that has been exchanged.*	☐	☐	☐	☐	☐
Usability	**1**	**2**	**3**	**4**	**5**

Appendix B: System Usability Scale (SUS) Questionnaire
Appendix C:

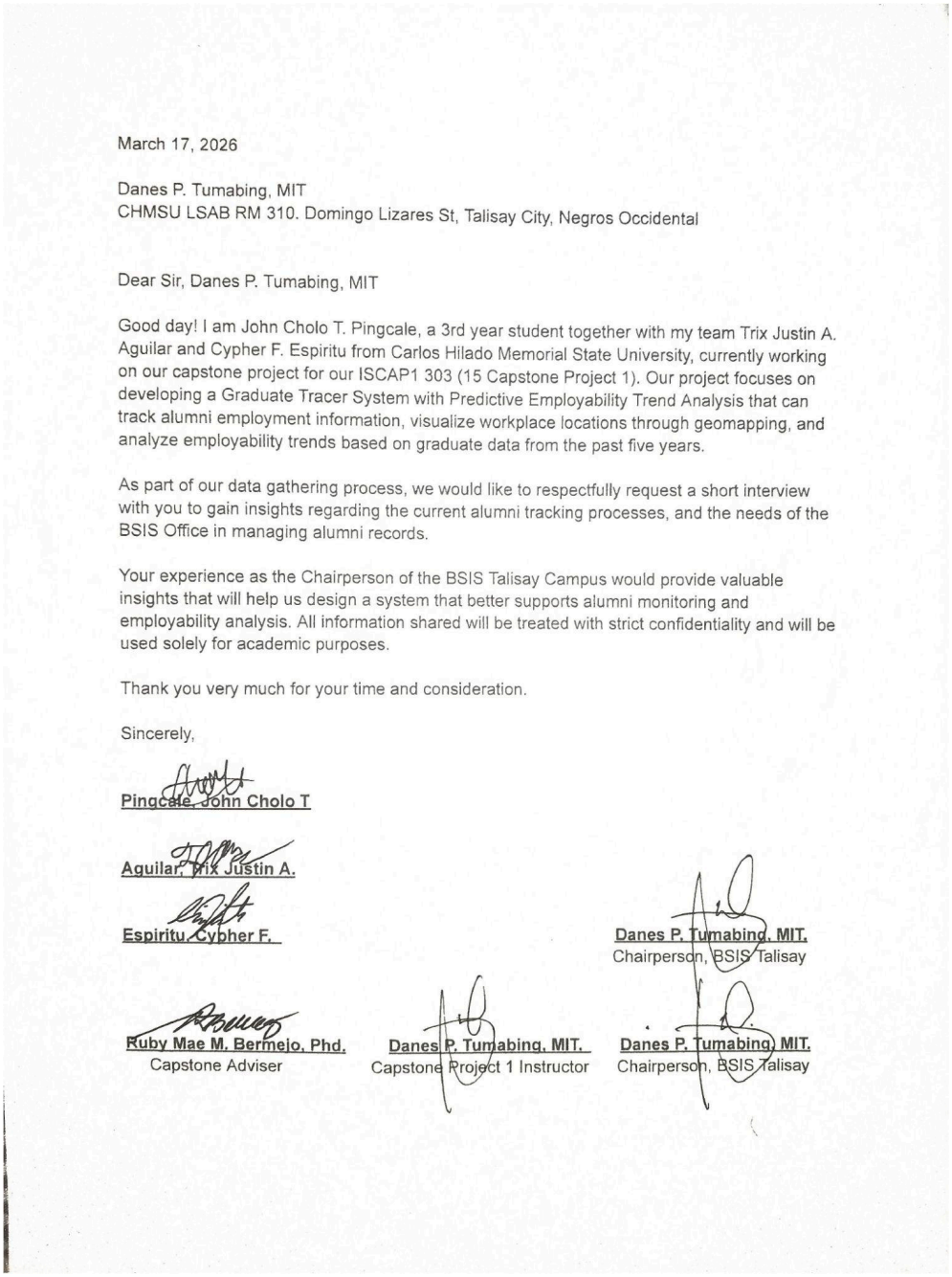
Appropriateness Recognizability *I can easily recognize whether the system is appropriate for my needs.*	☐	☐	☐	☐	☐
Learnability *The functions of the system can be learned and used within a reasonable amount of time.*	☐	☐	☐	☐	☐
Operability *The system has attributes that make it easy to operate and control.*	☐	☐	☐	☐	☐
User Error Protection *The system prevents users from making operation errors.*	☐	☐	☐	☐	☐
User Interface Aesthetics *The user interface enables pleasing and satisfying interaction for the user.*	☐	☐	☐	☐	☐
Accessibility *The system can be used by people with the widest range of characteristics and capabilities to achieve specified goals.*	☐	☐	☐	☐	☐
Reliability	**1**	**2**	**3**	**4**	**5**
Faultlessness *The system performs its specified functions without fault under normal operation.*	☐	☐	☐	☐	☐
Availability *The system is operational and accessible when required for use.*	☐	☐	☐	☐	☐
Fault Tolerance *The system operates as intended despite the presence of hardware or software faults.*	☐	☐	☐	☐	☐
Recoverability *In the event of an interruption or failure, the system can recover the data affected and re-establish the desired state.*	☐	☐	☐	☐	☐
Security	**1**	**2**	**3**	**4**	**5**
Confidentiality *The system ensures that data are accessible only to those authorized to have access.*	☐	☐	☐	☐	☐
Integrity *The system ensures that its data are protected from unauthorized modification or deletion.*	☐	☐	☐	☐	☐
Non-repudiation *Actions or events performed in the system can be proven to have taken place so they cannot be repudiated later.*	☐	☐	☐	☐	☐
Accountability *The actions of an entity in the system can be traced uniquely to that entity.*	☐	☐	☐	☐	☐
Authenticity *The identity of a subject or resource within the system can be proved to be the one claimed.*	☐	☐	☐	☐	☐
Maintainability	**1**	**2**	**3**	**4**	**5**
Modularity *The system is composed of discrete components such that a change to one component has minimal impact on other components.*	☐	☐	☐	☐	☐
Reusability *Parts of the system can be used as assets in more than one system, or in building other assets.*	☐	☐	☐	☐	☐
Analyzability *It is possible to assess the impact on the system of an intended change or to diagnose deficiencies or failures.*	☐	☐	☐	☐	☐
Modifiability	☐	☐	☐	☐	☐

The system can be effectively modified without introducing defects or degrading existing product quality.					
Testability *Test criteria can be established for the system, and tests can be performed to determine whether those criteria have been met.*	☐	☐	☐	☐	☐
Portability	**1**	**2**	**3**	**4**	**5**
Adaptability *The system can be adapted for different or evolving hardware, software, or other operational or usage environments.*	☐	☐	☐	☐	☐
Installability *The system can be successfully installed and/or uninstalled in a specified environment.*	☐	☐	☐	☐	☐
Replaceability *The system can replace another specified software product for the same purpose in the same environment.*	☐	☐	☐	☐	☐

Thank you and GOD Bless!

Appendix B

LETTER TO BENEFICIARY



March 17, 2026

Danes P. Tumabing, MIT
CHMSU LSAB RM 310. Domingo Lizares St, Talisay City, Negros Occidental

Dear Sir, Danes P. Tumabing, MIT

Good day! I am John Cholo T. Pingcale, a 3rd year student together with my team Trix Justin A. Aguilar and Cypher F. Espiritu from Carlos Hilado Memorial State University, currently working on our capstone project for our ISCAP1 303 (15 Capstone Project 1). Our project focuses on developing a Graduate Tracer System with Predictive Employability Trend Analysis that can track alumni employment information, visualize workplace locations through geomapping, and analyze employability trends based on graduate data from the past five years.

As part of our data gathering process, we would like to respectfully request a short interview with you to gain insights regarding the current alumni tracking processes, and the needs of the BSIS Office in managing alumni records.

Your experience as the Chairperson of the BSIS Talisay Campus would provide valuable insights that will help us design a system that better supports alumni monitoring and employability analysis. All information shared will be treated with strict confidentiality and will be used solely for academic purposes.

Thank you very much for your time and consideration.

Sincerely,

Pingcale, John Cholo T

Aguilar, Trix Justin A.

Espiritu, Cypher F.

Danes P. Tumabing, MIT.
Chairperson, BSIS Talisay

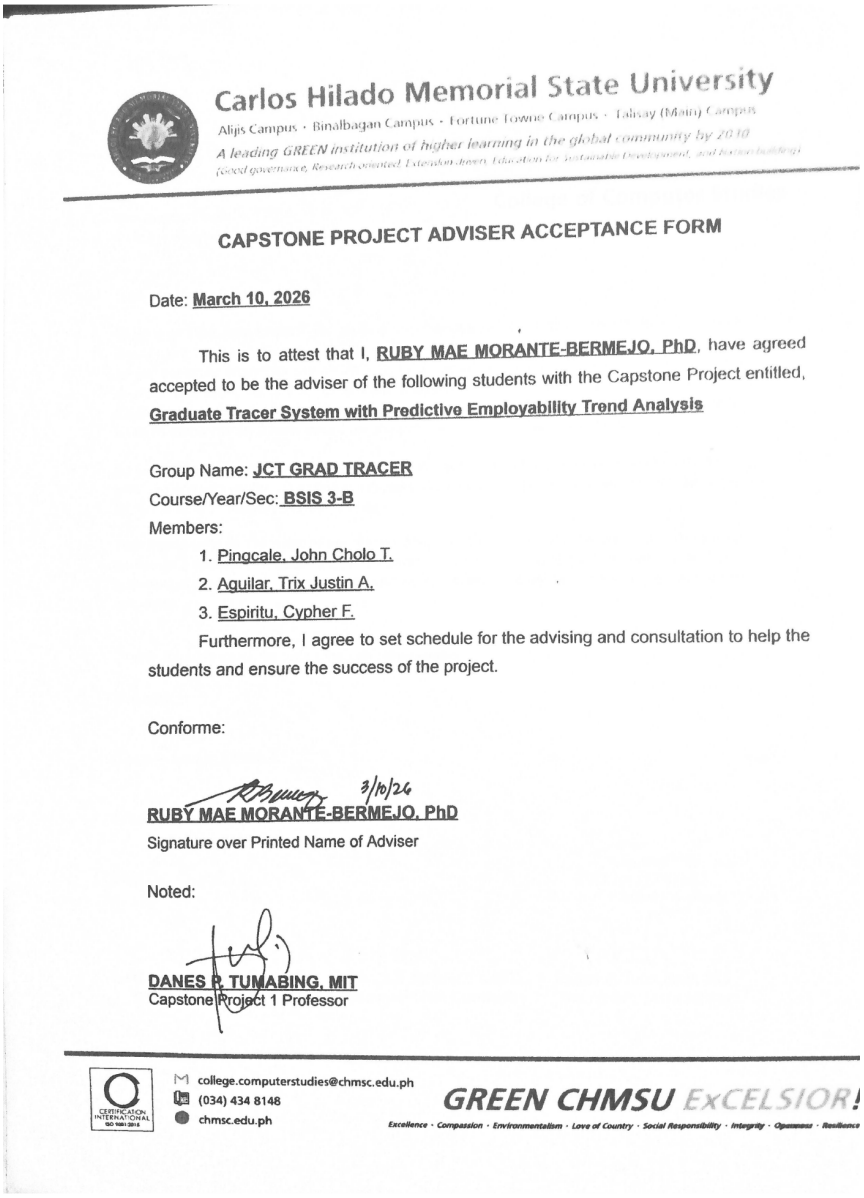
Ruby Mae M. Bermejo, Phd.
Capstone Adviser

Danes P. Tumabing, MIT.
Capstone Project 1 Instructor

Danes P. Tumabing, MIT.
Chairperson, BSIS Talisay

Appendix C

LETTER TO CAPSTONE PROJECT ADVISER



Carlos Hilado Memorial State University
Alijis Campus · Binalbagan Campus · Fortune Towne Campus · Talisay (Main) Campus
A leading GREEN institution of higher learning in the global community by 2030
(Good governance, Research oriented, Extension driven, Education for sustainable development, and Green buildings)

CAPSTONE PROJECT ADVISER ACCEPTANCE FORM

Date: **March 10, 2026**

This is to attest that I, **RUBY MAE MORANTE-BERMEJO, PhD**, have agreed accepted to be the adviser of the following students with the Capstone Project entitled, **Graduate Tracer System with Predictive Employability Trend Analysis**

Group Name: **JCT GRAD TRACER**
Course/Year/Sec: **BSIS 3-B**
Members:

1. Pingcale, John Cholo T.
2. Aguilar, Trix Justin A.
3. Espiritu, Cypher F.

Furthermore, I agree to set schedule for the advising and consultation to help the students and ensure the success of the project.

Conforme:

3/10/26
RUBY MAE MORANTE-BERMEJO, PhD
Signature over Printed Name of Adviser

Noted:

DANES P. TUMABING, MIT
Capstone Project 1 Professor

college.computerstudies@chmsc.edu.ph
(034) 434 8148
chmsc.edu.ph

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Appendix D

BLACK-BOX TESTING – ALPHA TESTING

Figure 28

Account Creation Step Displaying the Password Policy Fully Satisfied

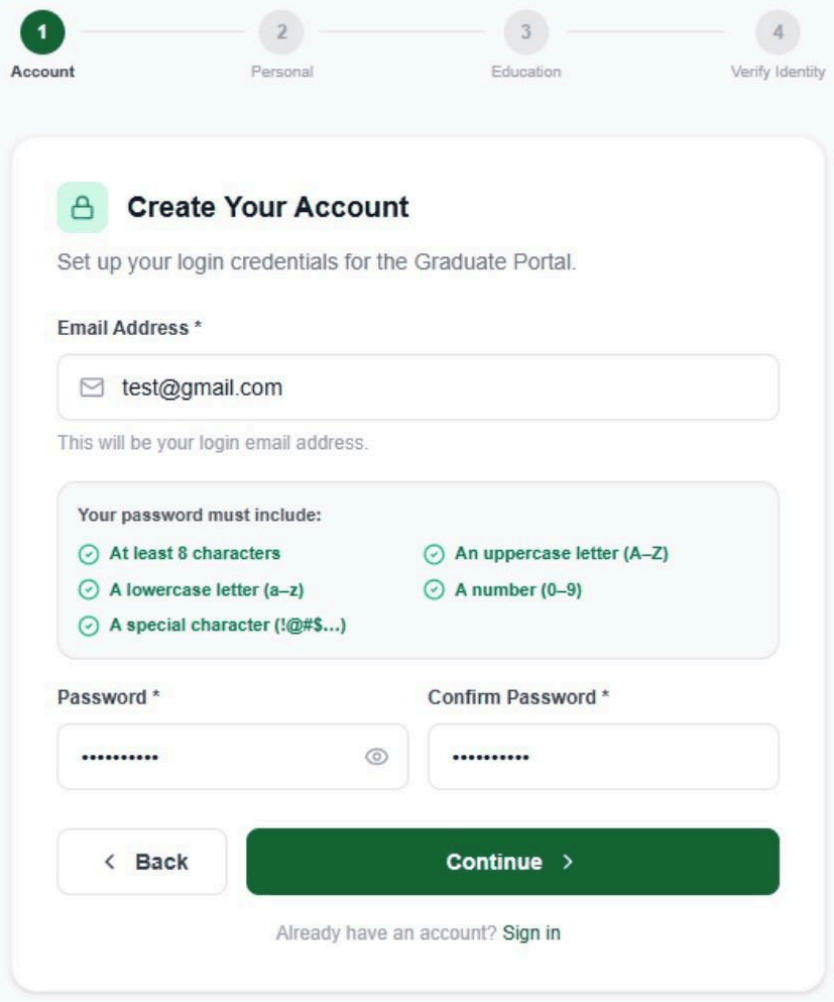


Figure 29

Account Creation Step Rejecting a Password That Fails the Stated Policy

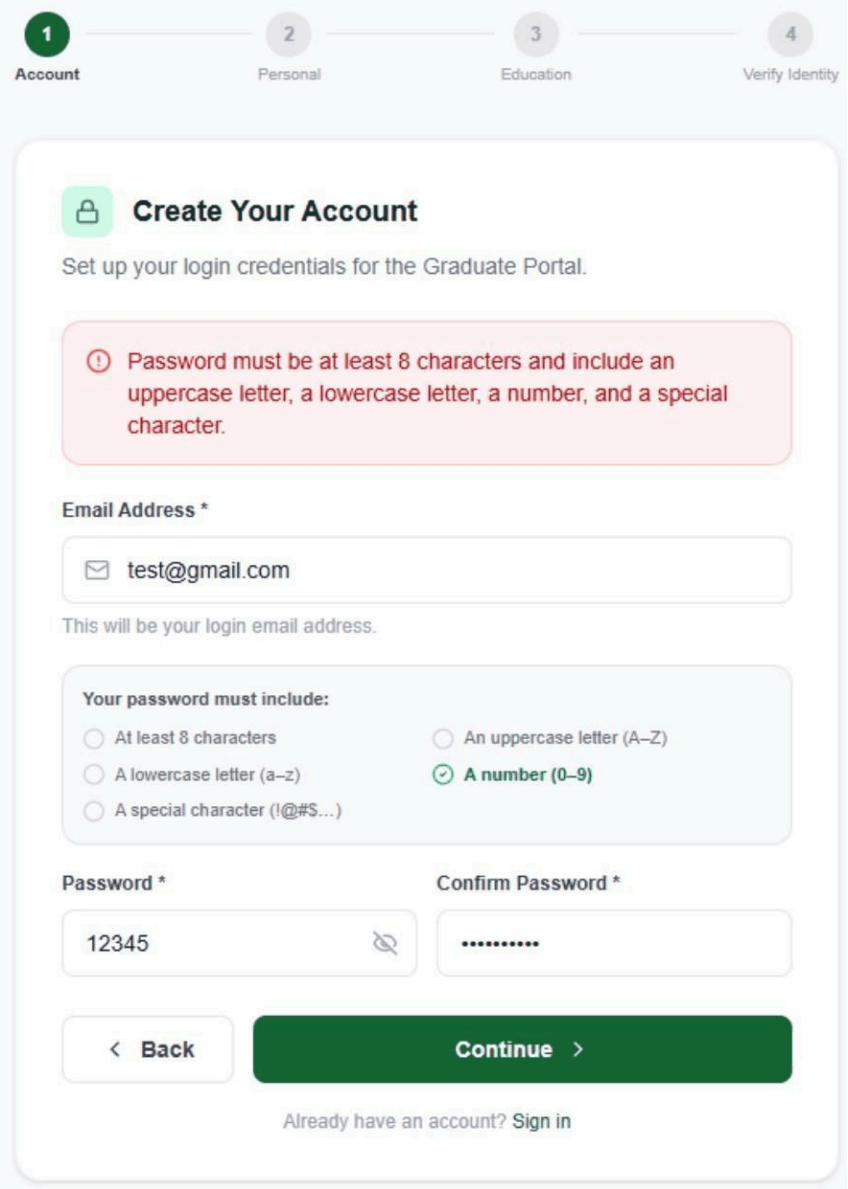


Figure 30

Personal Information Step Rejecting an Invalid Philippine Mobile Number

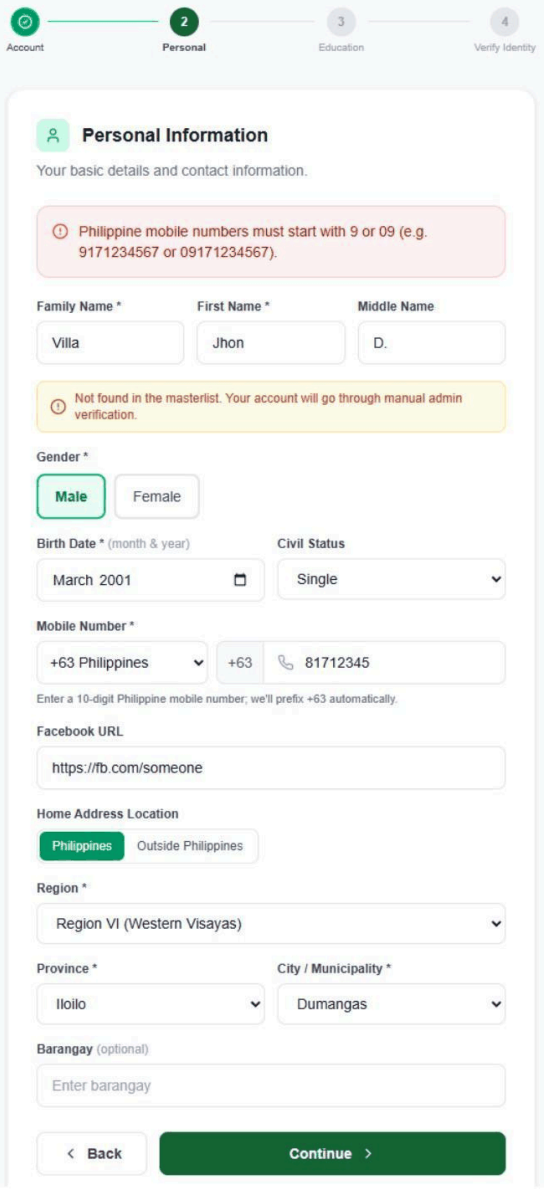


Figure 31

Personal Information Step Rejecting a Non-Facebook Profile Link and Showing the Cascading Address Selection

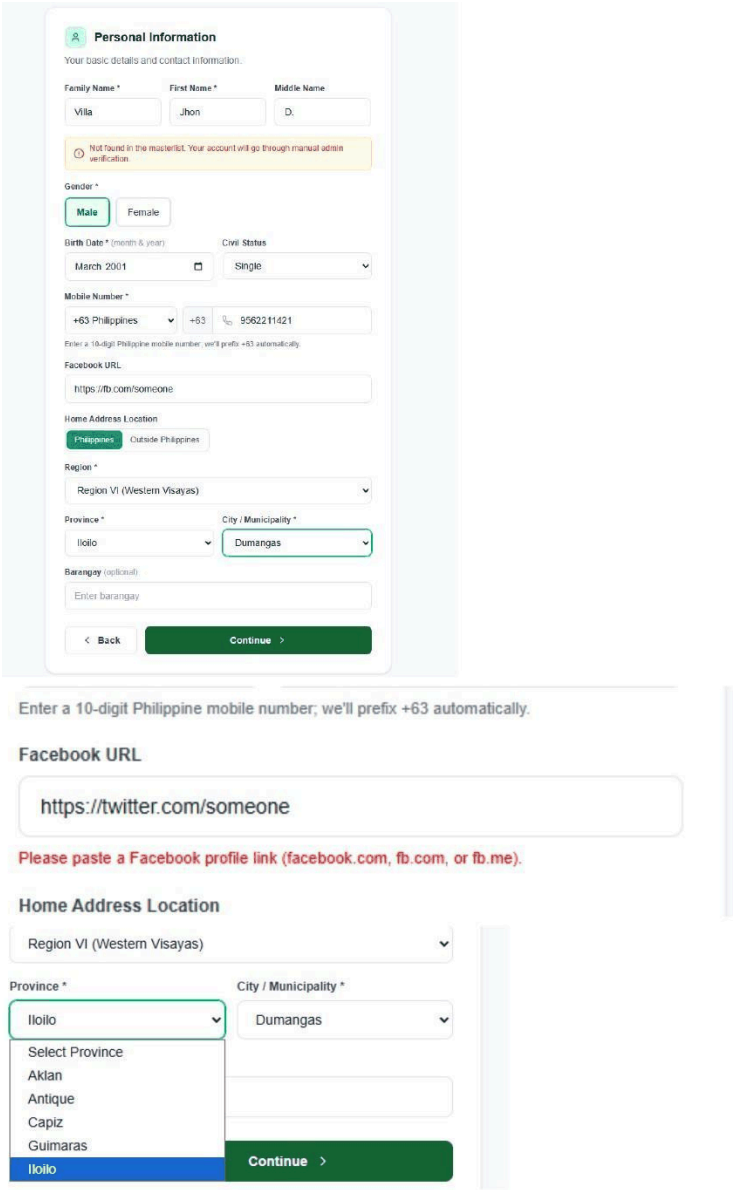


Figure 32

Identity Verification Step Executing the Liveness Challenges in Sequence

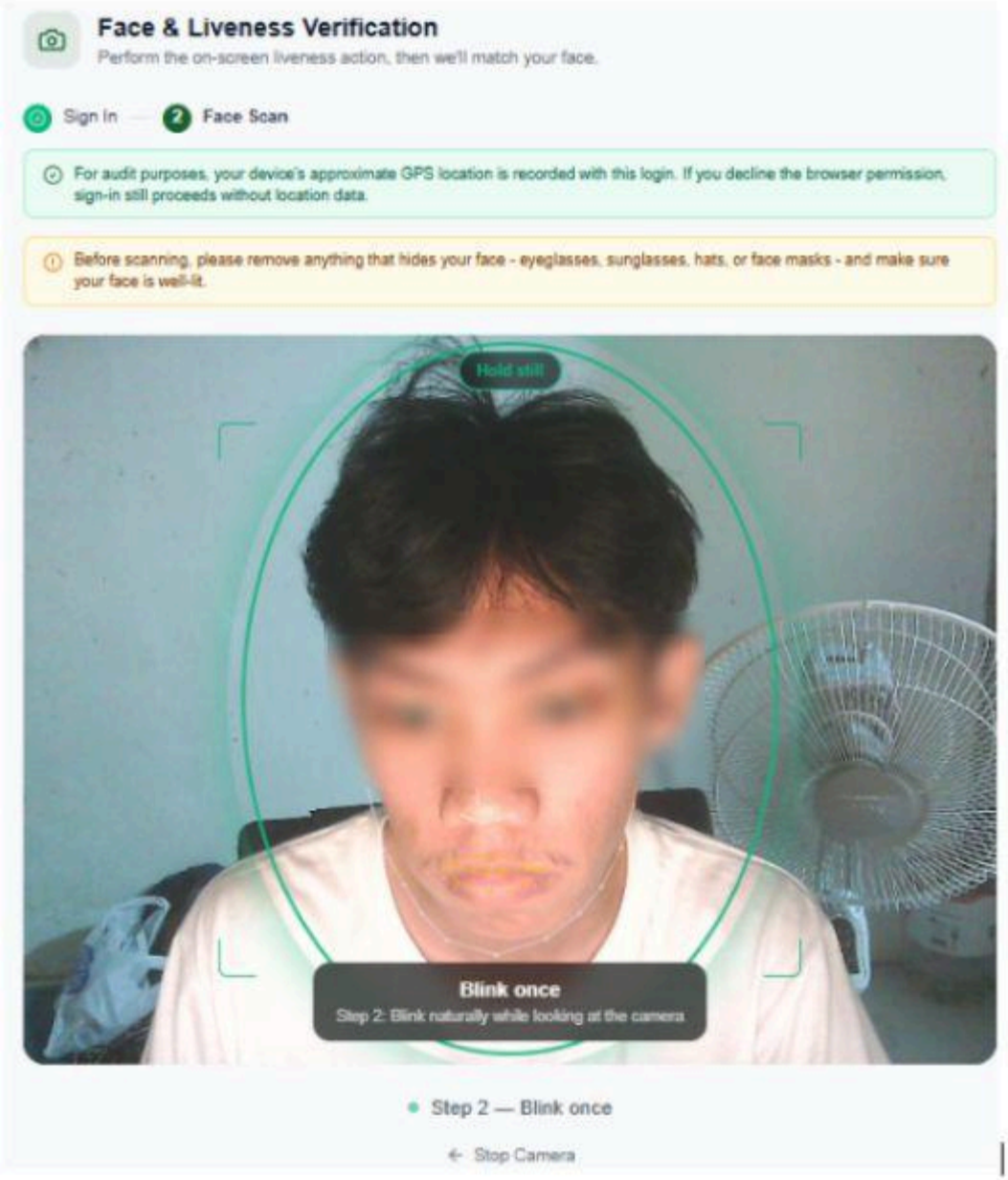


Figure 33

Identity Verification Step After All Liveness Challenges Were Passed

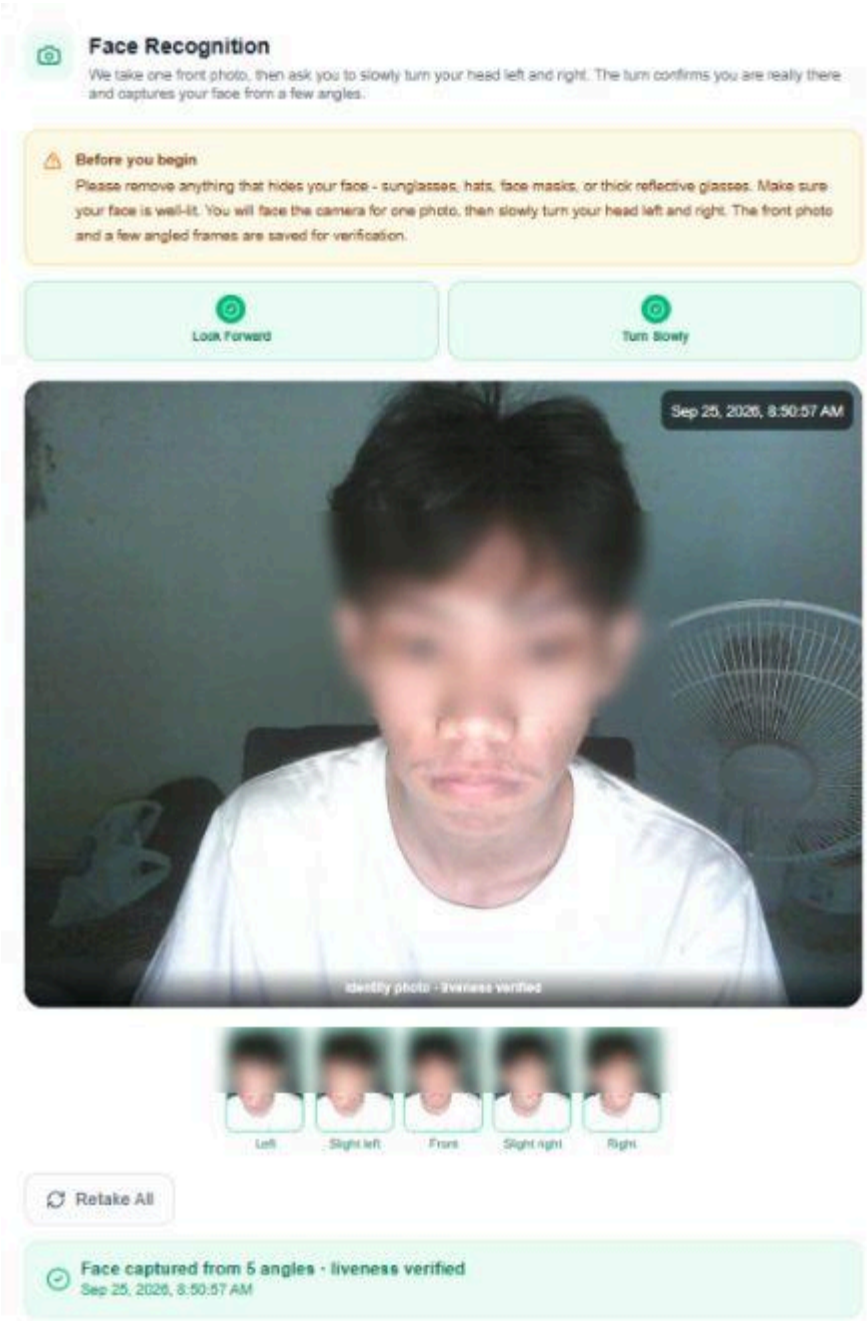


Figure 34

Selection of Technical Skills from the Controlled Reference List

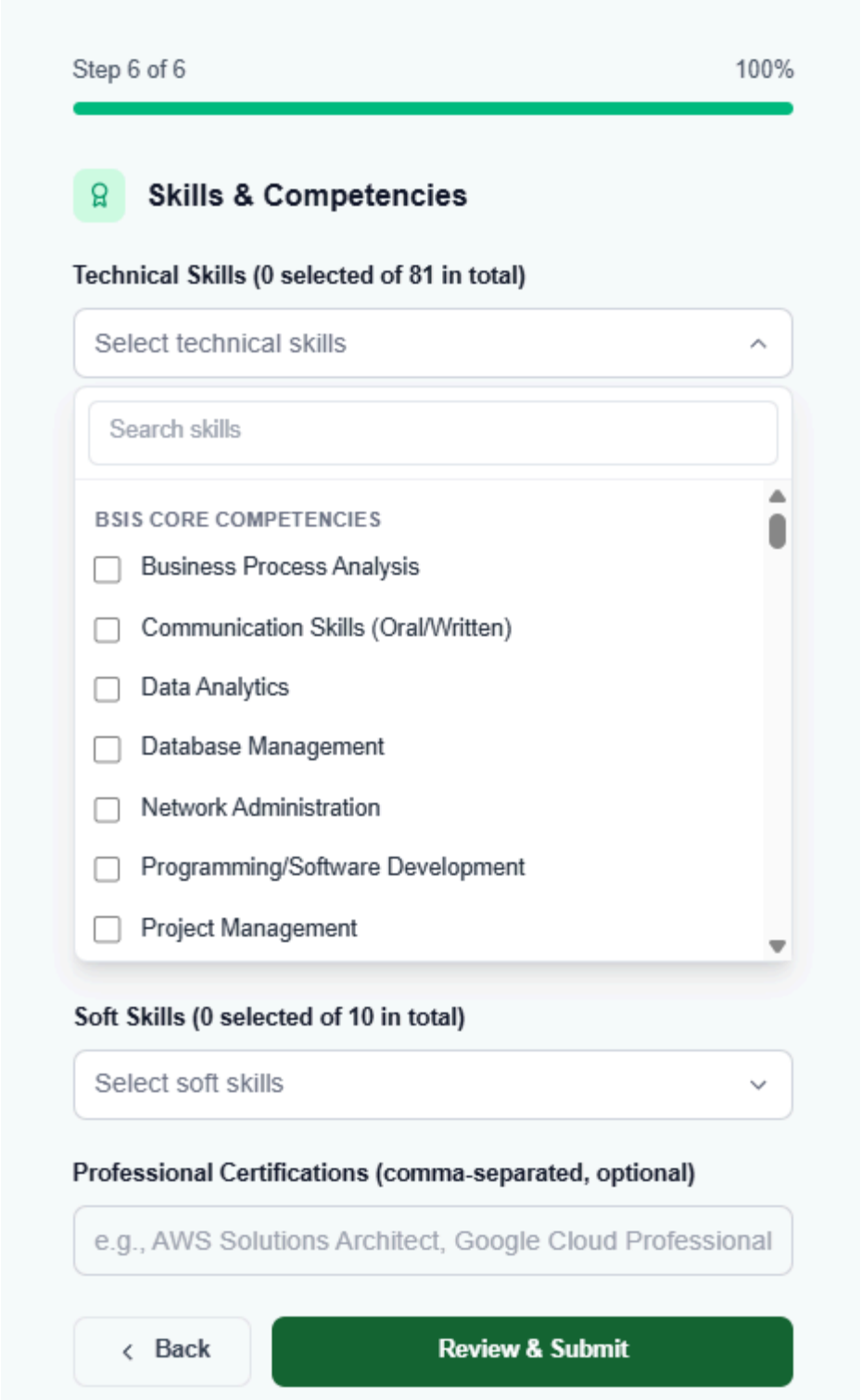


Figure 35

Skills and Competencies Step with the Maximum Selections Reached

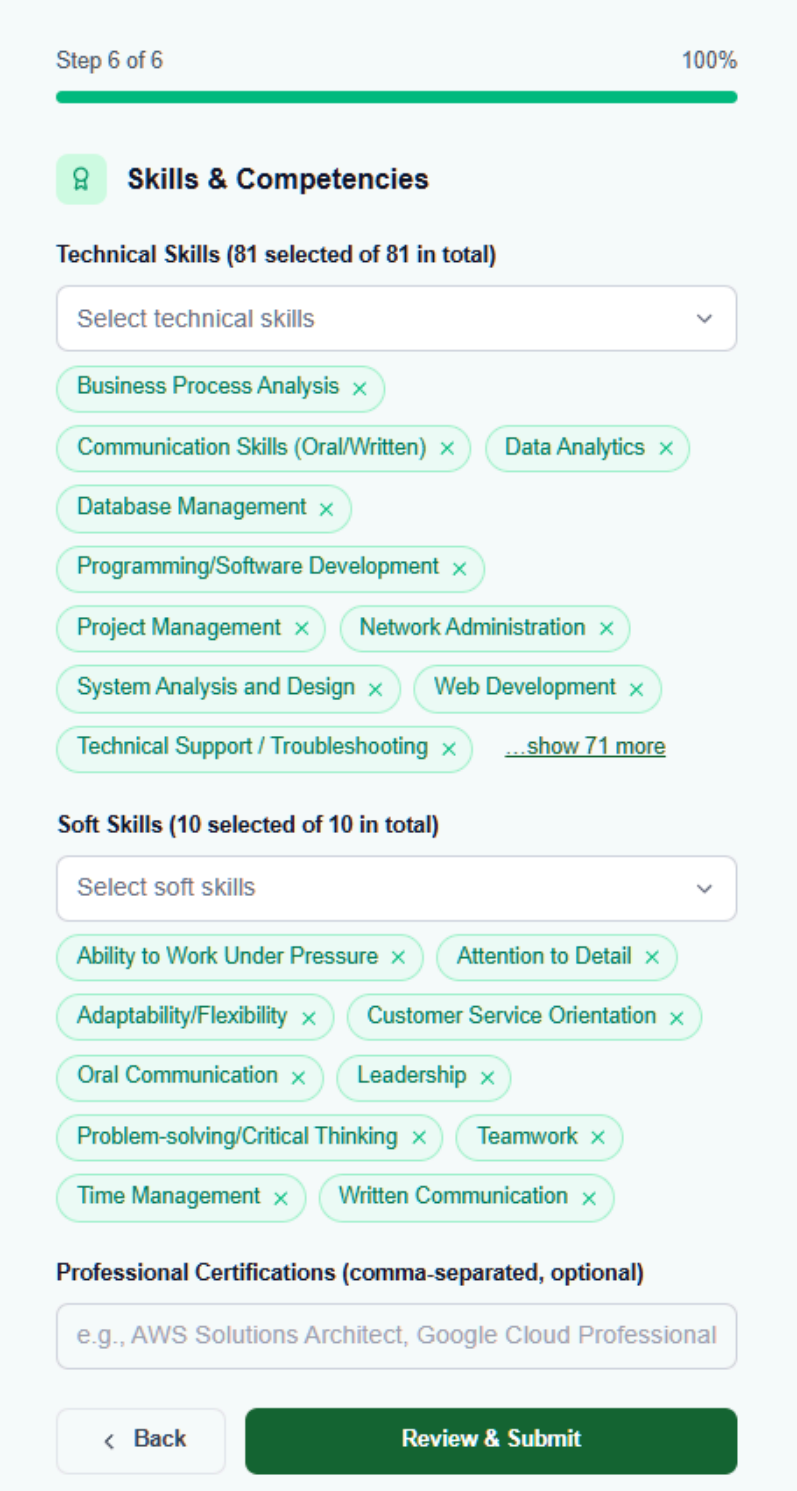


Figure 36

Terms and Data Privacy Consent Screen with the Optional Geomapping Consent

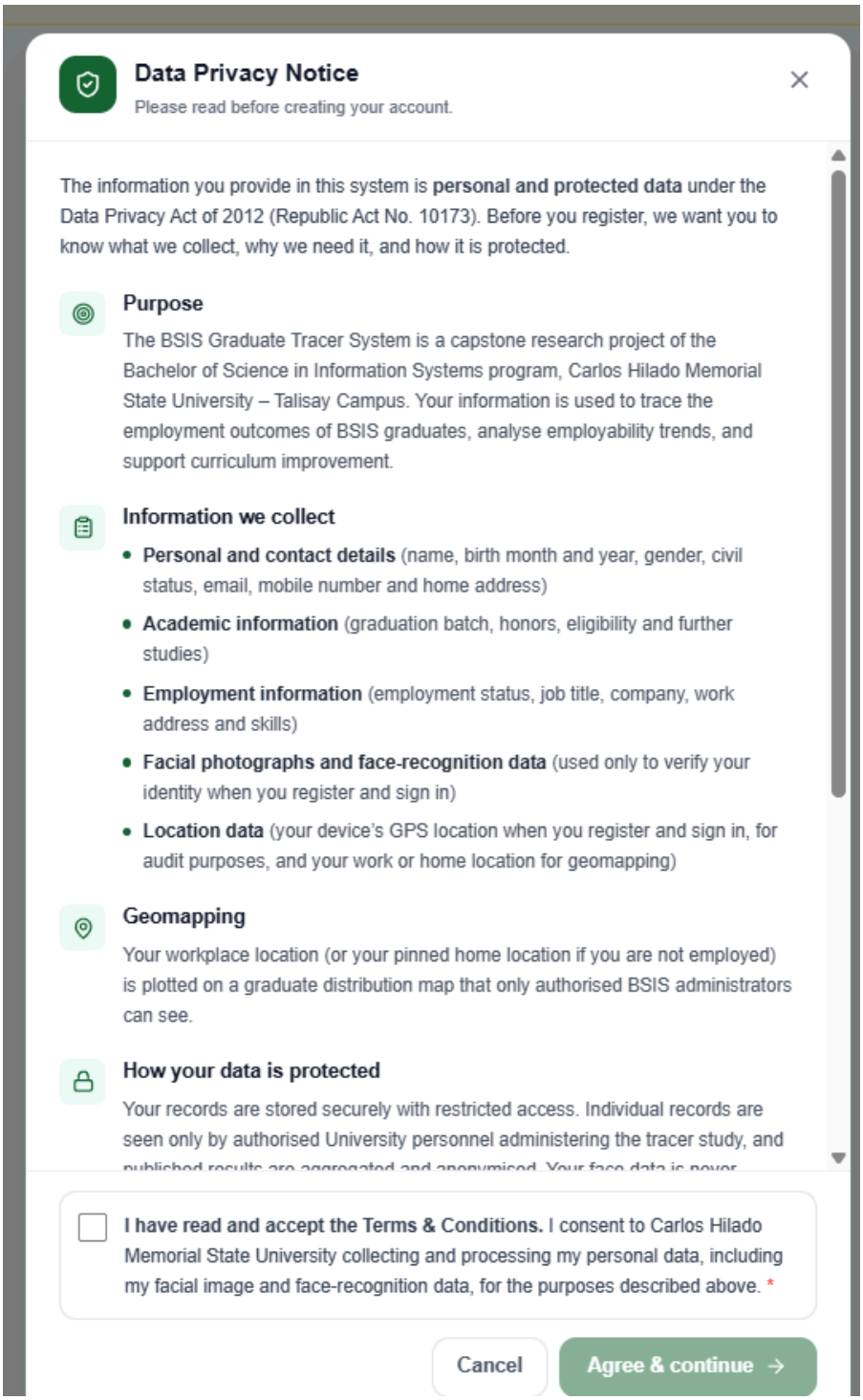


Figure 37

Sign-In Screen Rejecting an Unregistered Email Address

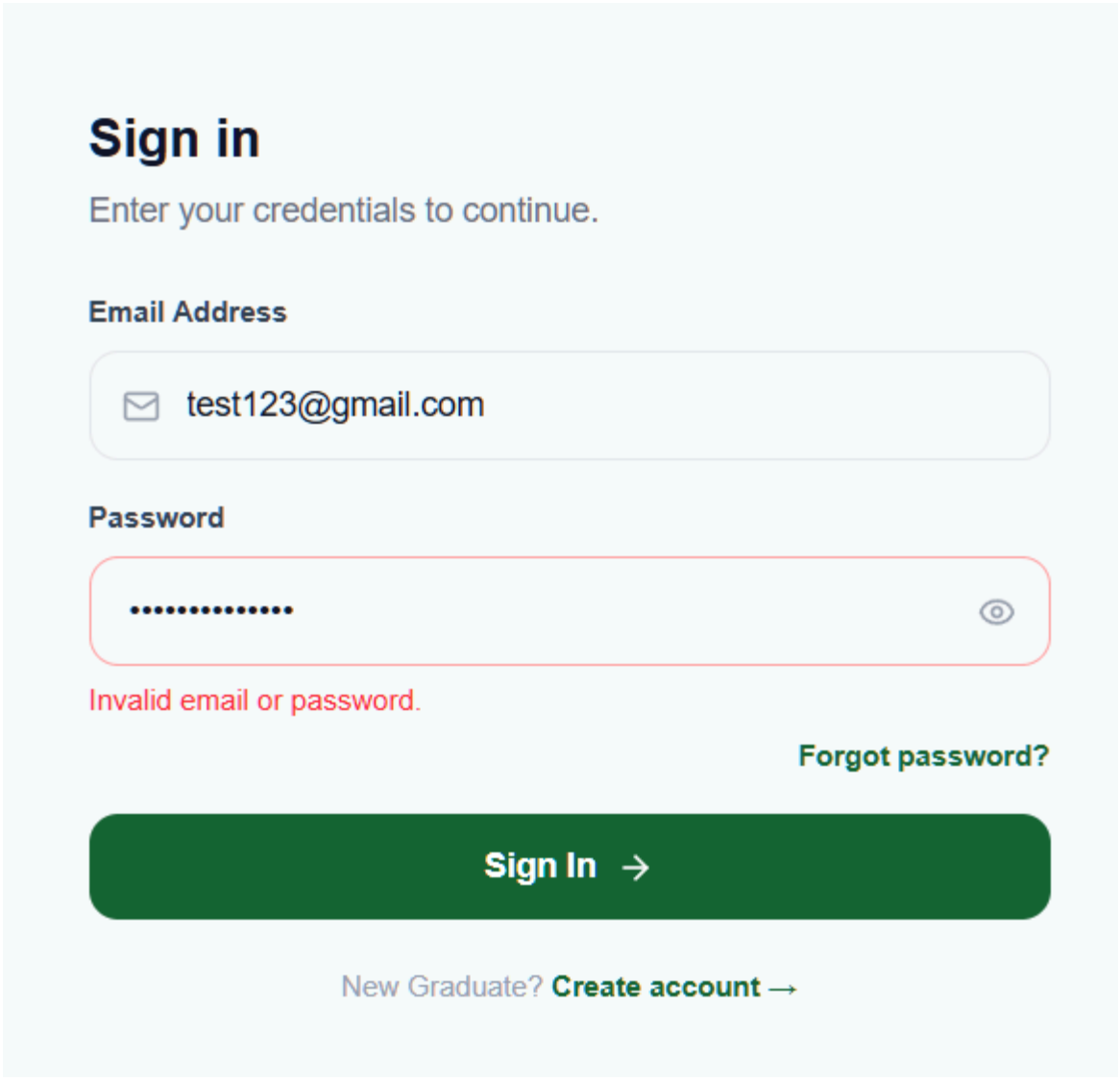


Figure 38

Face Scan at Sign-In Refusing an Unclear Capture

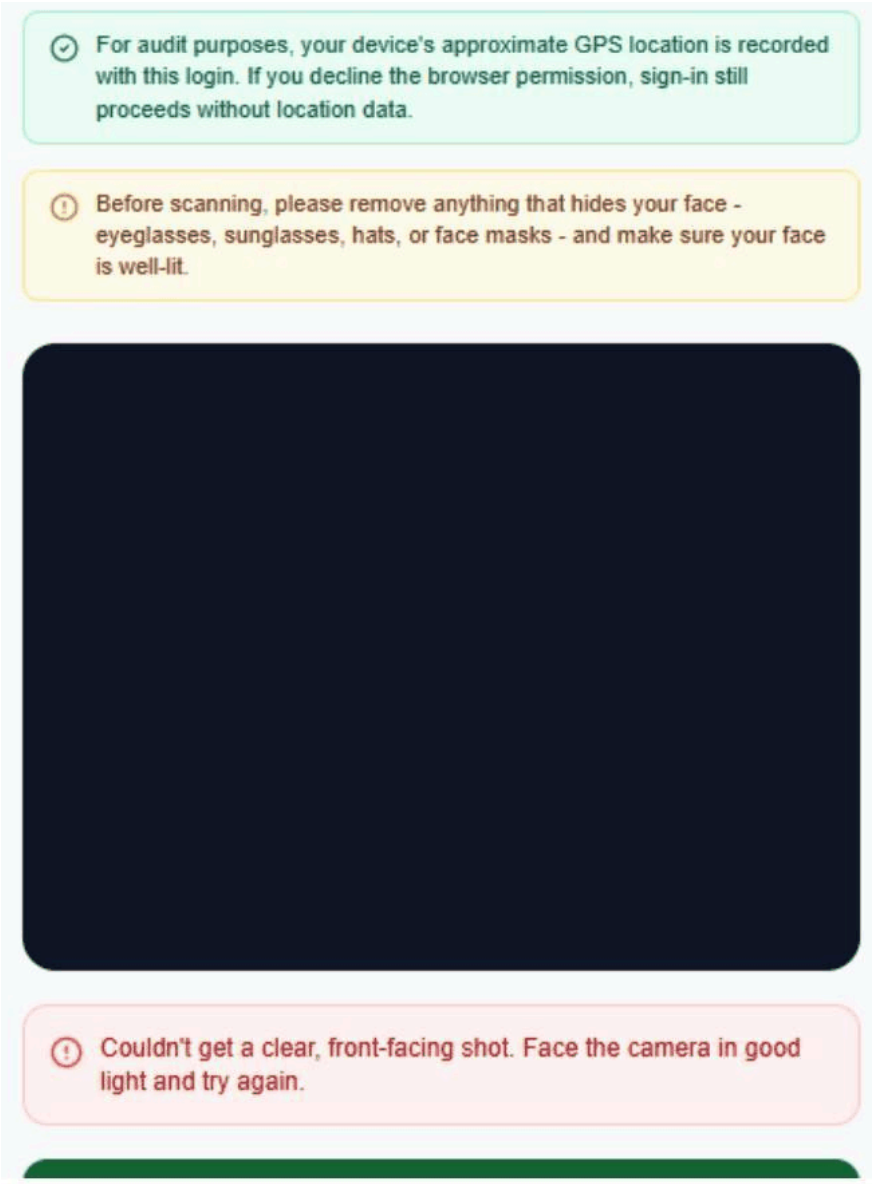


Figure 39

Graduate Dashboard Displaying the Verified Account Status

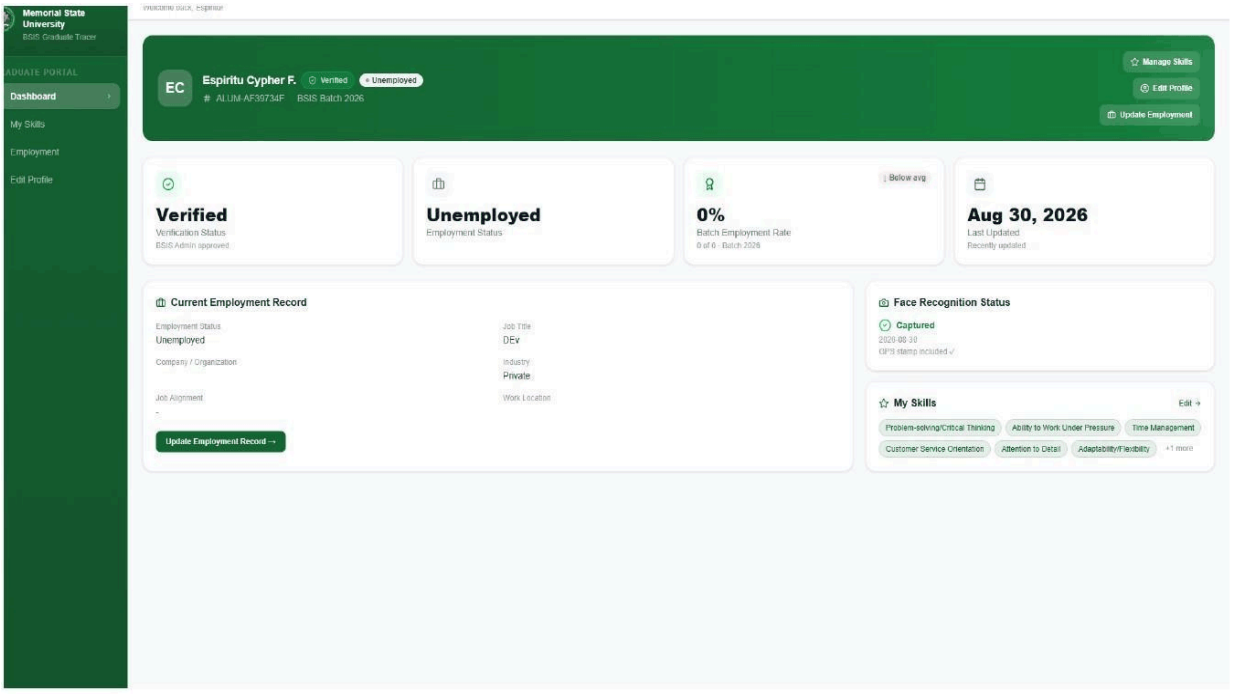


Figure 40

Graduate Dashboard Showing the Employment Record, the Face Recognition Status and the Recorded Skills

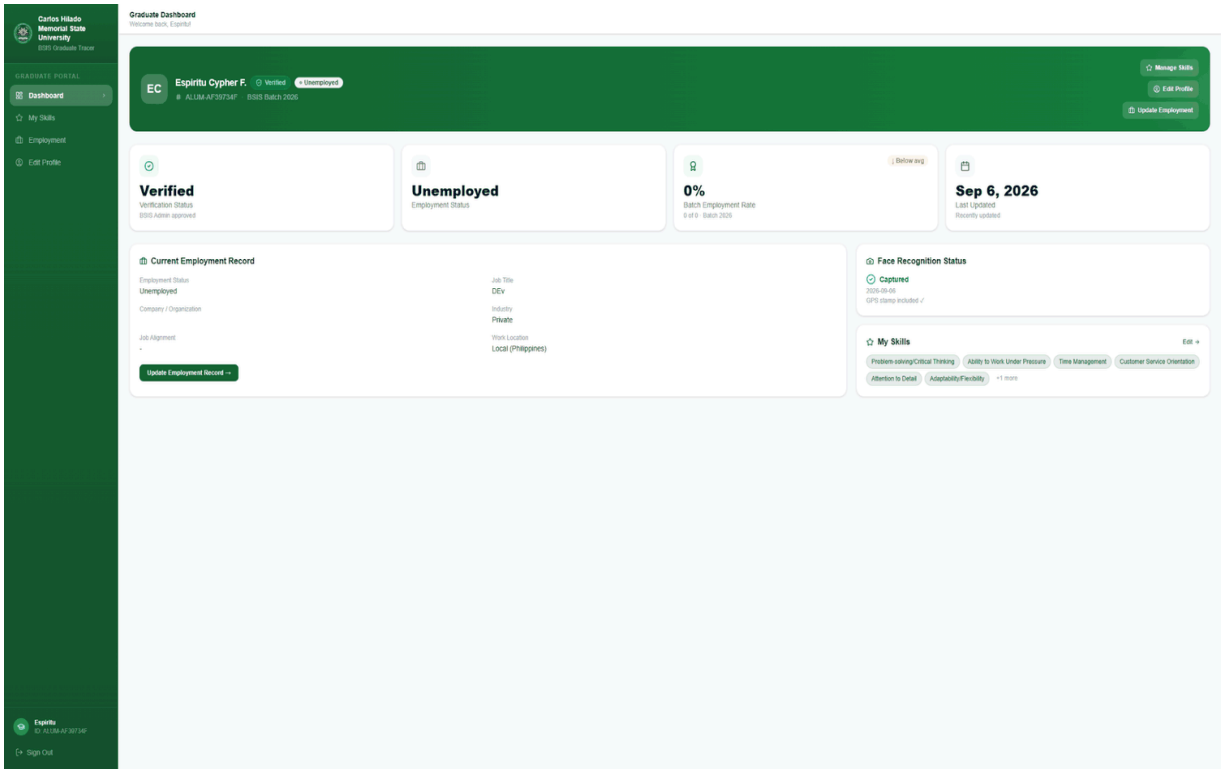


Figure 41

Graduate Generating a Single-Use Employer Verification Link From the Dashboard

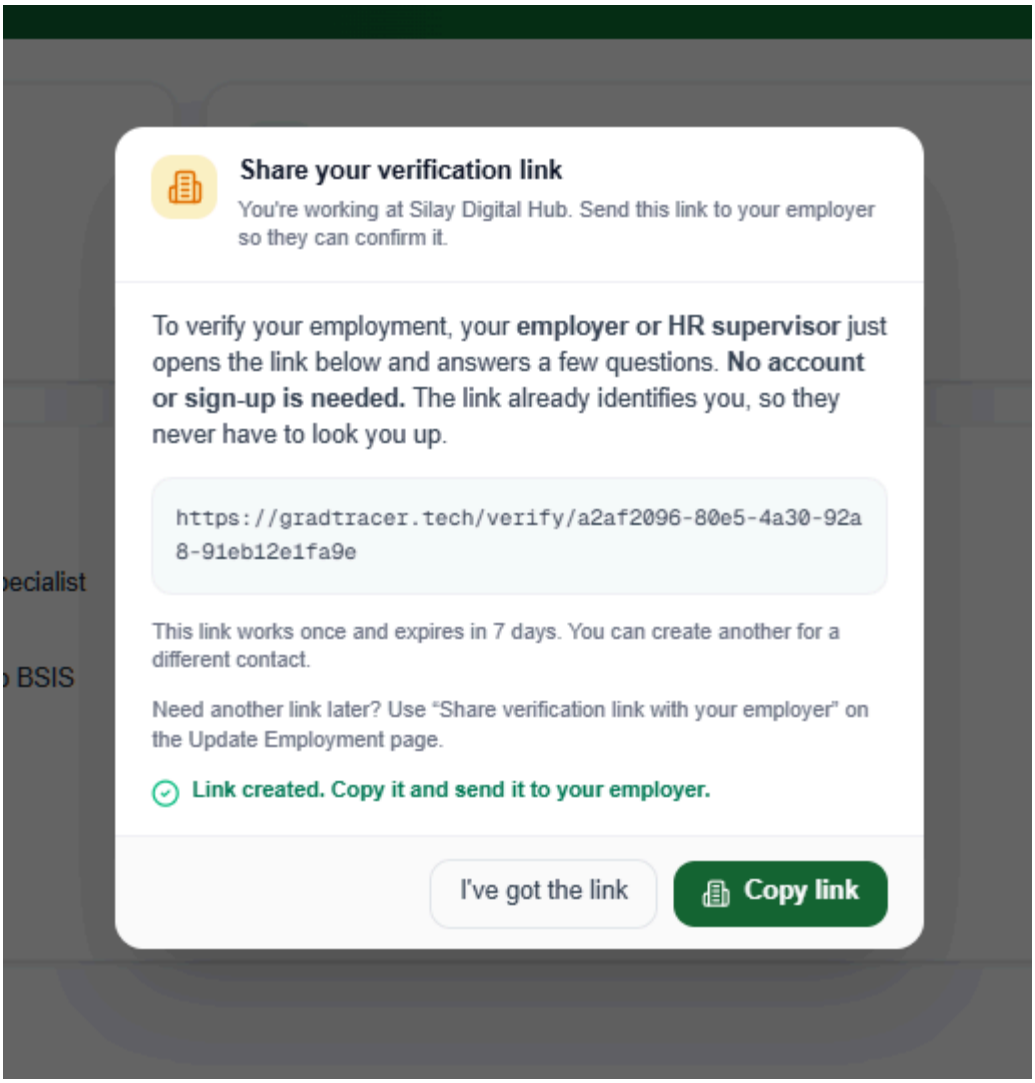


Figure 42

Employer Verification Page Opened Directly From the Link Without an Employer Account

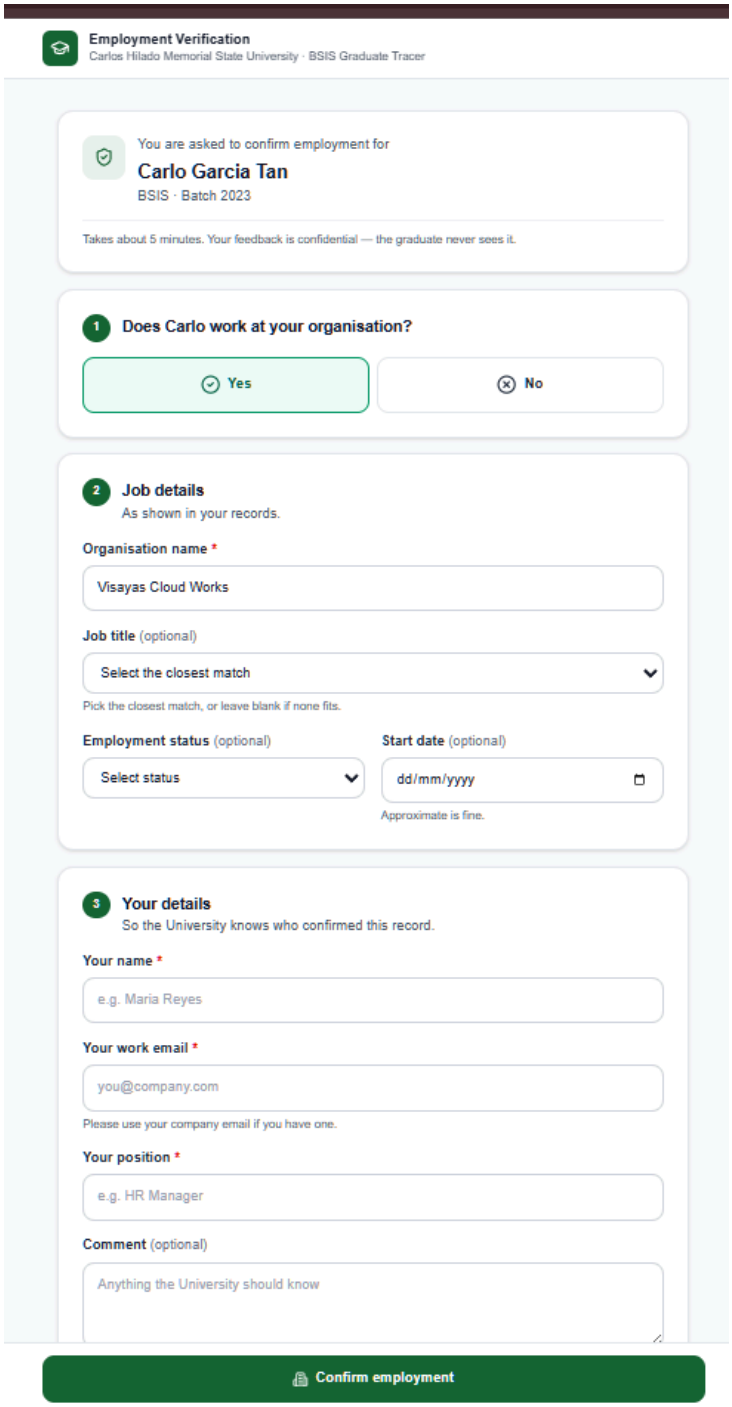


Figure 43

Employment Verification Form Completed by the Employer With the Verifier Identity Supplied

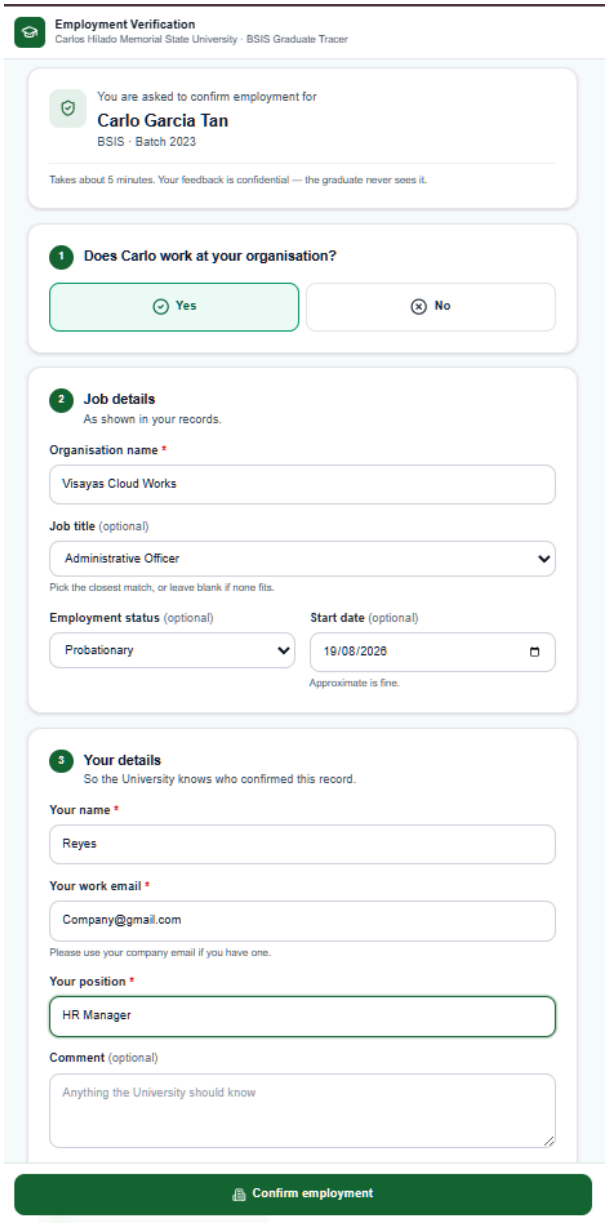


Figure 44

Confirmation Recorded After the Employer Submitted the Verification

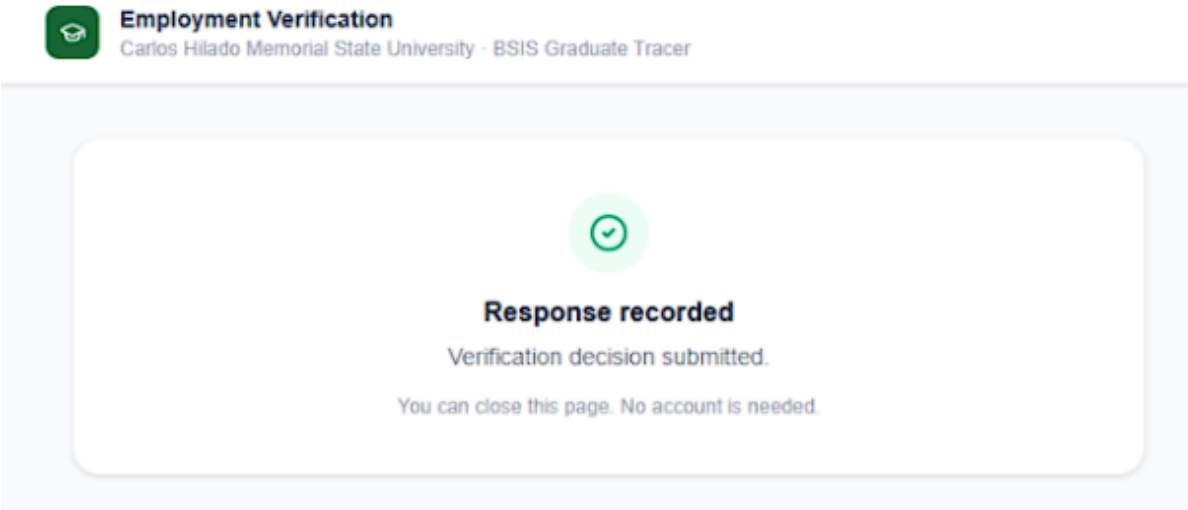


Figure 45

Refusal of a Verification Link That Had Already Been Used

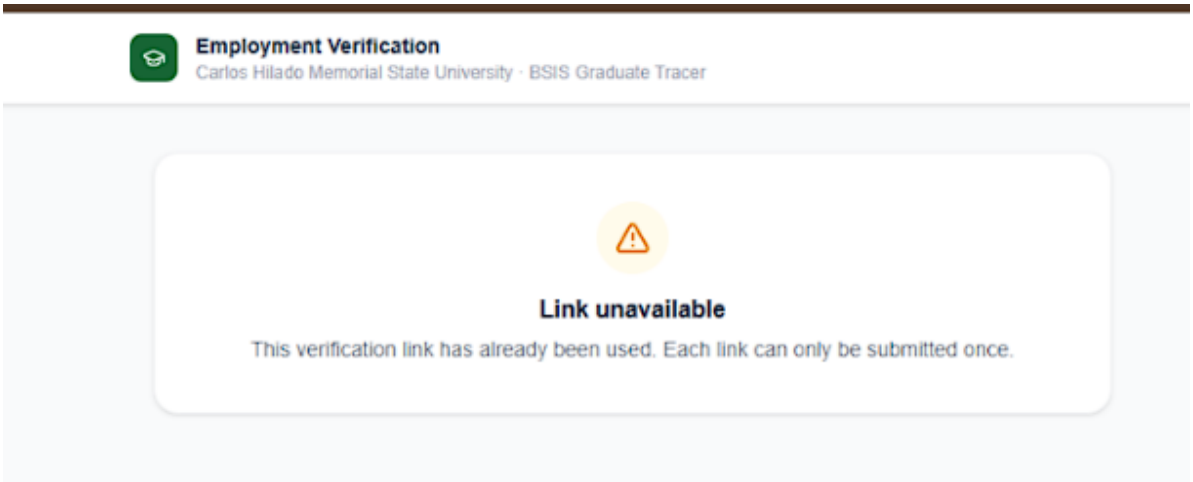


Figure 46

Plain Message Returned for an Invalid or Withdrawn Verification Link

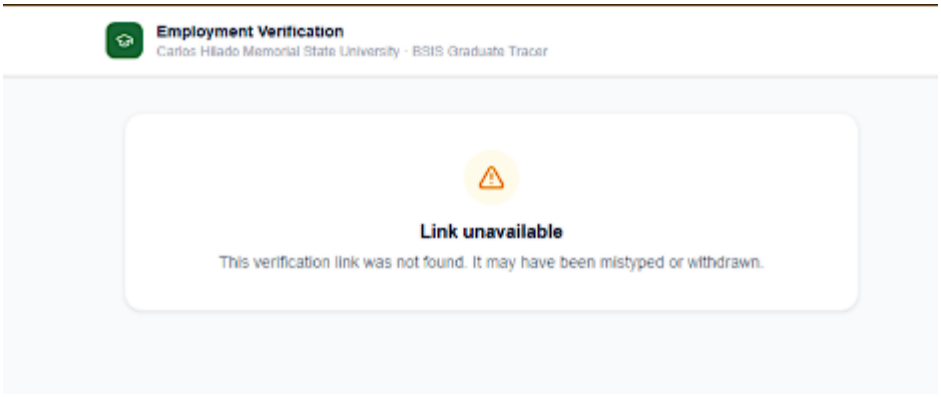


Figure 47

Administrator Sign-In Screen of the Graduate Tracer System

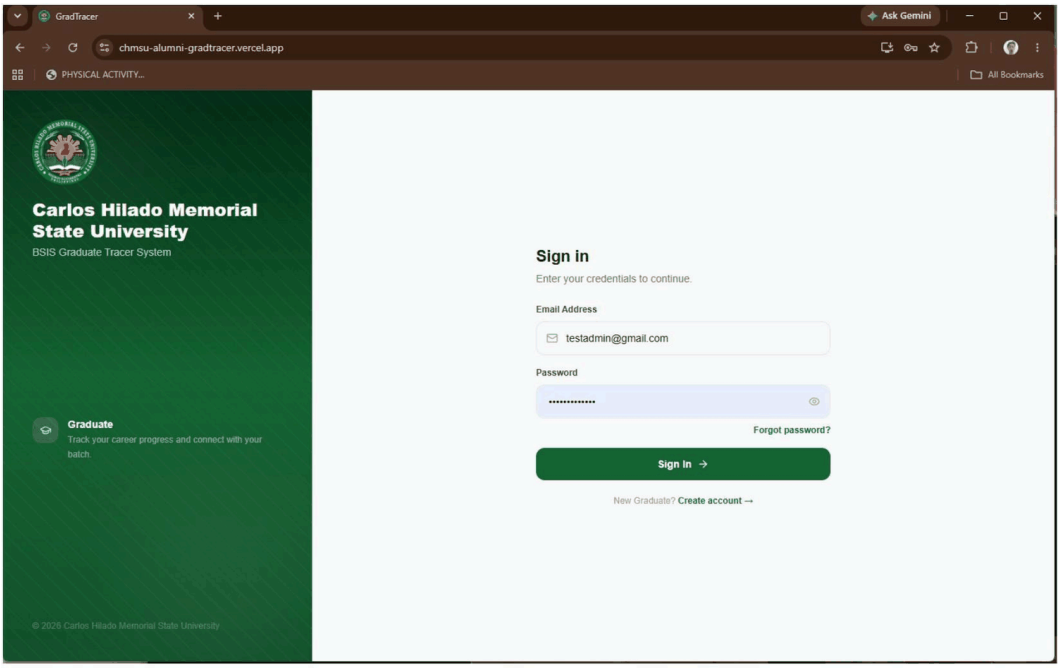


Figure 48

Administrator Dashboard Showing the Summary Indicators and the Curriculum Alignment

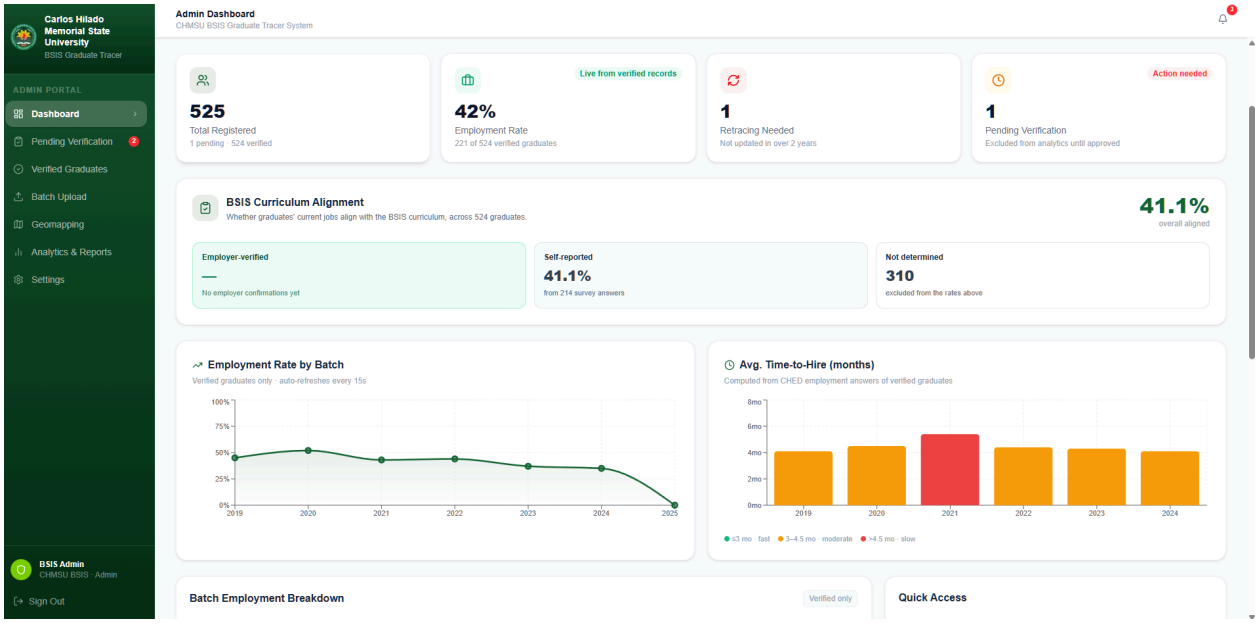


Figure 49

Administrator Dashboard Showing the Recent Profile Updates and the Top Skills

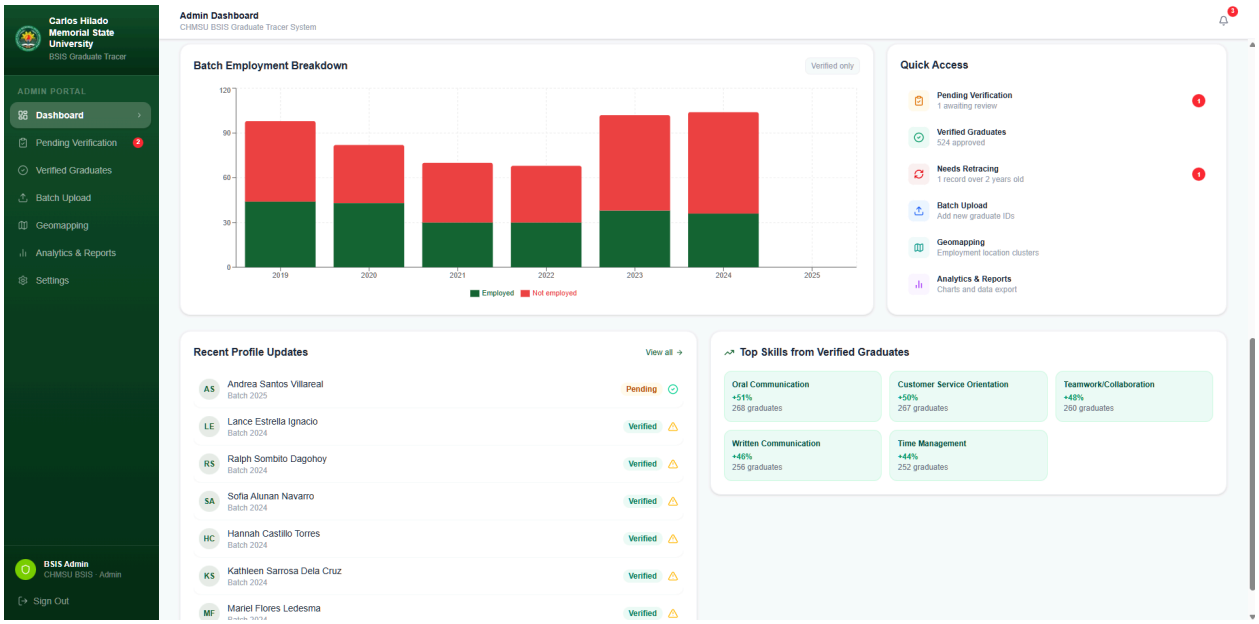


Figure 50

Administrator Rejecting a Registration with a Stated Reason

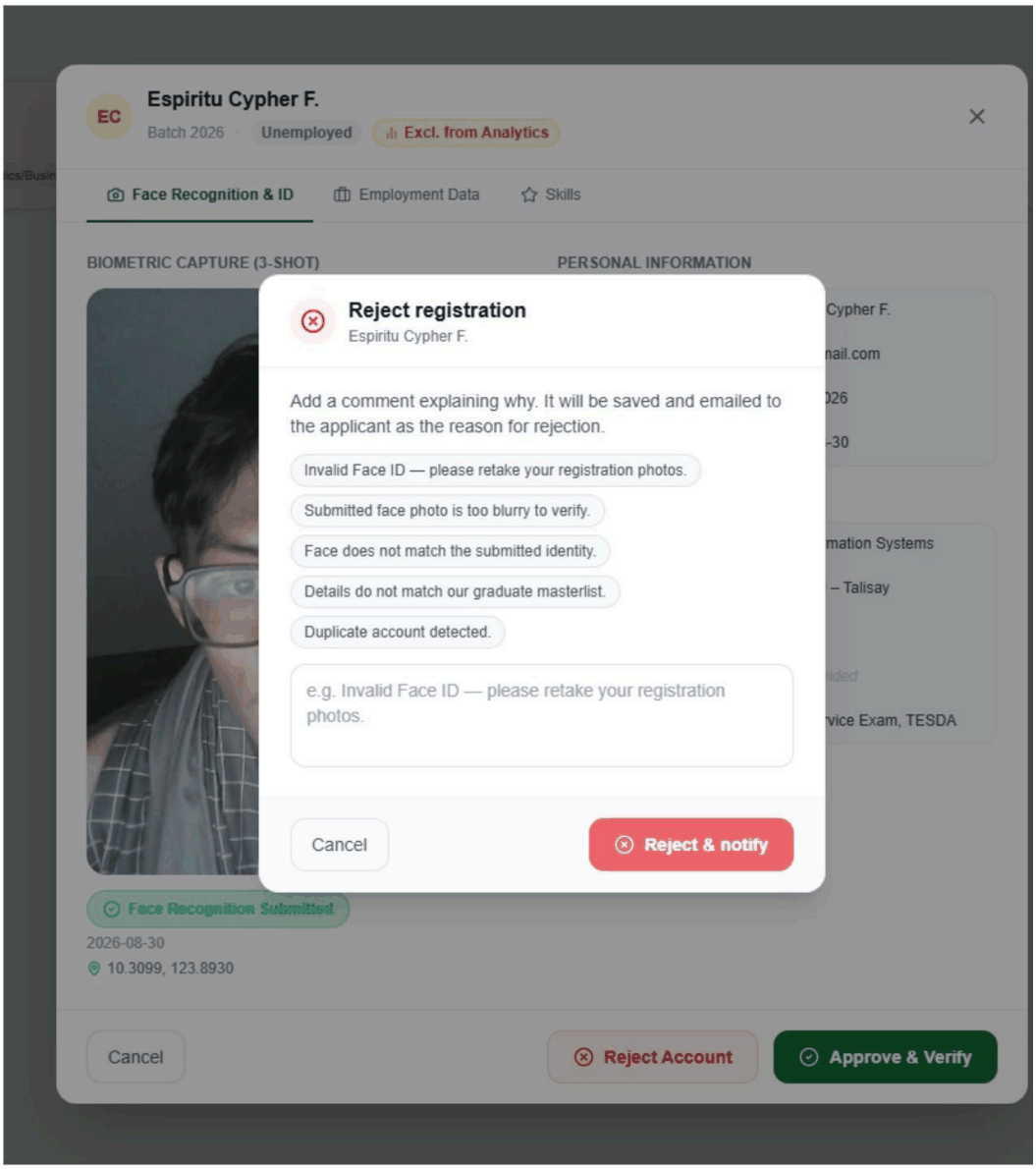


Figure 51

Verified Graduate Records Filtered by Graduation Batch

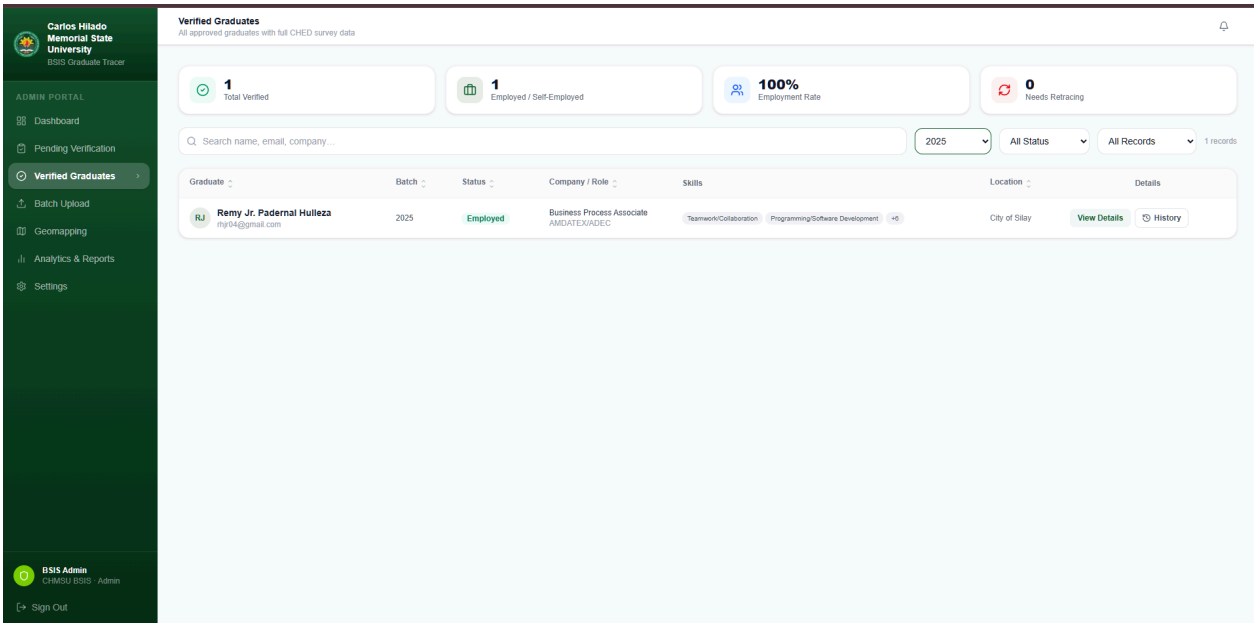


Figure 52

Verified Graduates Module with Filtering by Batch and Status

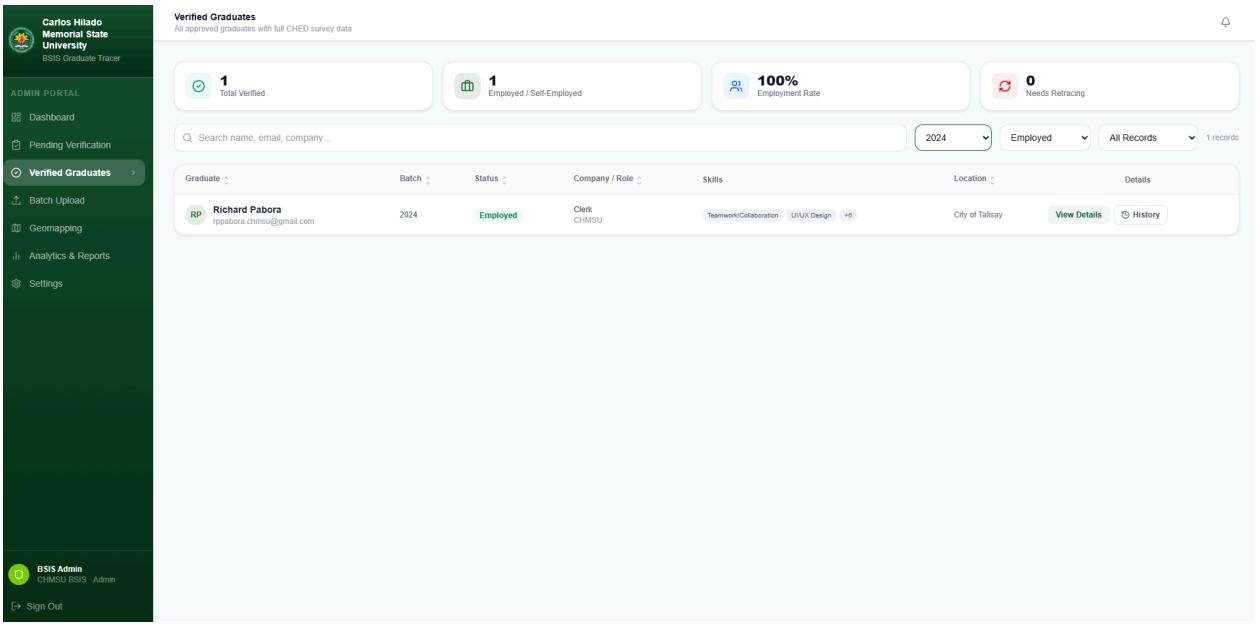


Figure 53

Batch Upload Module Showing the Master List Status

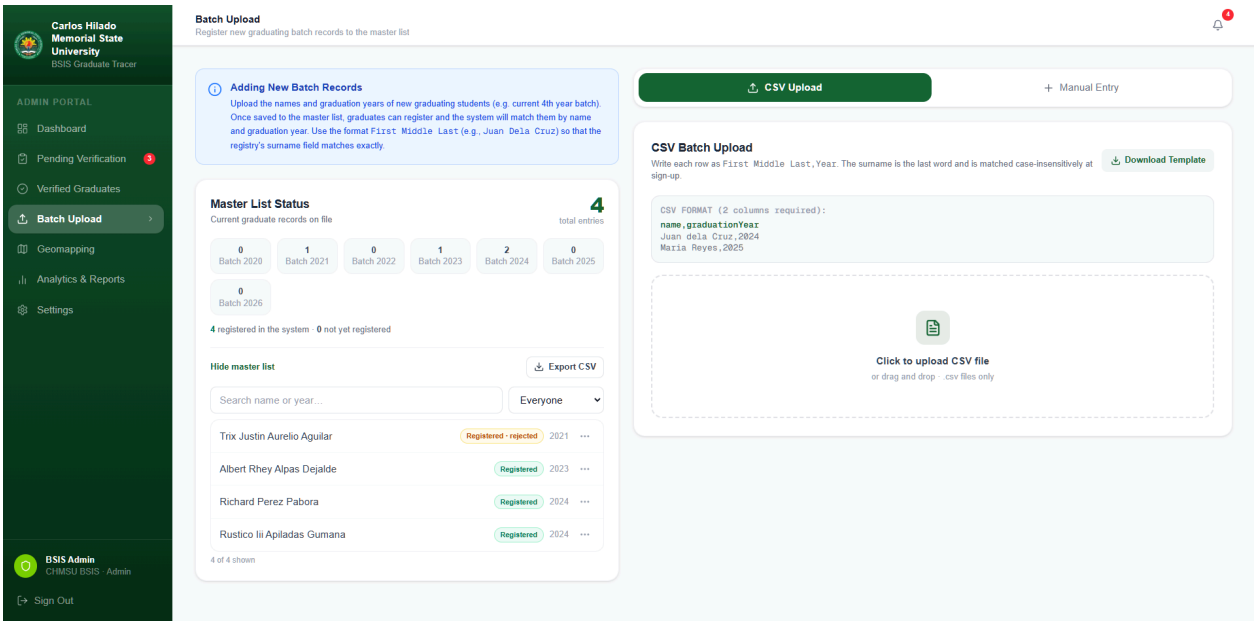


Figure 54

Batch Upload Reviewing the Parsed Rows Before Saving to the Master List

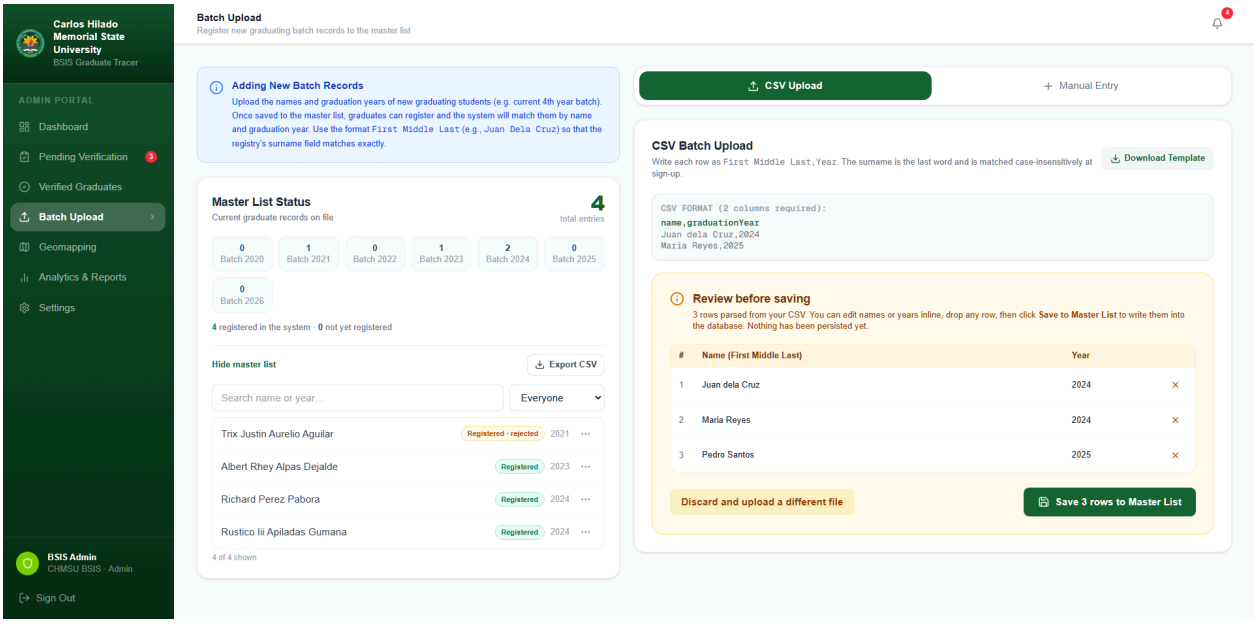


Figure 55

Batch Upload Result Reporting the New Records Added and the Duplicate Rows Skipped

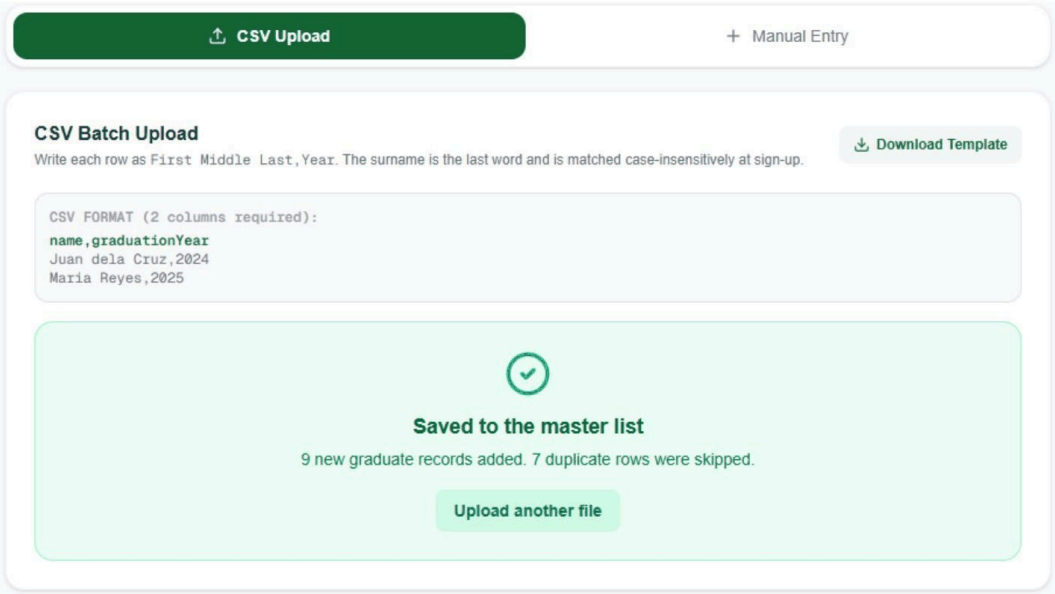


Figure 56

Second Batch Upload Confirming That Re-Uploaded Rows Create No Duplicates

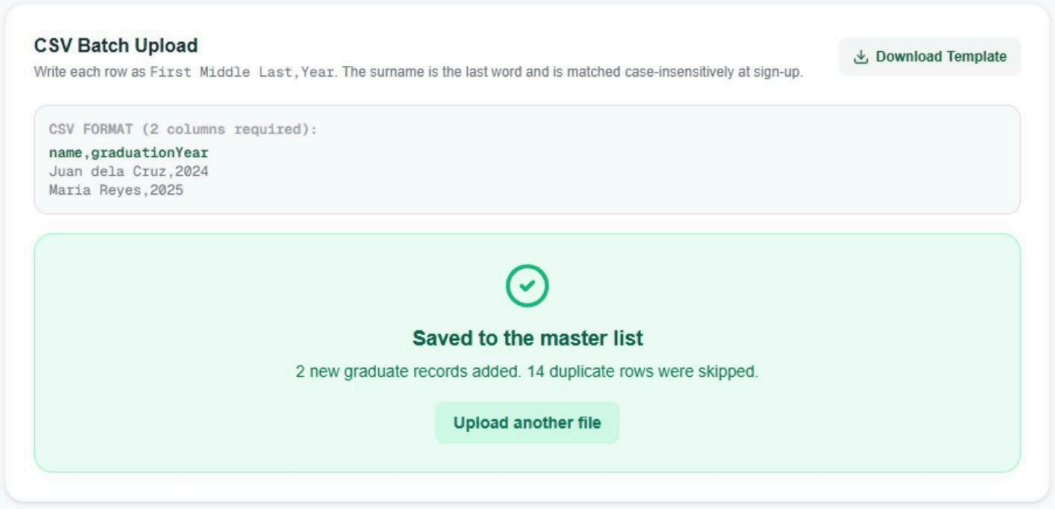


Figure 57

Geomapping View of the Graduate Employment Distribution

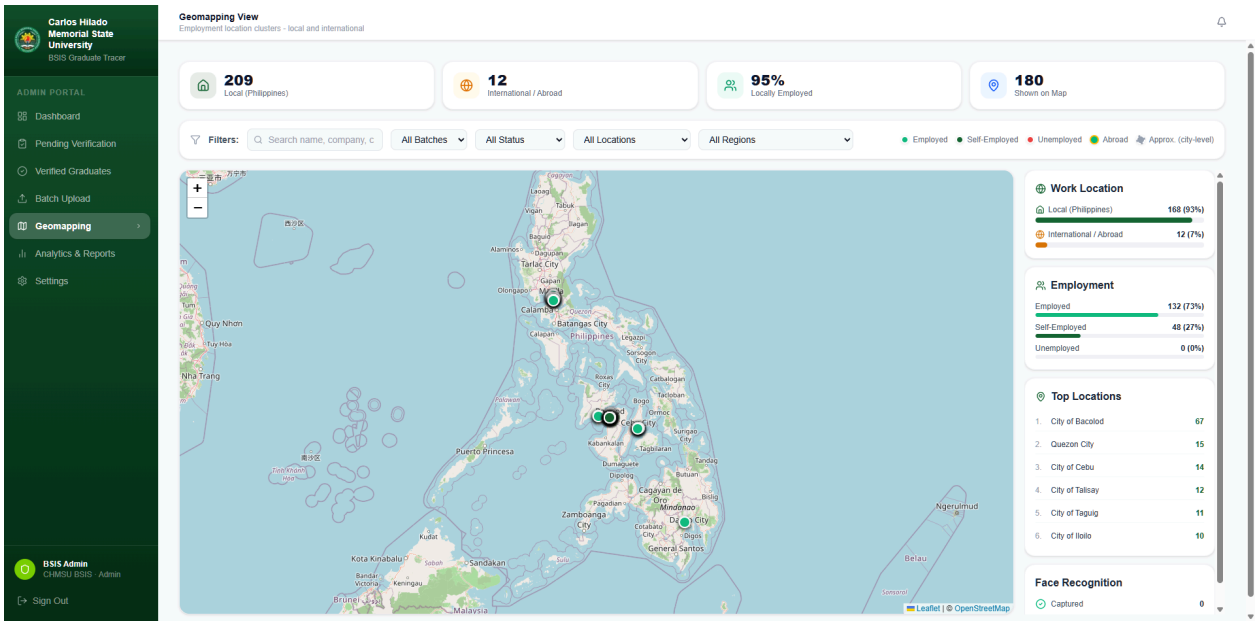


Figure 58

Analytics Module Presenting the Employability Prediction Across All Batches

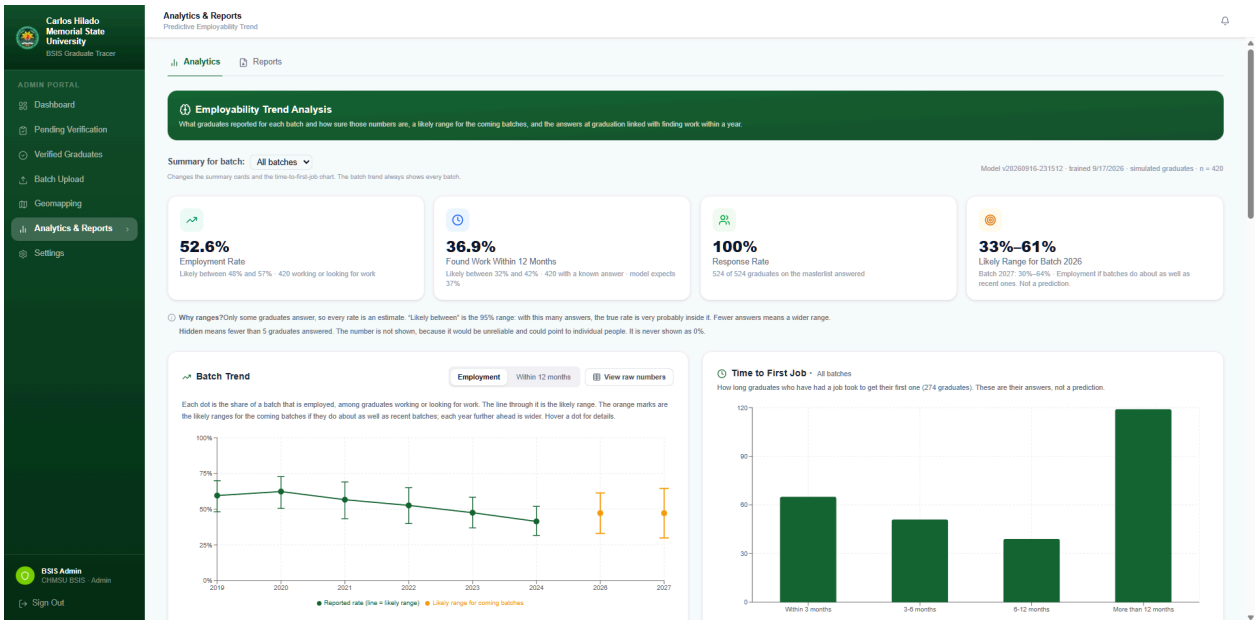


Figure 59

Analytics Module Presenting the Employability Prediction for a Selected Batch

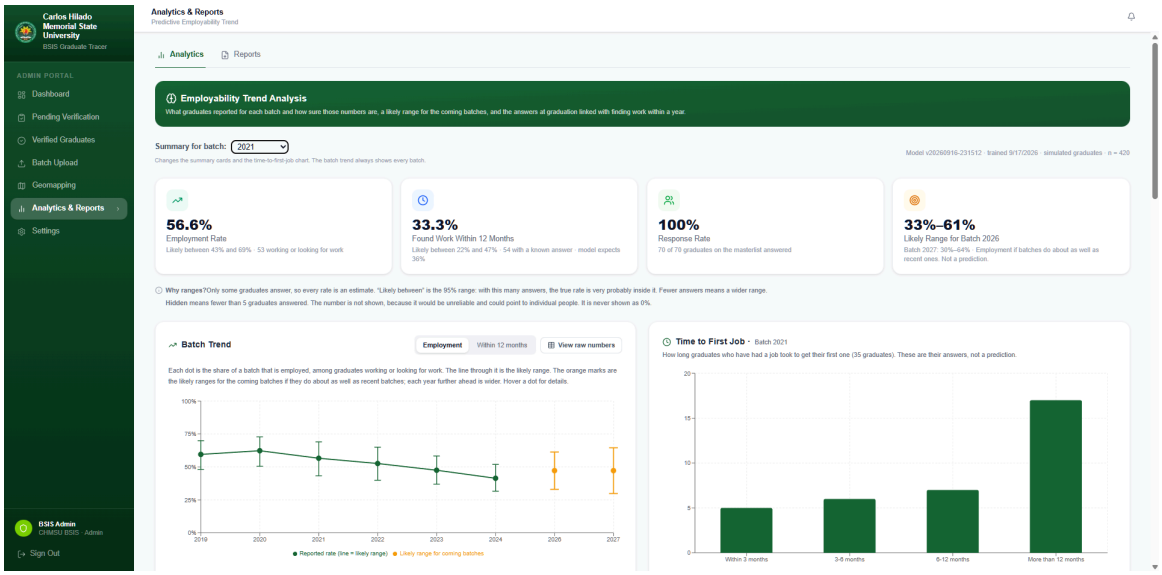


Figure 60

Analytics and Reports Module Showing the Forecast Summary Indicators

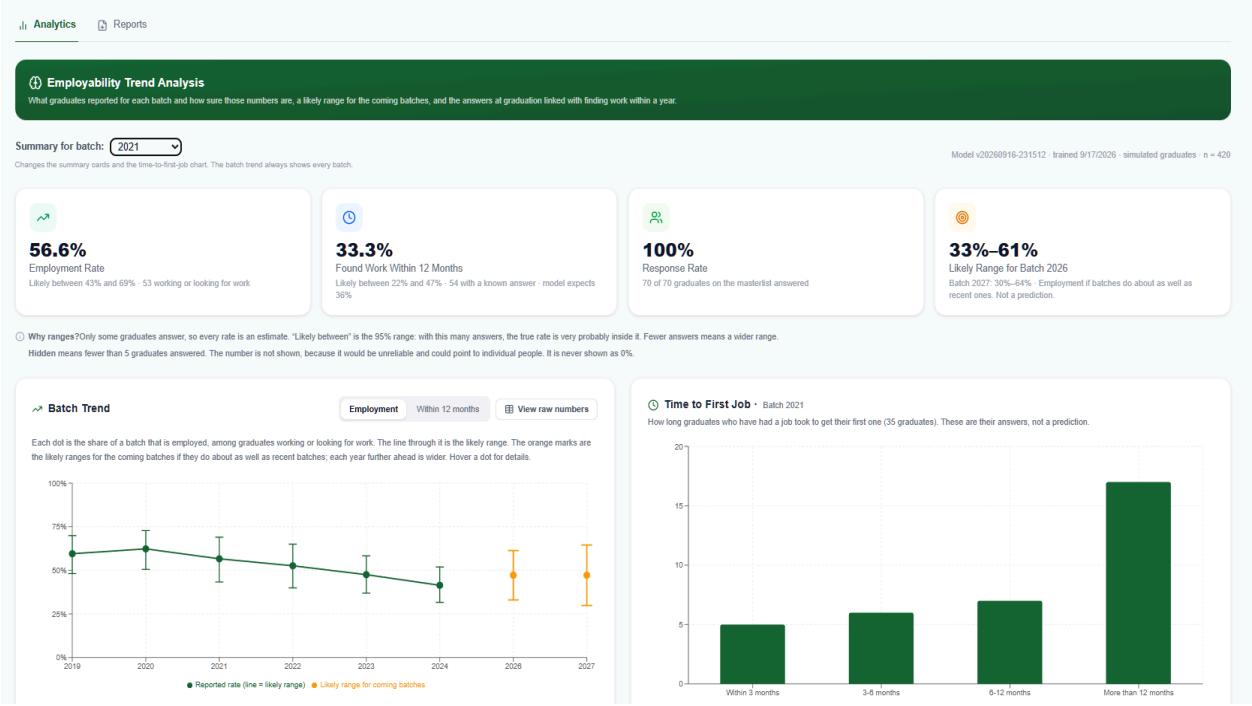


Figure 61

Reports Module Listing the Generated Report Types

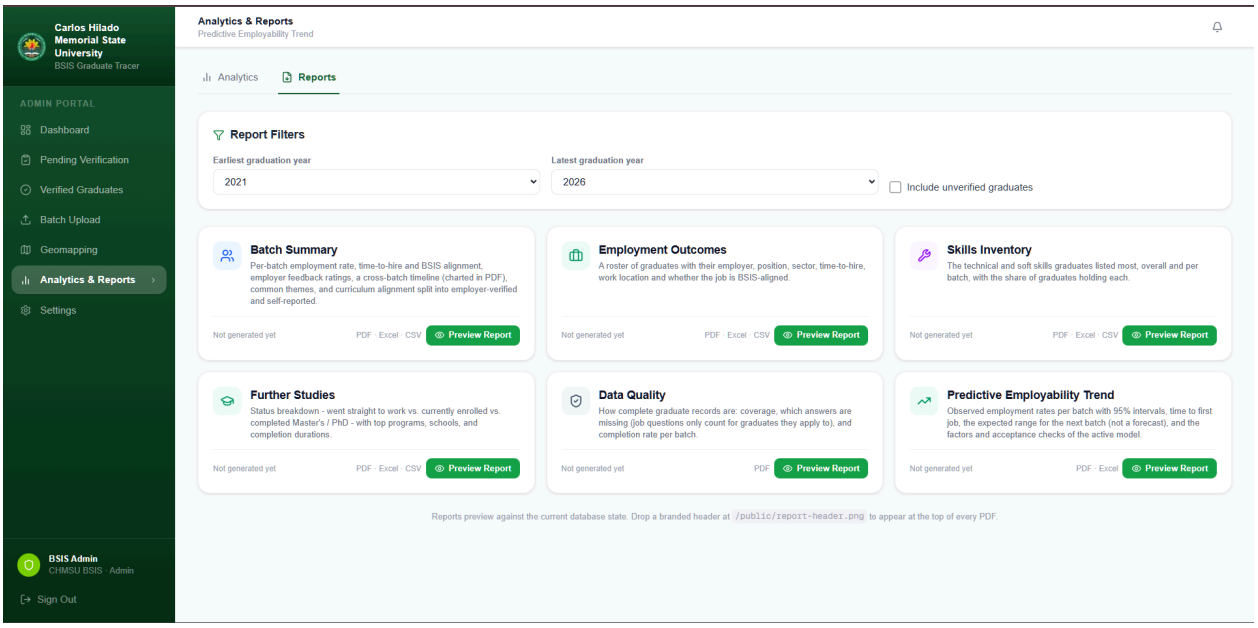


Figure 62

Reference Settings Module for the Management of Skills

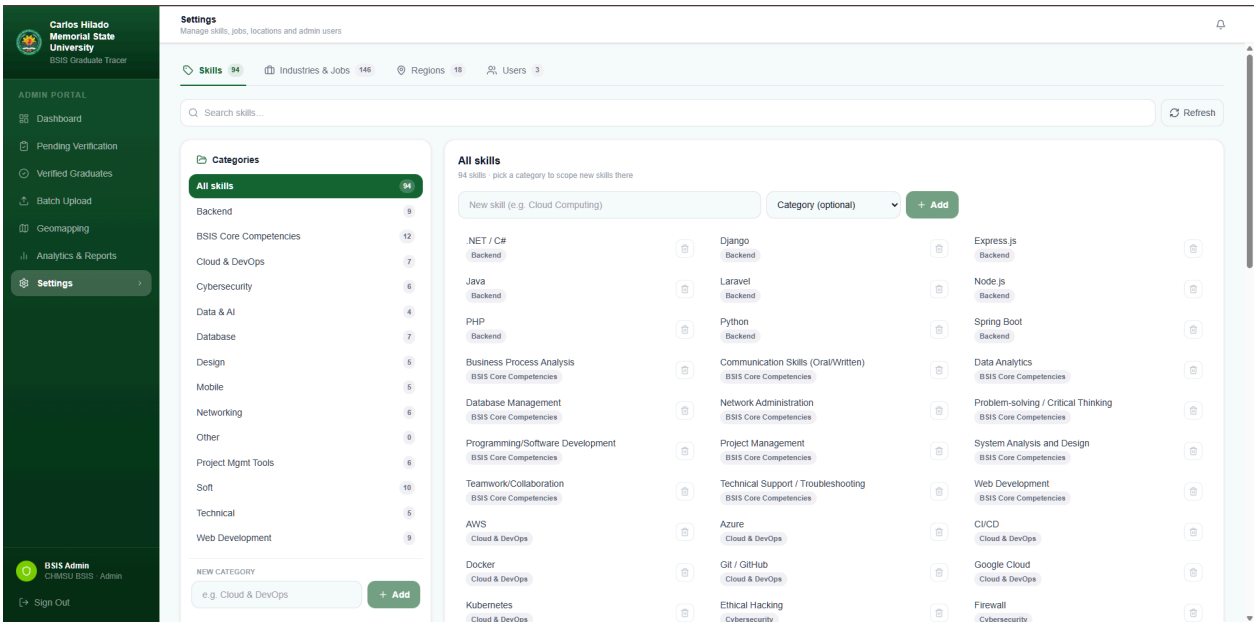


Figure 63

Reference Settings Module for the Management of Industries and Job Titles

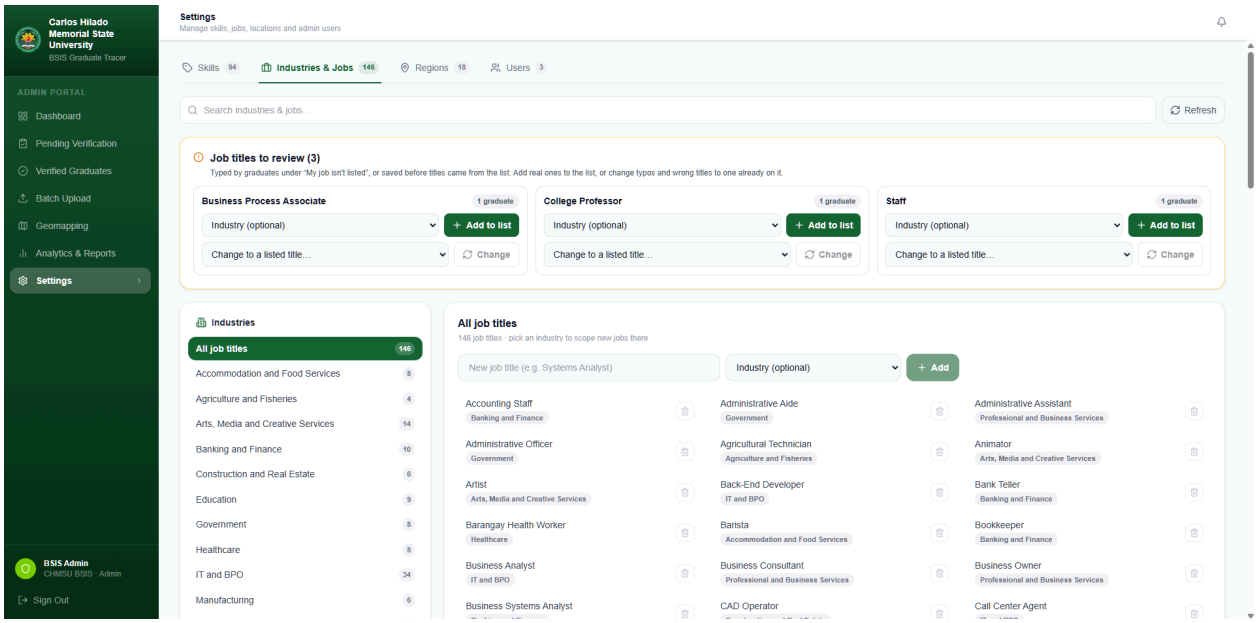


Figure 64

Reference Settings Module for the Management of Regions

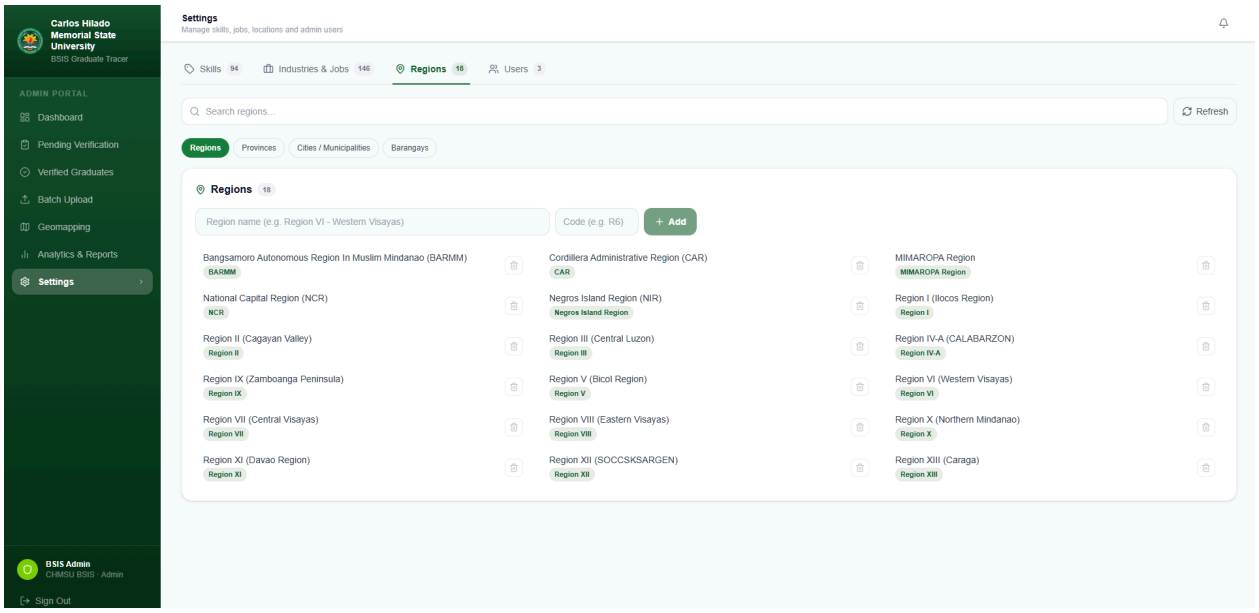


Figure 65

Reference Settings Module for the Management of Administrator Accounts

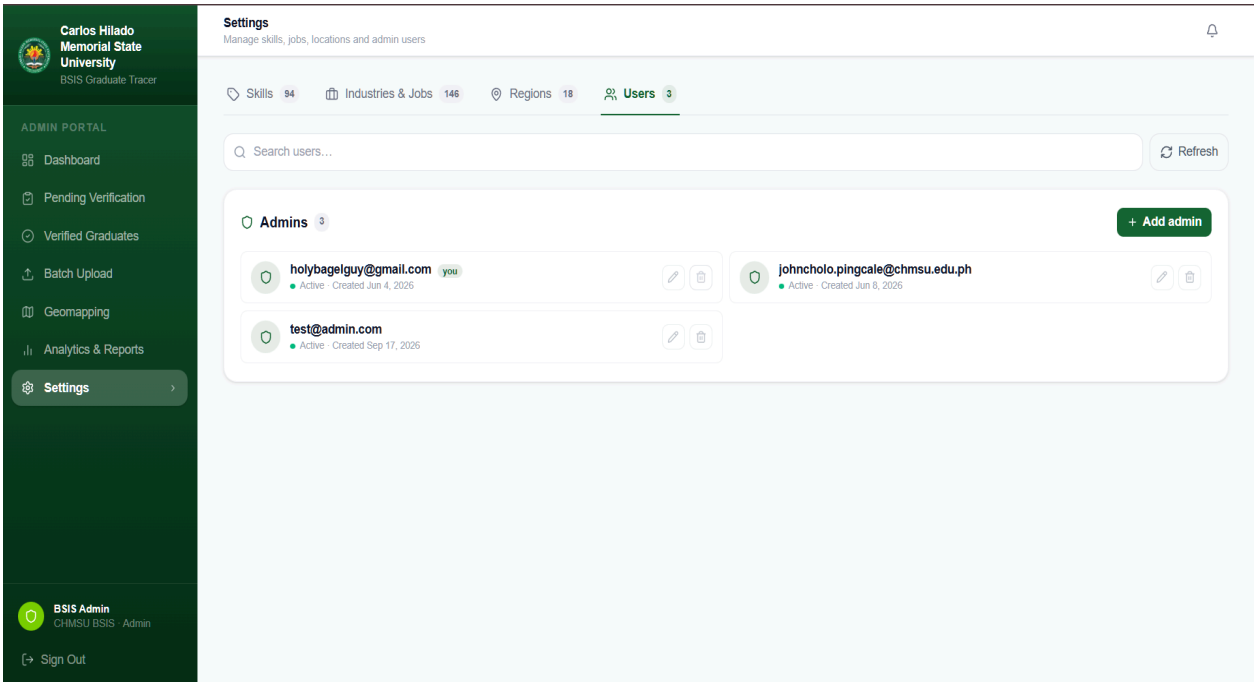


Figure 66

Pending Verification Queue with No Accounts Awaiting Review

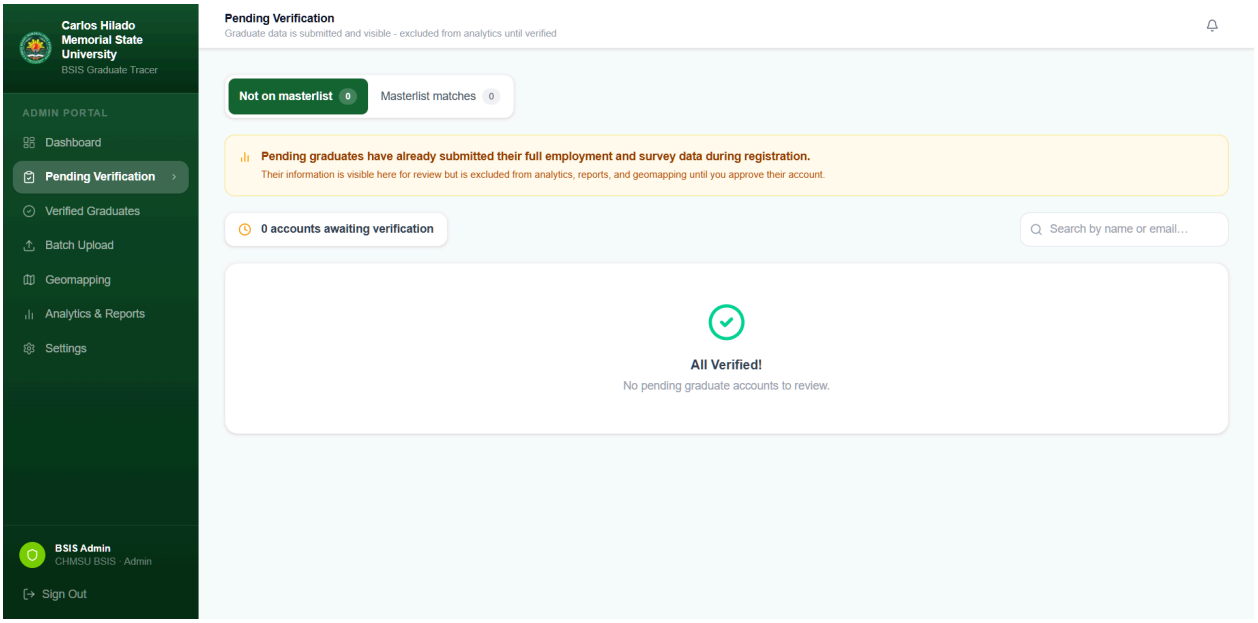


Figure 67

Employer Route Refused by Direct Address Entry, with the Pending Verification Queue Cleared

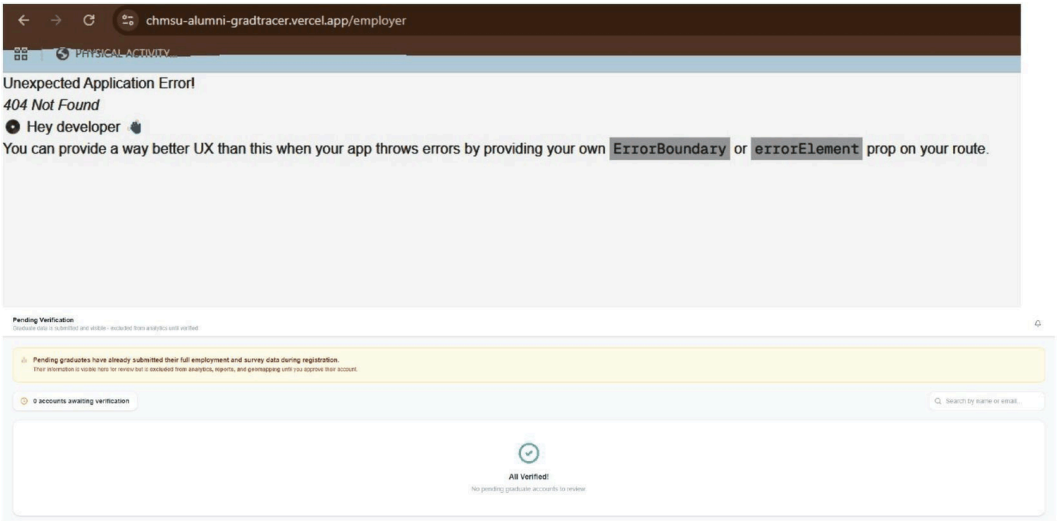


Figure 68

QASE Test Management – Black Box Testing Suite (BB-01 to BB-27)

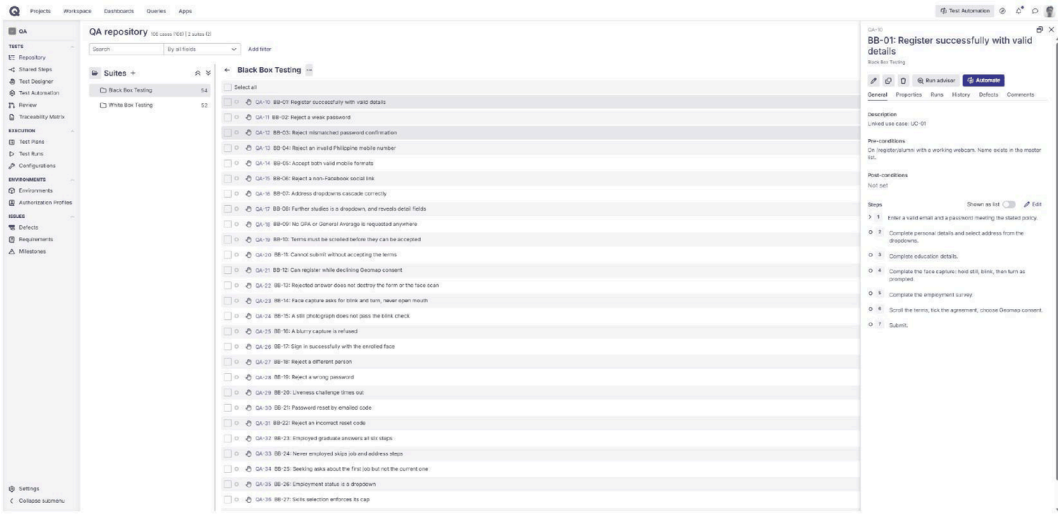


Figure 69

QASE Test Management – Black Box Testing Suite (BB-28 to BB-54)

Appendix E

BLACK-BOX TESTING – BETA TESTING

Figure 70

Account Creation Step Showing the Password Policy Satisfied

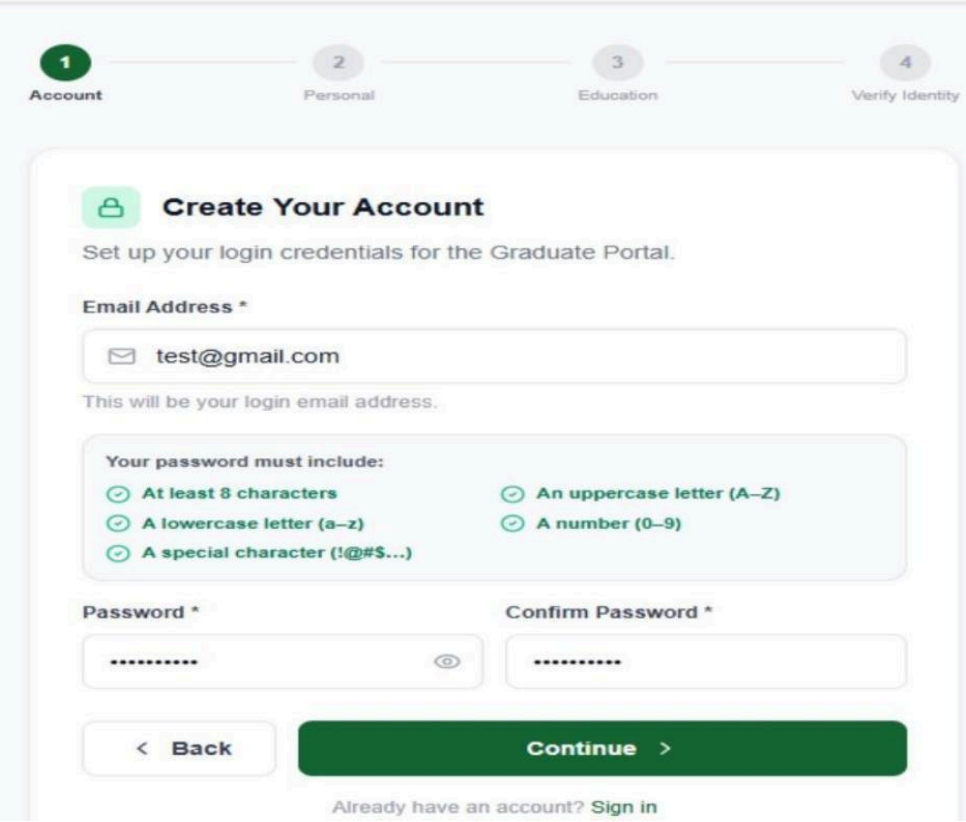


Figure 71

Account Creation Step Rejecting a Password That Does Not Meet the Policy

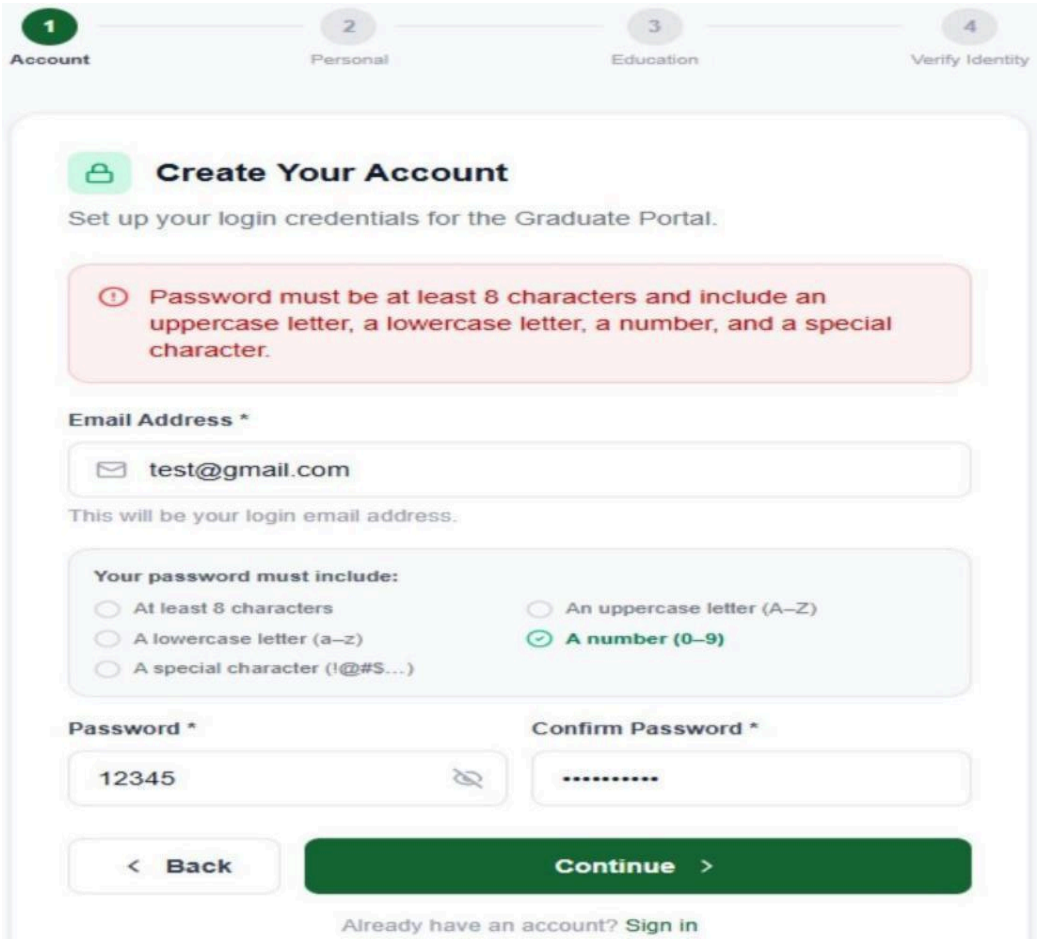


Figure 72

Personal Information Step Rejecting an Invalid Philippine Mobile Number

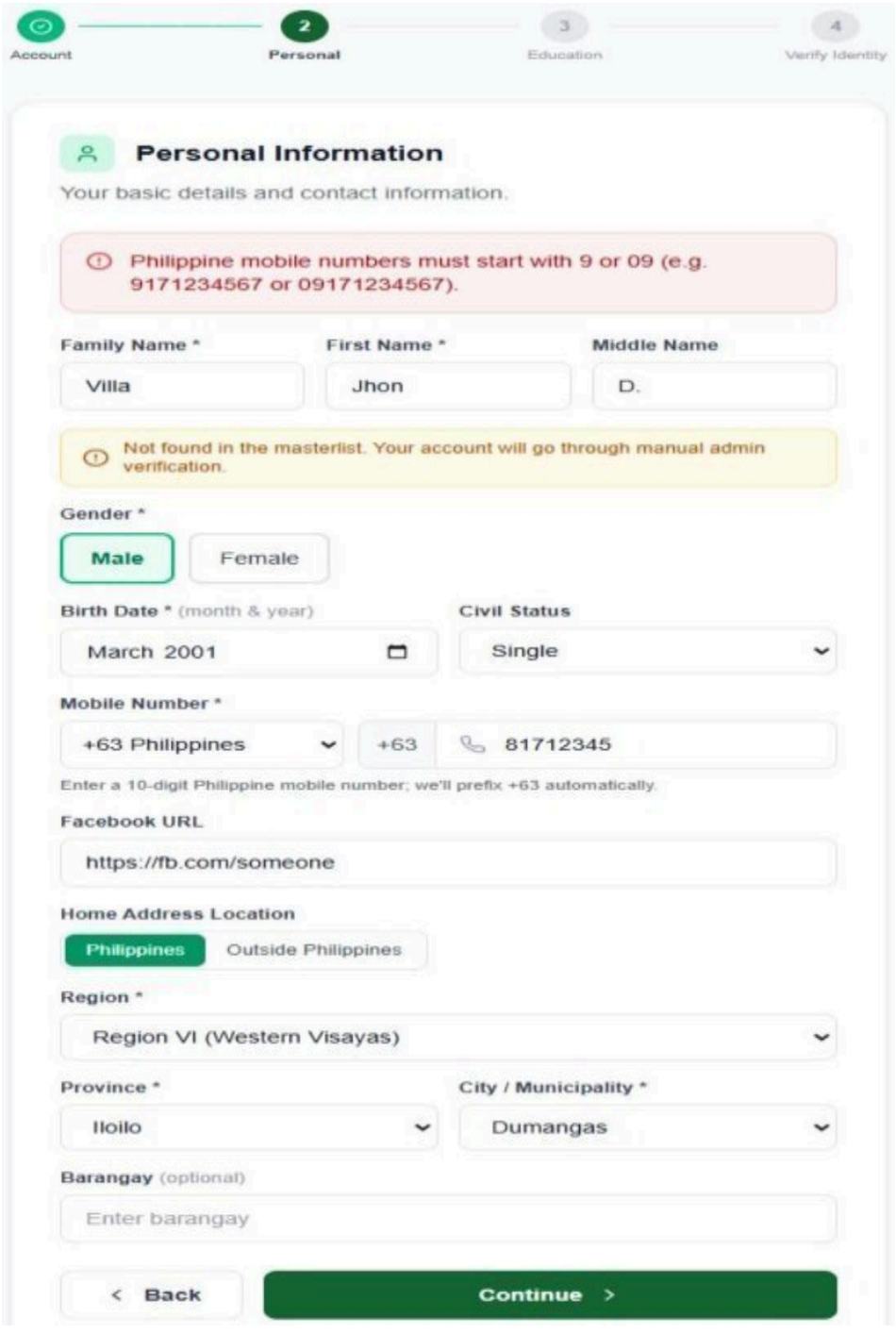


Figure 73

Rejection of a Non-Facebook Social Media Link and the Cascading Address Selection

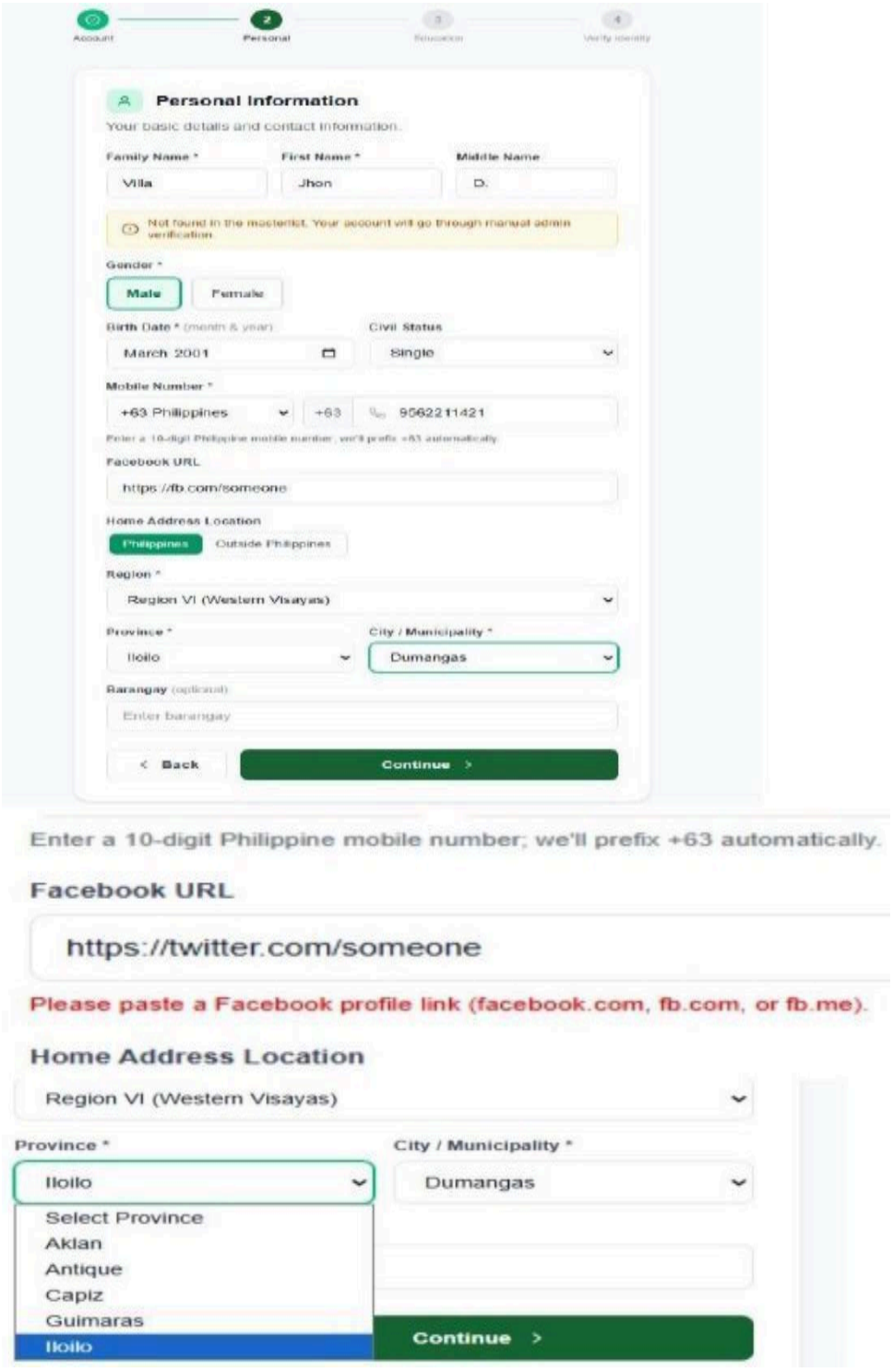


Figure 74

Terms and Data Privacy Consent Screen with the Geographic Mapping Option

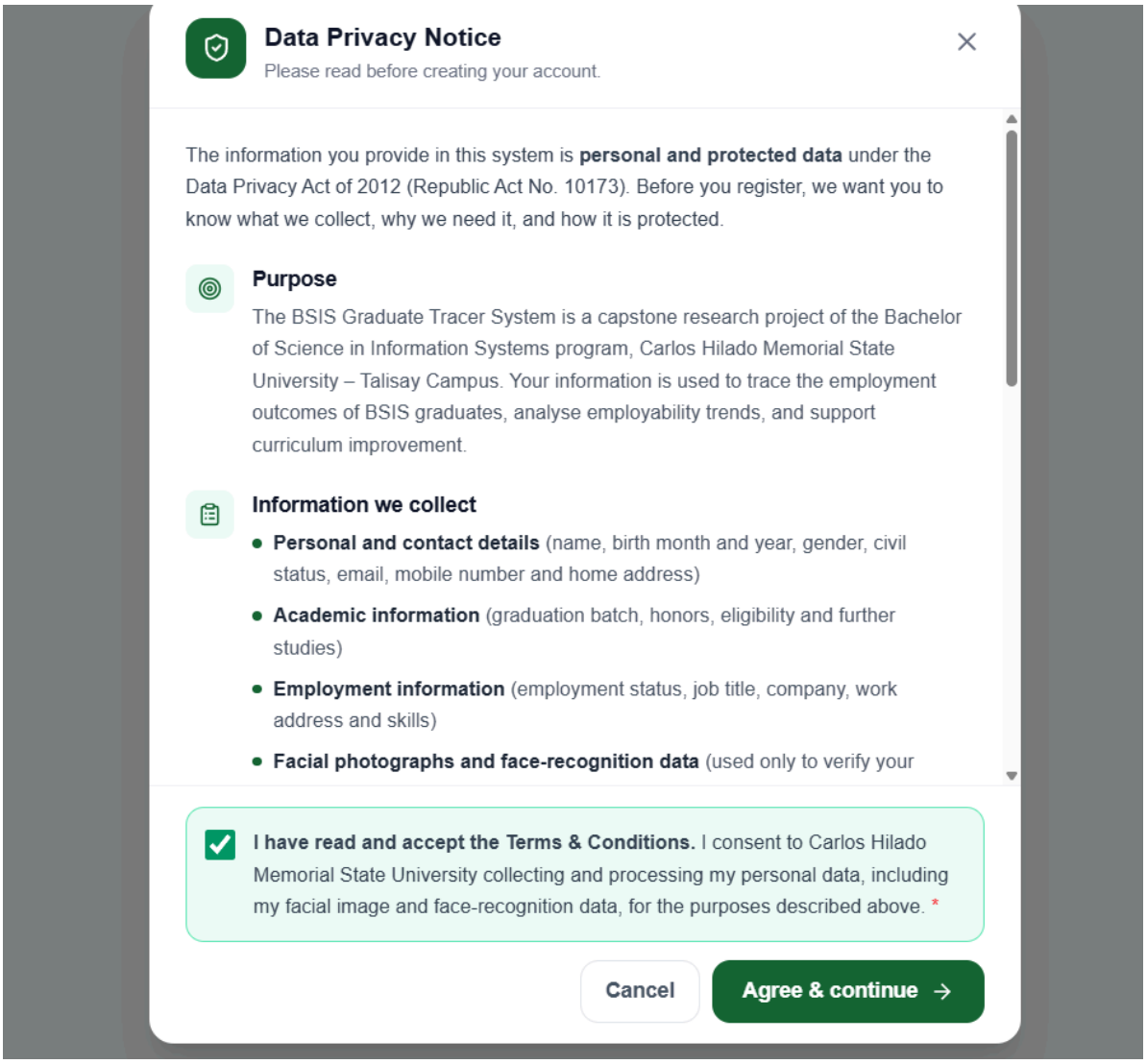


Figure 75

Face Recognition Step Presenting the Liveness Challenges in Sequence

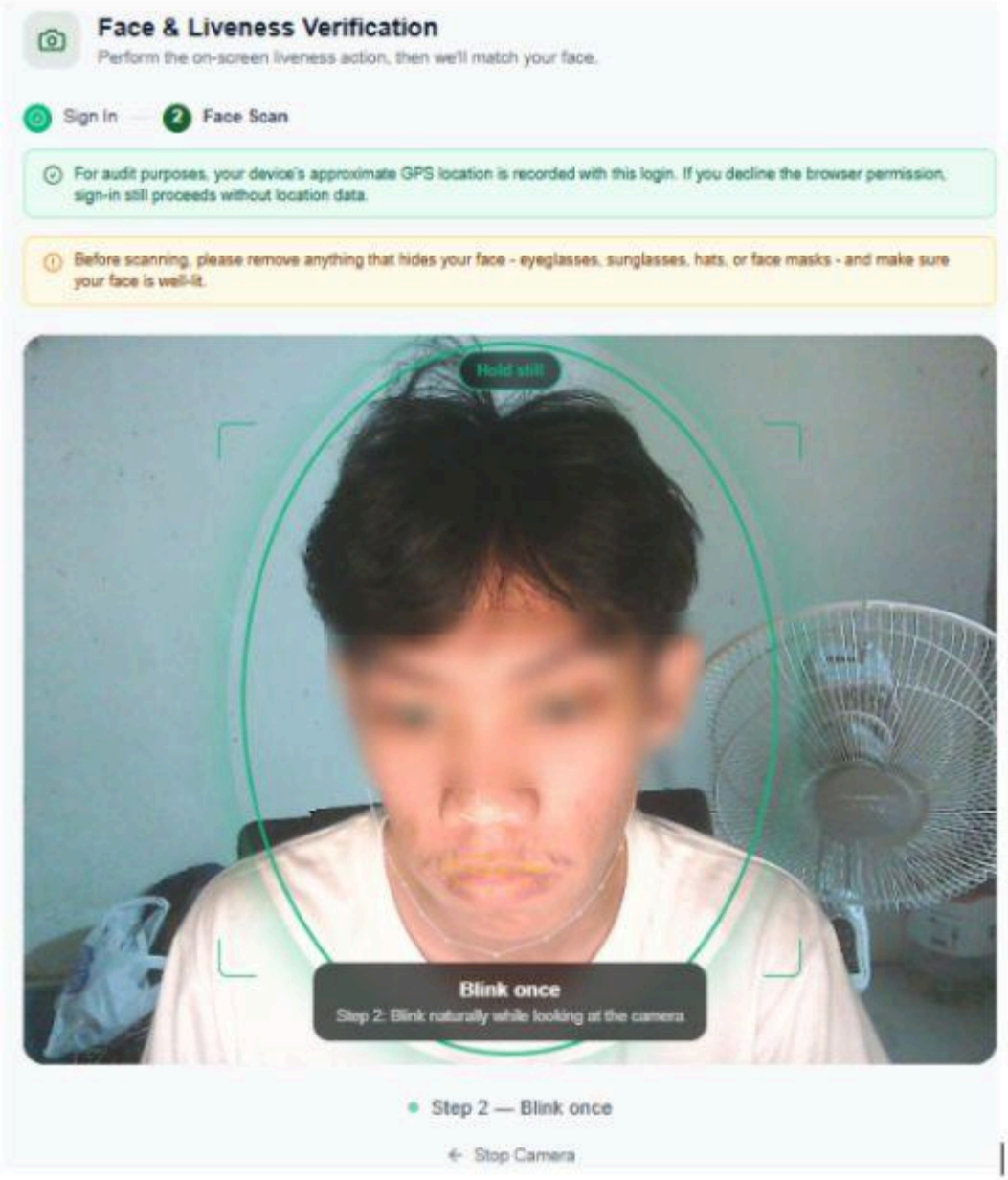


Figure 76

Face Recognition Step After All Liveness Challenges Were Passed

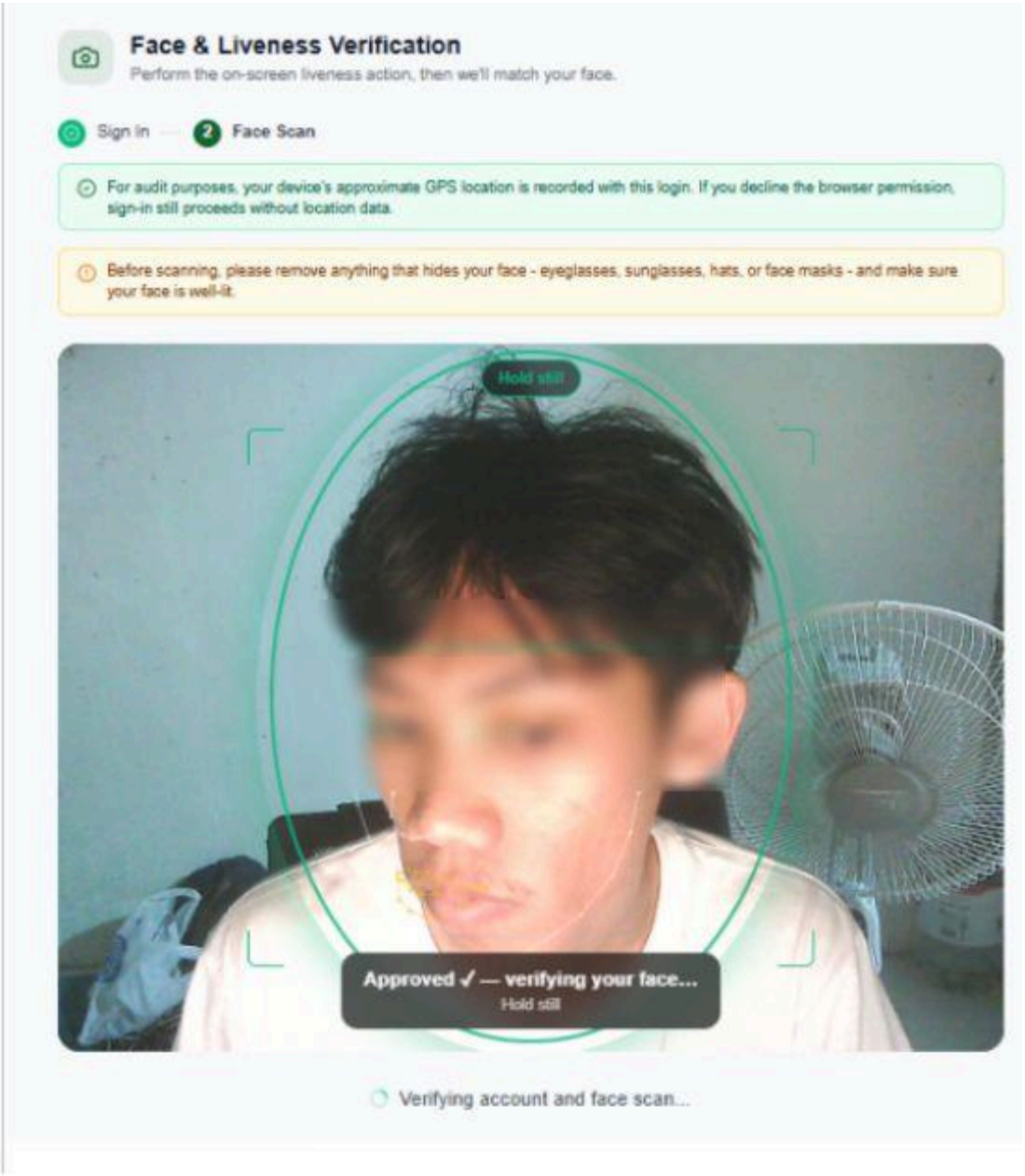


Figure 77

Sign-In Screen of the Graduate Tracer System

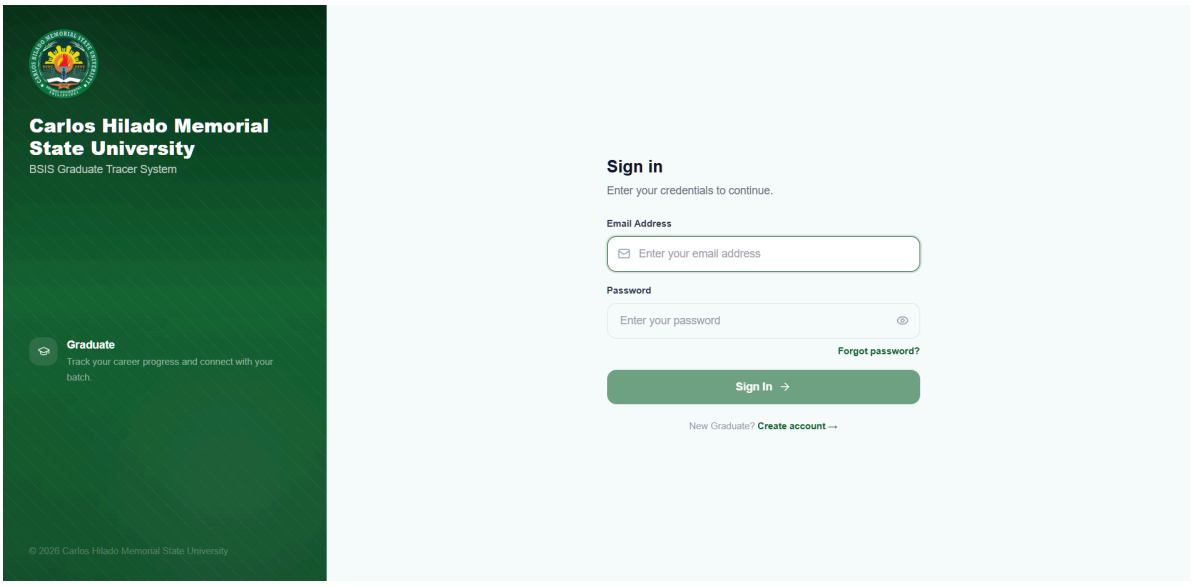


Figure 78

Sign-In Screen Accessed Through the Desktop Browser

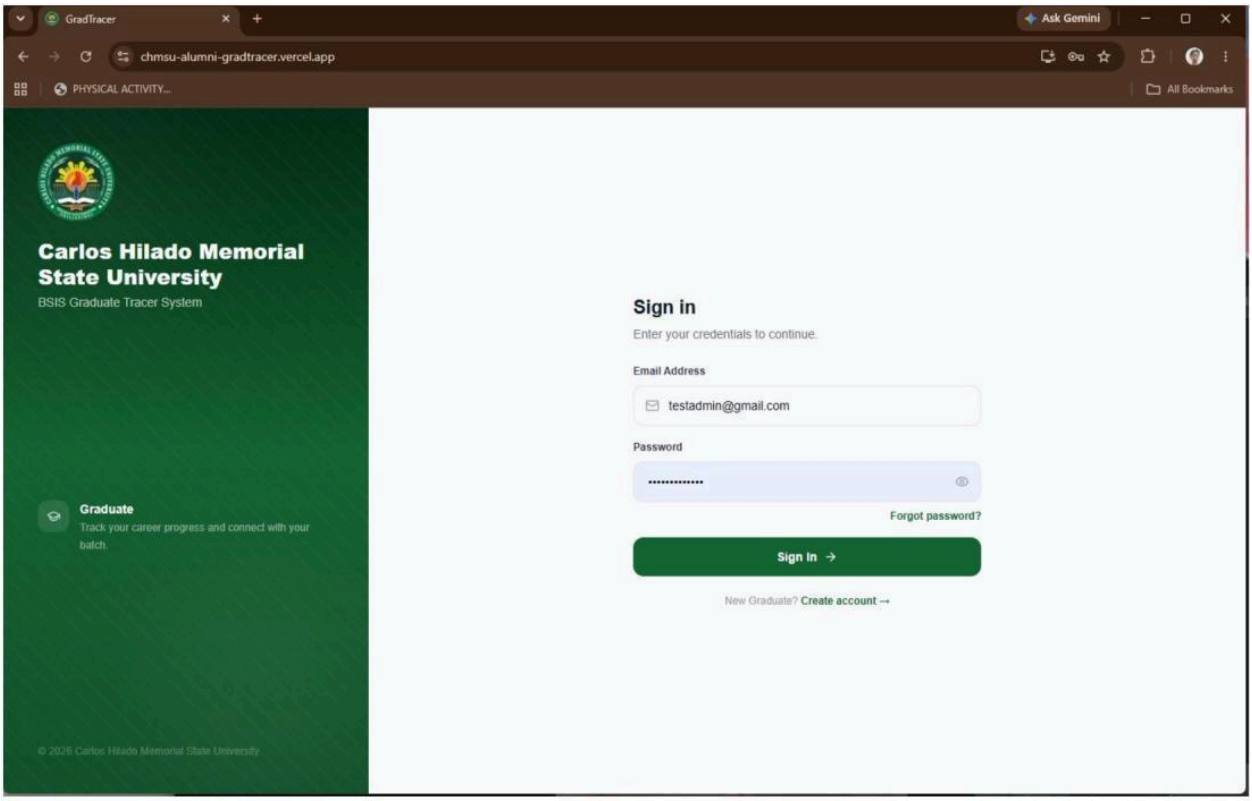


Figure 79

Skills and Competencies Step with All Selections Completed

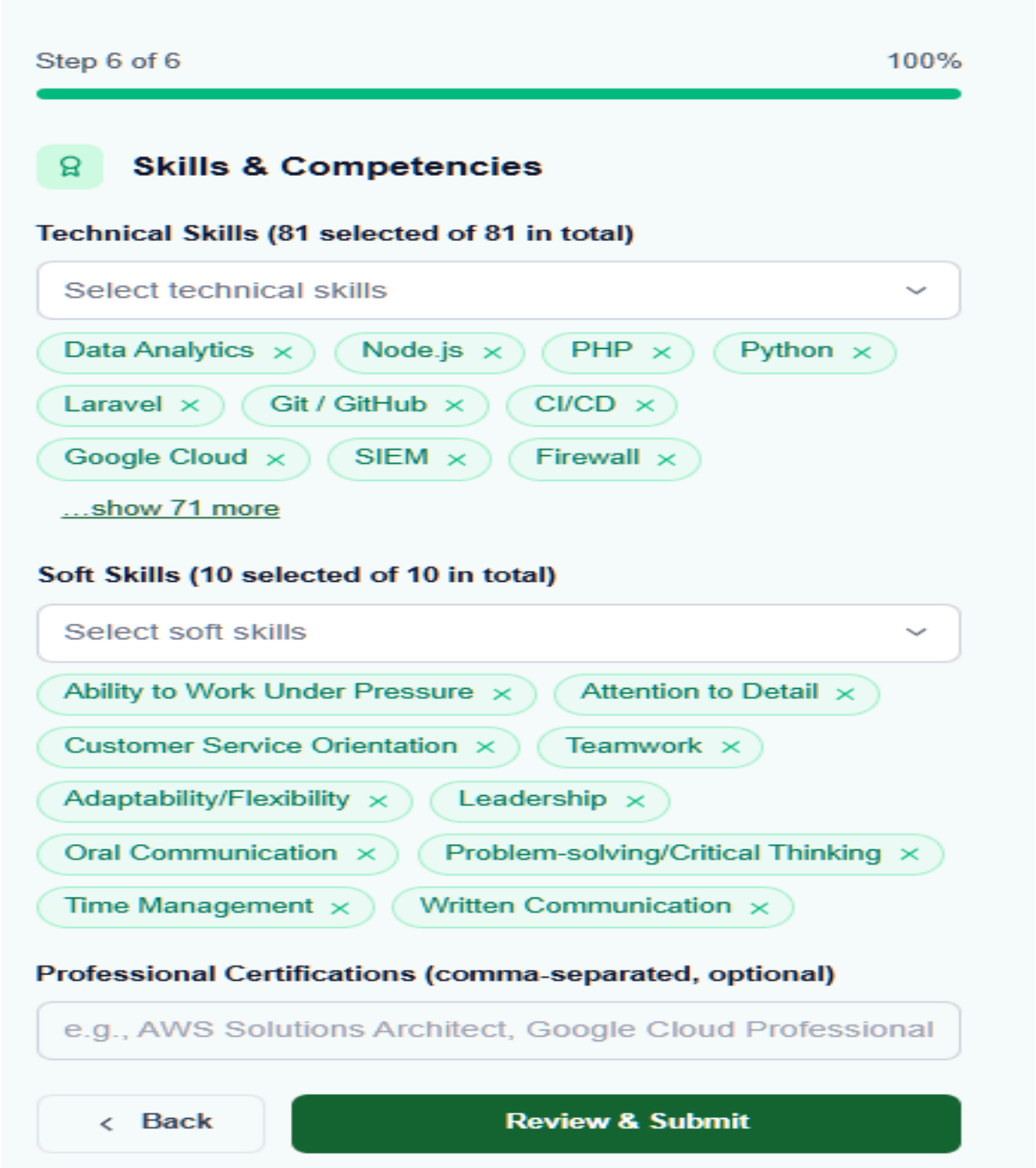


Figure 80

Selection of Technical Skills from the Controlled Reference List

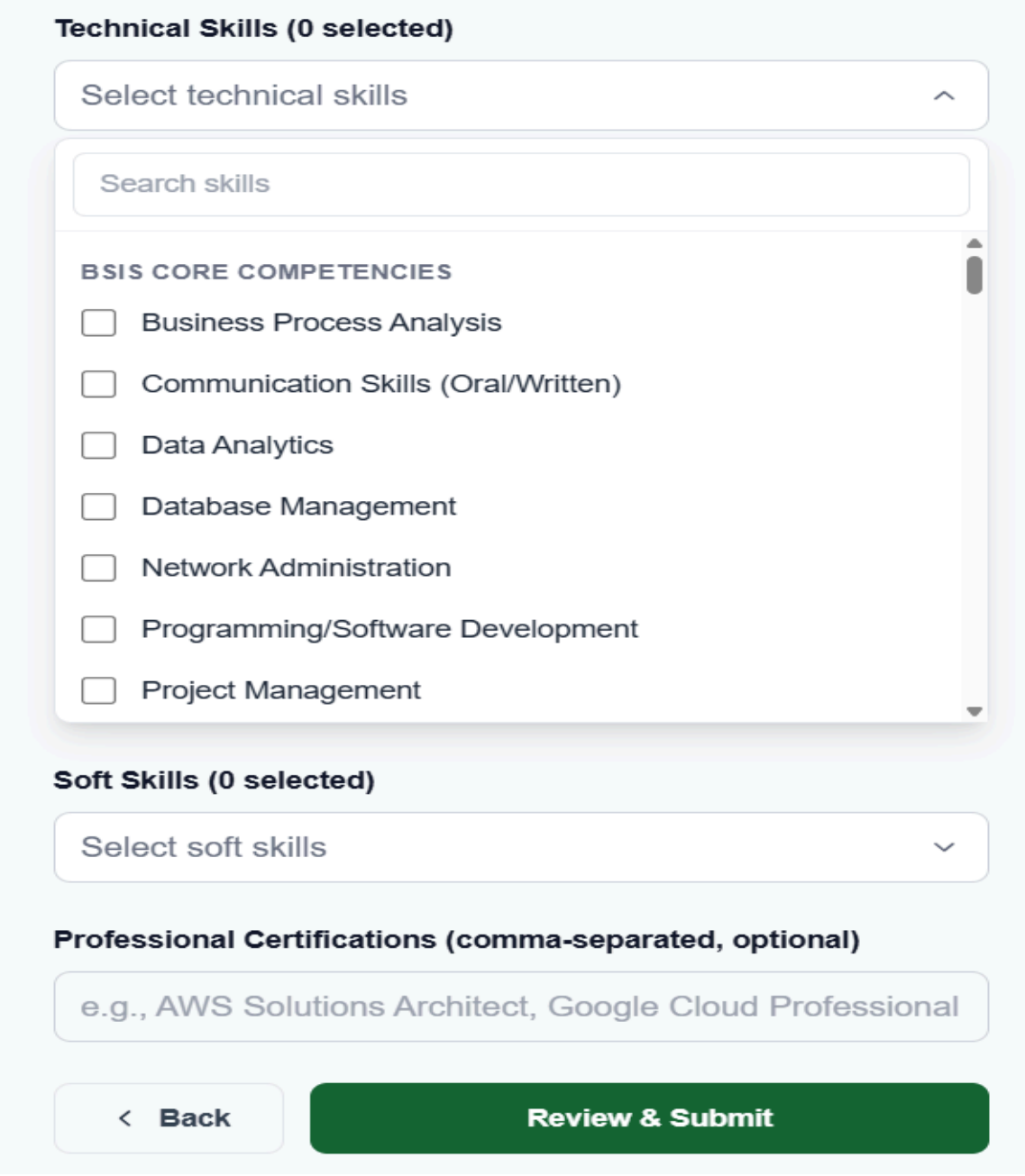


Figure 81

Graduate Dashboard Displaying the Verified Account Status

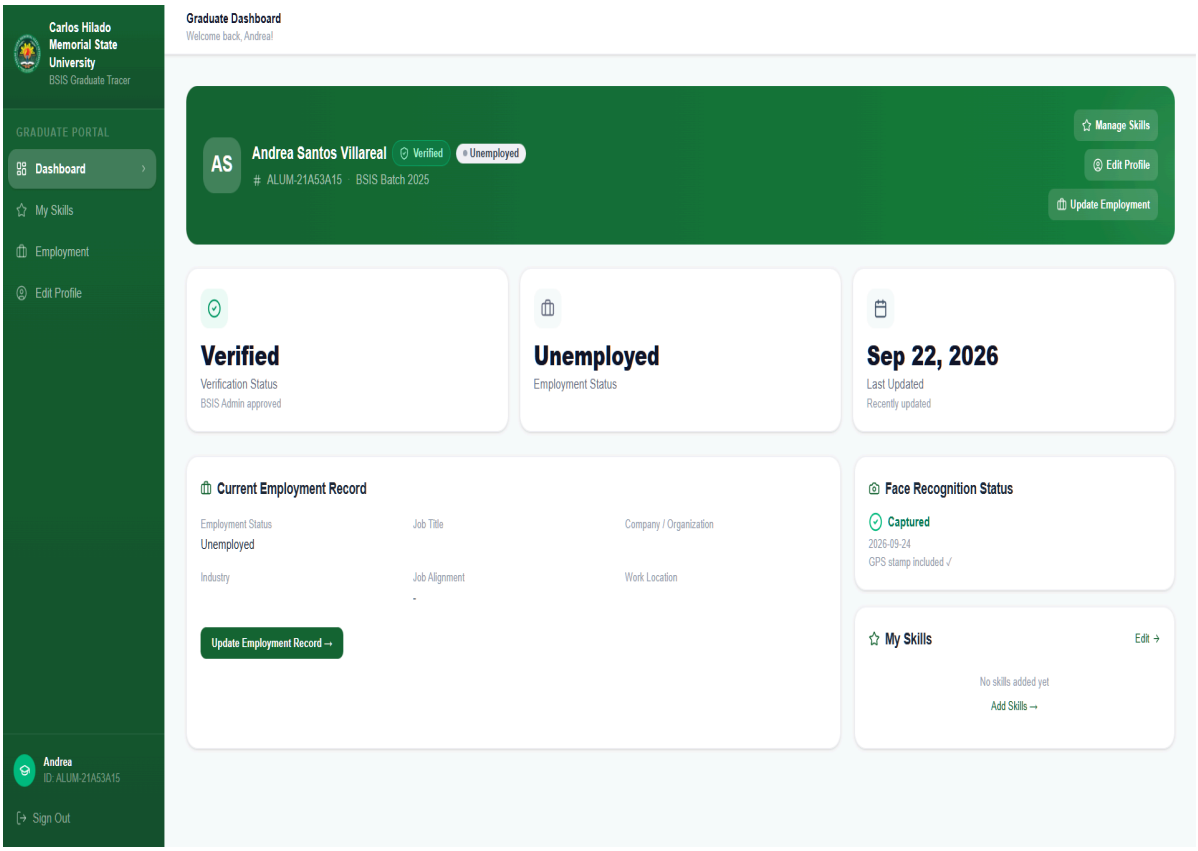


Figure 82

Graduate Dashboard Showing the Employment Record and Verification Status

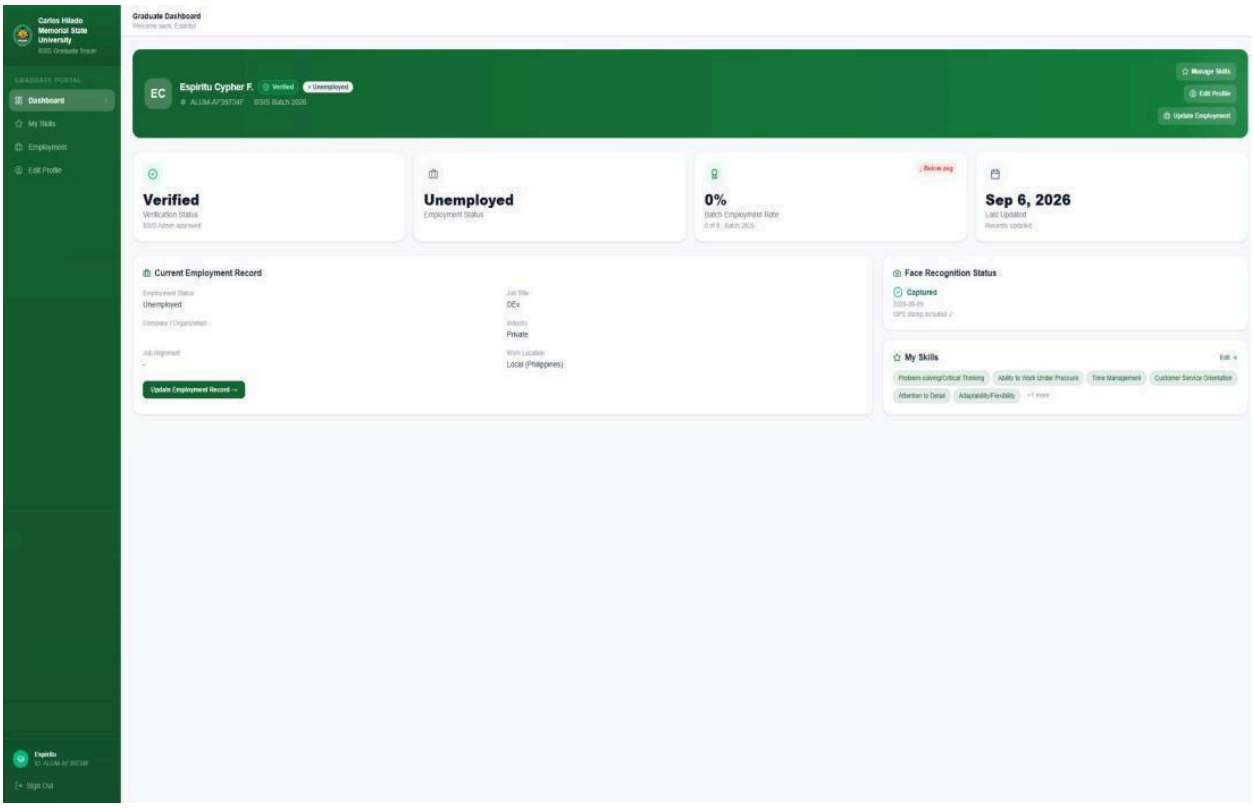


Figure 83

Graduate Generating a Single-Use Employer Verification Link From the Dashboard

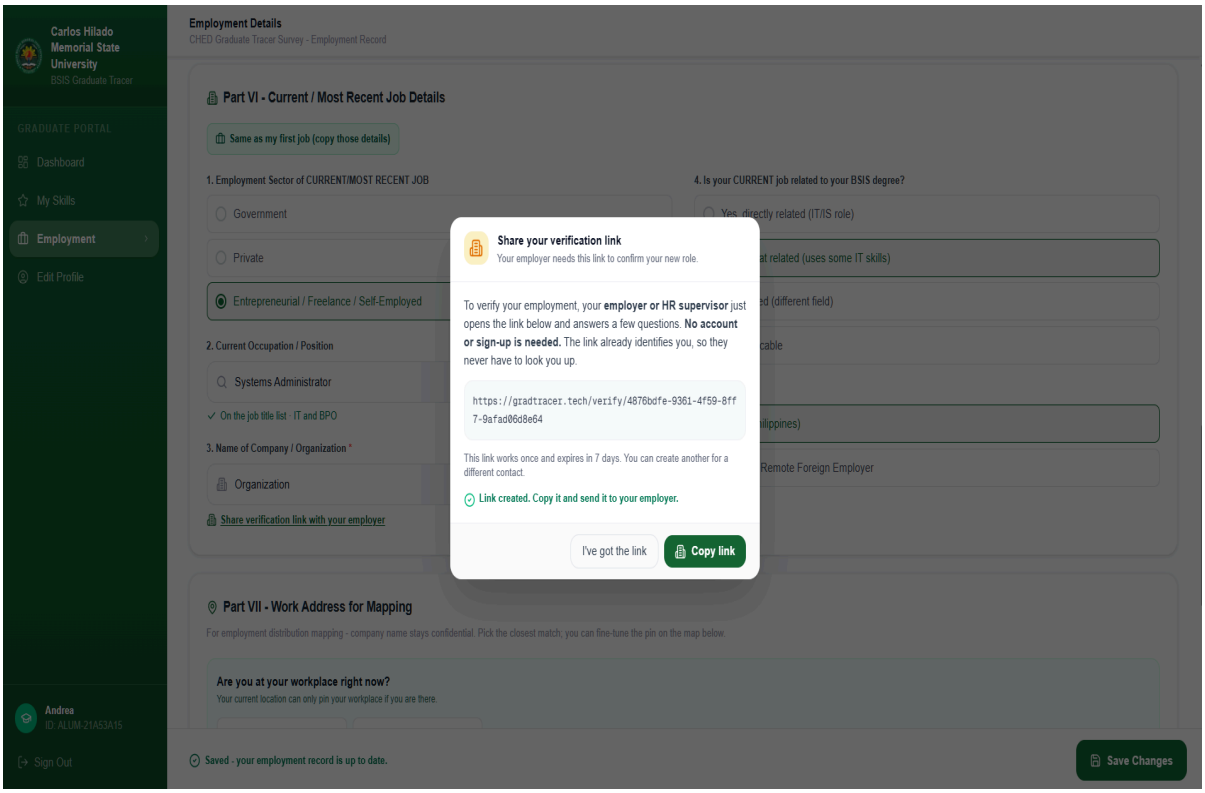


Figure 84

Employer Verification Page Opened Directly From the Link Without an Employer Account

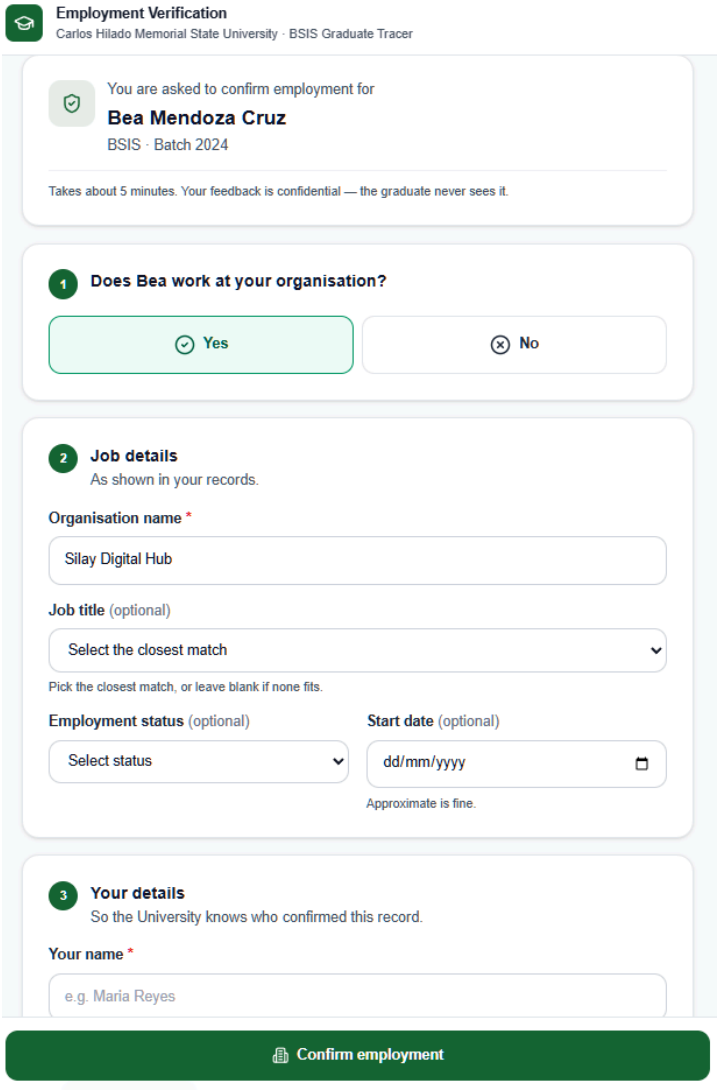


Figure 85

Employment Verification Form Completed by the Employer With the Verifier Identity Supplied

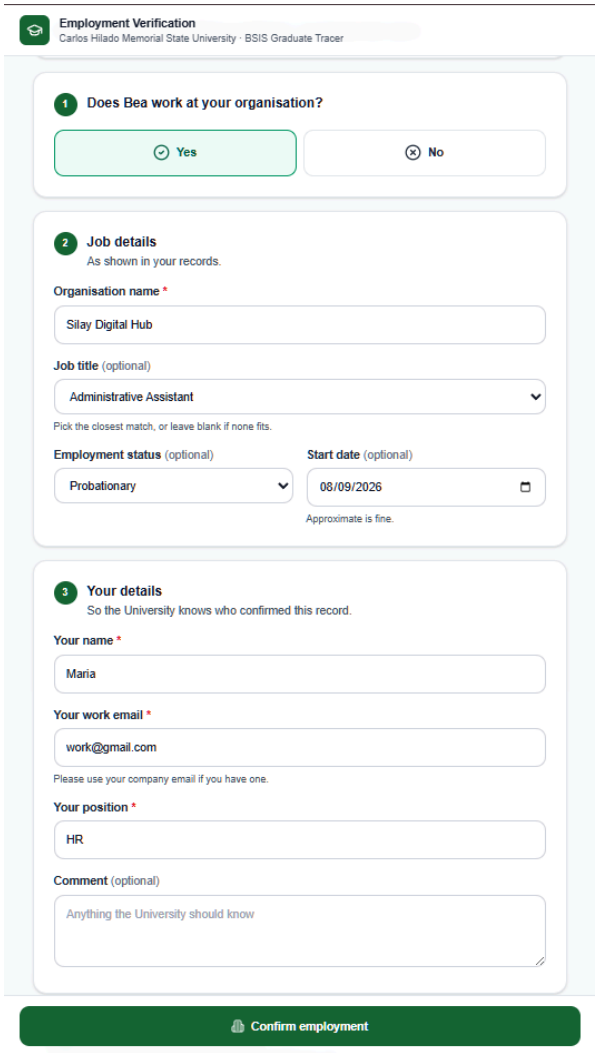


Figure 86

Confirmation Recorded After the Employer Submitted the Verification

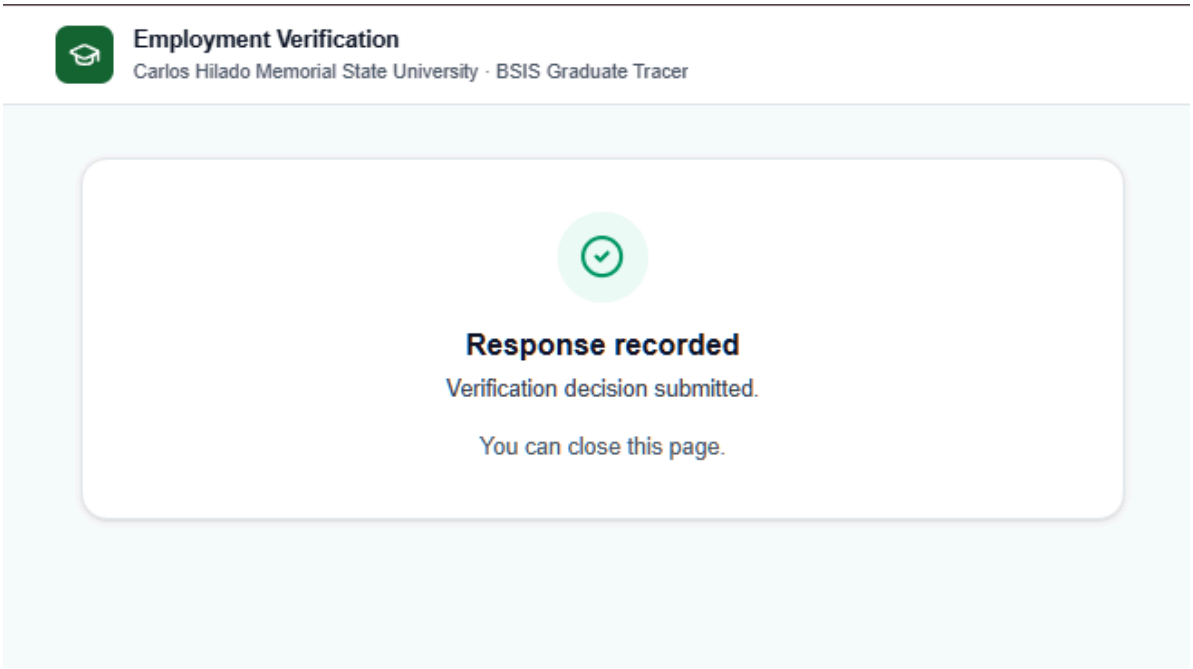


Figure 87

Refusal of a Verification Link That Had Already Been Used

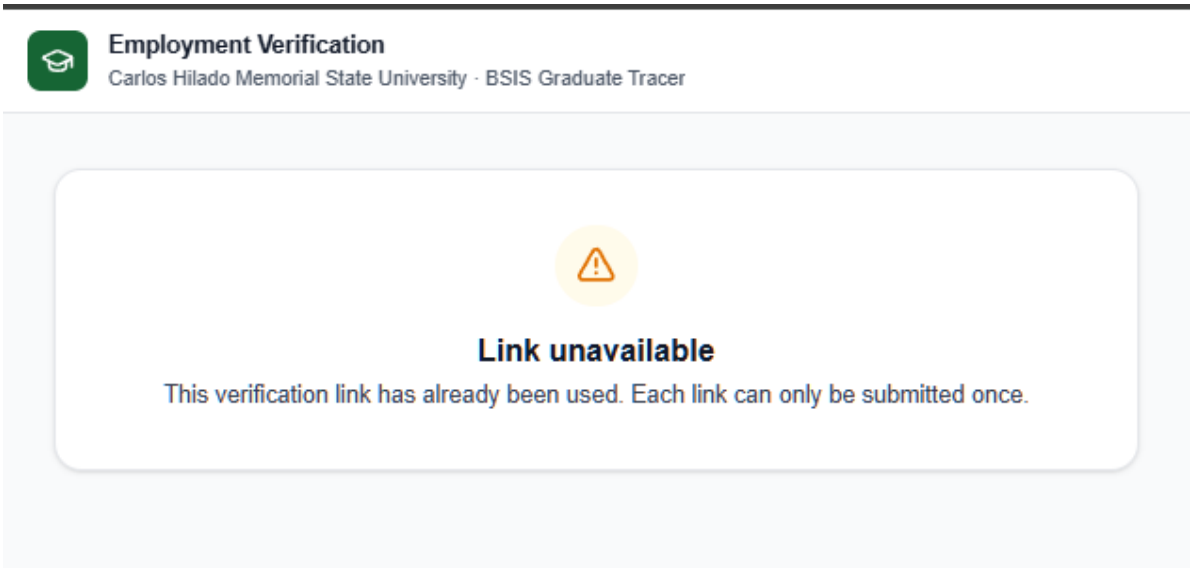


Figure 88

Plain Message Returned for an Invalid or Withdrawn Verification Link

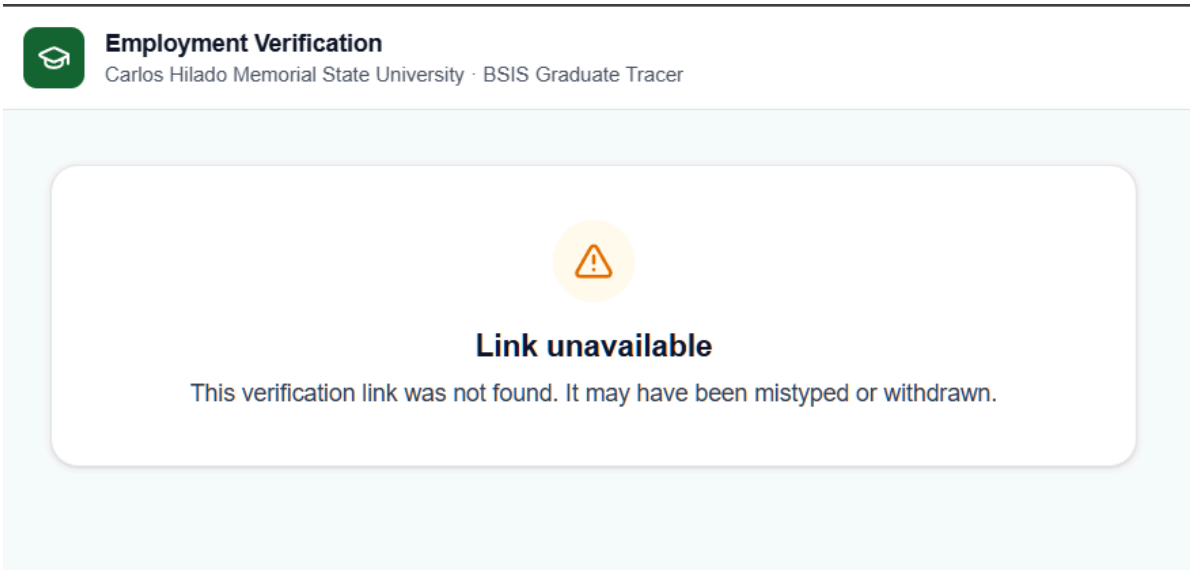


Figure 89

Administrator Rejecting a Registration with a Stated Reason

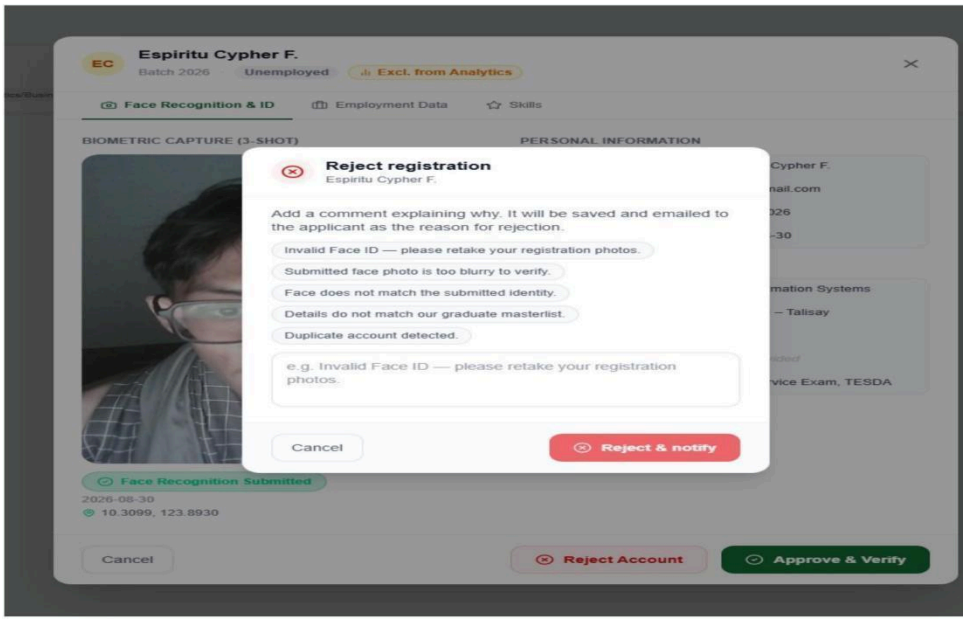


Figure 90

Verified Graduates List in the Administrator Portal

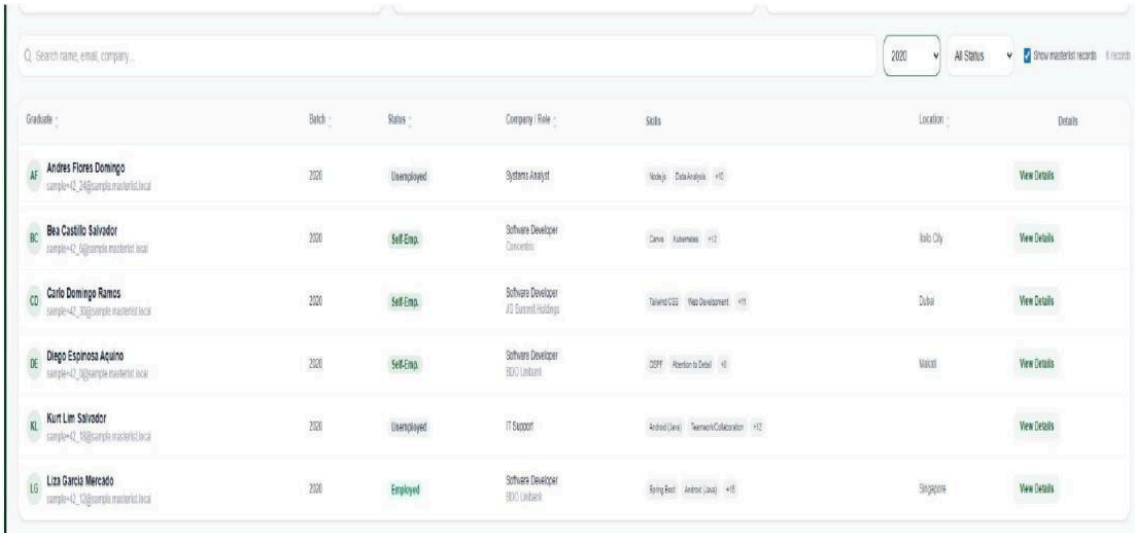


Figure 91

Administrator Analytics Dashboard Showing Batch Trends

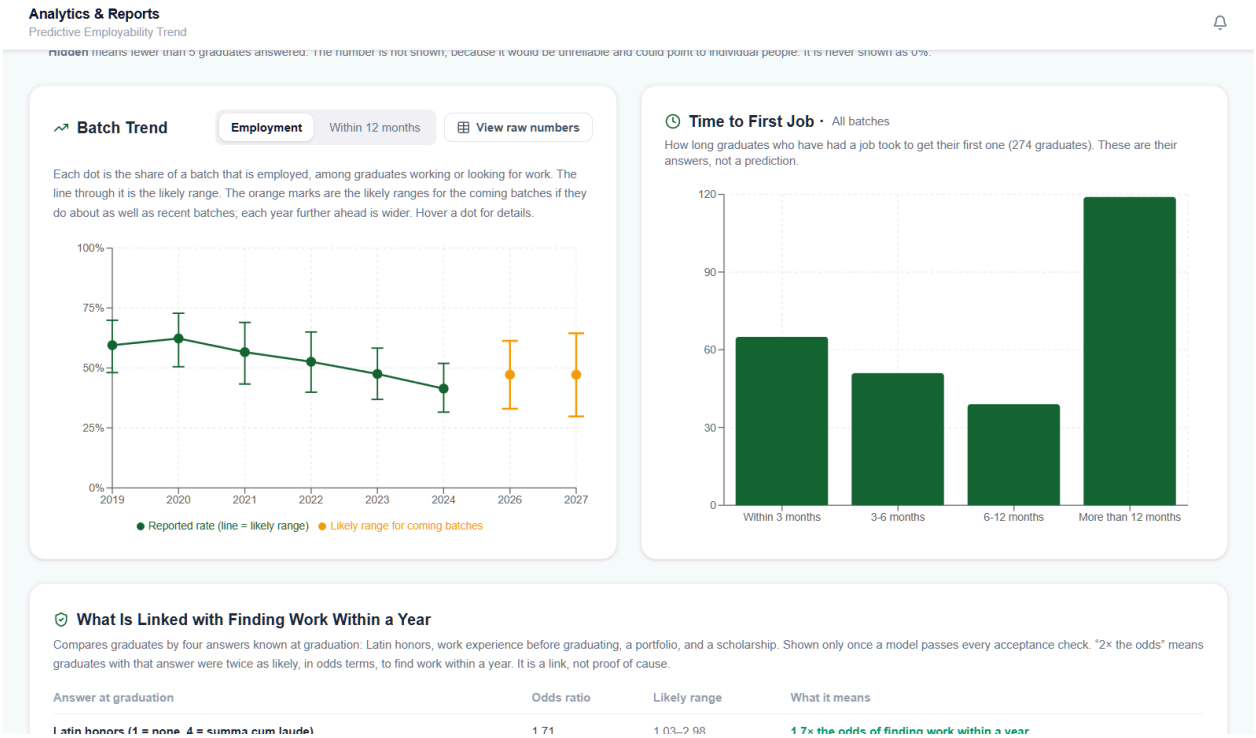
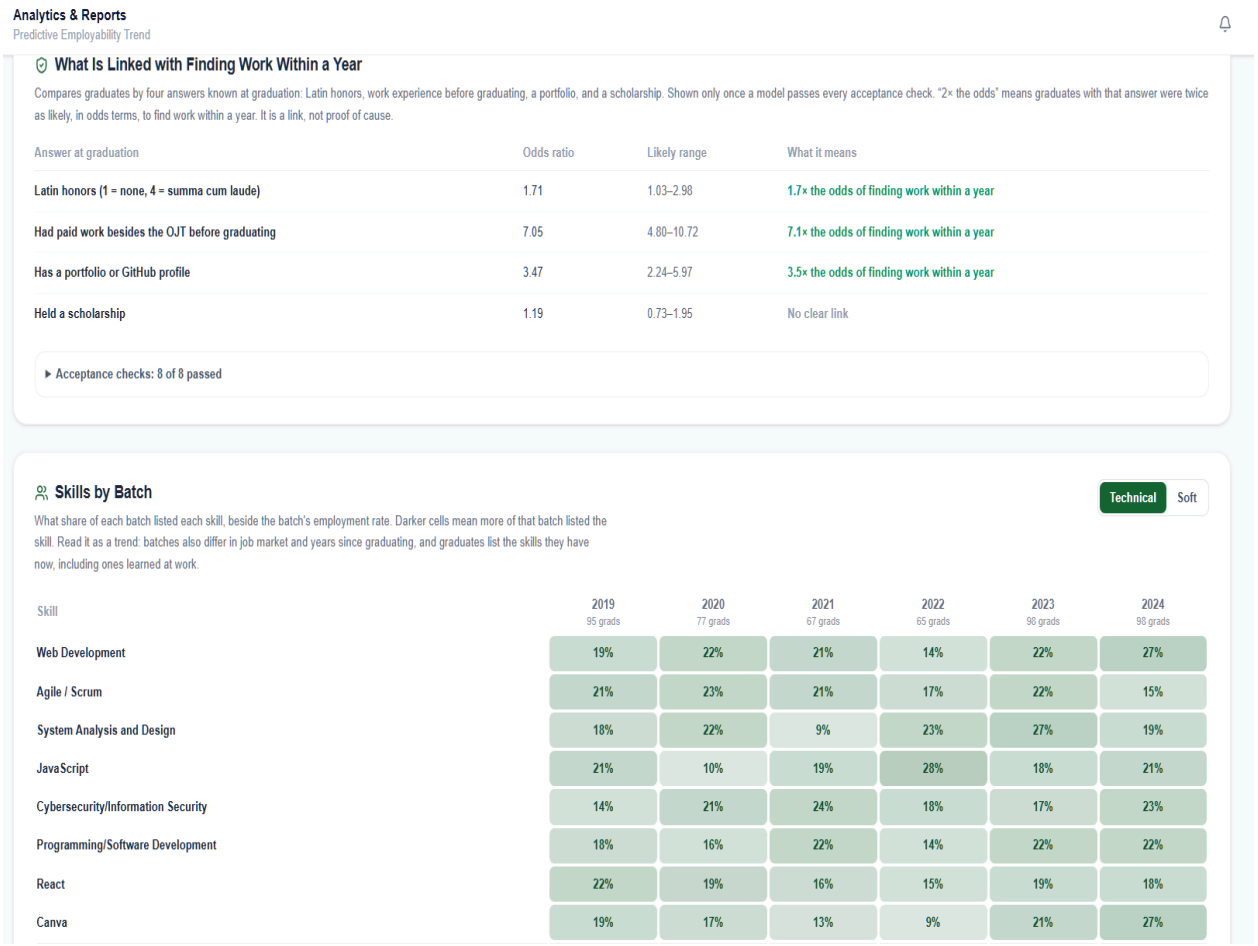


Figure 92

Factors Linked to Employment and Skills Distribution by Batch in the Analytics Module



Analytics & Reports

Predictive Employability Trend

What Is Linked with Finding Work Within a Year

Compares graduates by four answers known at graduation: Latin honors, work experience before graduating, a portfolio, and a scholarship. Shown only once a model passes every acceptance check. "2× the odds" means graduates with that answer were twice as likely, in odds terms, to find work within a year. It is a link, not proof of cause.

Answer at graduation	Odds ratio	Likely range	What it means
Latin honors (1 = none, 4 = summa cum laude)	1.71	1.03–2.98	1.7× the odds of finding work within a year
Had paid work besides the OJT before graduating	7.05	4.80–10.72	7.1× the odds of finding work within a year
Has a portfolio or GitHub profile	3.47	2.24–5.97	3.5× the odds of finding work within a year
Held a scholarship	1.19	0.73–1.95	No clear link

▸ Acceptance checks: 8 of 8 passed

Skills by Batch

Technical | Soft

What share of each batch listed each skill, beside the batch's employment rate. Darker cells mean more of that batch listed the skill. Read it as a trend: batches also differ in job market and years since graduating, and graduates list the skills they have now, including ones learned at work.

Skill	2019 (95 grads)	2020 (77 grads)	2021 (67 grads)	2022 (65 grads)	2023 (98 grads)	2024 (98 grads)
Web Development	19%	22%	21%	14%	22%	27%
Agile / Scrum	21%	23%	21%	17%	22%	15%
System Analysis and Design	18%	22%	9%	23%	27%	19%
JavaScript	21%	10%	19%	28%	18%	21%
Cybersecurity/Information Security	14%	21%	24%	18%	17%	23%
Programming/Software Development	18%	16%	22%	14%	22%	22%
React	22%	19%	16%	15%	19%	18%
Canva	19%	17%	13%	9%	21%	27%

Figure 93

Batch Upload Reviewing the Parsed Rows Before Saving to the Master List

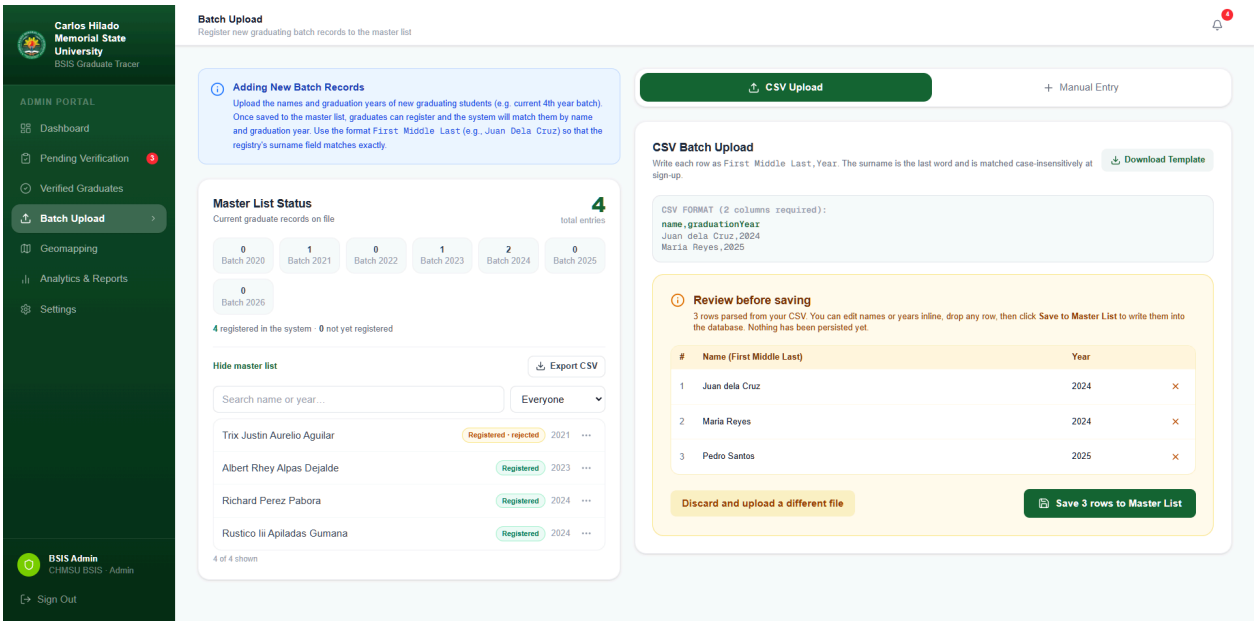


Figure 94

Batch Upload Result Reporting New Records Added and Duplicate Rows Skipped

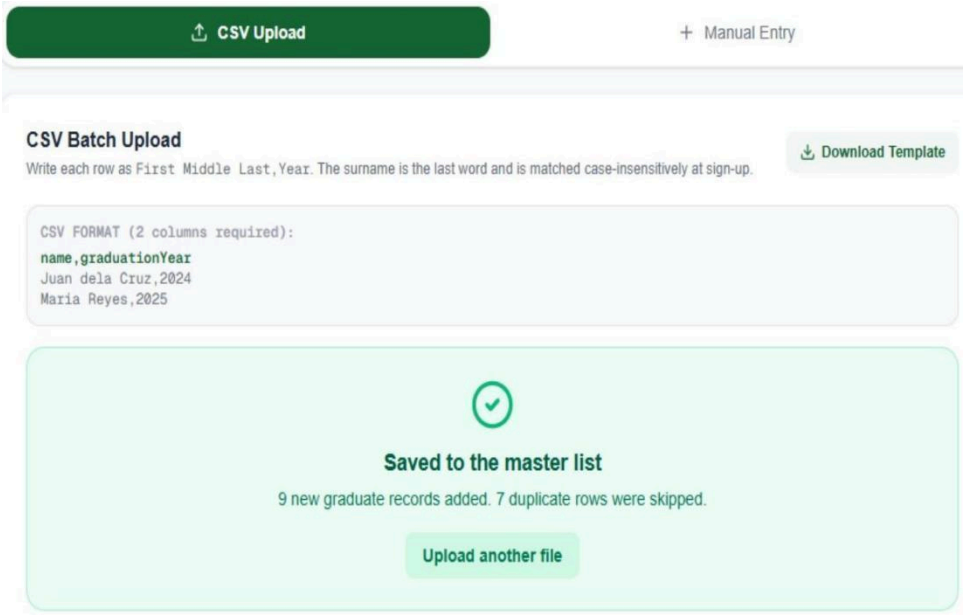


Figure 95

Second Batch Upload Confirming That Re-Uploaded Rows Create No Duplicates

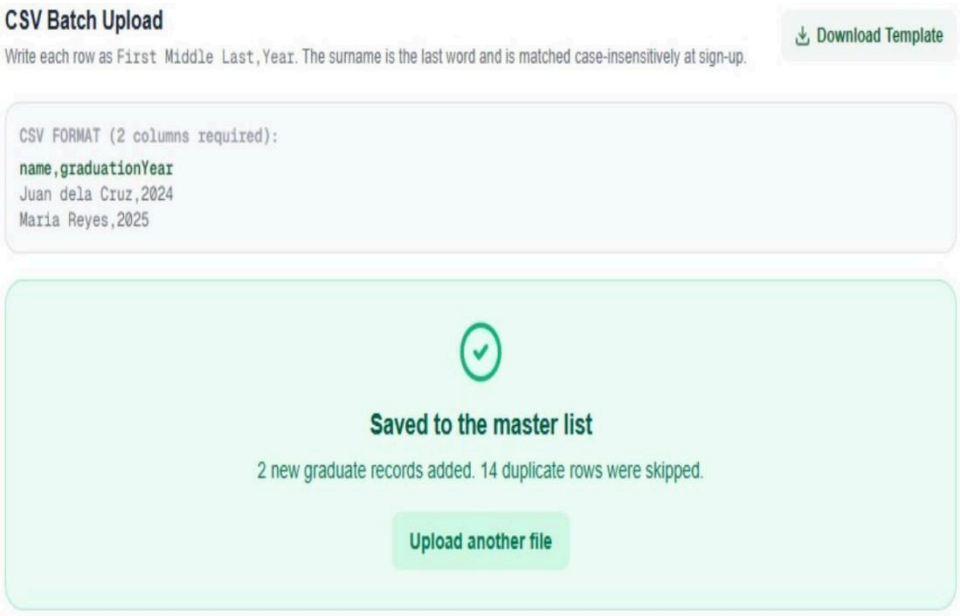


Figure 96

Administrator Dashboard Summarizing Registration and Verification Counts

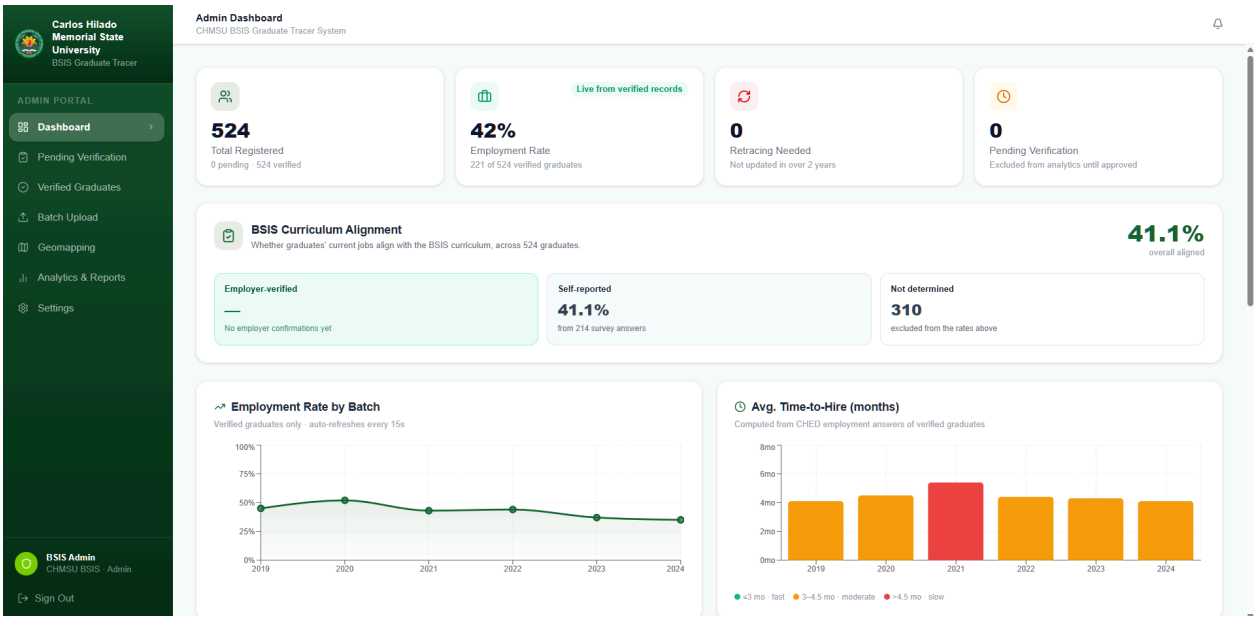


Figure 97

Recent Profile Updates and Top Skills Panels of the Administrator Dashboard

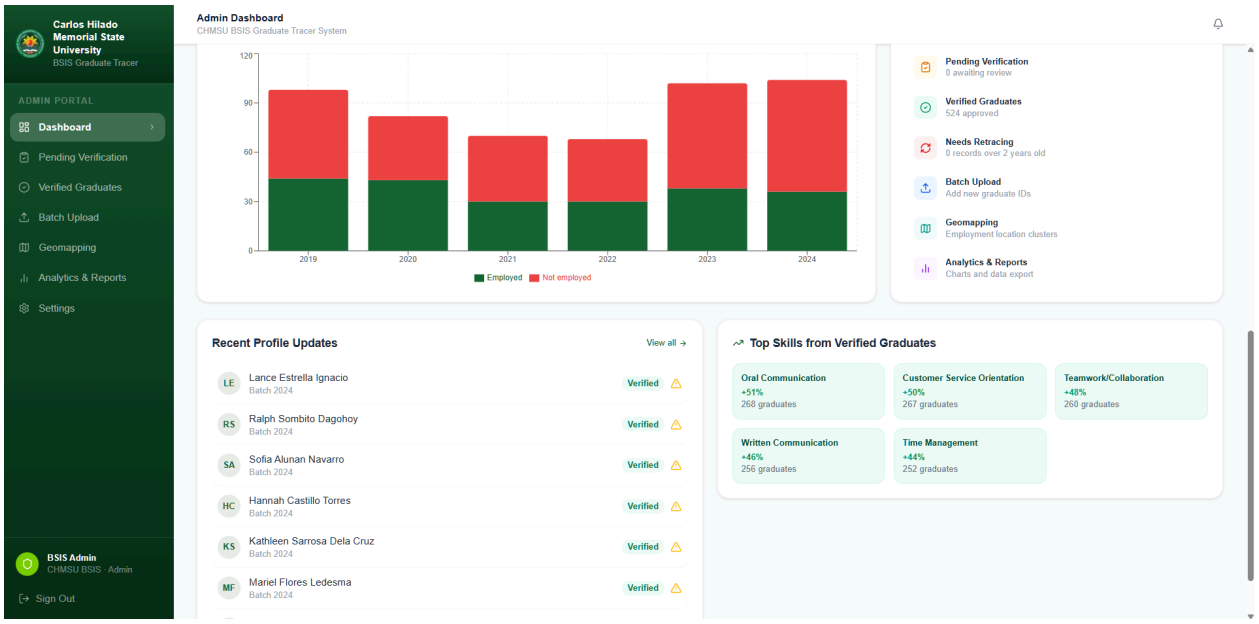


Figure 98

Verified Graduates Module with Filtering by Batch and Status

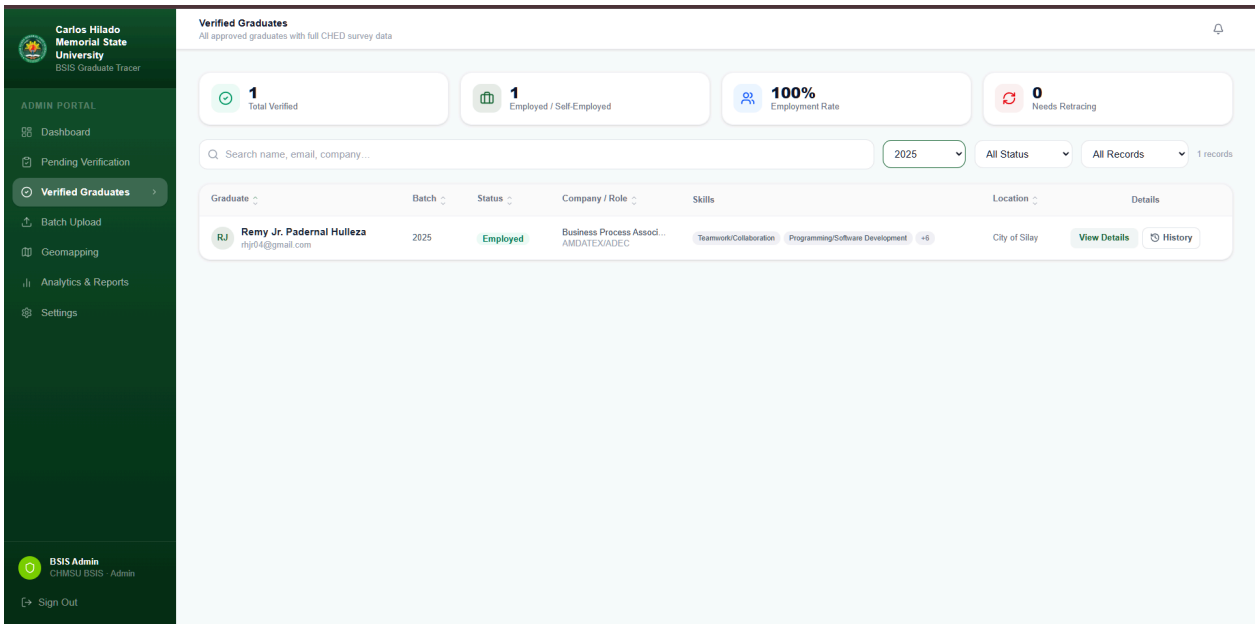


Figure 99

Batch Upload Module Showing the Master List Status

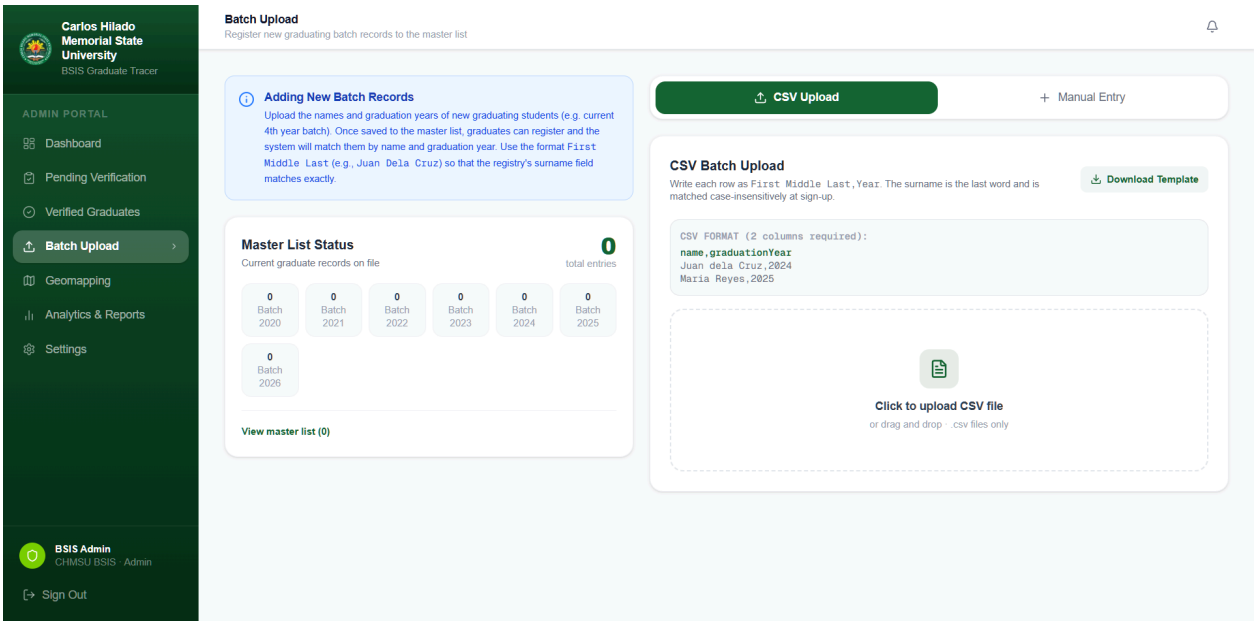


Figure 100

Geomapping View of the Graduate Employment Distribution

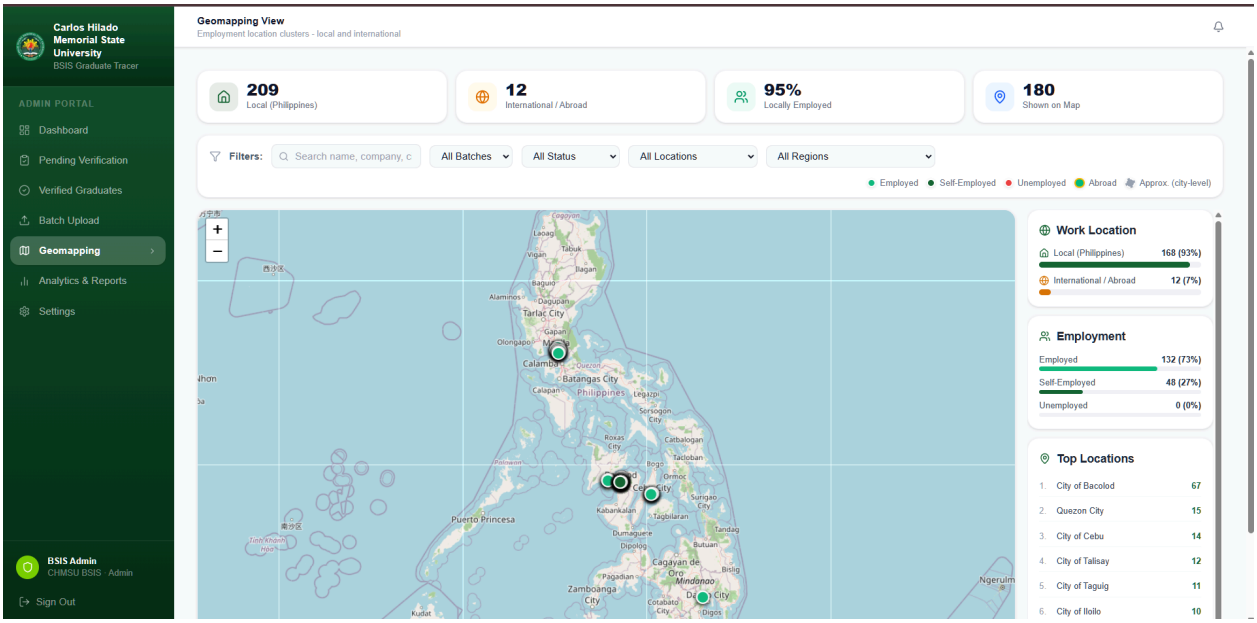


Figure 101

Analytics Module Presenting the Employability Prediction and Forecast

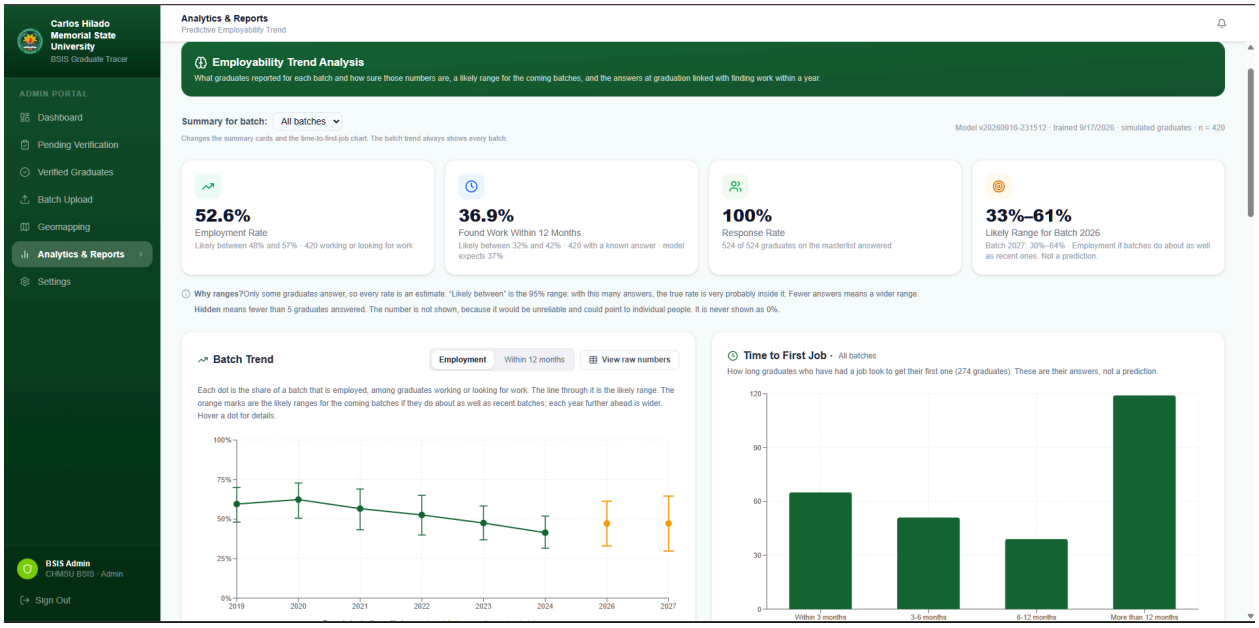


Figure 102

Reports Module Listing the Generated Report Types

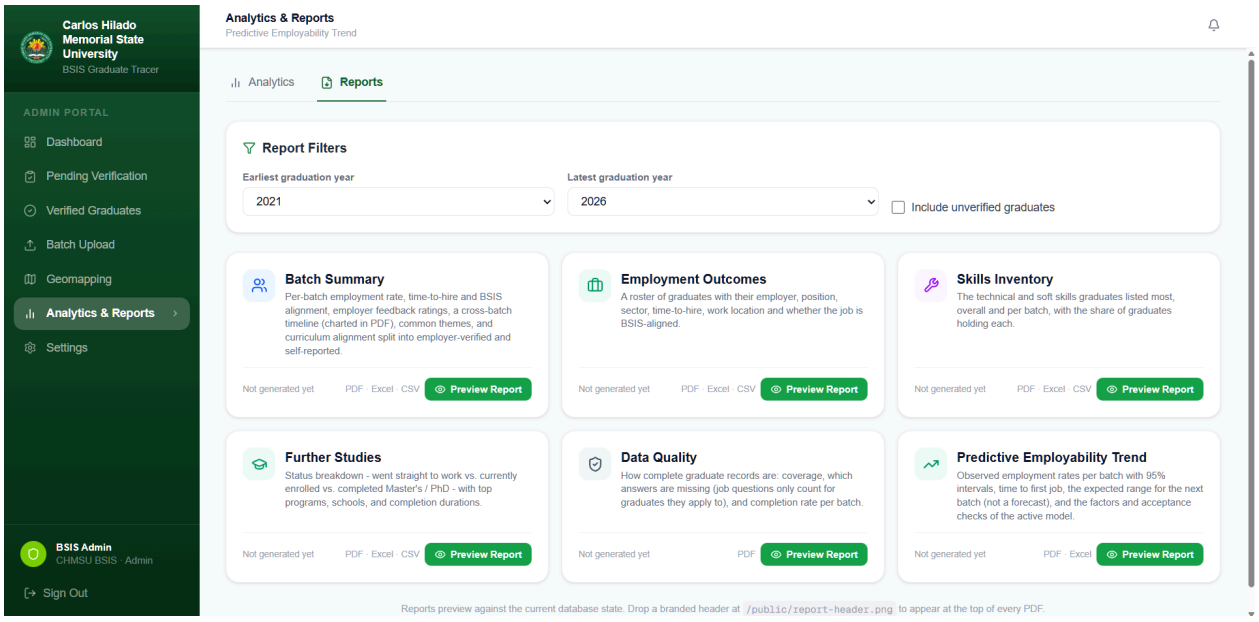


Figure 103

Reference Settings Module for the Management of Skills

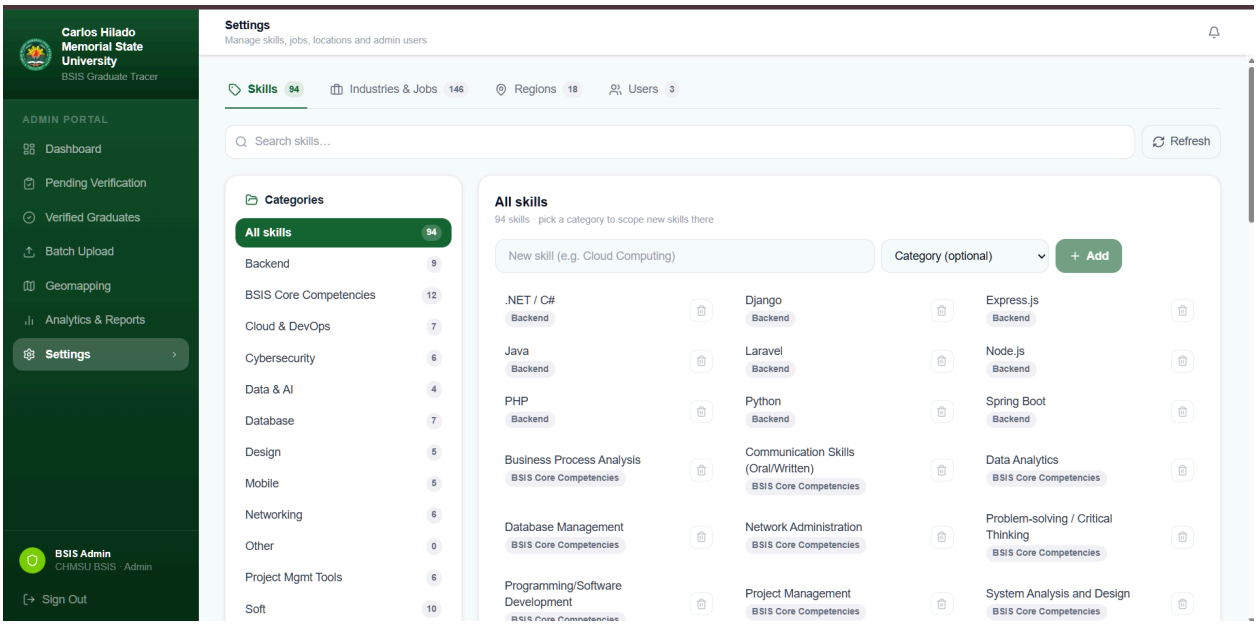


Figure 104

Reference Settings Module for the Management of Industries and Job Titles

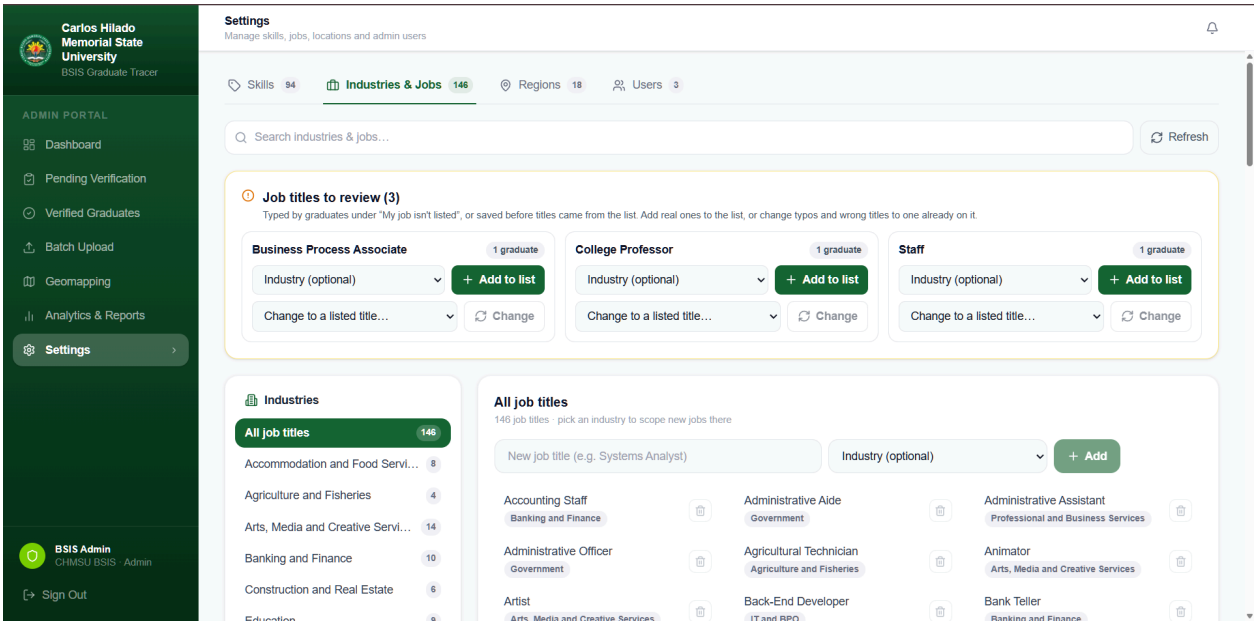


Figure 105

Reference Settings Module for the Management of Regions

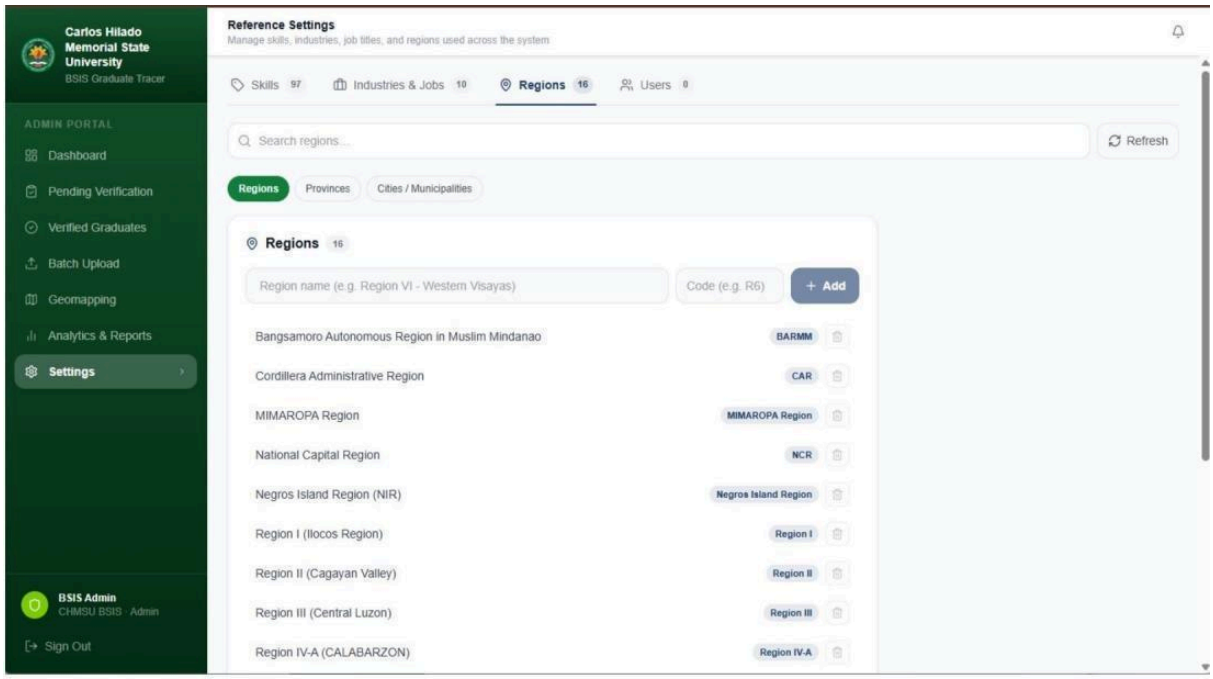


Figure 106

Reference Settings Module for the Management of Administrator Accounts

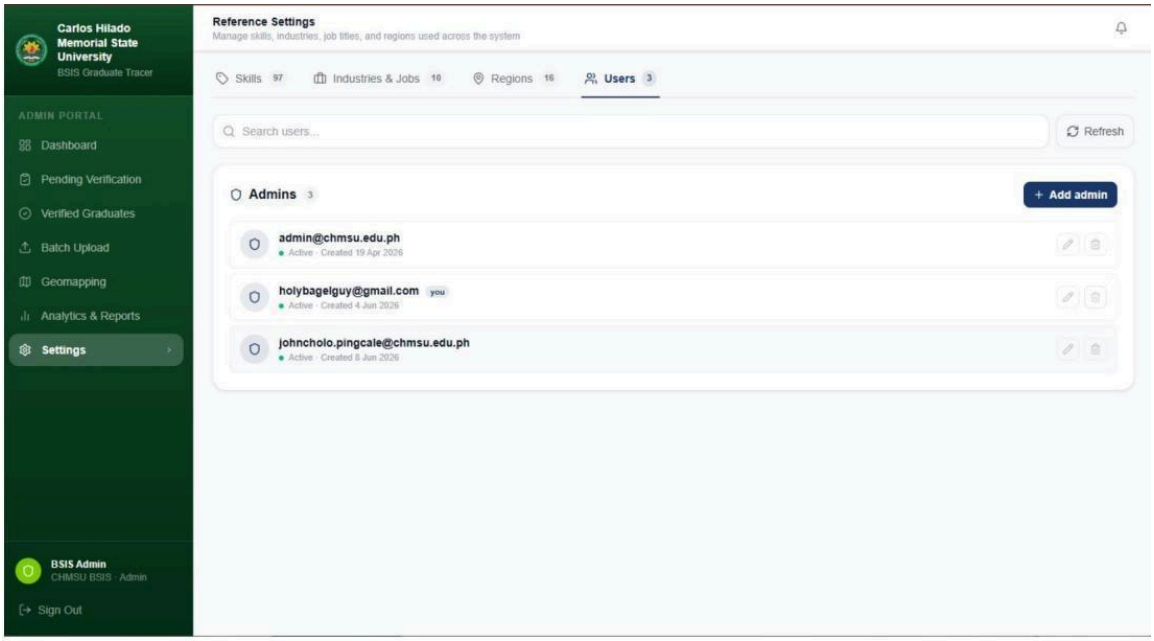


Figure 107

Pending Verification Queue with No Accounts Awaiting Review

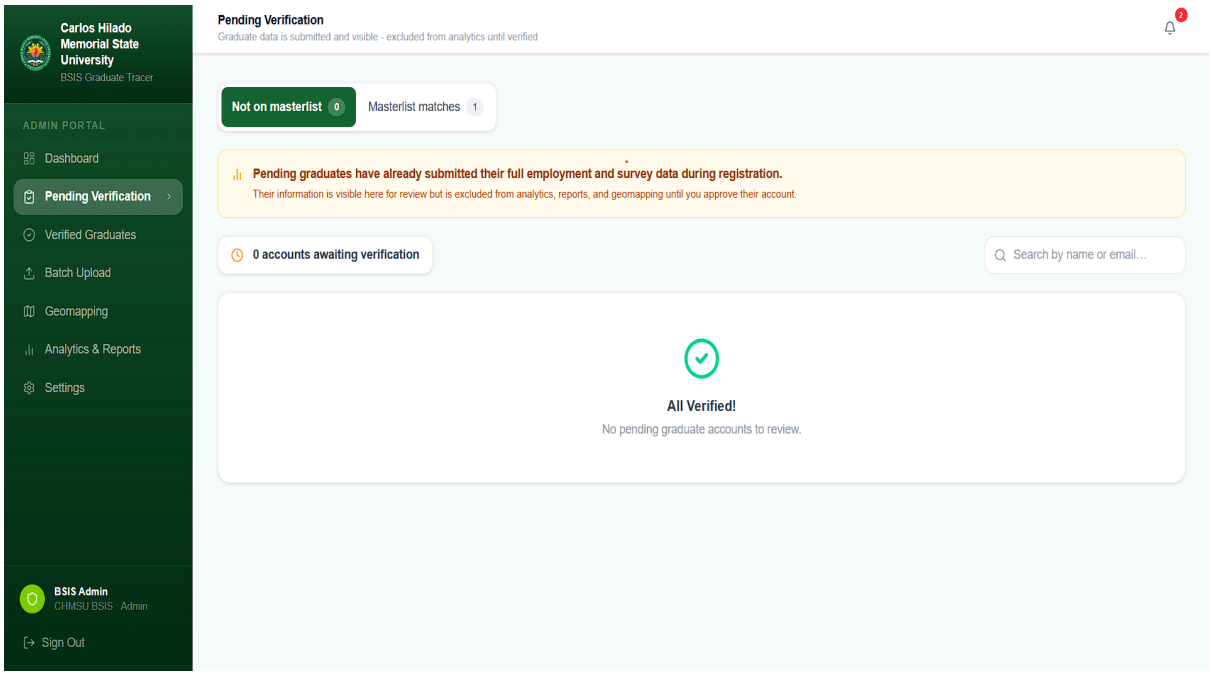
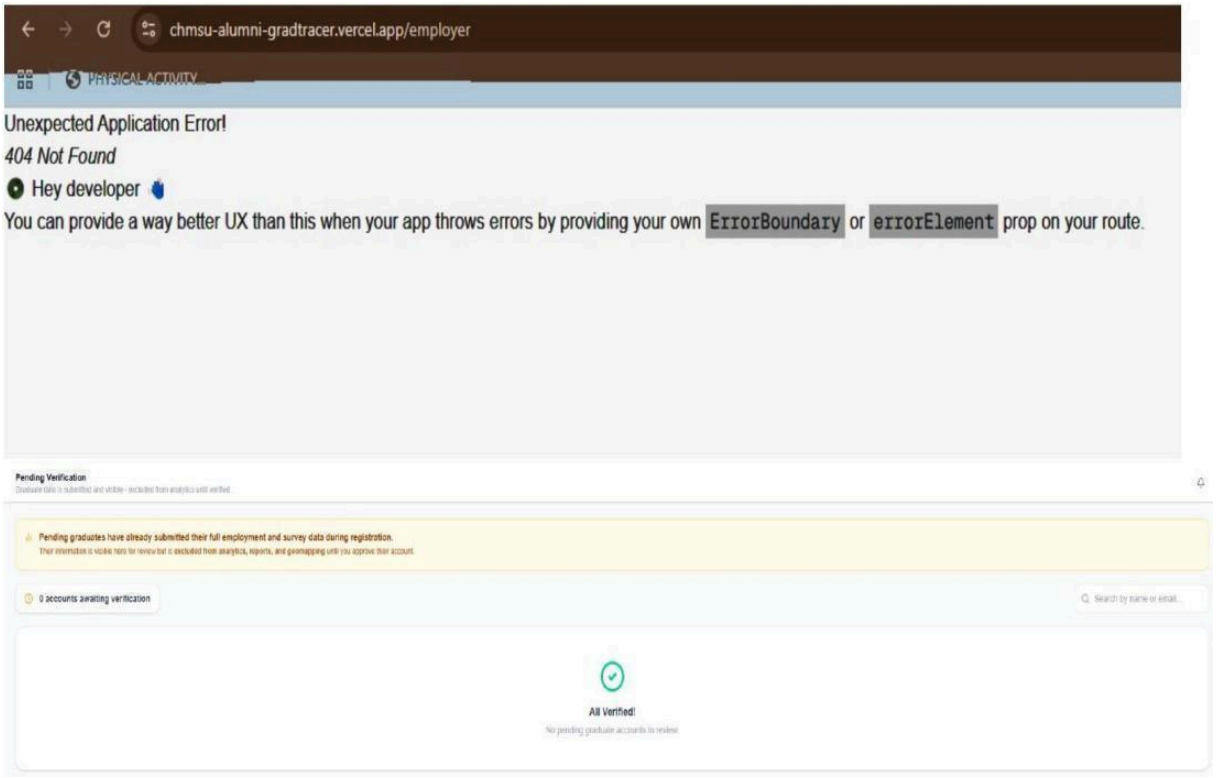


Figure 108

Employer Route Refused by Direct Address Entry, with the Pending Verification Queue Cleared



Appendix F

WHITE-BOX TESTING – ALPHA TESTING

Figure 109

Source Code Review – validators.py: Birth Date and Mobile Number Validation

Whitebox test

```
        })

    # Validate birth date format and age
    if data.get('birth_date'):
        try:
            birth_date = datetime.fromisoformat(data['birth_date']).date() if isinstance(data['birth_date'], str) else data['birth_date']
            age = (datetime.now().date() - birth_date).days // 365
            if not (FieldValidationRules.AGE_RANGE[0] <= age <= FieldValidationRules.AGE_RANGE[1]):
                self.warnings.append({
                    'section': 'personal_information',
                    'field': 'birth_date',
                    'warning': f'Age {age} outside typical range {FieldValidationRules.AGE_RANGE}'
                })
        except (ValueError, AttributeError):
            self.errors.append({
                'section': 'personal_information',
                'field': 'birth_date',
                'error': 'Invalid date format. Use YYYY-MM-DD'
            })

    # Validate mobile format (basic check)
    if data.get('mobile') and not (len(data['mobile']) >= 10):
        self.errors.append({
            'section': 'personal_information',
            'field': 'mobile',
            'error': 'Mobile number too short'
```

Figure 110

Source Code Review – validators.py: Comprehensive Survey Validation Functions

```
    # Validate educational background
    if 'educational_background' in survey_data:
        self._validate_educational_background(survey_data['educational_background'])

    # Validate academic & pre-employment
    if 'academic_preemployment' in survey_data:
        self._validate_academic_preemployment(survey_data['academic_preemployment'])

    # Validate employment status
    if 'employment_status' in survey_data:
        self._validate_employment_status(survey_data['employment_status'])

    # Validate first job details
    if 'first_job_details' in survey_data:
        self._validate_first_job_details(survey_data['first_job_details'])

    # Validate current job details
    if 'current_job_details' in survey_data:
        self._validate_current_job_details(survey_data['current_job_details'])

    # Validate work address
    if 'work_address' in survey_data:
        self._validate_work_address(survey_data['work_address'])

    # Validate competency assessment
    if 'competency_assessment' in survey_data:
        self._validate_competency_assessment(survey_data['competency_assessment'])

    # Validate logical consistency across sections
    self._validate_consistency(survey_data)

    # Calculate metrics
    status = self._determine_status()
    completeness = self._calculate_completeness(survey_data)

    return {
        'status': status.value,
        'is_valid': len(self.errors) == 0,
        'completeness_score': completeness,
        'errors': self.errors,
        'warnings': self.warnings,
        'field_completeness': self.field_completeness
    }

def _validate_personal_info(self, data: Dict):
    """Validate personal information fields"""
    required_fields = ['first_name', 'last_name', 'gender', 'birth_date', 'mobile', 'city', 'province']

    for field in required_fields:
        if not data.get(field):
            self.errors.append({
                'section': 'personal_information',
                'field': field,
                'error': f'{field} is required'
            })
        else:
            self.field_completeness[field] = True

    # Validate gender
    if data.get('gender') and data['gender'].upper() not in {g.upper() for g in FieldValidationRules.GENDER_CHOICES}:
        self.errors.append({
            'section': 'personal_information',
            'field': 'gender',
            'error': f"Invalid gender: {data['gender']}"
        })
```

Figure 111

Source Code Review – FieldValidationRules Constants and SurveyDataValidator Class

```
    # Count Ranges
    TECHNICAL_SKILL_COUNT_RANGE = (0, 12)
    SOFT_SKILL_COUNT_RANGE = (0, 10)

    # Philippines Regions
    REGIONS_VALID = {
        'NCR', 'Region I', 'Region II', 'Region III', 'Region IV-A', 'Region IV-B',
        'Region V', 'Region VI', 'Region VII', 'Region VIII', 'Region IX', 'Region X',
        'Region XI', 'Region XII', 'Region XIII', 'CAR', 'BARMM', 'Abroad'
    }


class SurveyDataValidator:
    """Comprehensive survey data validation"""

    def __init__(self):
        self.errors: List[Dict[str, Any]] = []
        self.warnings: List[Dict[str, Any]] = []
        self.field_completeness: Dict[str, bool] = {}

    def validate_comprehensive_survey(self, survey_data: Dict) -> Dict:
        """
        Validate complete survey response

        Args:
            survey_data: Dictionary containing all survey sections

        Returns:
            Validation result with status, errors, warnings, completeness score
        """
        self.errors = []
        self.warnings = []
        self.field_completeness = {}

        # Validate personal information section
        if 'personal_information' in survey_data:
            self._validate_personal_info(survey_data['personal_information'])

        # Validate educational background
        if 'educational_background' in survey_data:
            self._validate_educational_background(survey_data['educational_background'])

        # Validate academic & pre-employment
        if 'academic_preemployment' in survey_data:
            self._validate_academic_preemployment(survey_data['academic_preemployment'])

        # Validate employment status
        if 'employment_status' in survey_data:
            self._validate_employment_status(survey_data['employment_status'])

        # Validate first job details
        if 'first_job_details' in survey_data:
            self._validate_first_job_details(survey_data['first_job_details'])

        # Validate current job details
        if 'current_job_details' in survey_data:
            self._validate_current_job_details(survey_data['current_job_details'])

        # Validate work address
        if 'work_address' in survey_data:
            self._validate_work_address(survey_data['work_address'])

        # Validate competency assessment
        if 'competency_assessment' in survey_data:
            self._validate_competency_assessment(survey_data['competency_assessment'])
```

Figure 112

API Endpoint Testing – Administrator Login POST Request (200 OK)

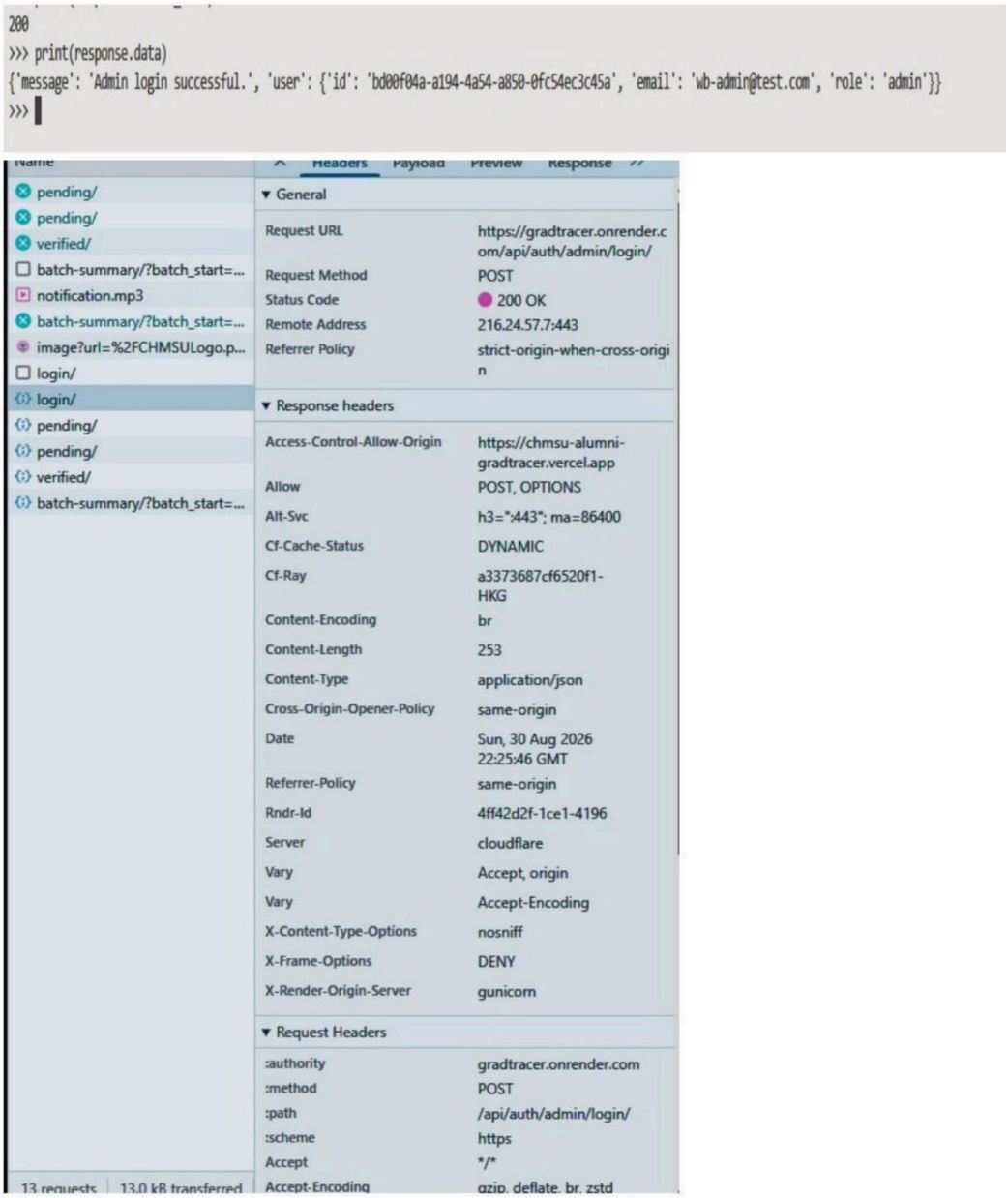


Figure 113

Source Code Inspection – PowerShell Get-Content tracer\validators.py

```
(venv) PS C:\Users\Spirit\NEW-Capstone\backend> Get-Content tracer\validators.py | Select-Object -First 180
"""
Data Validation Framework for Survey Responses

Ensures all survey data is clean before storage and model training.
All inputs MUST be clean data - this framework validates against encoding rules,
field constraints, and logical consistency requirements.

Reference: EyeOfTheTiger/Data_Validation_Rules.md
"""

from datetime import datetime
from enum import Enum
from typing import Dict, List, Optional, Tuple, Any
from django.core.exceptions import ValidationError


class ValidationStatus(Enum):
    """Validation result status"""
    VALID = "valid"
    INVALID = "invalid"
    WARNINGS = "warnings"


class FieldValidationRules:
    """Define validation rules per field based on PredictiveModelContext.md"""

    # Binary Fields
    GENDER_CHOICES = {'M', 'F', 'Male', 'Female'}

    # Ordinal Fields (must be in valid range)
    GPA_RANGE_VALID = {0, 1, 2, 3, 4, 5}  # None allowed separately
    HONORS_VALID = {1, 2, 3, 4}
    OJT_RELEVANCE_VALID = {0, 1, 2, 3}
    JOB_APPLICATIONS_VALID = {1, 2, 3, 4}

    # Categorical Fields
    JOB_SECTOR_VALID = {'government', 'private', 'entrepreneurial'}
    JOB_STATUS_VALID = {'regular', 'probationary', 'contractual', 'self_employed'}
    JOB_SOURCE_VALID = {
        'personal_network', 'online_portal', 'career_fair', 'walk_in',
        'social_media', 'entrepreneurship', 'other'
    }
    EMPLOYMENT_STATUS_VALID = {
        'employed_full_time', 'employed_part_time', 'self_employed',
        'seeking', 'not_seeking', 'never_employed'
    }

    # Continuous/Range Fields
    TIME_TO_HIRE_VALID = {1, 3, 4.5, 9, 18, 30}
    BATCH_RANGE = (2020, 2025)
    AGE_RANGE = (18, 70)

    # Count Ranges
    TECHNICAL_SKILL_COUNT_RANGE = (0, 12)
    SOFT_SKILL_COUNT_RANGE = (0, 10)

    # Philippines Regions
    REGIONS_VALID = {
        'NCR', 'Region I', 'Region II', 'Region III', 'Region IV-A', 'Region IV-B',
        'Region V', 'Region VI', 'Region VII', 'Region VIII', 'Region IX', 'Region X',
        'Region XI', 'Region XII', 'Region XIII', 'CAR', 'BARMM', 'Abroad'
    }
```

Figure 114

Unit Testing – SurveyDataValidator Execution with Traceback Output

```
tracer\api.py:2198:        artifacts = _load_ml_artifacts()
tracer\reports_api.py:875:        from .api import _aggregate_for_batch, _load_ml_artifacts
tracer\reports_api.py:884:        artifacts = _load_ml_artifacts()
tracer\api.py:1801:def _build_live_df():
tracer\api.py:2219:            live_df = _build_live_df()
tracer\api.py:2222:            _log.warning("_build_live_df() failed: %s", _exc)
tracer\api.py:1798:_EMPLOYED_STATUSES: frozenset[str] = frozenset({"employed_full_time", "employed_part_time", "self_employed"})
tracer\api.py:1852:            row["employment_status"]        = 1 if emp.employment_status in _EMPLOYED_STATUSES else 0
tracer\api.py:1887:def _aggregate_for_batch(artifacts: dict, batch: int | None) -> dict:
tracer\api.py:2236:        overall = _aggregate_for_batch(local_artifacts, batch)
tracer\api.py:2239:            entry = _aggregate_for_batch(local_artifacts, int(c))
tracer\reports_api.py:875:        from .api import _aggregate_for_batch, _load_ml_artifacts
tracer\reports_api.py:895:        overall = _aggregate_for_batch(artifacts, None)
tracer\reports_api.py:907:            agg = _aggregate_for_batch(artifacts, c)
```

```
>>> from tracer.validators import *
>>> [name for name in dir() if 'Validator' in name]
['SurveyDataValidator']
>>> validator = SurveyDataValidator()
>>> flat_dict = {
...     'employment_status': 'employed_full_time',
...     'field1': 'value1'
... }
>>> validator.validate_comprehensive_survey(flat_dict)
Traceback (most recent call last):
  File "<console>", line 1, in <module>
  File "C:\Users\Spirit\NEW-Capstone\backend\tracer\validators.py", line 101, in validate_comprehensive_survey
    self._validate_employment_status(survey_data['employment_status'])
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Spirit\NEW-Capstone\backend\tracer\validators.py", line 252, in _validate_employment_status
    if data.get('employment_status'):
       ^^^^^^^^
```

Figure 115

API Endpoint Testing – Django REST Framework Region List API (GET /api/reference/regions/)

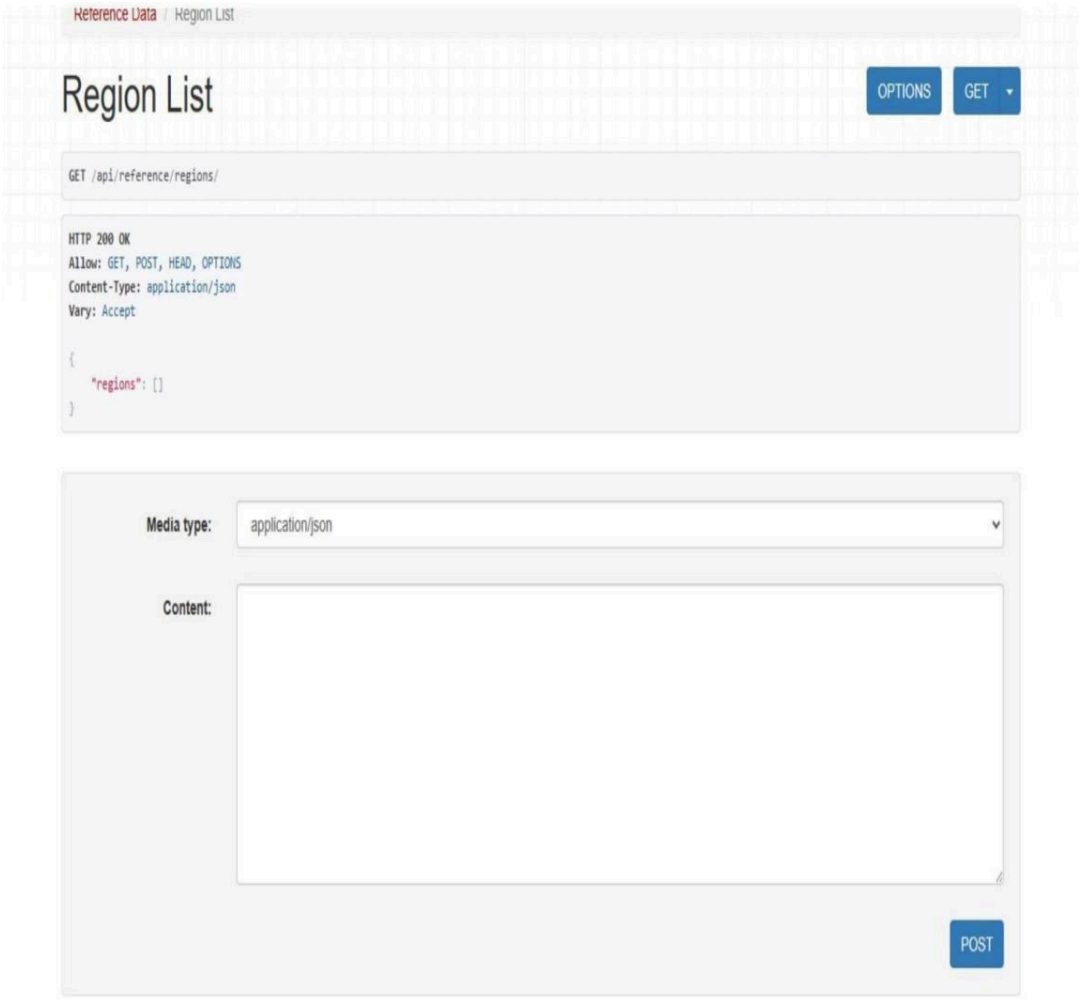


Figure 116

API Endpoint Testing – Django REST Framework Region List API (continued)

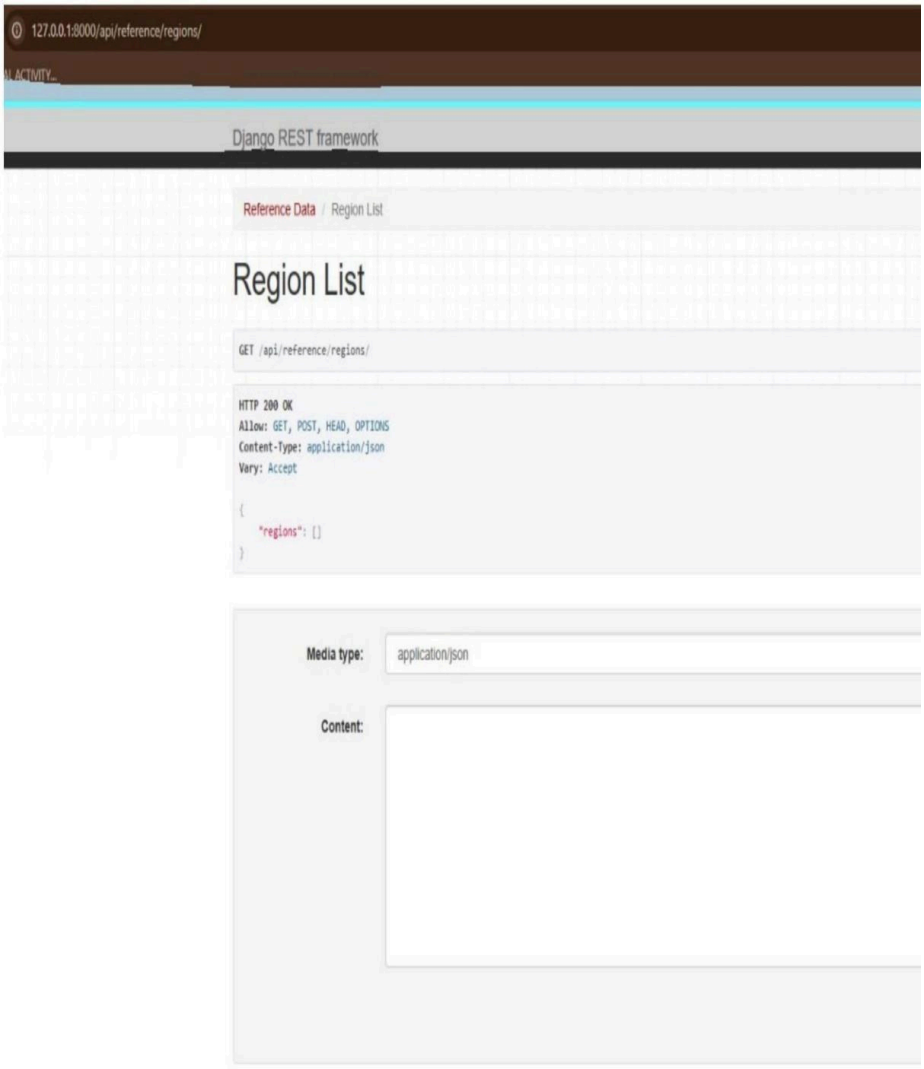


Figure 117

API Endpoint Testing – Browser Console fetch() POST and PATCH Operations

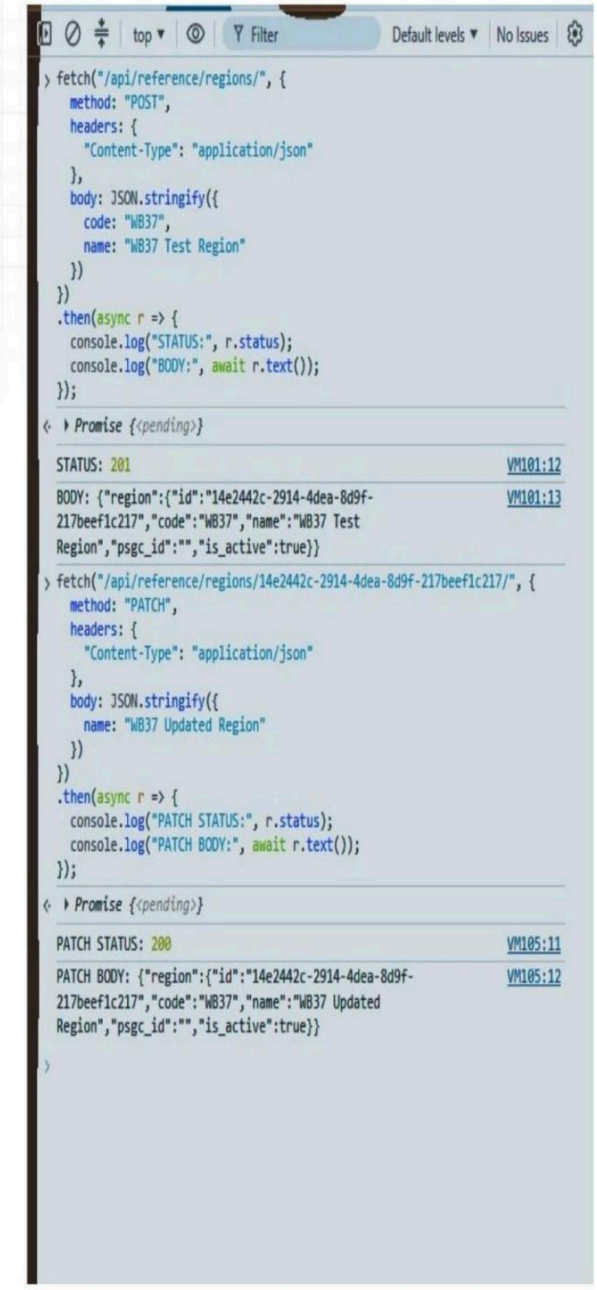


Figure 118

API Endpoint Testing – Network Tab GET /api/admin/alumni/verified/ (200 OK)

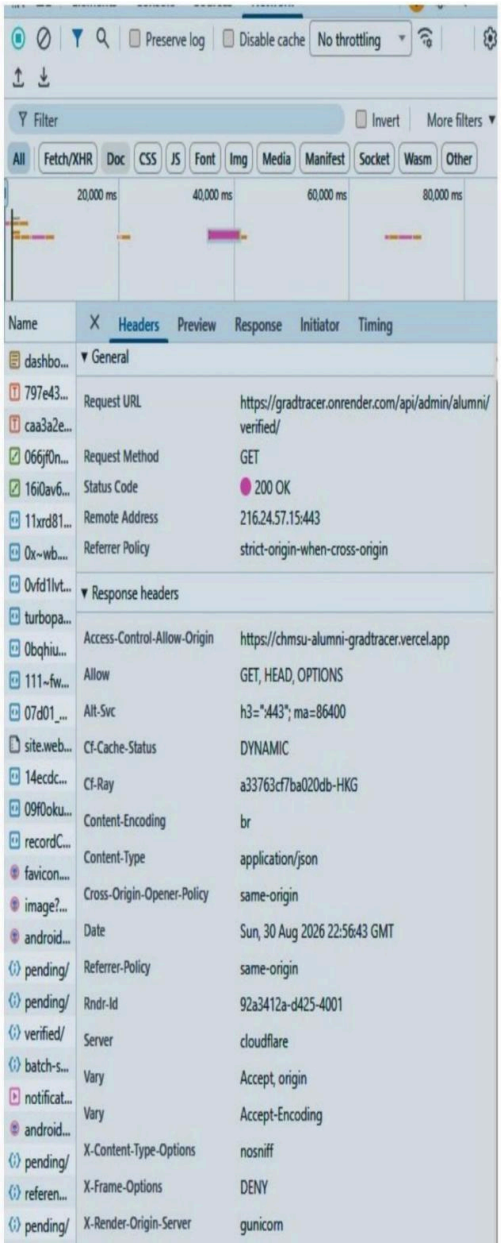


Figure 119

API Endpoint Testing – Network Tab OPTIONS CORS Headers for /api/reference/cities/

Figure 120

API Endpoint Testing – Network Tab GET /api/reference/ (200 OK)

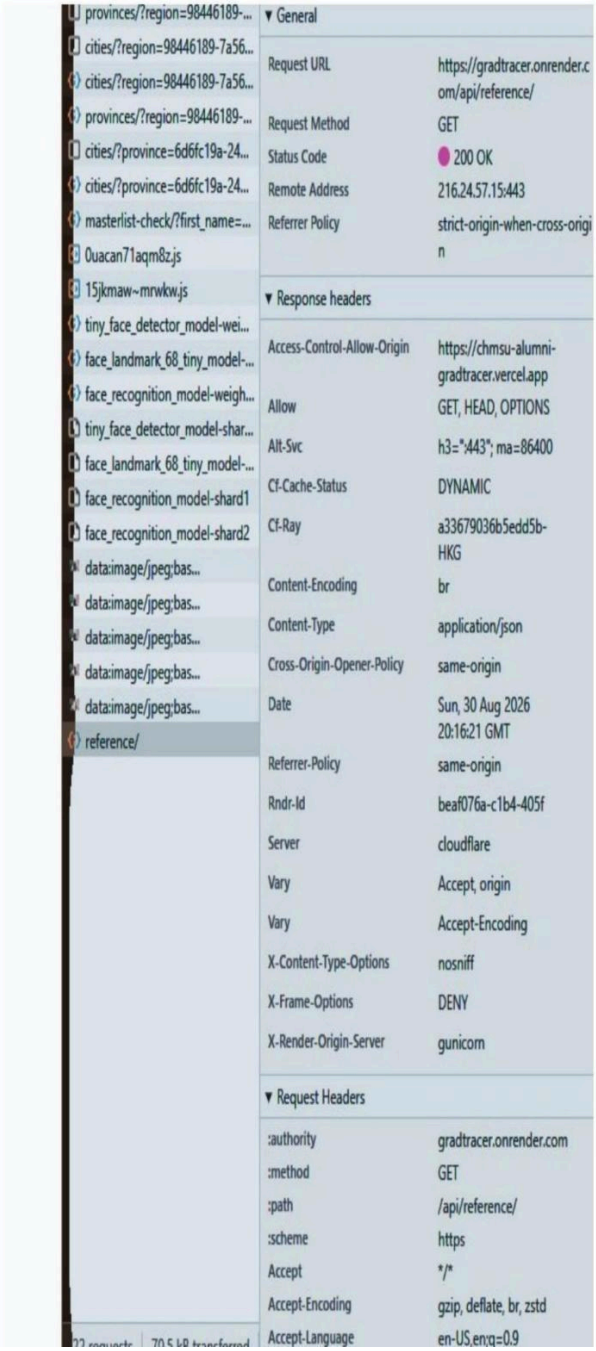


Figure 121

API Endpoint Testing – Network Tab POST /api/auth/alumni/register/ (201 Created)

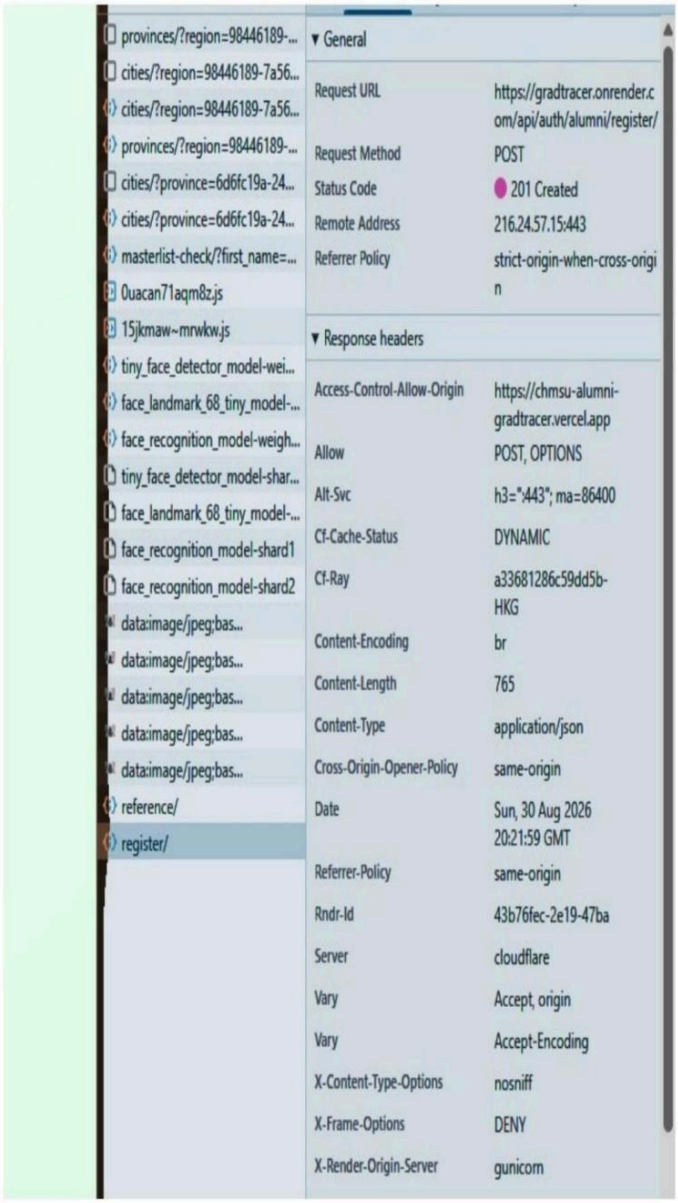


Figure 122

Unit Test Execution – Django Test Runner (38 Tests Passed) and Vitest (21 Tests Passed)

```
----------------------------------------------------------------------
Ran 38 tests in 11.563s

OK
```

```
WB-39 reducer NEEDS_CORRECTION preserves all state > returns to the owning form...
WB-40 reducer SET_ERROR on a non-validation failure > routes to the error stage...
WB-41 classifySubmitFailure discriminates on field_errors presence > ...
 [... all 14 cases, 21 assertions ...]
WB-52 null-safe formatters render a dash, never a false zero > months(null)...

Test Files  1 passed (1)
     Tests  21 passed (21)
```

Figure 123

API Endpoint Testing – Network Tab POST Reject Endpoint (200 OK)

Figure 124

QASE Test Management – White Box Testing Suite (WB-01 to WB-27)

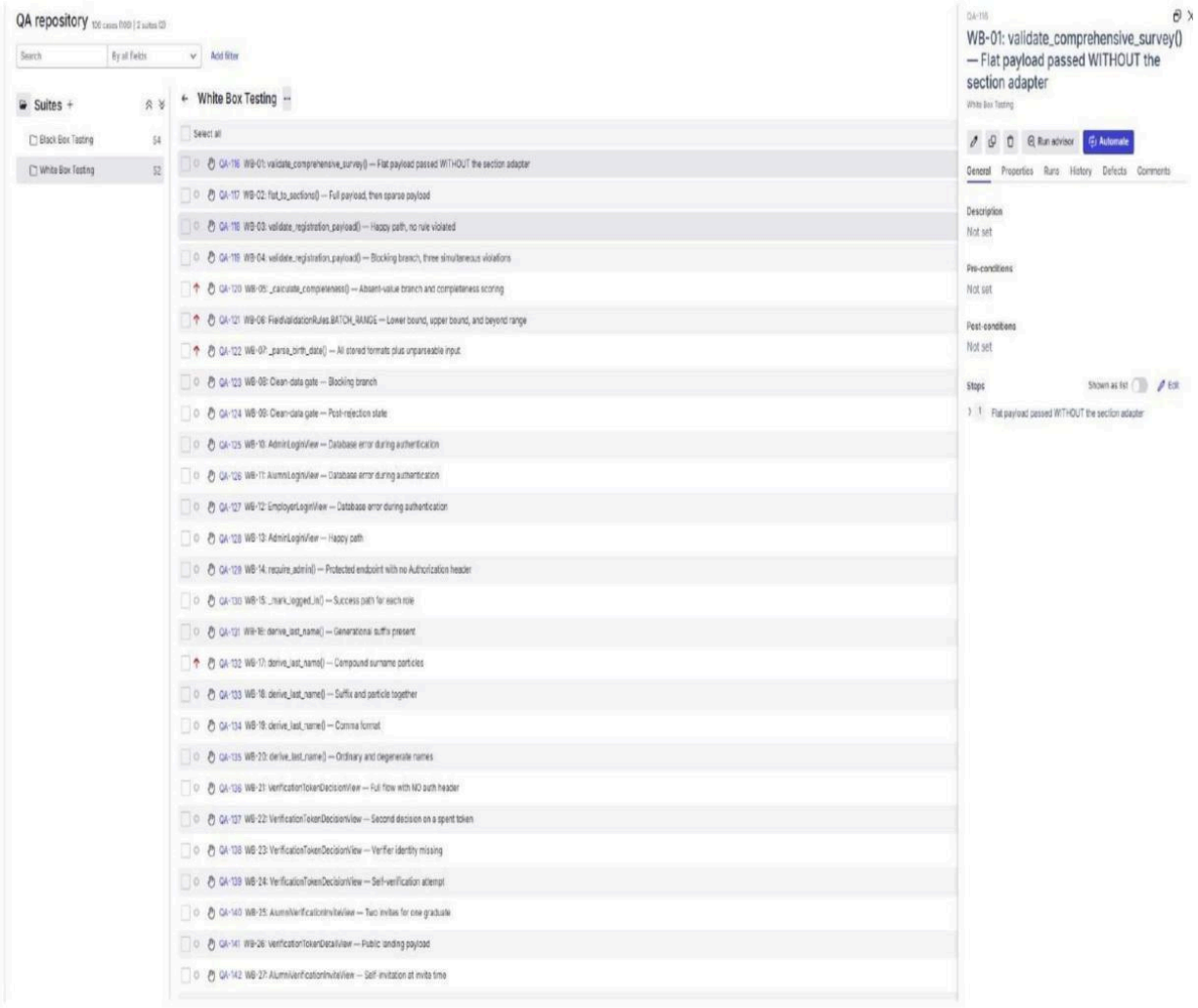
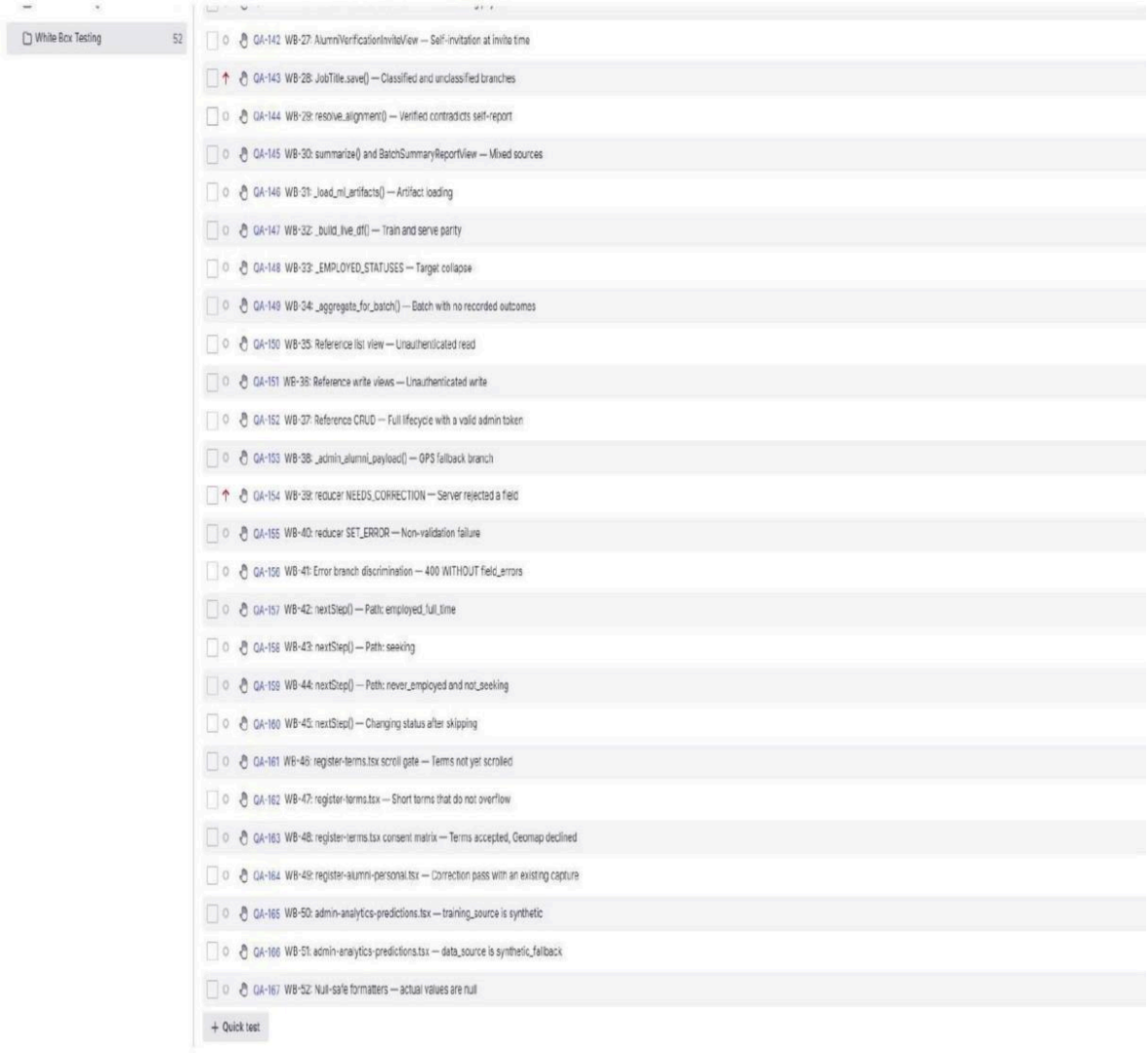


Figure 125

QASE Test Management – White Box Testing Suite (WB-28 to WB-52)



Appendix G

WHITE-BOX TESTING – BETA TESTING

Figure 126

Validation of the Birth Date Format and the Mobile Number Format in the Source Code

Whitebox test

```
        })

    # Validate birth date format and age
    if data.get('birth_date'):
        try:
            birth_date = datetime.fromisoformat(data['birth_date']).date() if isinstance(data['birth_date'], str) else data['birth_date']
            age = (datetime.now().date() - birth_date).days // 365
            if not (FieldValidationRules.AGE_RANGE[0] <= age <= FieldValidationRules.AGE_RANGE[1]):
                self.warnings.append({
                    'section': 'personal_information',
                    'field': 'birth_date',
                    'warning': f'Age {age} outside typical range {FieldValidationRules.AGE_RANGE}'
                })
        except (ValueError, AttributeError):
            self.errors.append({
                'section': 'personal_information',
                'field': 'birth_date',
                'error': 'Invalid date format. Use YYYY-MM-DD'
            })

    # Validate mobile format (basic check)
    if data.get('mobile') and not (len(data['mobile']) >= 10):
        self.errors.append({
            'section': 'personal_information',
            'field': 'mobile',
            'error': 'Mobile number too short'
```

Figure 127

Validation Methods for the Educational, Employment and Competency Sections

```
        # Validate educational background
        if 'educational_background' in survey_data:
            self._validate_educational_background(survey_data['educational_background'])

        # Validate academic & pre-employment
        if 'academic_preemployment' in survey_data:
            self._validate_academic_preemployment(survey_data['academic_preemployment'])

        # Validate employment status
        if 'employment_status' in survey_data:
            self._validate_employment_status(survey_data['employment_status'])

        # Validate first job details
        if 'first_job_details' in survey_data:
            self._validate_first_job_details(survey_data['first_job_details'])

        # Validate current job details
        if 'current_job_details' in survey_data:
            self._validate_current_job_details(survey_data['current_job_details'])

        # Validate work address
        if 'work_address' in survey_data:
            self._validate_work_address(survey_data['work_address'])

        # Validate competency assessment
        if 'competency_assessment' in survey_data:
            self._validate_competency_assessment(survey_data['competency_assessment'])

        # Validate logical consistency across sections
        self._validate_consistency(survey_data)

        # Calculate metrics
        status = self._determine_status()
        completeness = self._calculate_completeness(survey_data)

        return {
            'status': status.value,
            'is_valid': len(self.errors) == 0,
            'completeness_score': completeness,
            'errors': self.errors,
            'warnings': self.warnings,
            'field_completeness': self.field_completeness
        }

    def _validate_personal_info(self, data: Dict):
        """Validate personal information fields"""
        required_fields = ['first_name', 'last_name', 'gender', 'birth_date', 'mobile', 'city', 'province']

        for field in required_fields:
            if not data.get(field):
                self.errors.append({
                    'section': 'personal_information',
                    'field': field,
                    'error': f'{field} is required'
                })
            else:
                self.field_completeness[field] = True

        # Validate gender
        if data.get('gender') and data['gender'].upper() not in {g.upper() for g in FieldValidationRules.GENDER_CHOICES}:
            self.errors.append({
                'section': 'personal_information',
                'field': 'gender',
                'error': f"Invalid gender: {data['gender']}"
            })
```

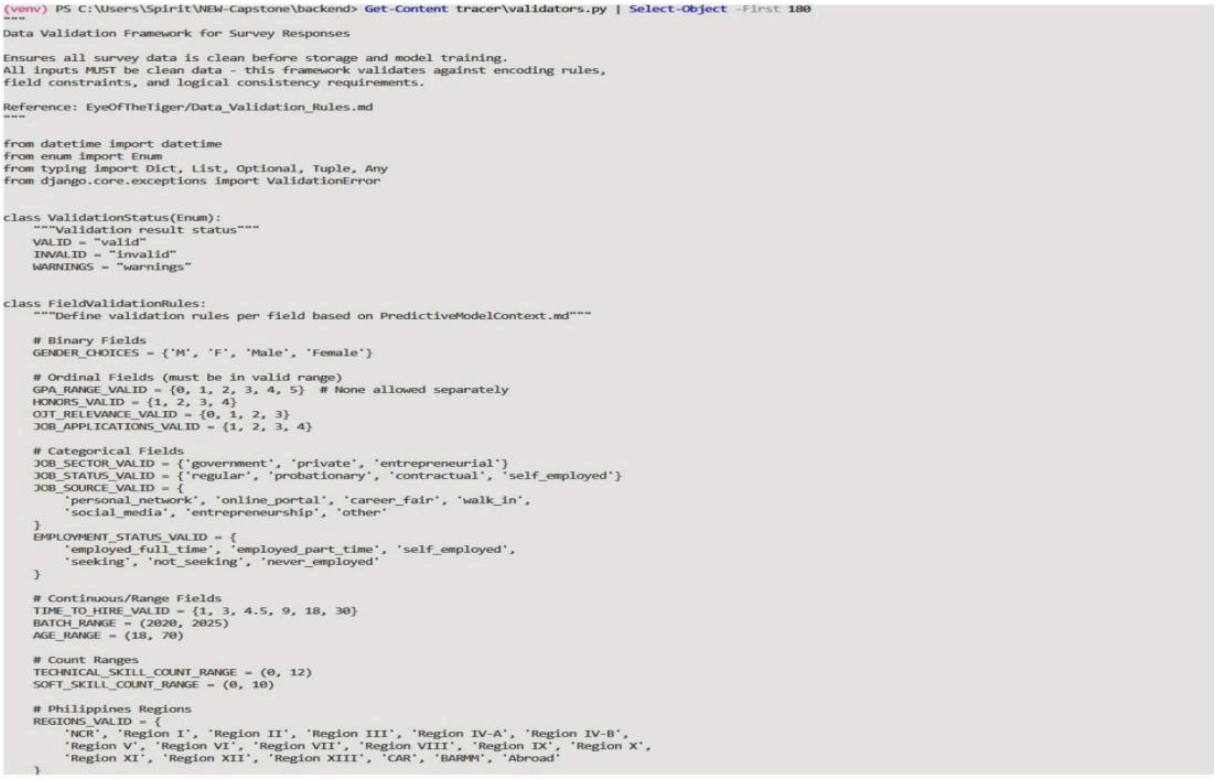
Figure 128

Survey Data Validator Class with the Region and Skill Count Constants

```
    # Count Ranges
    TECHNICAL_SKILL_COUNT_RANGE = (0, 12)
    SOFT_SKILL_COUNT_RANGE = (0, 10)

    # Philippines Regions
    REGIONS_VALID = {
        'NCR', 'Region I', 'Region II', 'Region III', 'Region IV-A', 'Region IV-B',
        'Region V', 'Region VI', 'Region VII', 'Region VIII', 'Region IX', 'Region X',
        'Region XI', 'Region XII', 'Region XIII', 'CAR', 'BARMM', 'Abroad'
    }

class SurveyDataValidator:
    """Comprehensive survey data validation"""

    def __init__(self):
        self.errors: List[Dict[str, Any]] = []
        self.warnings: List[Dict[str, Any]] = []
        self.field_completeness: Dict[str, bool] = {}

    def validate_comprehensive_survey(self, survey_data: Dict) -> Dict:
        """
        Validate complete survey response

        Args:
            survey_data: Dictionary containing all survey sections

        Returns:
            Validation result with status, errors, warnings, completeness score
        """
        self.errors = []
        self.warnings = []
        self.field_completeness = {}

        # Validate personal information section
        if 'personal_information' in survey_data:
            self._validate_personal_info(survey_data['personal_information'])

        # Validate educational background
        if 'educational_background' in survey_data:
            self._validate_educational_background(survey_data['educational_background'])

        # Validate academic & pre-employment
        if 'academic_preemployment' in survey_data:
            self._validate_academic_preemployment(survey_data['academic_preemployment'])

        # Validate employment status
        if 'employment_status' in survey_data:
            self._validate_employment_status(survey_data['employment_status'])

        # Validate first job details
        if 'first_job_details' in survey_data:
            self._validate_first_job_details(survey_data['first_job_details'])

        # Validate current job details
        if 'current_job_details' in survey_data:
            self._validate_current_job_details(survey_data['current_job_details'])

        # Validate work address
        if 'work_address' in survey_data:
            self._validate_work_address(survey_data['work_address'])

        # Validate competency assessment
        if 'competency_assessment' in survey_data:
            self._validate_competency_assessment(survey_data['competency_assessment'])
```

Figure 129

Validation Framework Module Defining the Field Validation Rules



```
(venv) PS C:\Users\Spirit\NEW-Capstone\backend> Get-Content tracer\validators.py | Select-Object -First 180
"""
Data Validation Framework for Survey Responses

Ensures all survey data is clean before storage and model training.
All inputs MUST be clean data - this framework validates against encoding rules,
field constraints, and logical consistency requirements.

Reference: EyeOfTheTiger/Data_Validation_Rules.md
"""

from datetime import datetime
from enum import Enum
from typing import Dict, List, Optional, Tuple, Any
from django.core.exceptions import ValidationError


class ValidationStatus(Enum):
    """Validation result status"""
    VALID = "valid"
    INVALID = "invalid"
    WARNINGS = "warnings"


class FieldValidationRules:
    """Define validation rules per field based on PredictiveModelContext.md"""

    # Binary Fields
    GENDER_CHOICES = {'M', 'F', 'Male', 'Female'}

    # Ordinal Fields (must be in valid range)
    GPA_RANGE_VALID = {0, 1, 2, 3, 4, 5}  # None allowed separately
    HONORS_VALID = {1, 2, 3, 4}
    OJT_RELEVANCE_VALID = {0, 1, 2, 3}
    JOB_APPLICATIONS_VALID = {1, 2, 3, 4}

    # Categorical Fields
    JOB_SECTOR_VALID = {'government', 'private', 'entrepreneurial'}
    JOB_STATUS_VALID = {'regular', 'probationary', 'contractual', 'self_employed'}
    JOB_SOURCE_VALID = {
        'personal_network', 'online_portal', 'career_fair', 'walk_in',
        'social_media', 'entrepreneurship', 'other'
    }
    EMPLOYMENT_STATUS_VALID = {
        'employed_full_time', 'employed_part_time', 'self_employed',
        'seeking', 'not_seeking', 'never_employed'
    }

    # Continuous/Range Fields
    TIME_TO_HIRE_VALID = {1, 3, 4.5, 9, 18, 30}
    BATCH_RANGE = (2020, 2025)
    AGE_RANGE = (18, 70)

    # Count Ranges
    TECHNICAL_SKILL_COUNT_RANGE = (0, 12)
    SOFT_SKILL_COUNT_RANGE = (0, 10)

    # Philippines Regions
    REGIONS_VALID = {
        'NCR', 'Region I', 'Region II', 'Region III', 'Region IV-A', 'Region IV-B',
        'Region V', 'Region VI', 'Region VII', 'Region VIII', 'Region IX', 'Region X',
        'Region XI', 'Region XII', 'Region XIII', 'CAR', 'BARMM', 'Abroad'
    }
```

Figure 130

Interactive Shell Session Driving the Comprehensive Survey Validator

```
tracer\api.py:2198:        artifacts = _load_ml_artifacts()
tracer\reports_api.py:875:        from .api import _aggregate_for_batch, _load_ml_artifacts
tracer\reports_api.py:884:        artifacts = _load_ml_artifacts()
tracer\api.py:1801:def _build_live_df():
tracer\api.py:2219:            live_df = _build_live_df()
tracer\api.py:2222:            _log.warning("_build_live_df() failed: %s", _exc)
tracer\api.py:1798:_EMPLOYED_STATUSES: frozenset[str] = frozenset({"employed_full_time", "employed_part_time", "self_employed"})
tracer\api.py:1852:            row["employment_status"]        = 1 if emp.employment_status in _EMPLOYED_STATUSES else 0
tracer\api.py:1887:def _aggregate_for_batch(artifacts: dict, batch: int | None) -> dict:
tracer\api.py:2236:        overall = _aggregate_for_batch(local_artifacts, batch)
tracer\api.py:2239:            entry = _aggregate_for_batch(local_artifacts, int(c))
tracer\reports_api.py:875:        from .api import _aggregate_for_batch, _load_ml_artifacts
tracer\reports_api.py:895:        overall = _aggregate_for_batch(artifacts, None)
tracer\reports_api.py:907:            agg = _aggregate_for_batch(artifacts, c)
>>> from tracer.validators import *
>>> [name for name in dir() if 'Validator' in name]
['SurveyDataValidator']
>>> validator = SurveyDataValidator()
>>> flat_dict = {
...     'employment_status': 'employed_full_time',
...     'field1': 'value1'
... }
>>> validator.validate_comprehensive_survey(flat_dict)
Traceback (most recent call last):
  File "<console>", line 1, in <module>
  File "C:\Users\Spirit\NEW-Capstone\backend\tracer\validators.py", line 101, in validate_comprehensive_survey
    self._validate_employment_status(survey_data['employment_status'])
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Spirit\NEW-Capstone\backend\tracer\validators.py", line 252, in _validate_employment_status
    if data.get('employment_status'):
       ^^^^^^^^
```

Figure 131

Network Inspection of the Registration Request Returning the Created Response

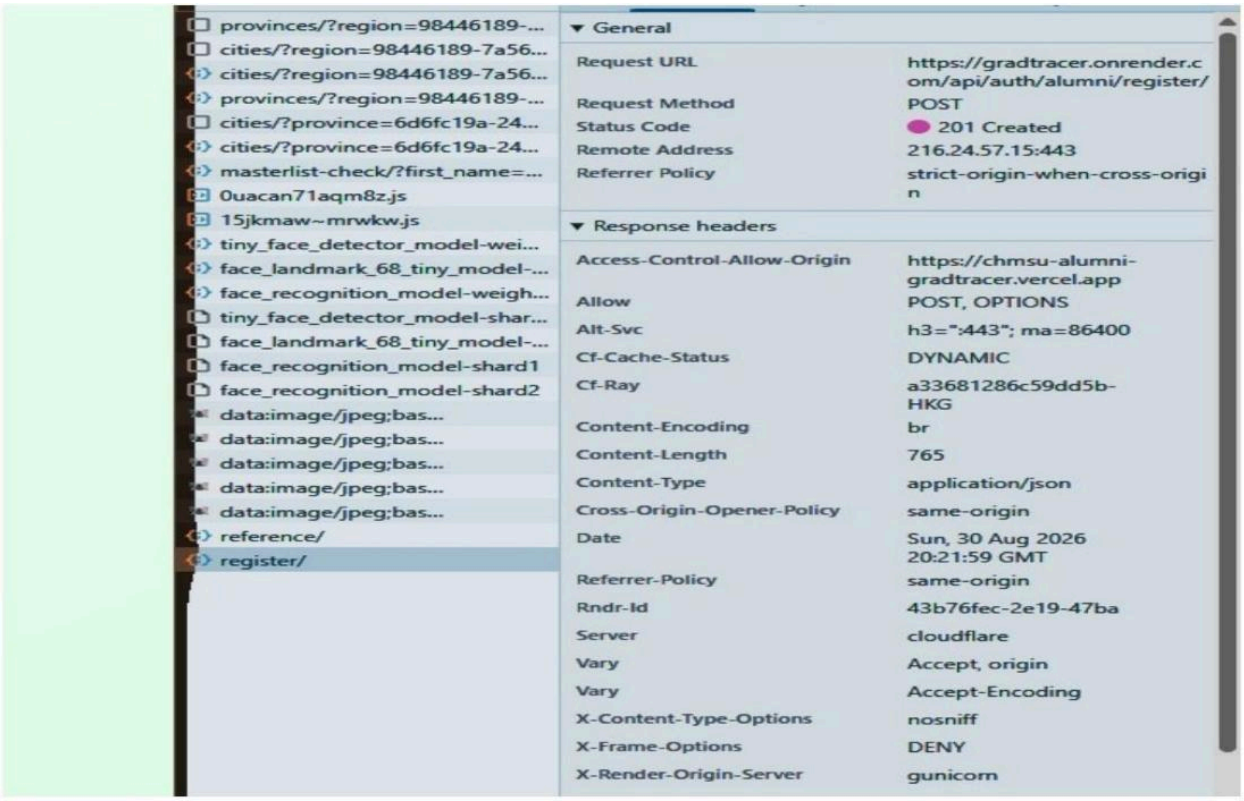


Figure 132

Network Inspection of the Authentication Request Issued at Sign-In

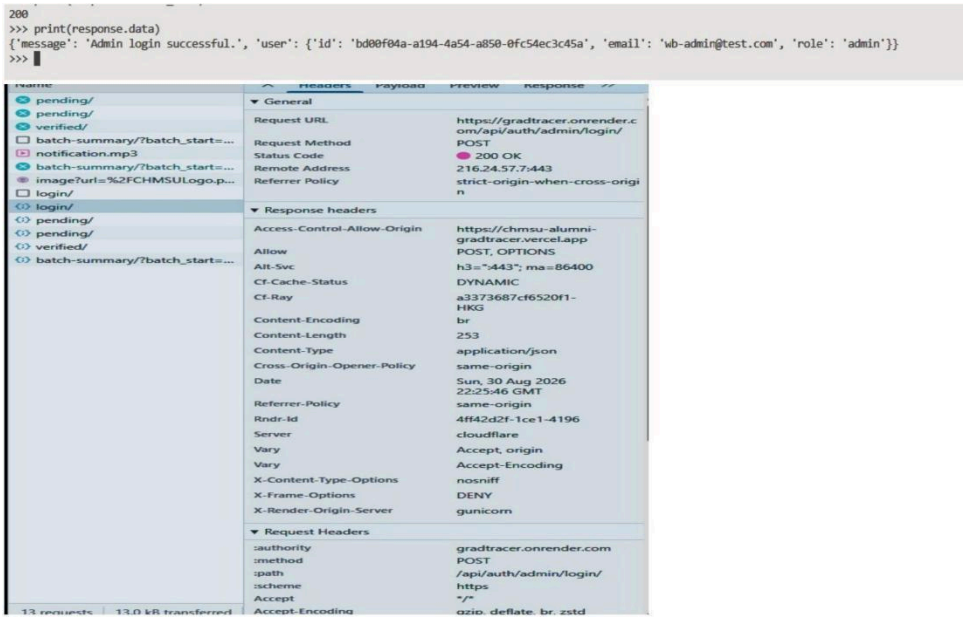


Figure 133

Interactive Shell Session Driving the Batch Aggregation Function

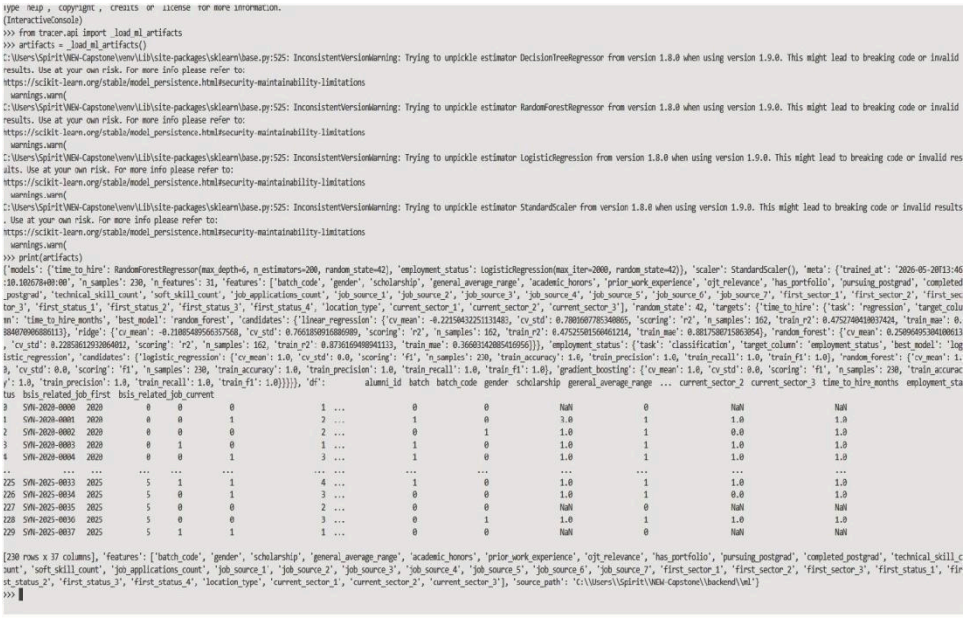


Figure 134

Verification of the Employed Status Set Used by the Analytics Module

Figure 135

Network Inspection of the Batch Summary Report Request

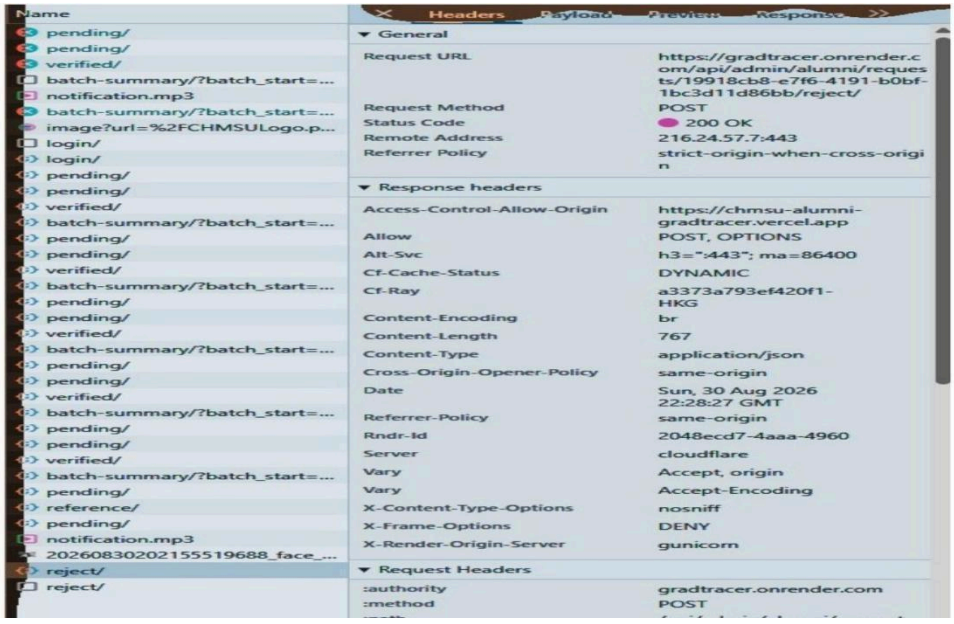


Figure 136

Browsable Application Programming Interface for the Reference Region Endpoint

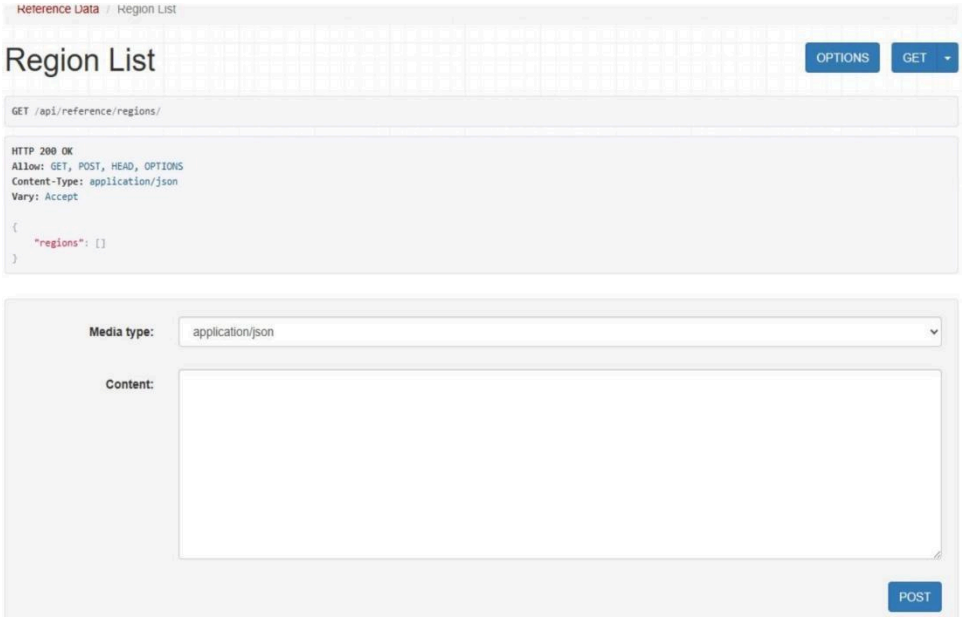


Figure 137

Browsable Application Programming Interface Accessed on the Local Server

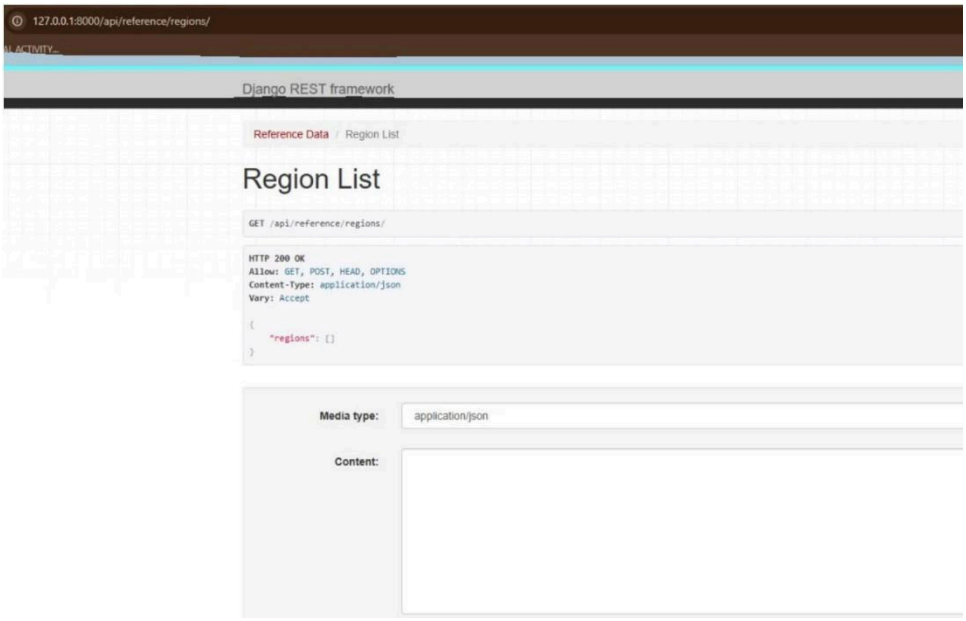


Figure 138

Console-Driven Create and Update Operations on the Reference Region Endpoint

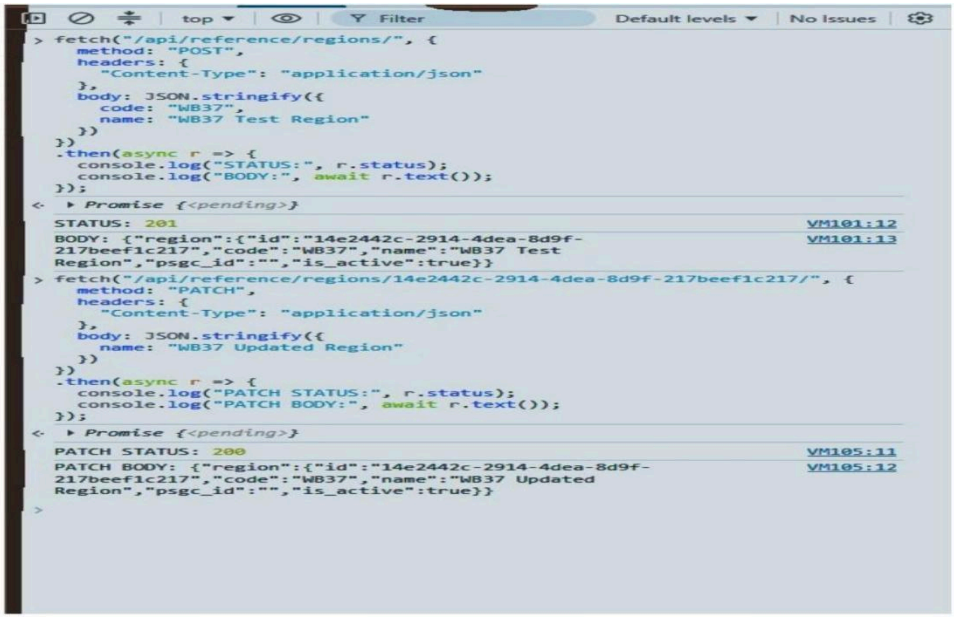


Figure 139

Network Inspection of the Graduate Record Retrieval Request

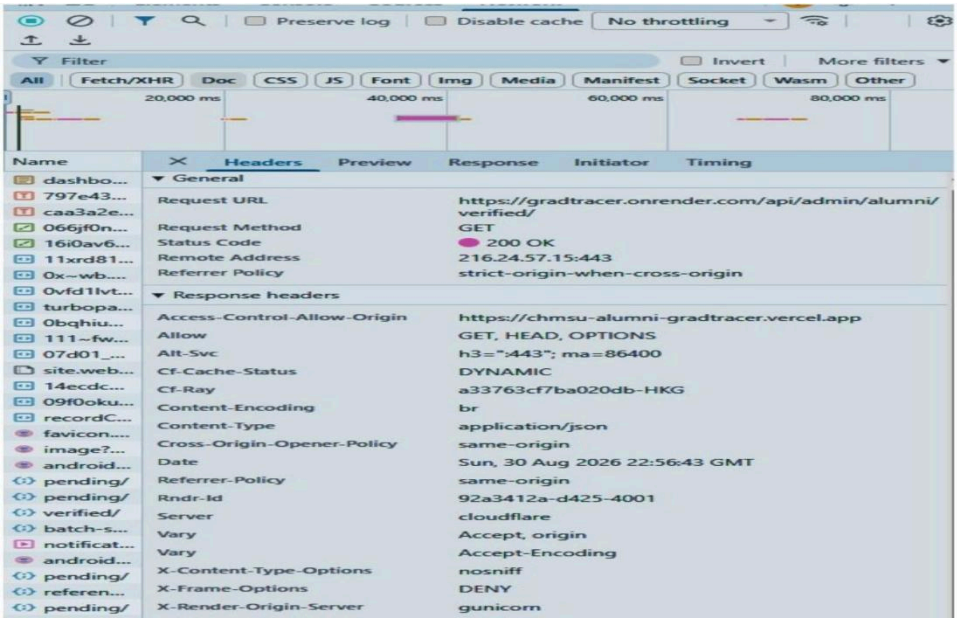


Figure 140

Network Inspection of the Cascading Address Endpoint and Its Cross-Origin Headers

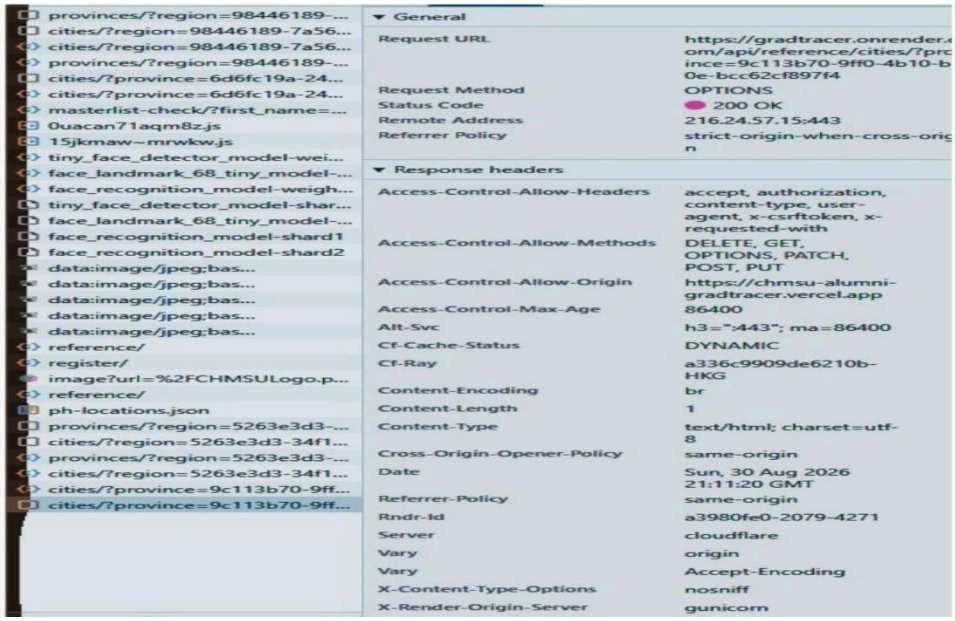


Figure 141

Network Inspection of the Reference Data Endpoint and Its Permitted Methods

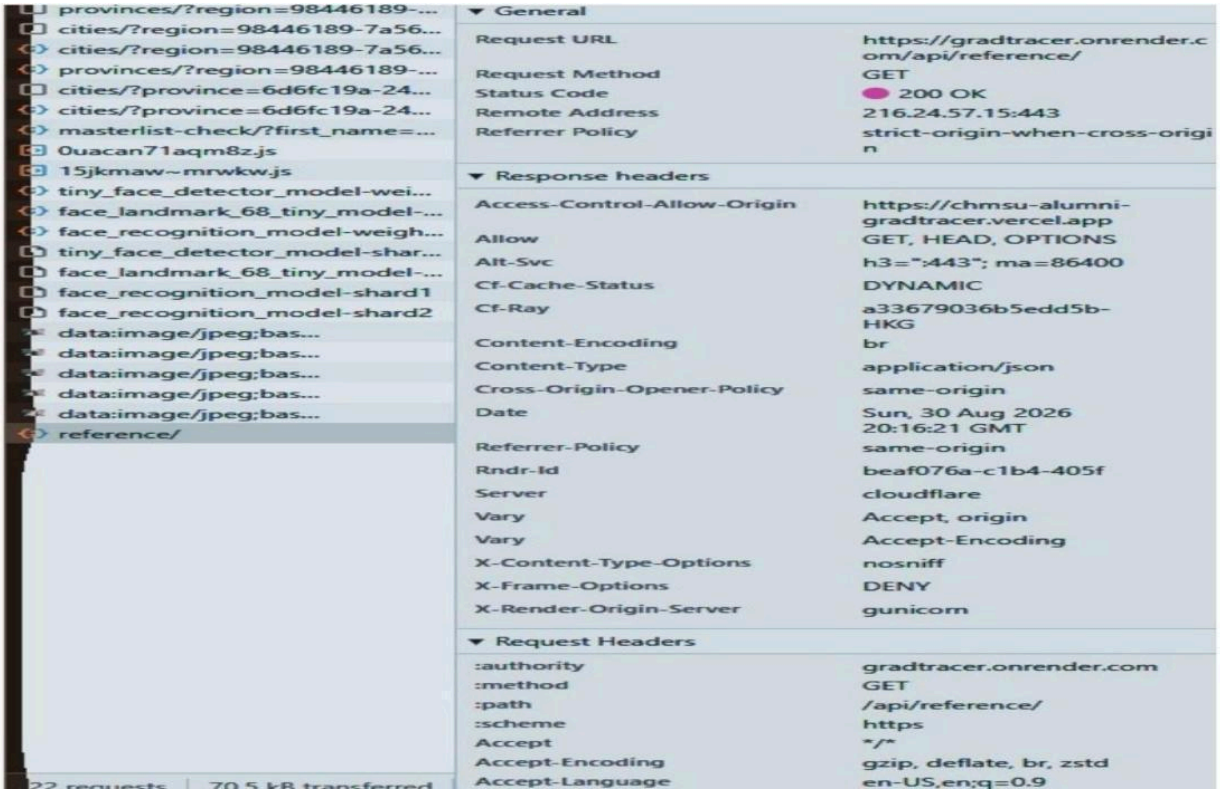


Figure 142

Automated Test Runs of the Server-Side and Client-Side White-Box Suites

```
----------------------------------------------------------------------
Ran 38 tests in 11.563s

OK
```

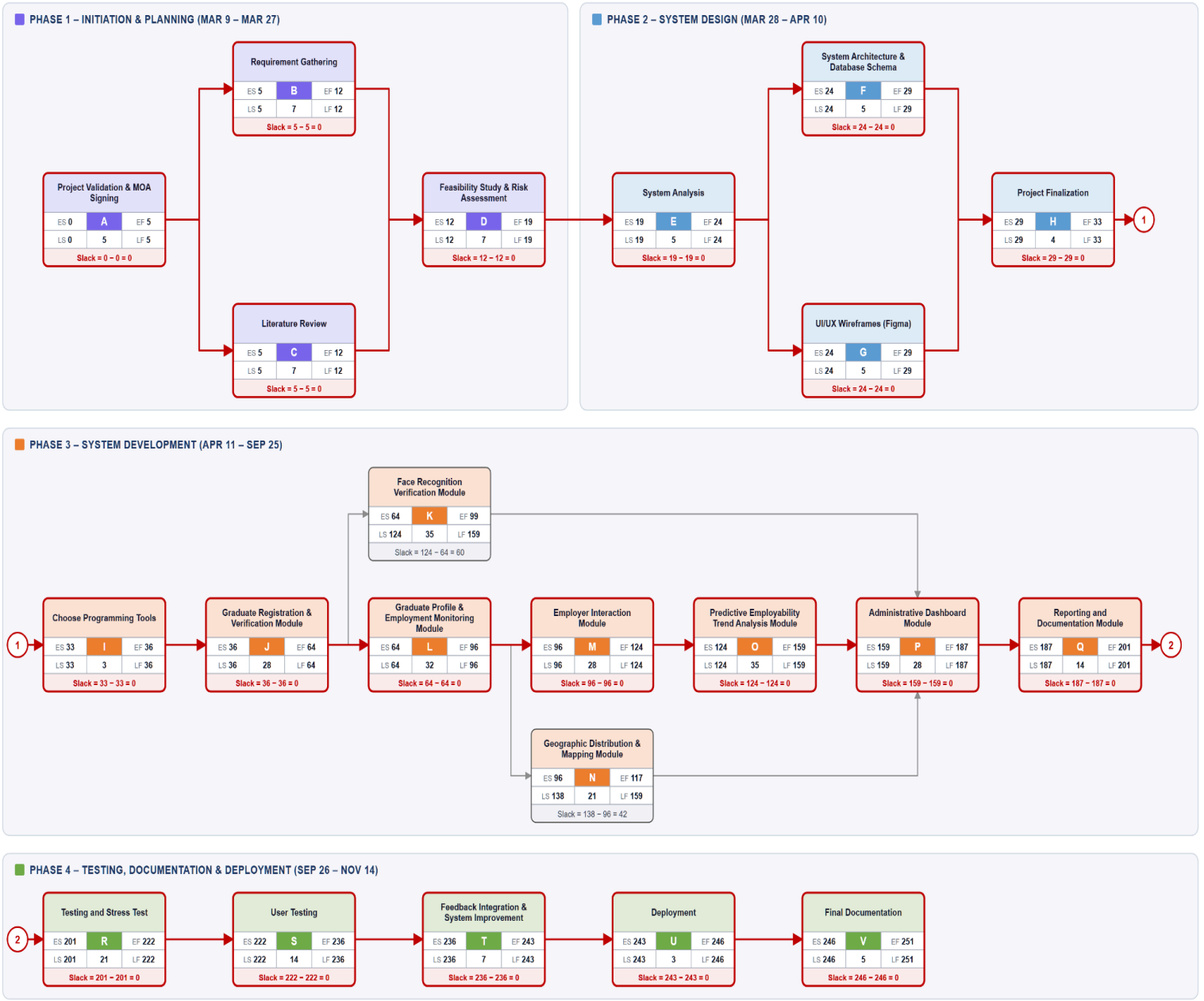
```
WB-39 reducer NEEDS_CORRECTION preserves all state > returns to the owning form...
WB-40 reducer SET_ERROR on a non-validation failure > routes to the error stage...
WB-41 classifySubmitFailure discriminates on field_errors presence > ...
 [... all 14 cases, 21 assertions ...]
WB-52 null-safe formatters render a dash, never a false zero > months(null)...

Test Files  1 passed (1)
     Tests  21 passed (21)
```

Appendix H

PERT – CPM

Figure 143

PERT Chart of the Development of the Graduate Tracer System

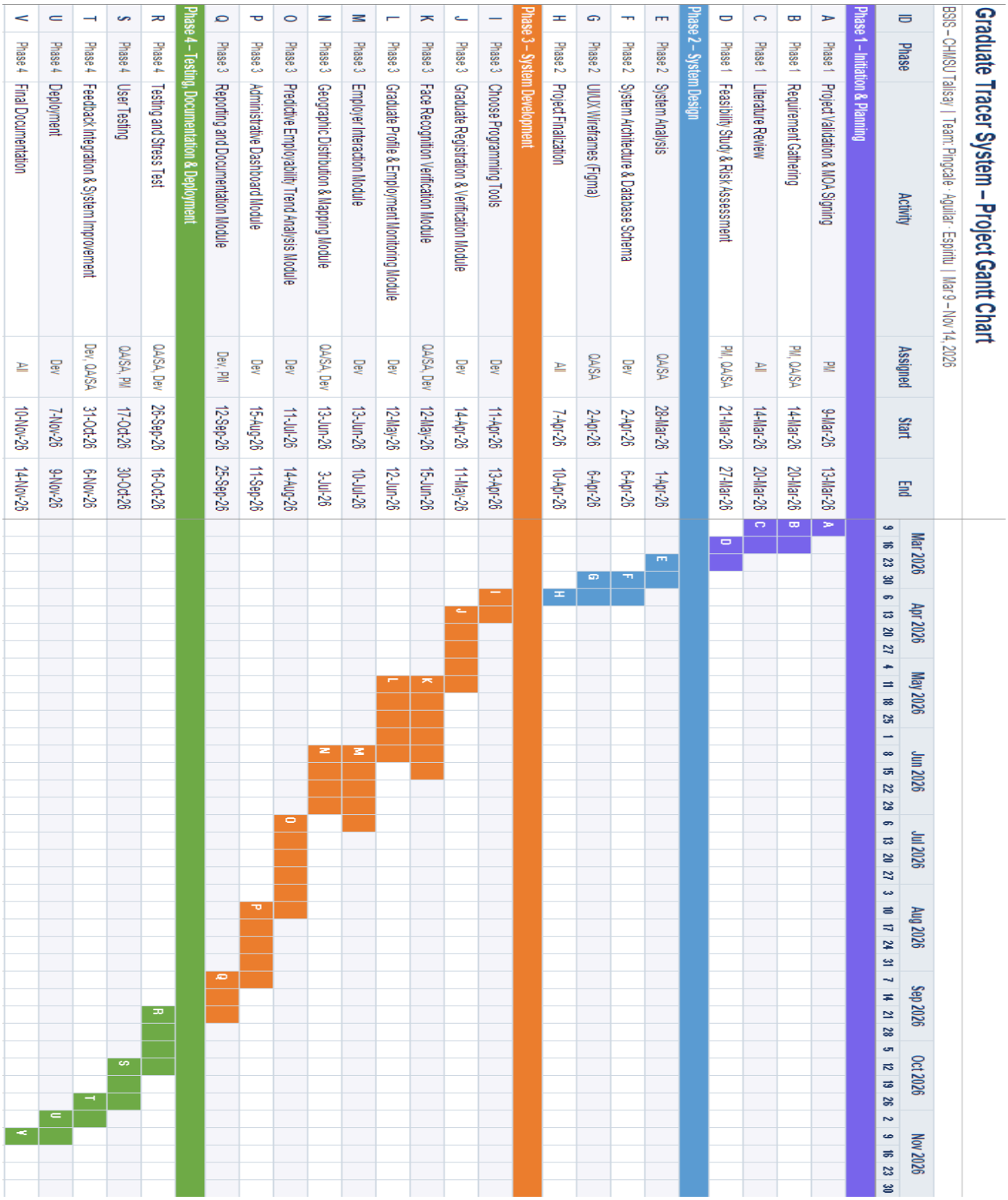


Appendix I

GANTT CHART

Figure 144

Gantt Chart of the Development of the Graduate Tracer System

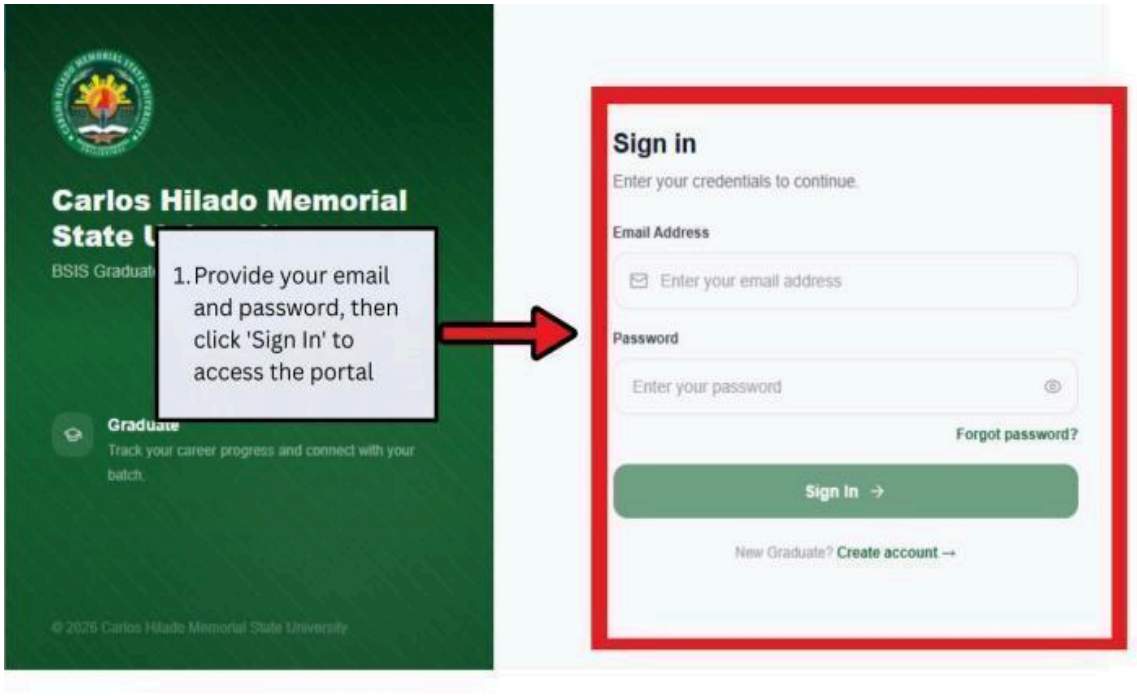


Appendix J

USER'S GUIDE

The User's Guide for the Graduate Tracer System with Predictive Employability Trend Analysis is presented on the pages that follow. It provides step-by-step instructions for the three intended user roles – Graduate, Employer, and System Administrator – covering the tasks that each role performs in the system, from account creation enrollment to the review of pending graduate registrations and the interpretation of the analytics and reporting features.

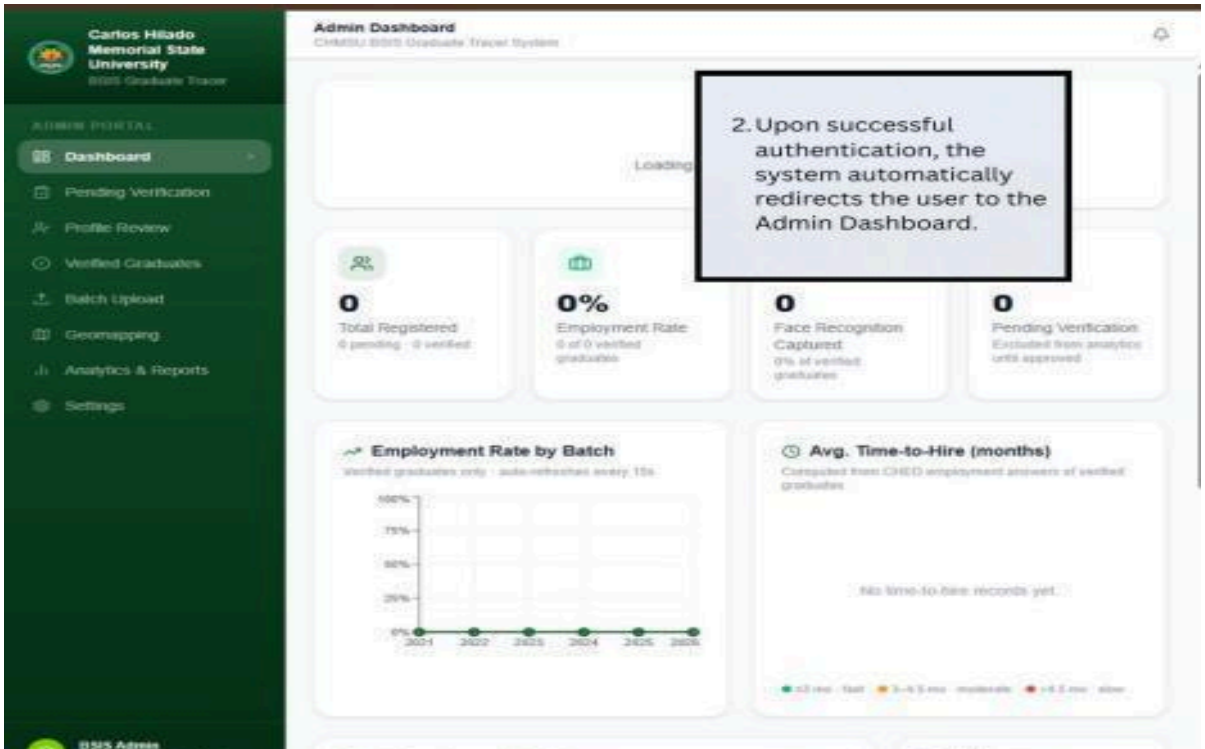
Administrator Sign In



1. Provide your email and password, then click “Sign In” to access the portal.

2. Upon successful authentication, the system automatically redirects the user to the Admin Dashboard.

Admin Dashboard

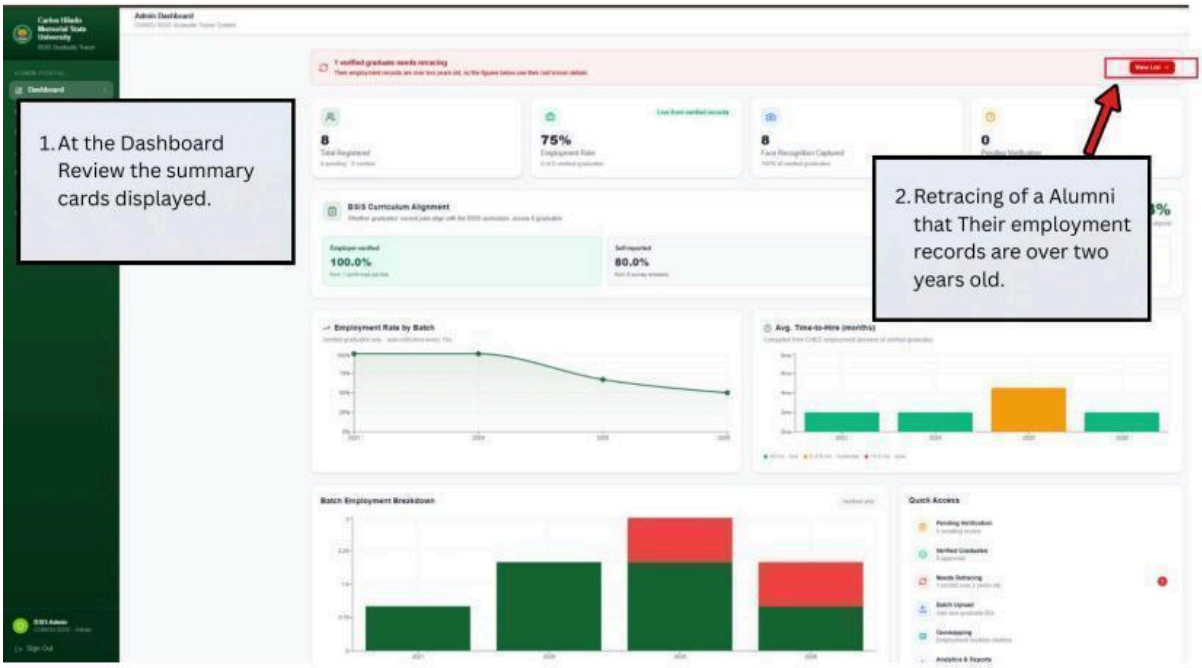


1. Upon successful authentication, the system automatically redirects the user to the Admin Dashboard.

2. Review the summary cards showing Total Registered, Employment Rate, Face Recognition Captured, and Pending Verification.

3. View the Employment Rate by Batch chart and the Average Time-to-Hire chart for a quick overview of graduate employment outcomes.

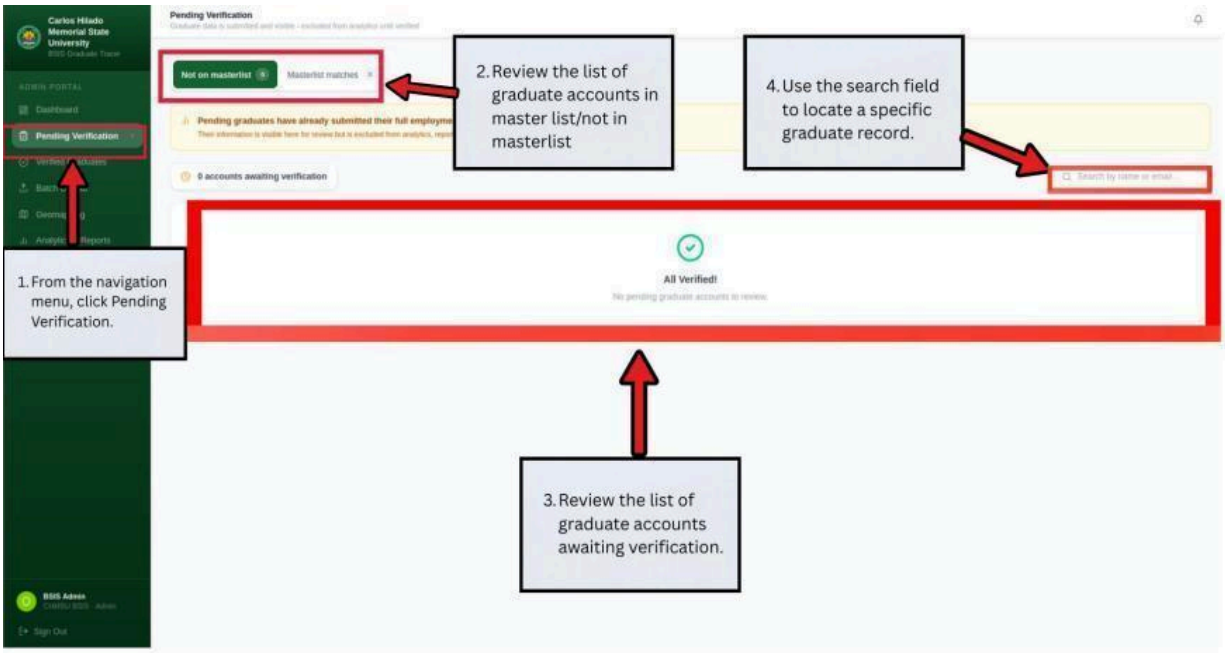
Admin Dashboard – Retracing Notification



1. At the Dashboard, review the summary cards displayed.

2. Retracing of alumni whose employment records are over two years old. Click “View List” to review the affected records.

Pending Verification



1. From the navigation menu, click Pending Verification.

2. Review the list of graduate accounts in the master list / not in master list tabs.

3. Review the list of graduate accounts awaiting verification.

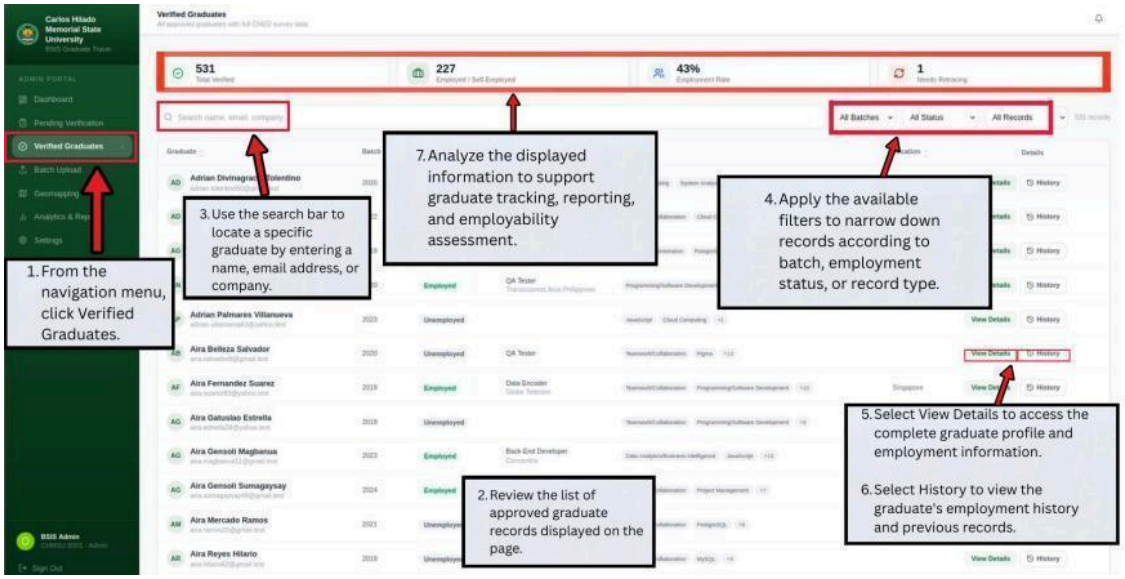
4. Use the search field to locate a specific graduate record.

5. Examine the submitted graduate information.

6. Verify the account based on the provided information and institutional records.

7. Once approved, the graduate record becomes part of the verified graduate database.

Verified Graduates



1. From the navigation menu, click Verified Graduates.

2. Review the list of approved graduate records displayed on the page.

3. Use the search bar to locate a specific graduate by entering a name, email address, or company.

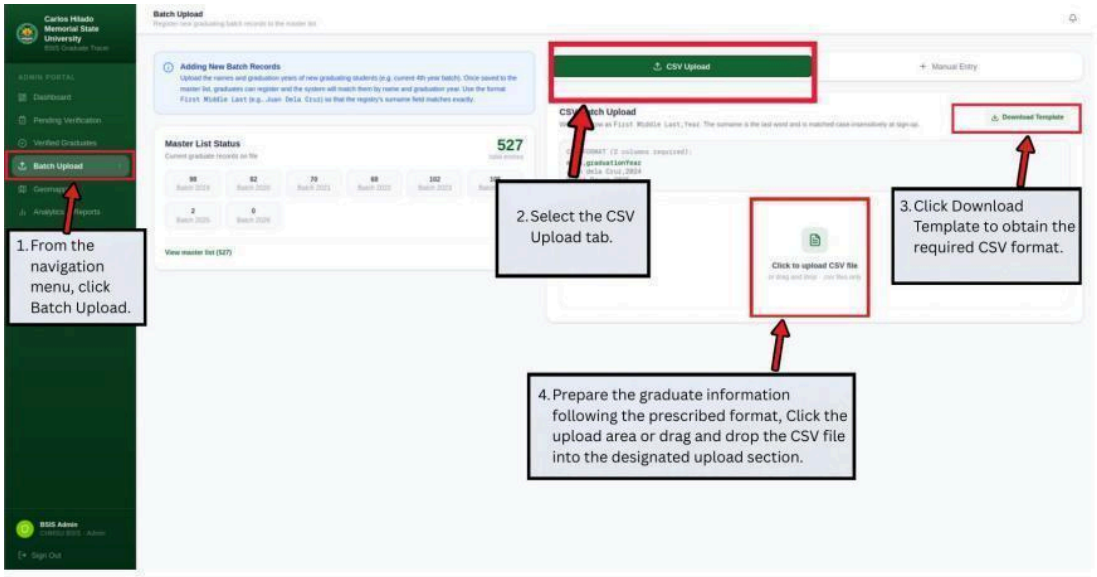
4. Apply the available filters to narrow down records according to batch, employment status, or record type.

5. Select View Details to access the complete graduate profile and employment information.

6. Select History to view the graduate's employment history and previous records.

7. Analyze the displayed information to support graduate tracking, reporting, and employability assessment.

Batch Upload



1. From the navigation menu, click Batch Upload.

2. Select the CSV Upload tab.

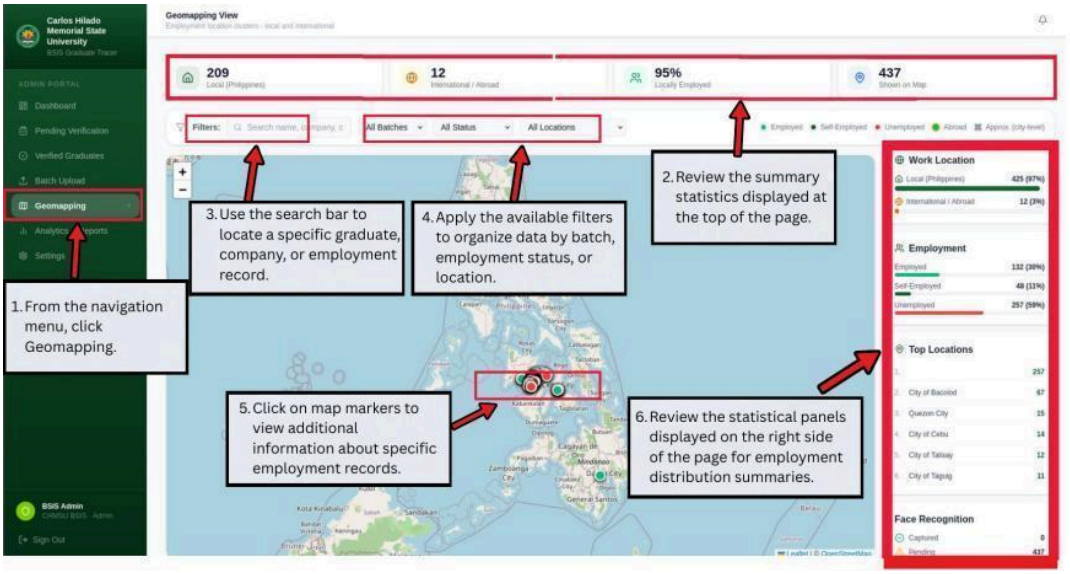
3. Click Download Template to obtain the required CSV format.

4. Prepare the graduate information following the prescribed format. Click the upload area or drag and drop the CSV file into the designated upload section.

5. Wait for the system to validate and process the uploaded file.

6. Verify that the uploaded records have been successfully added to the master list.

Geomapping



1. From the navigation menu, click Geomapping.

2. Review the summary statistics displayed at the top of the page.

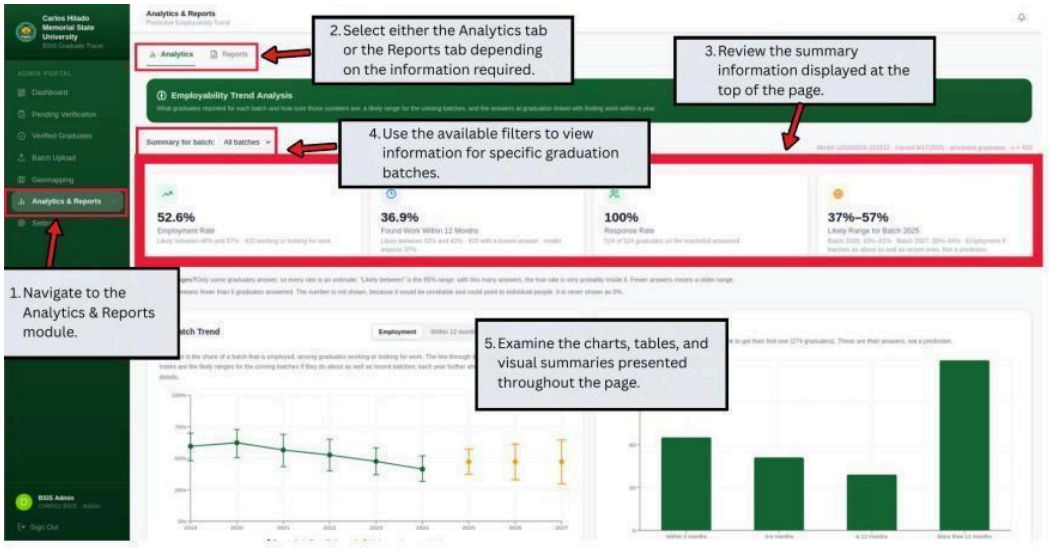
3. Use the search bar to locate a specific graduate, company, or employment record.

4. Apply the available filters to organize data by batch, employment status, or location.

5. Click on map markers to view additional information about specific employment records.

6. Review the statistical panels displayed on the right side of the page for employment distribution summaries.

Analytics – Predictive Employability Trend



1. From the navigation menu, select Analytics & Reports.

2. Select either the Analytics tab or the Reports tab depending on the information required.

3. Review the summary information displayed at the top of the page.

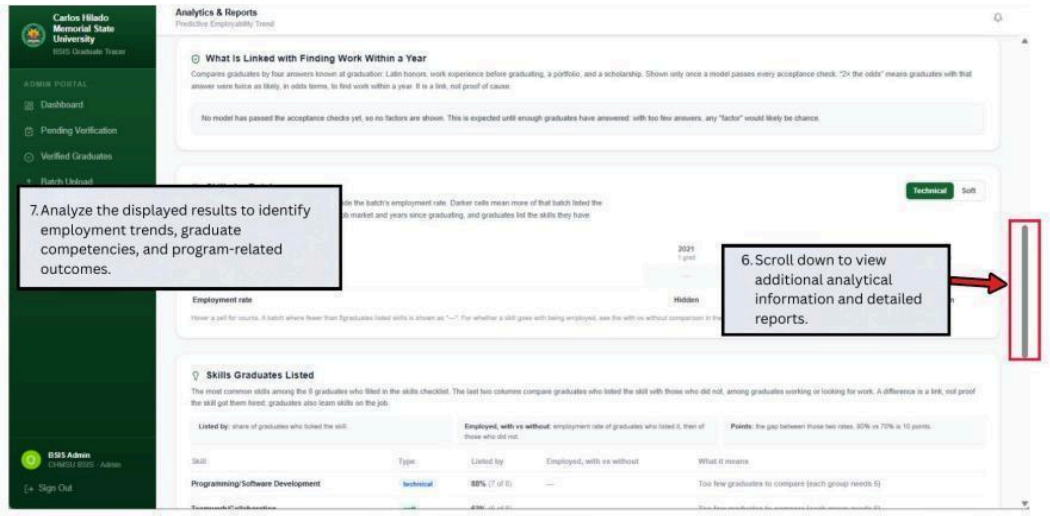
4. Use the available filters to view information for specific graduation batches.

5. Examine the charts, tables, and visual summaries presented throughout the page.

6. Scroll down to view additional analytical information and detailed reports.

7. Analyze the displayed results to identify employment trends, graduate competencies, and program-related outcomes.

Analytics – Skills and Competency Analysis



1. From the navigation menu, select Analytics & Reports.

2. Select either the Analytics tab or the Reports tab depending on the information required.

3. Review the summary information displayed at the top of the page.

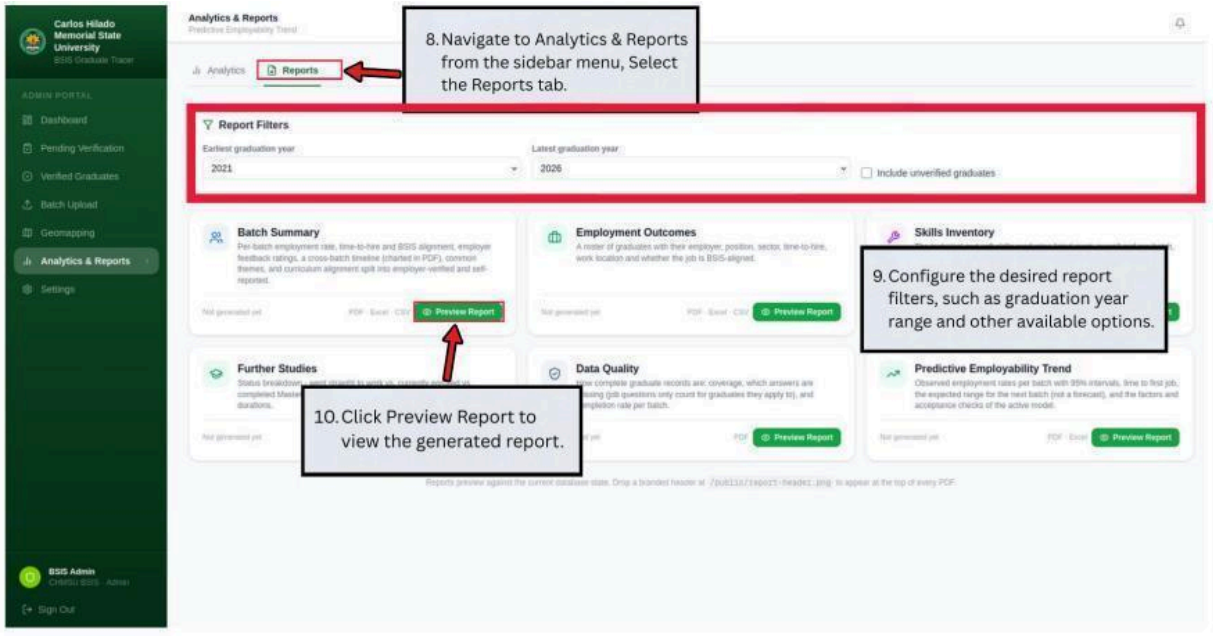
4. Use the available filters to view information for specific graduation batches.

5. Examine the charts, tables, and visual summaries presented throughout the page.

6. Scroll down to view additional analytical information and detailed reports.

7. Analyze the displayed results to identify employment trends, graduate competencies, and program-related outcomes.

Reports – Report Generation



1. Navigate to Analytics & Reports from the sidebar menu. Select the Reports tab.

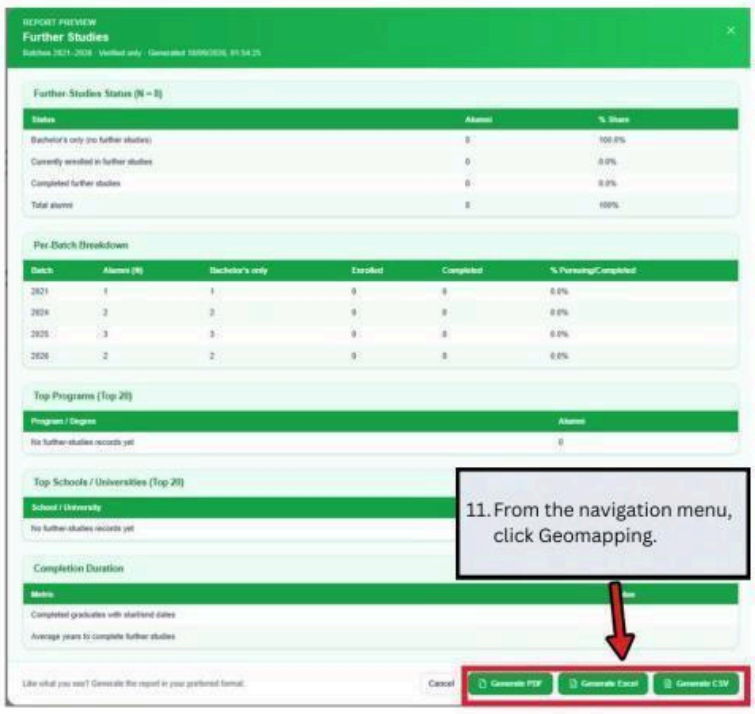
2. Configure the desired report filters, such as graduation year range and other available options.

3. Click Preview Report to view the generated report.

4. Export the report using the available export formats when needed.

5. Use the generated reports for analysis, documentation, and institutional reporting purposes.

Reports – Report Preview and Export



1. Navigate to Analytics & Reports from the sidebar menu. Select the Reports tab.

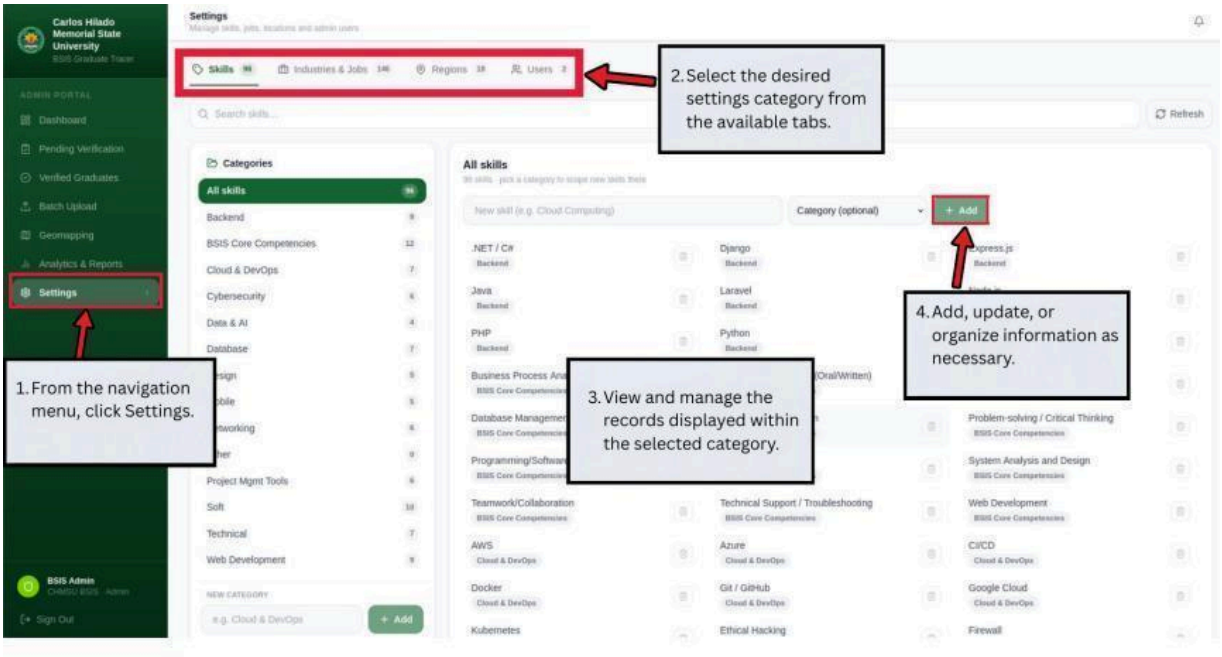
2. Configure the desired report filters, such as graduation year range and other available options.

3. Click Preview Report to view the generated report.

4. Review the report content displayed in the preview window.

5. Export the report using the available export formats (PDF, Excel, CSV) when needed.

Settings – Reference Data Management



1. From the navigation menu, click Settings.

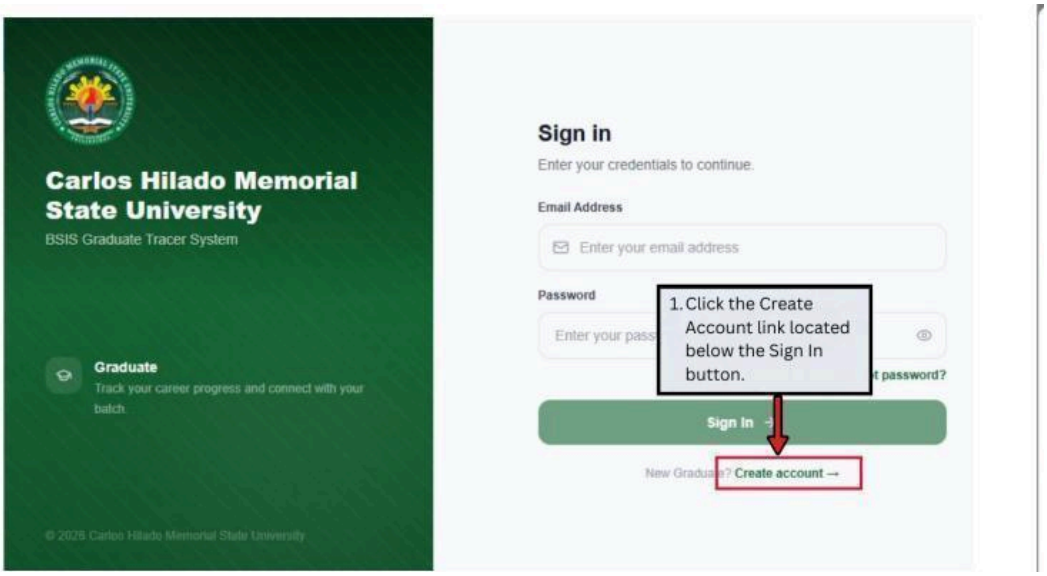
2. Select the desired settings category from the available tabs (Skills, Industries & Jobs, Regions, Users).

3. View and manage the records displayed within the selected category.

4. Add, update, or organize information as necessary.

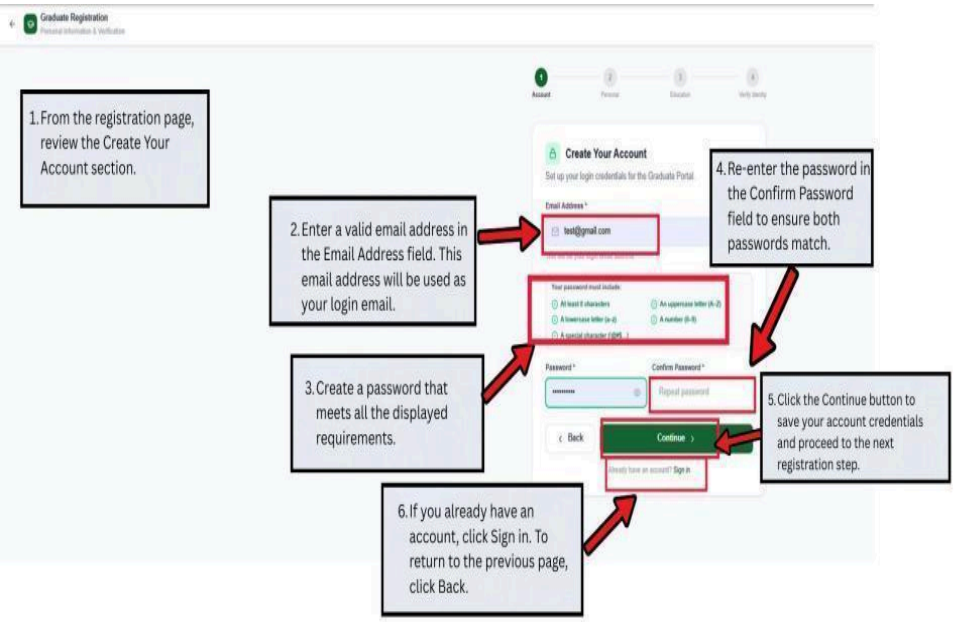
Graduate User Guide

Graduate Registration – Create Account



1. Click the Create Account link located below the Sign In button.
2. A Data Privacy Notice window will appear.
3. Click I Understand, Continue to proceed.

Graduate Registration – Account Credentials



1. From the registration page, review the Create Your Account section.

2. Enter a valid email address in the Email Address field. This email address will be used as your login email.

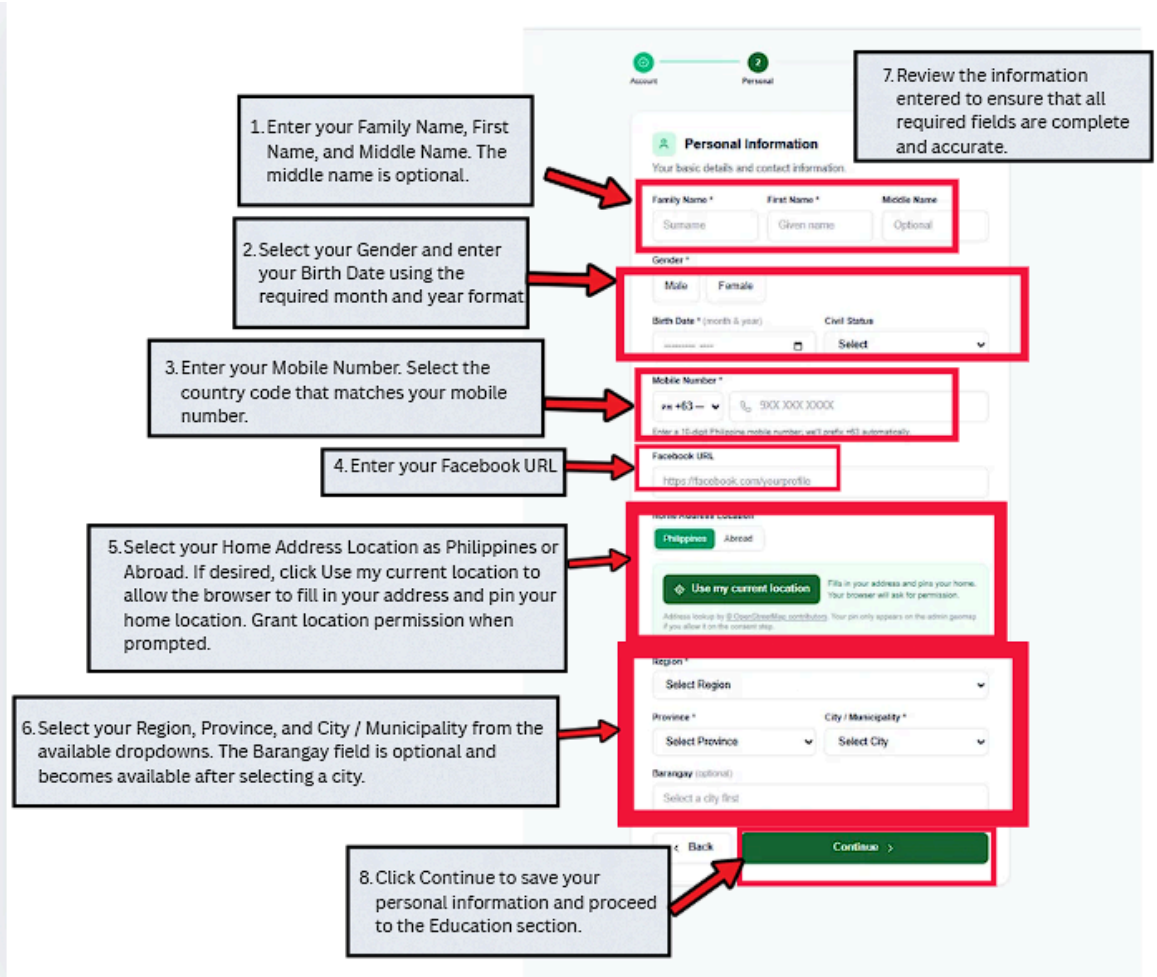
3. Create a password that meets all the displayed requirements.

4. Re-enter the password in the Confirm Password field to ensure both passwords match.

5. Click the Continue button to save your account credentials and proceed to the next registration step.

6. If you already have an account, click Sign in. To return to the previous page, click Back.

Graduate Registration – Personal Information



1. Enter your Family Name, First Name, and Middle Name. The middle name is optional.

2. Select your Gender and enter your Birth Date using the required month and year format.

3. Select your Civil Status from the available options.

4. Enter your Mobile Number. Select the country code that matches your mobile number.

5. Enter your Facebook URL if applicable.

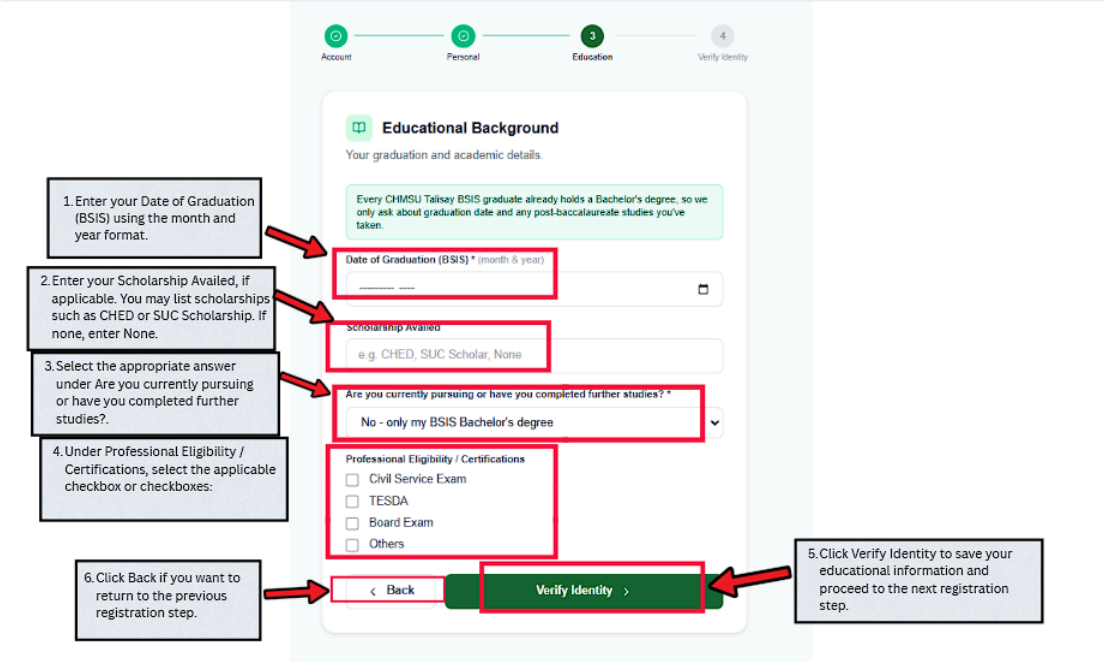
6. Select your Home Address Location as Philippines or Abroad. If desired, click Use my current location to allow the browser to fill in your address and pin your home location. Grant location permission when prompted.

7. Select your Region, Province, and City / Municipality from the available dropdowns. The Barangay field is optional and becomes available after selecting a city.

8. Review the information entered to ensure that all required fields are complete and accurate.

9. Click Continue to save your personal information and proceed to the Education section.

Graduate Registration – Educational Background



1. Enter your Date of Graduation (BSIS) using the month and year format.

2. Enter your Scholarship Availed, if applicable. You may list scholarships such as CHED or SUC Scholarship. If none, enter None.

3. Select the appropriate answer under Are you currently pursuing or have you completed further studies?.

4. Under Professional Eligibility / Certifications, select the applicable checkbox or checkboxes.

5. Click Verify Identity to save your educational information and proceed to the next registration step.

6. Click Back if you want to return to the previous registration step.

Graduate Registration – Face Recognition

1. Review the instructions under Before you begin. Remove anything that covers your face, such as sunglasses, hats, face masks, or thick reflective glasses. Make sure your face is well-lit.

2. Click the Start Camera button to activate your device's camera. Allow camera access when prompted.

3. Position your face within the guide on the screen and look directly at the camera.

4. Complete the Look Forward step. The system will capture a front-facing photo of your face.

5. Follow the Turn Slowly instruction. Slowly turn your head to the left and right as requested by the system.

6. Keep your face within the guide while the system captures images from different angles. The progress indicators show the required face positions.

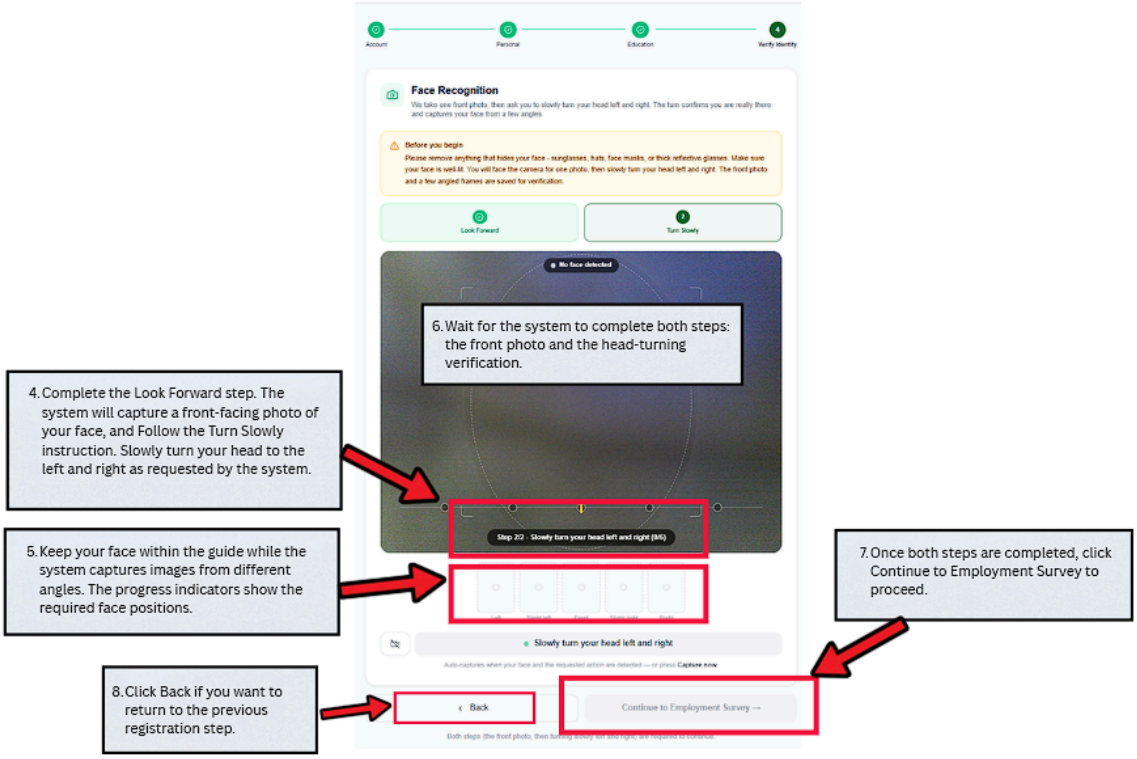
7. Wait for the system to complete both steps: the front photo and the head-turning verification.

8. Review the captured images, if displayed, and ensure your face is clearly visible.

9. Once both steps are completed, click Continue to Employment Survey to proceed.

10. Click Back if you want to return to the previous registration step.

Graduate Registration – Face Recognition (continued)



1. Complete the Look Forward step. The system will capture a front-facing photo of your face, and follow the Turn Slowly instruction. Slowly turn your head to the left and right as requested by the system.

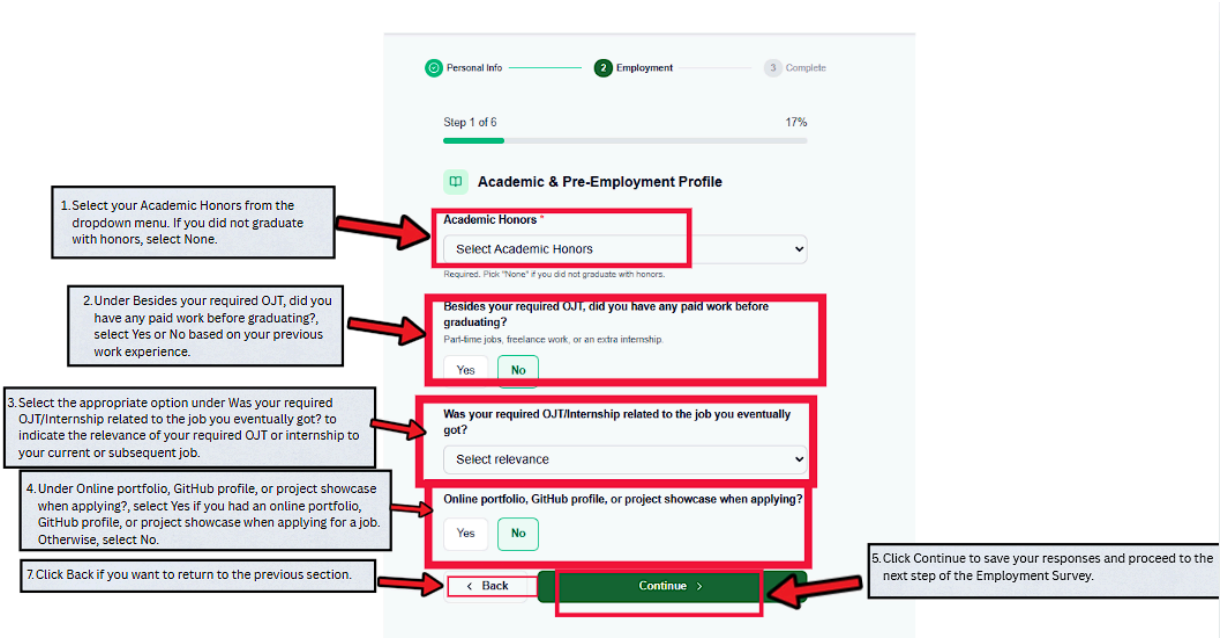
2. Keep your face within the guide while the system captures images from different angles. The progress indicators show the required face positions.

3. Wait for the system to complete both steps: the front photo and the head-turning verification.

4. Once both steps are completed, click Continue to Employment Survey to proceed.

5. Click Back if you want to return to the previous registration step.

Graduate Registration – Employment Survey (Academic & Pre-Employment Profile)



1. Select your Academic Honors from the dropdown menu. If you did not graduate with honors, select None.

2. Under Besides your required OJT, did you have any paid work before graduating?, select Yes or No based on your previous work experience.

3. Select the appropriate option under Was your required OJT/Internship related to the job you eventually got? to indicate the relevance of your required OJT or internship to your current or subsequent job.

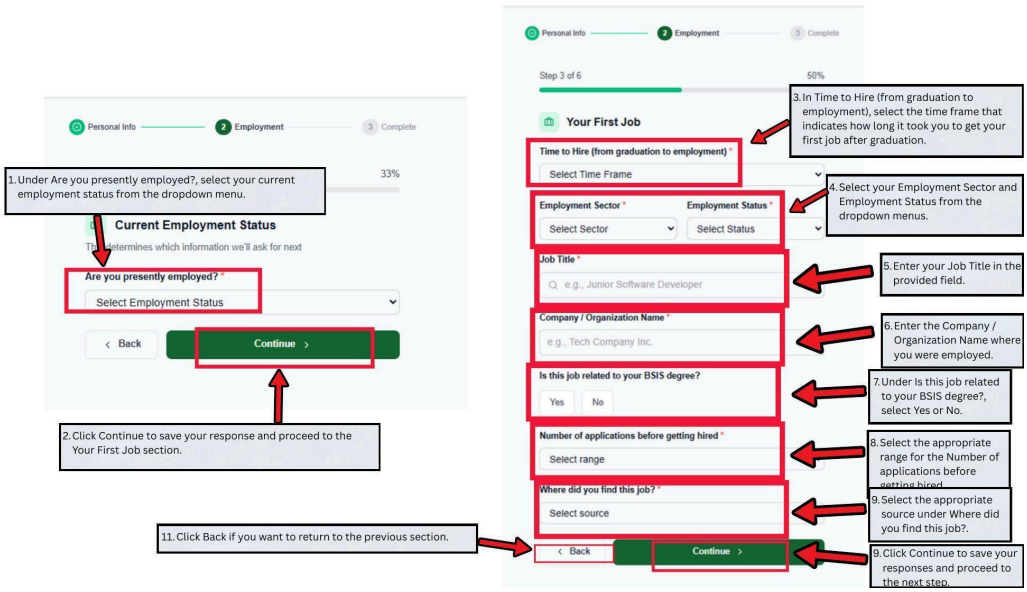
4. Under Online portfolio, GitHub profile, or project showcase when applying?, select Yes if you had an online portfolio, GitHub profile, or project showcase when applying for a job. Otherwise, select No.

5. Review your answers to ensure that all required information is complete and accurate.

6. Click Continue to save your responses and proceed to the next step of the Employment Survey.

7. Click Back if you want to return to the previous section.

Graduate Registration – Employment Survey (Current Employment Status and First Job)



1. Under Are you presently employed?, select your current employment status from the dropdown menu.

2. Click Continue to save your response and proceed to the Your First Job section.

3. In Time to Hire (from graduation to employment), select the time frame that indicates how long it took you to get your first job after graduation.

4. Select your Employment Sector and Employment Status from the dropdown menus.

5. Enter your Job Title in the provided field.

6. Enter the Company / Organization Name where you were employed.

7. Under Is this job related to your BSIS degree?, select Yes or No.

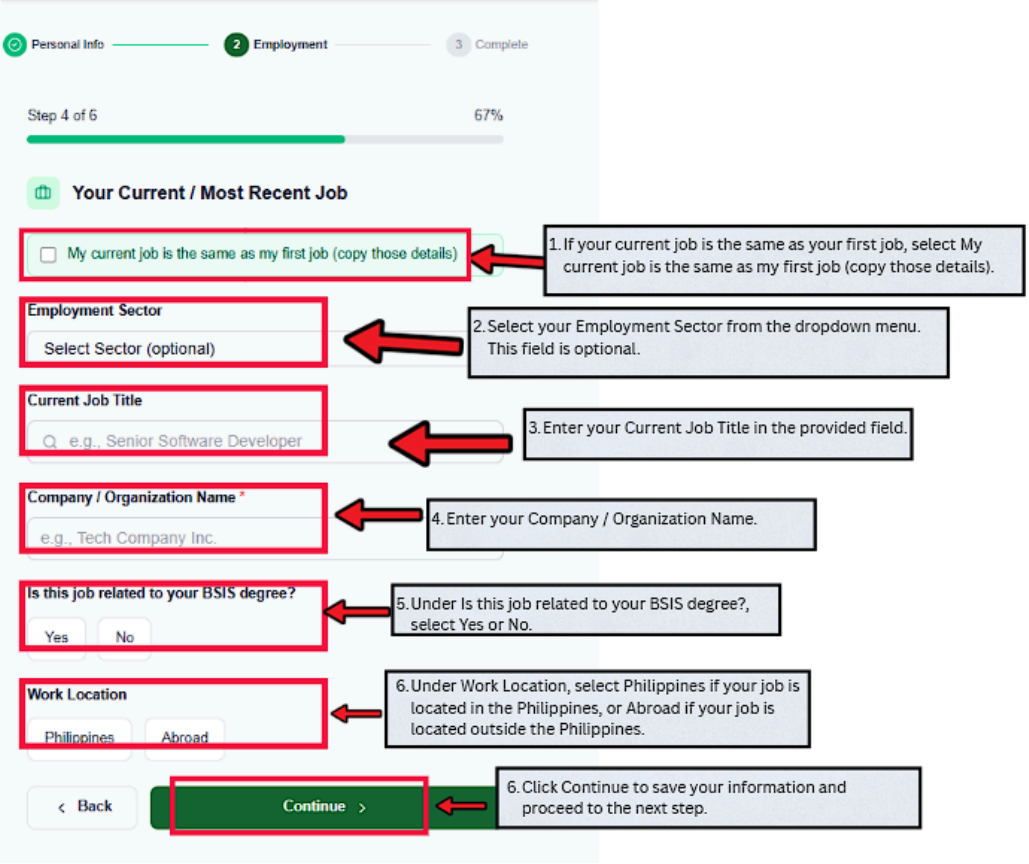
8. Select the appropriate range for the Number of applications before getting hired.

9. Select the appropriate source under Where did you find this job?.

10. Click Continue to save your responses and proceed to the next step.

11. Click Back if you want to return to the previous section.

Graduate Registration – Employment Survey (Current / Most Recent Job)



1. If your current job is the same as your first job, select My current job is the same as my first job (copy those details).

2. Select your Employment Sector from the dropdown menu. This field is optional.

3. Enter your Current Job Title in the provided field.

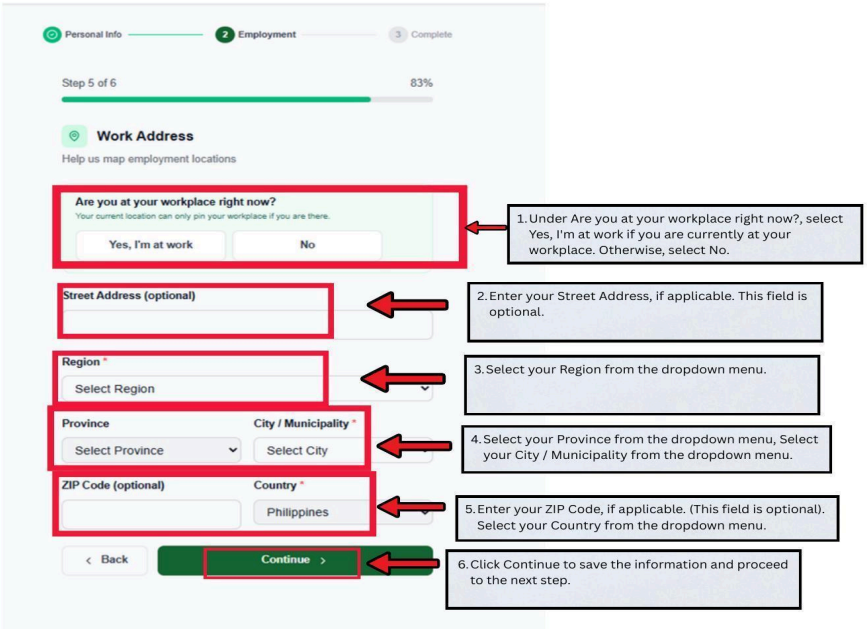
4. Enter your Company / Organization Name.

5. Under Is this job related to your BSIS degree?, select Yes or No.

6. Under Work Location, select Philippines if your job is located in the Philippines, or Abroad if your job is located outside the Philippines.

7. Click Continue to save your information and proceed to the next step.

Graduate Registration – Employment Survey (Work Address)



1. Under Are you at your workplace right now?, select Yes, I'm at work if you are currently at your workplace. Otherwise, select No.

2. Enter your Street Address, if applicable. This field is optional.

3. Select your Region from the dropdown menu.

4. Select your Province from the dropdown menu, Select your City / Municipality from the dropdown menu.

5. Enter your ZIP Code, if applicable. (This field is optional). Select your Country from the dropdown menu.

6. Click Continue to save the information and proceed to the next step.

Graduate Registration – Employment Survey (Skills & Competencies)

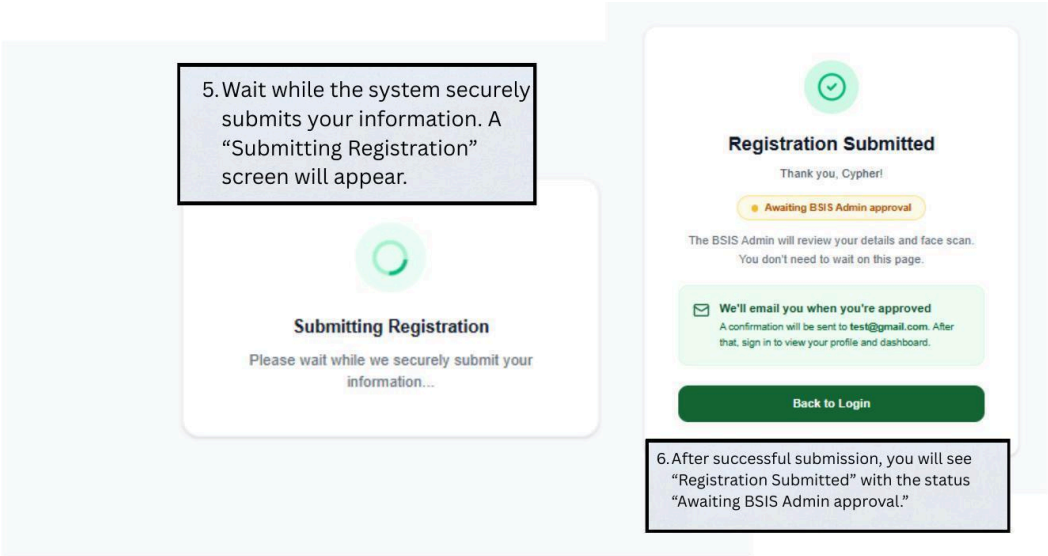
1. Under Technical Skills, select the technical skills that match your knowledge and experience. Select all that apply; there is no limit.

2. Under Soft Skills, select the soft skills that best describe your abilities. Select all that apply; there is no limit.

3. Enter your Professional Certifications, if applicable. Separate multiple certifications with commas.

4. Click Complete to submit your employment survey and finish the registration process.

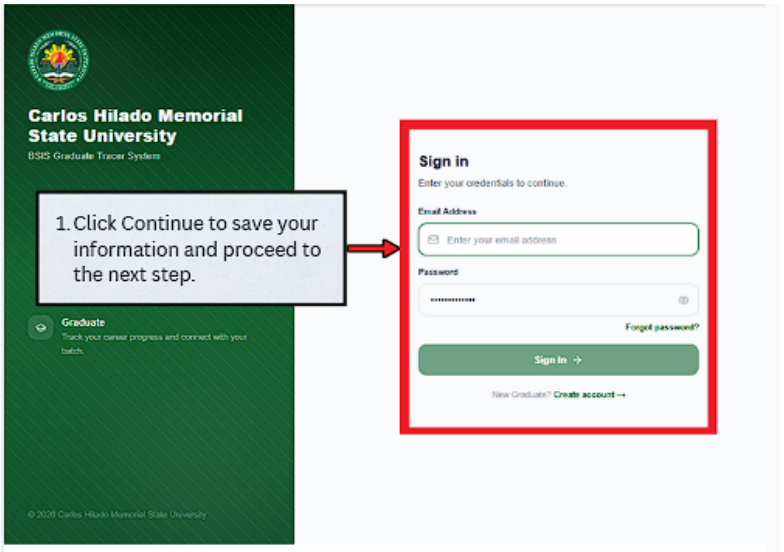
Graduate Registration – Registration Submitted

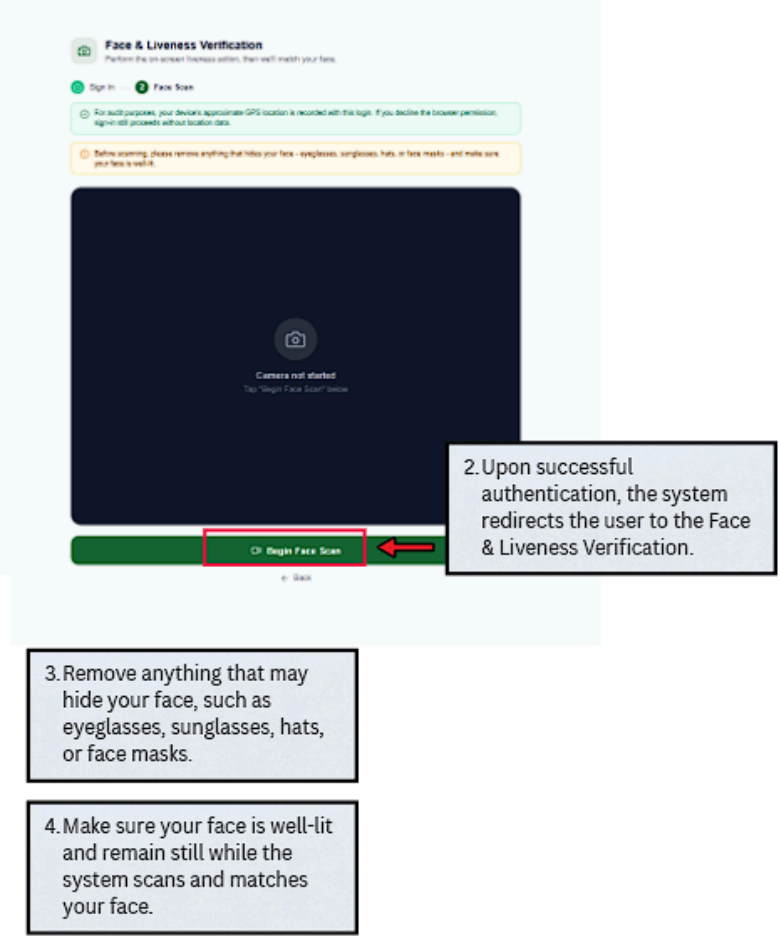


1. Wait while the system securely submits your information. A "Submitting Registration" screen will appear.

2. After successful submission, you will see "Registration Submitted" with the status "Awaiting BSIS Admin approval."

Graduate Sign In – Face & Liveness Verification





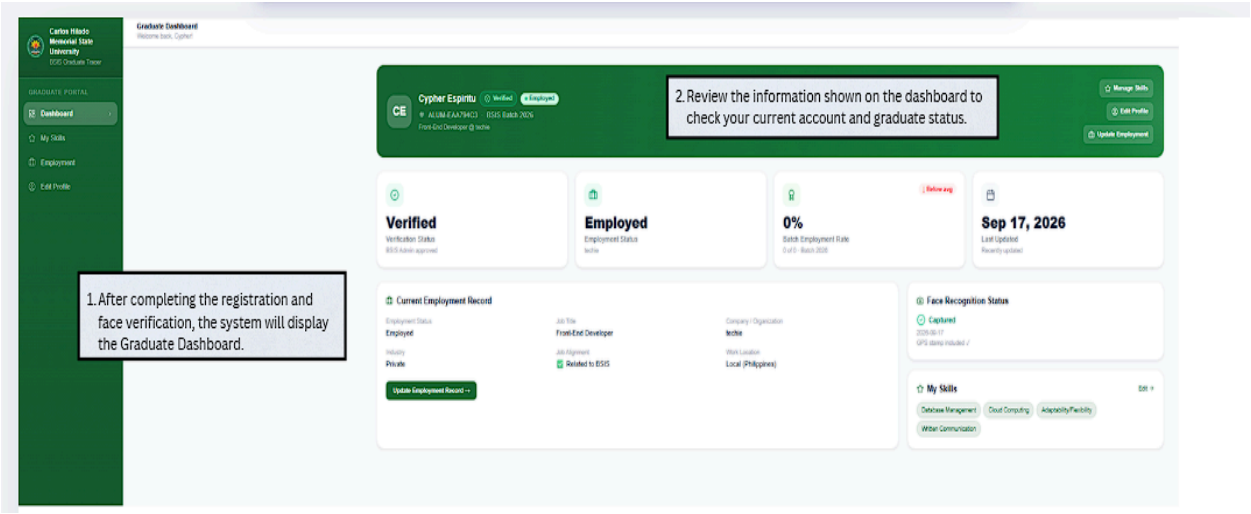
1. Provide your email and password, then click 'Sign In' to access the portal.

2. Upon successful authentication, the system redirects the user to the Face & Liveness Verification.

3. Remove anything that may hide your face, such as eyeglasses, sunglasses, hats, or face masks.

4. Make sure your face is well-lit and remain still while the system scans and matches your face.

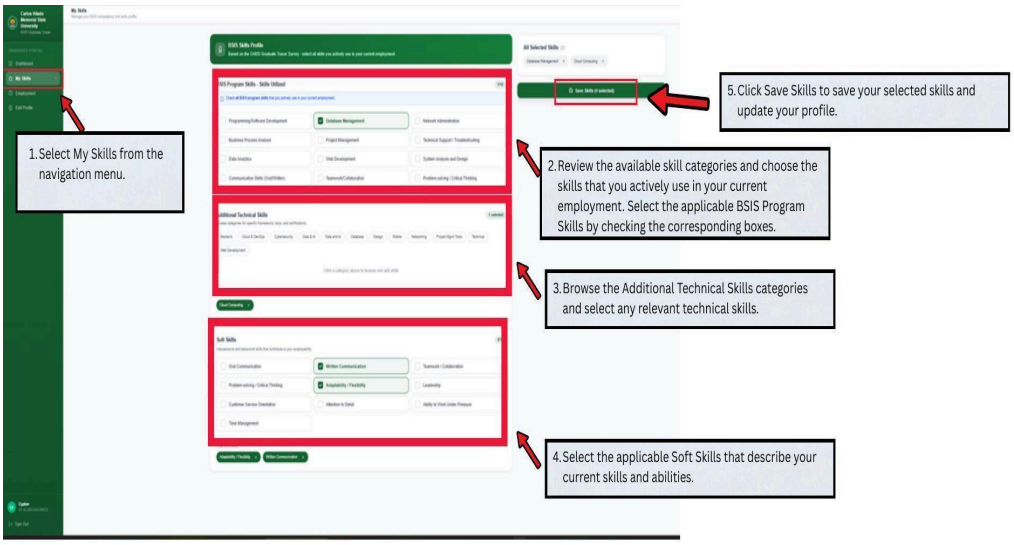
Graduate Dashboard



1. After completing the required steps, the system will display the Graduate Dashboard.

2. Review the information shown on the dashboard to check your current account and graduate status.

My Skills



1. Select My Skills from the navigation menu.

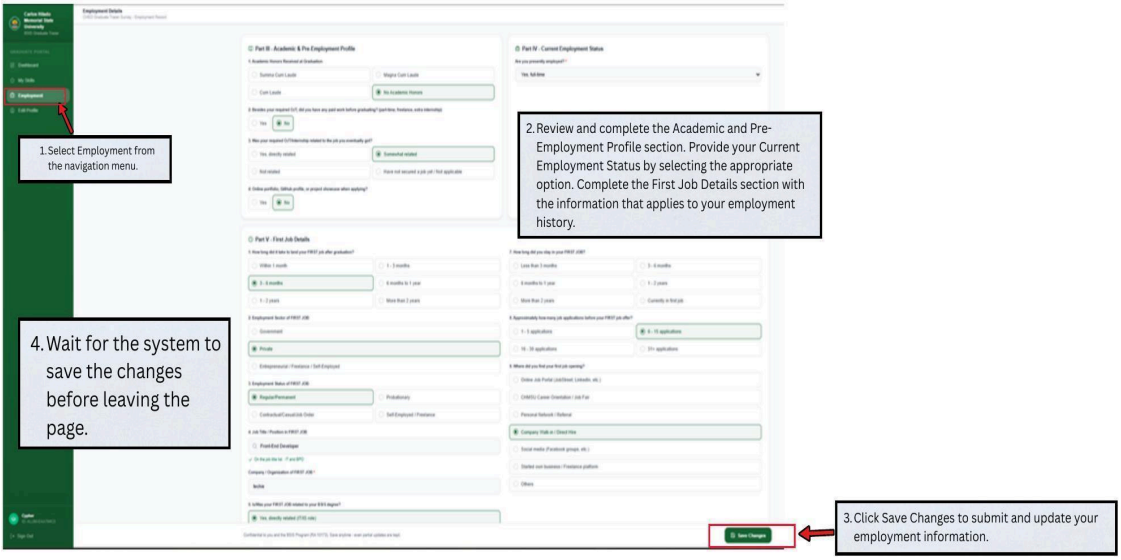
2. Review the available skill categories and choose the skills that you actively use in your current employment. Select the applicable BSIS Program Skills by checking the corresponding boxes.

3. Browse the Additional Technical Skills categories and select any relevant technical skills.

4. Select the applicable Soft Skills that describe your current skills and abilities.

5. Click Save Skills to save your selected skills and update your profile.

Employment – Edit Employment Record



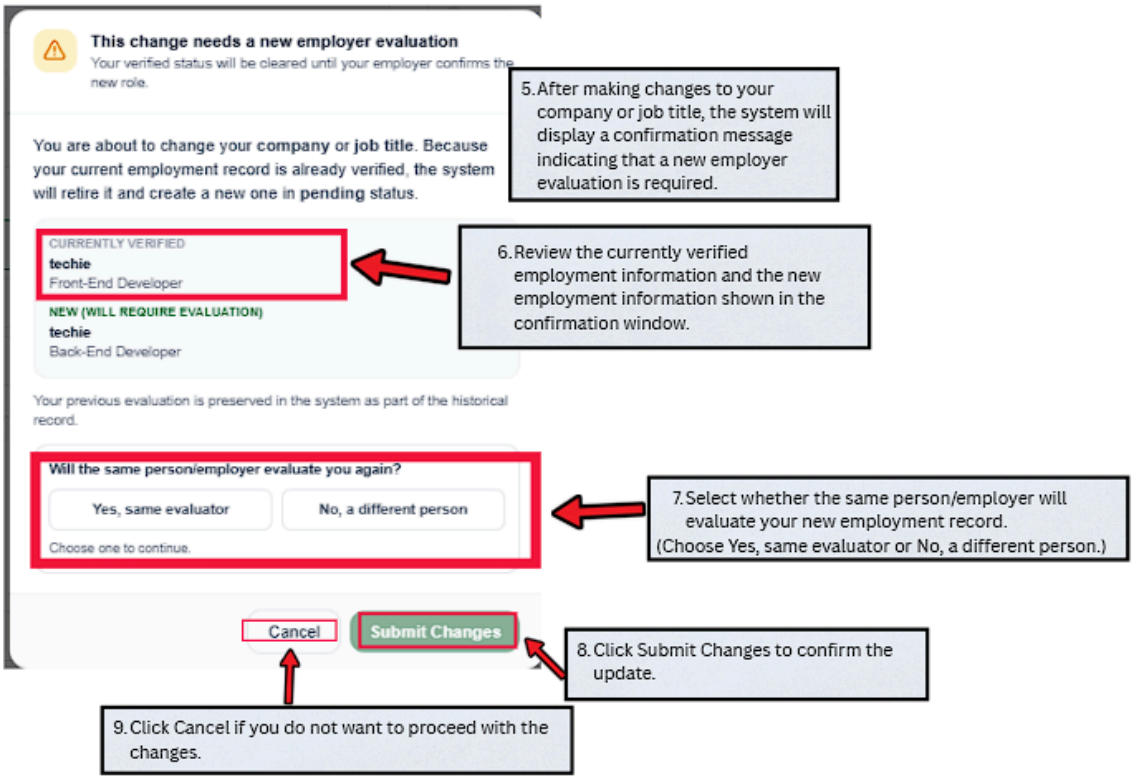
1. Select Employment from the navigation menu.

2. Review and complete the Academic and Pre-Employment Profile section. Provide your Current Employment Status by selecting the appropriate option. Complete the First Job Details section with the information that applies to your employment history.

3. Click Save Changes to submit and update your employment information.

4. Wait for the system to save the changes before leaving the page.

Employment – Employer Re-Evaluation Confirmation



1. After making changes to your company or job title, the system will display a confirmation message indicating that a new employer evaluation is required.

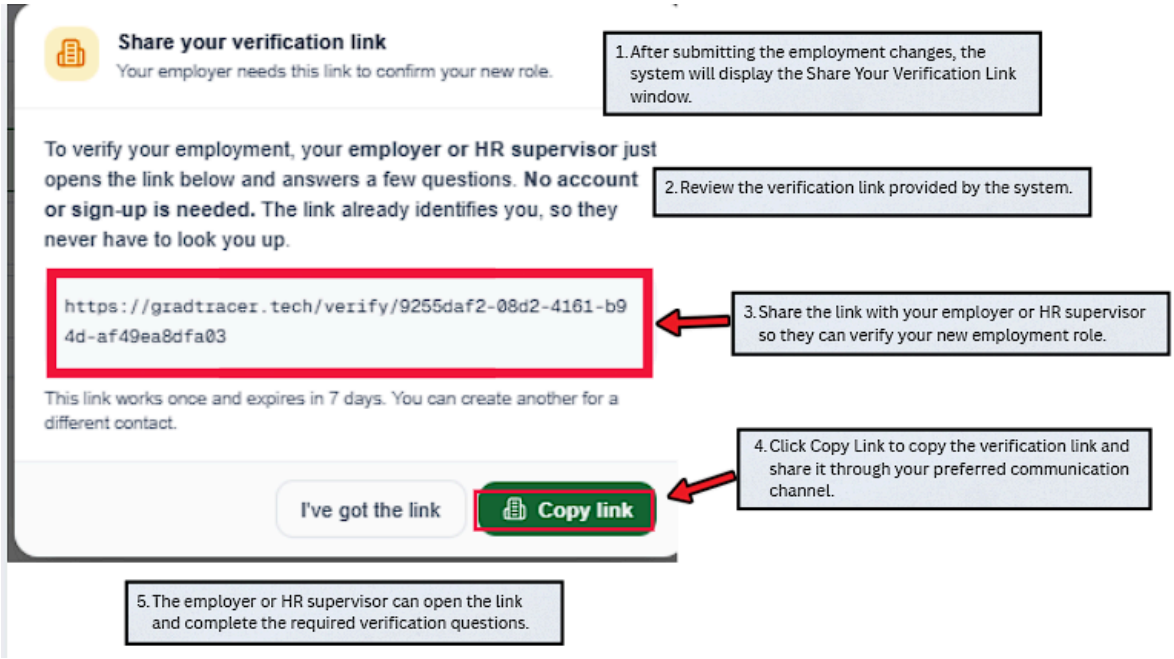
2. Review the currently verified employment information and the new employment information shown in the confirmation window.

3. Select whether the same person/employer will evaluate your new employment record. (Choose Yes, same evaluator or No, a different person.)

4. Click Submit Changes to confirm the update.

5. Click Cancel if you do not want to proceed with the changes.

Employment – Share Verification Link



1. After submitting the employment changes, the system will display the Share Your Verification Link window.

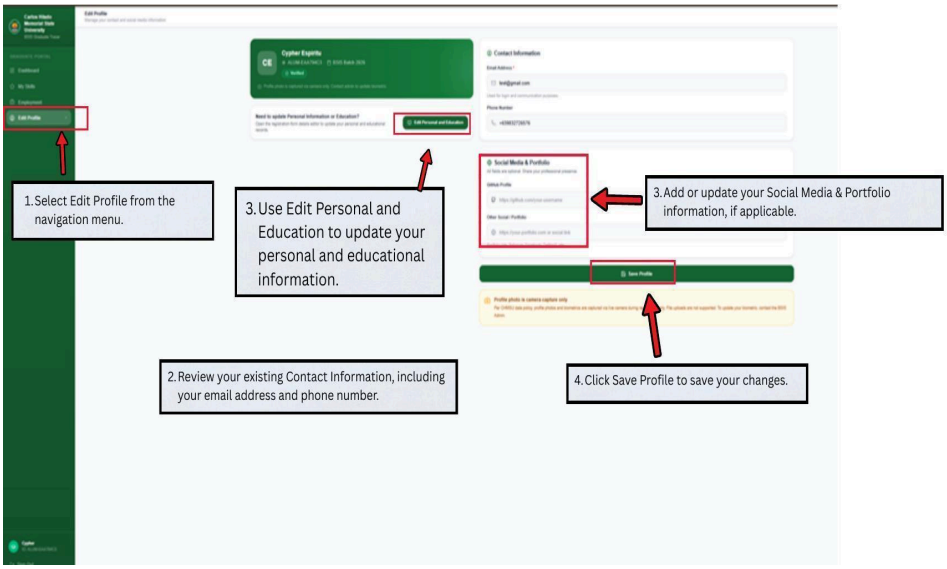
2. In the Employer or HR email field, enter the work email address of your employer or HR supervisor.

3. Click Send verification link. The system emails your employer a one-time link that is valid for seven days and shows “Link sent to …” once it has gone out. Your dashboard then shows Employer verification pending until your employer answers.

4. To send the link yourself instead, click Or create a link to copy and send yourself, then click Copy link and share it through your preferred communication channel.

5. The employer or HR supervisor opens the link, confirms or denies your employment, gives their name, position, and work email, and may complete the optional evaluation form.

Edit Profile



1. Select Edit Profile from the navigation menu.

2. Review your existing Contact Information, including your email address and phone number.

3. Add or update your Social Media & Portfolio information, if applicable.

4. Use Edit Personal and Education to update your personal and educational information.

5. Click Save Profile to save your changes.